GRADUAÇÃO EM FISIOTERAPIA

IMUNOLOGIA

IMUNO BIOLÓGICOS

SOROS E VACINAS

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Farmacêutico e Doutor em Ciências da Saúde – PUCPR

Especialista em Análises Clínicas – UFPR

Mestrado Integrado em Ciências Farmacêuticas – Universidade do Porto

Pós-Doutorado Université de Bourgogne, France e University of Galway, Ireland

ETAPAS DA AULA

01

HISTÓRIA DA VACINAÇÃO

02

IMUNO BIOLÓGICOS

03

P&D FASES

04

DESAFIOS E DEBATES ATUAIS

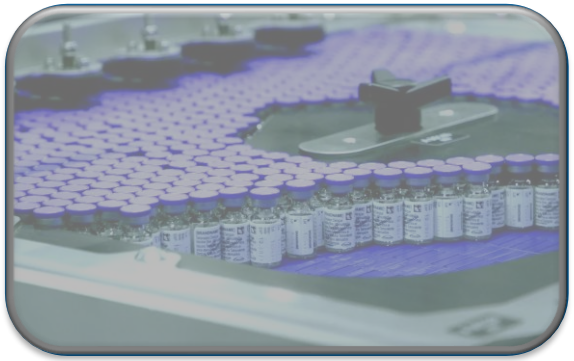
ETAPAS DA AULA

01

HISTÓRIA DA VACINAÇÃO

02

IMUNO BIOLÓGICOS



03

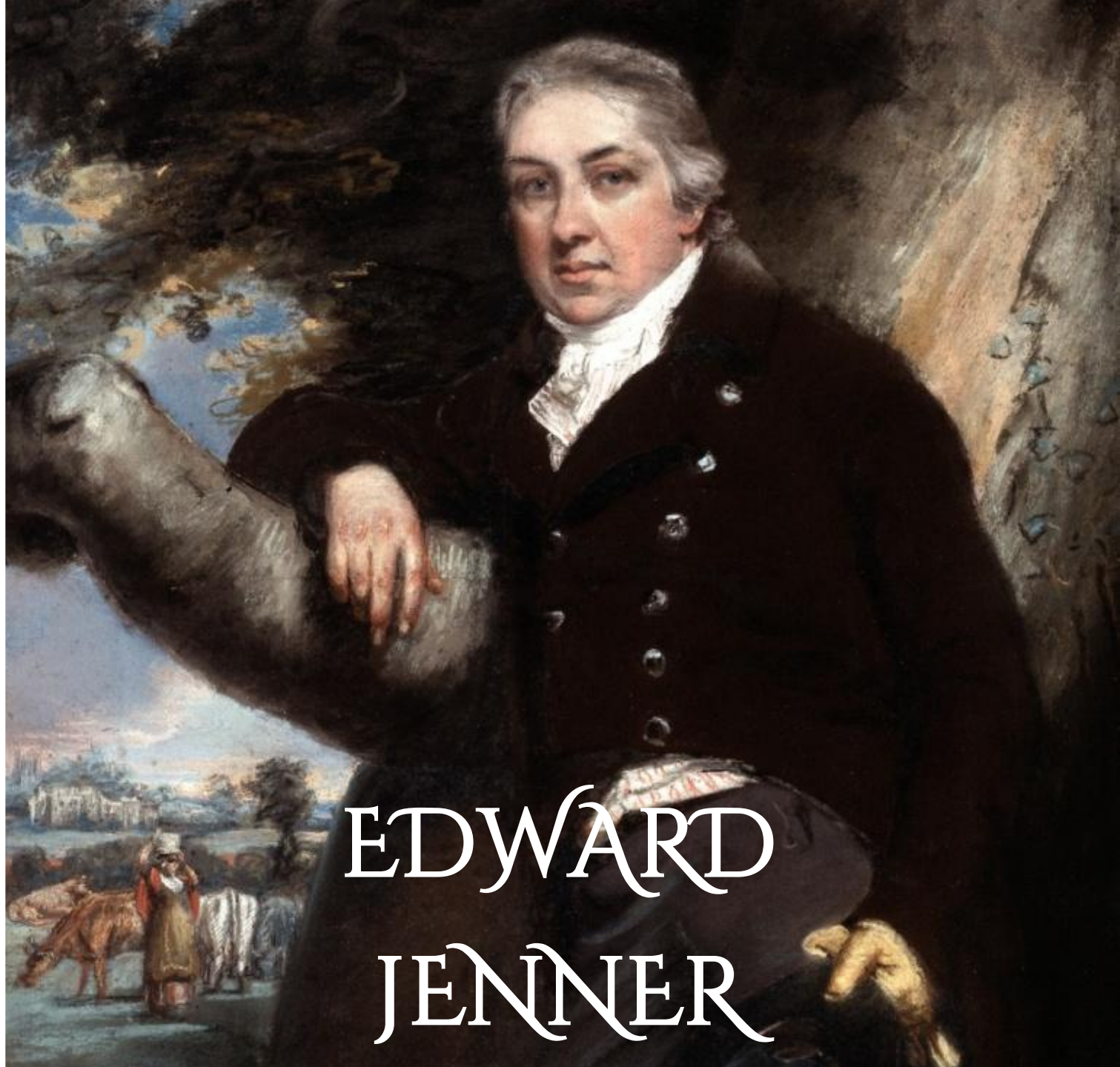
P&D FASES

04

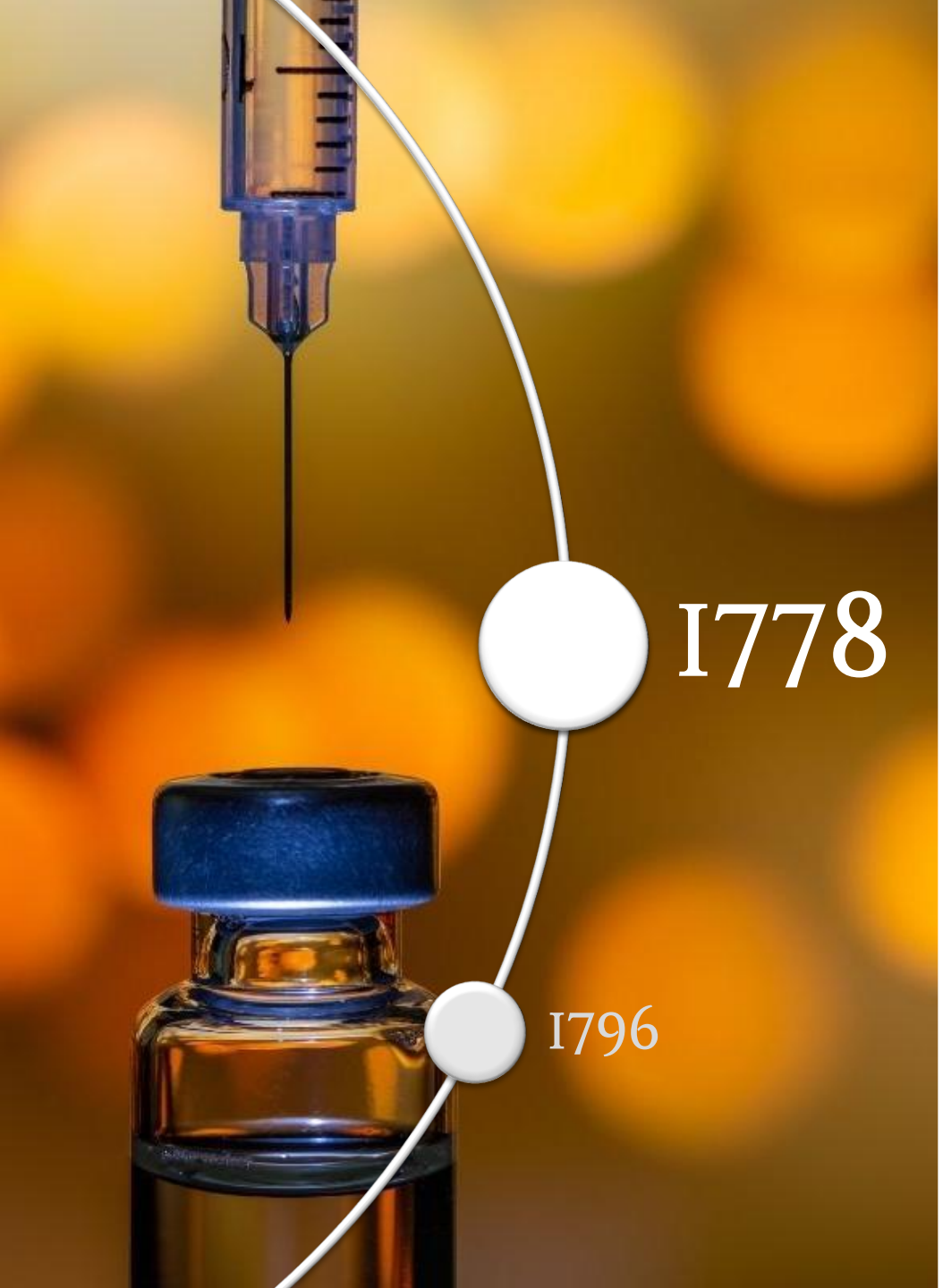
DESAFIOS E DEBATES ATUAIS

História da vacinação

1778



EDWARD JENNER



1778

1796

EDWARD JENNER

OBSERVOU QUE MULHERES QUE HAVIAM CONTRAÍDO *COWPOX* (VARÍOLA BOVINA) PARECIAM NÃO DESENVOLVER A VARÍOLA HUMANA (*SMALLPOX*)

1778

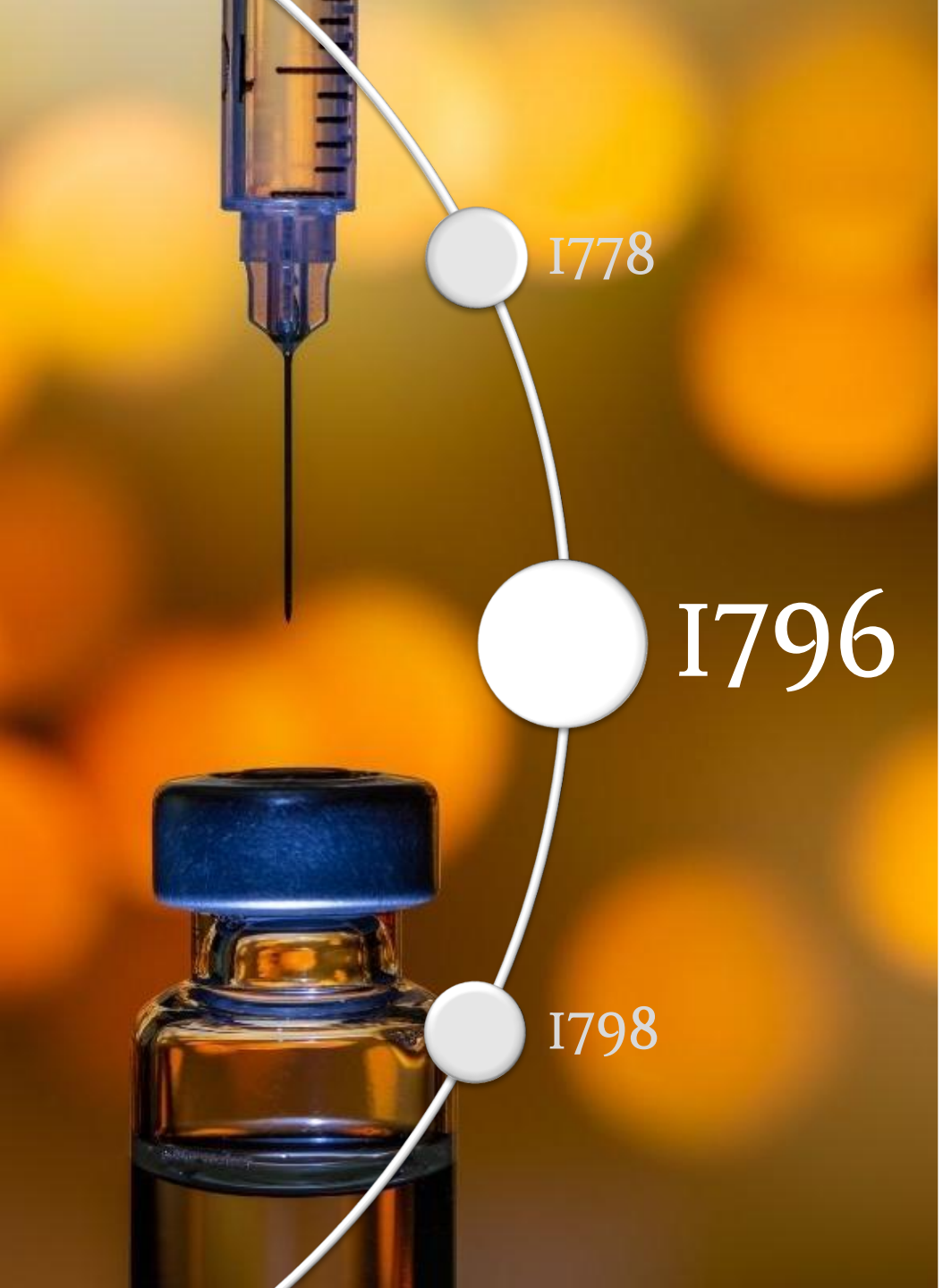
1796

1798

EDWARD JENNER

14/05/1796

REALIZOU O PRIMEIRO EXPERIMENTO DE VACINAÇÃO DOCUMENTADO, UTILIZANDO O VÍRUS DA VARÍOLA BOVINA PARA PROTEGER CONTRA A VARÍOLA HUMANA.


1778
1796
1798

1778

1796

1798

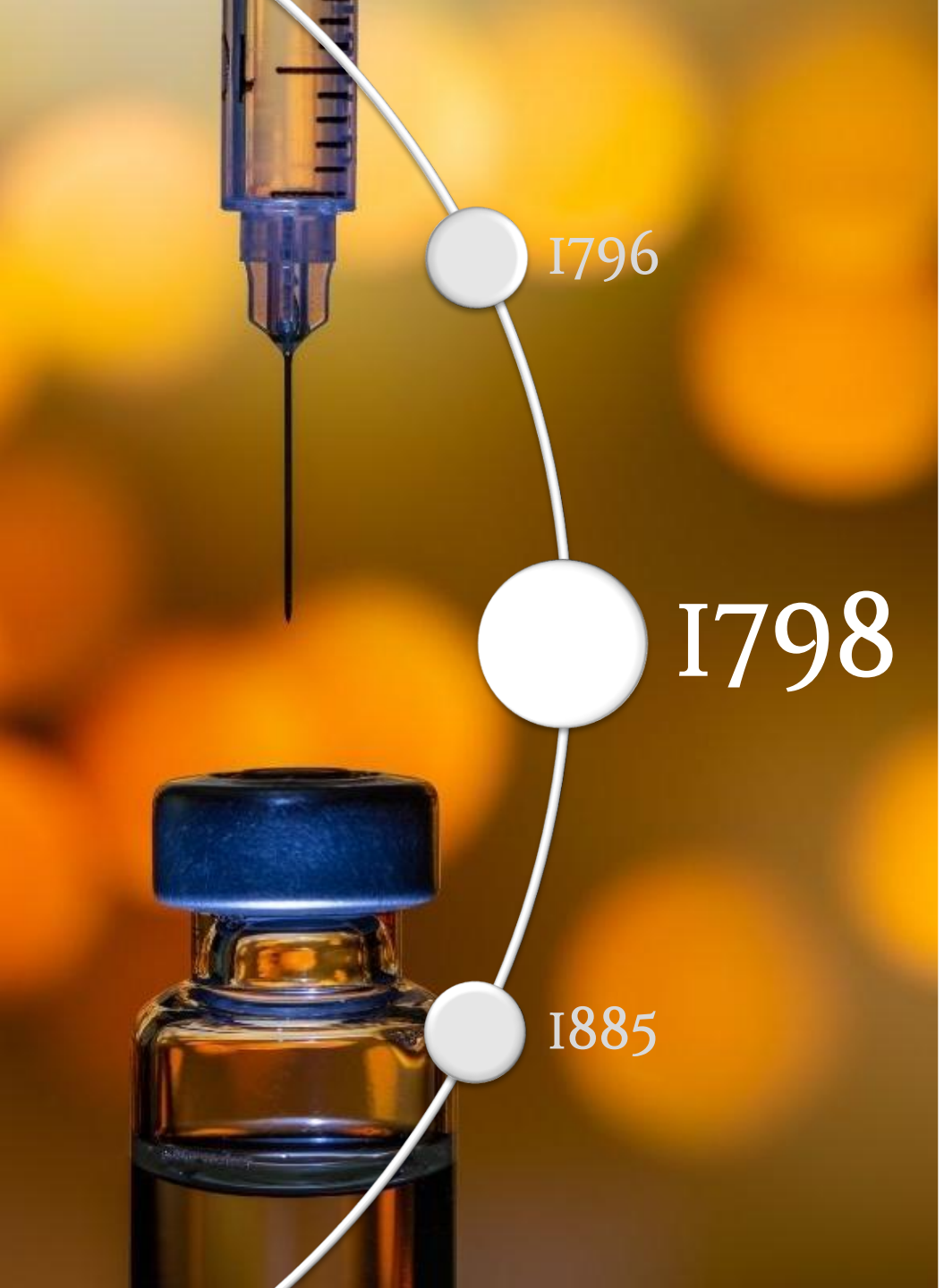
COLETOU MATERIAL DAS LESÕES DE SARAH NELMES, UMA ORDENHADORA INFECTADA COM COWPOX, E INOCULOU O MENINO JAMES PHIPPS, DE 8 ANOS.

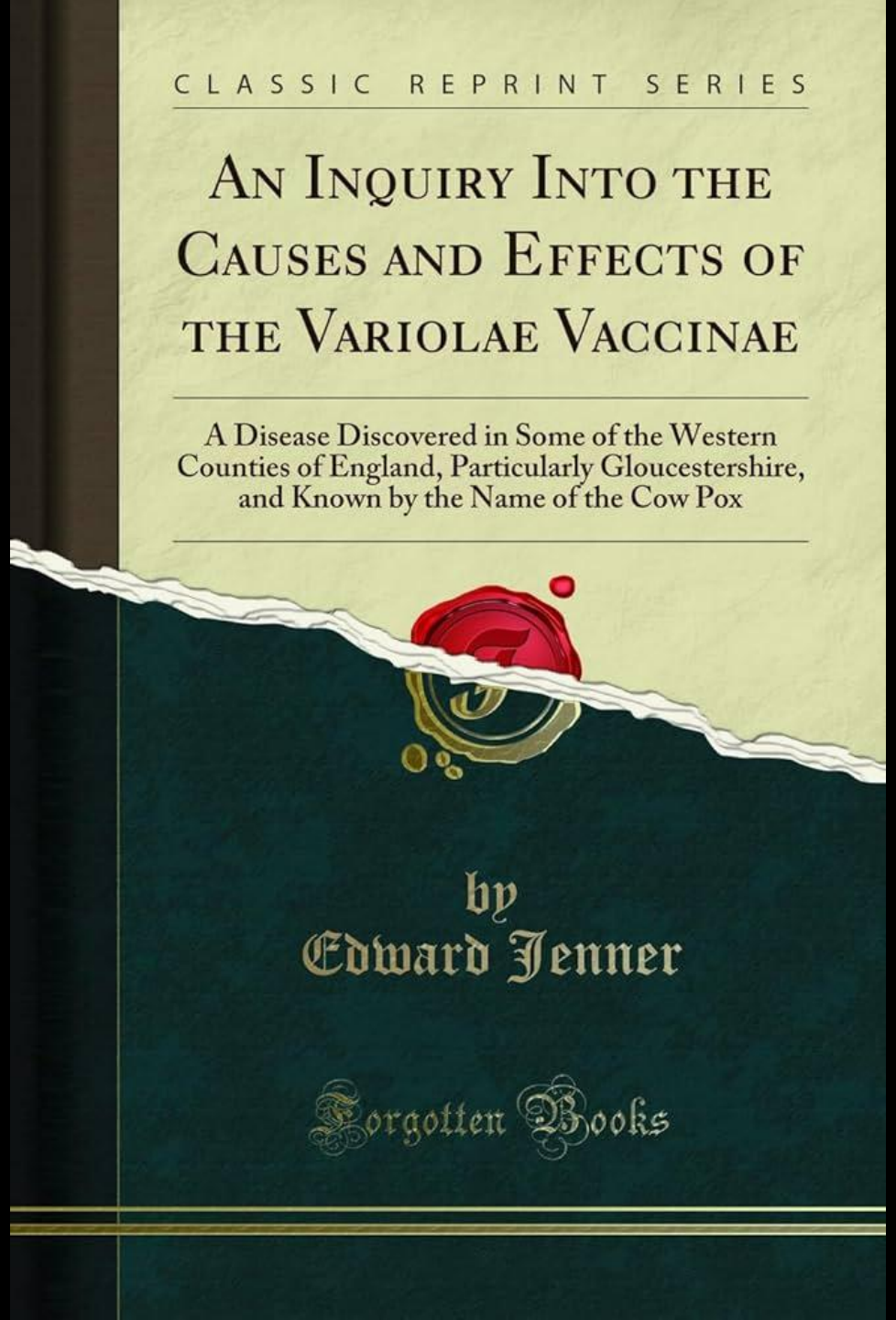
1778

1796

1798

POSTERIORMENTE, EXPÔS O MENINO AO VÍRUS DA VARÍOLA HUMANA, E ELE NÃO DESENVOLVEU A DOENÇA, DEMONSTRANDO PROTEÇÃO.

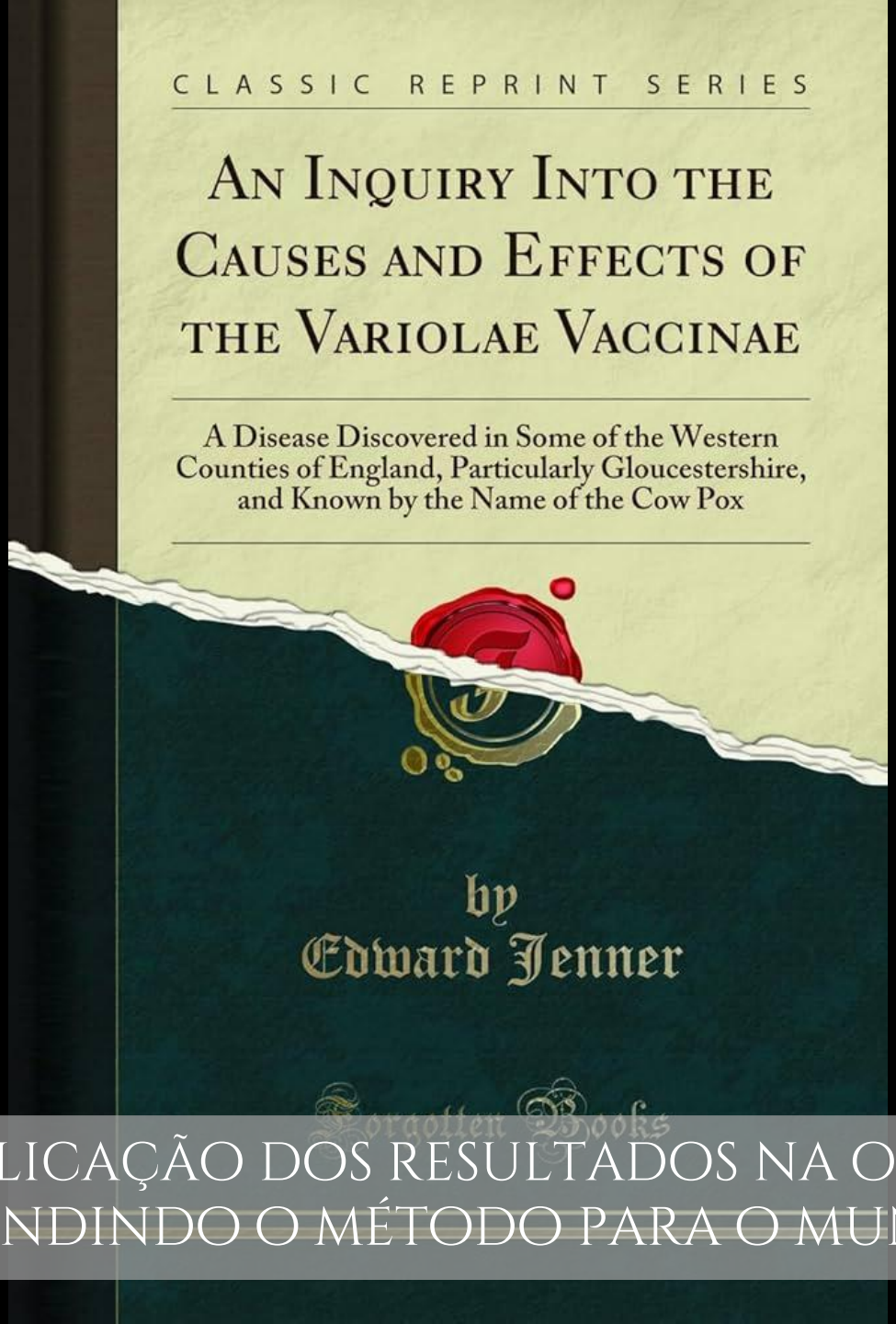

1796
1798
1885


CLASSIC REPRINT SERIES
An Inquiry Into the Causes and Effects of the Variolae Vaccinae
A Disease Discovered in Some of the Western Counties of England, Particularly Gloucestershire, and Known by the Name of the Cow Pox
by
Edward Jenner
Forgotten Books

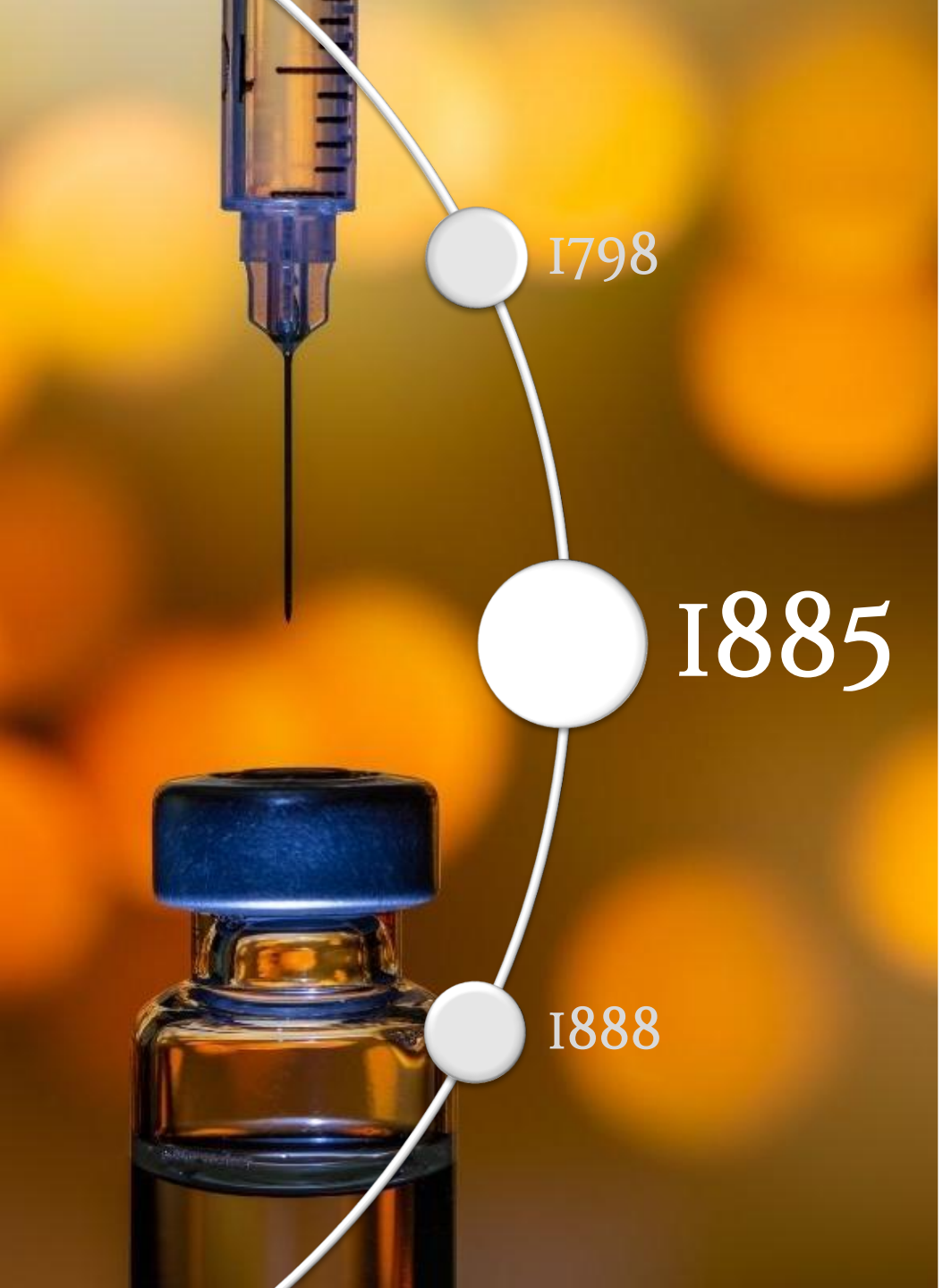
1796

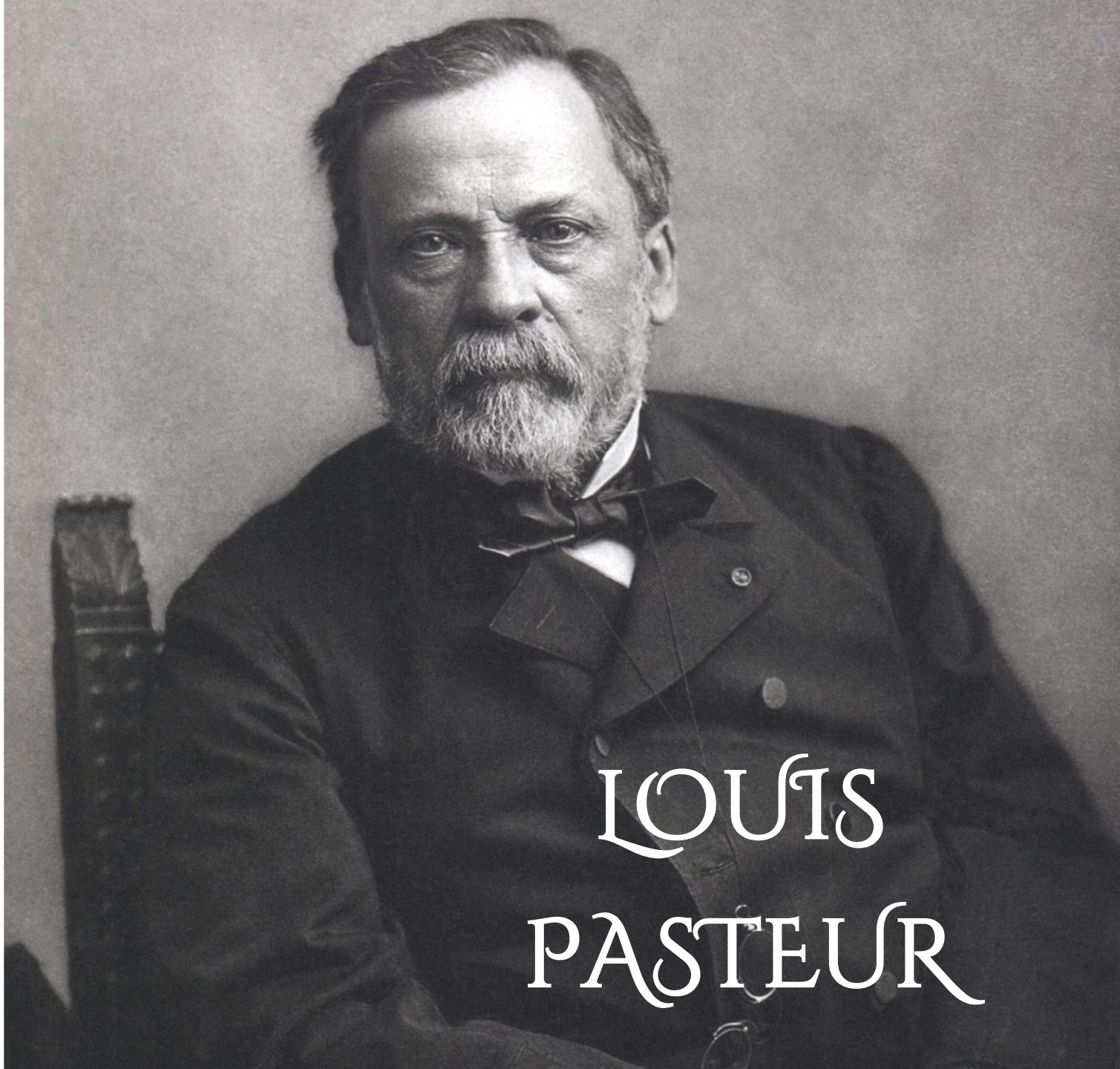
1798

1885



PUBLICAÇÃO DOS RESULTADOS NA OBRA, DIFUNDINDO O MÉTODO PARA O MUNDO.

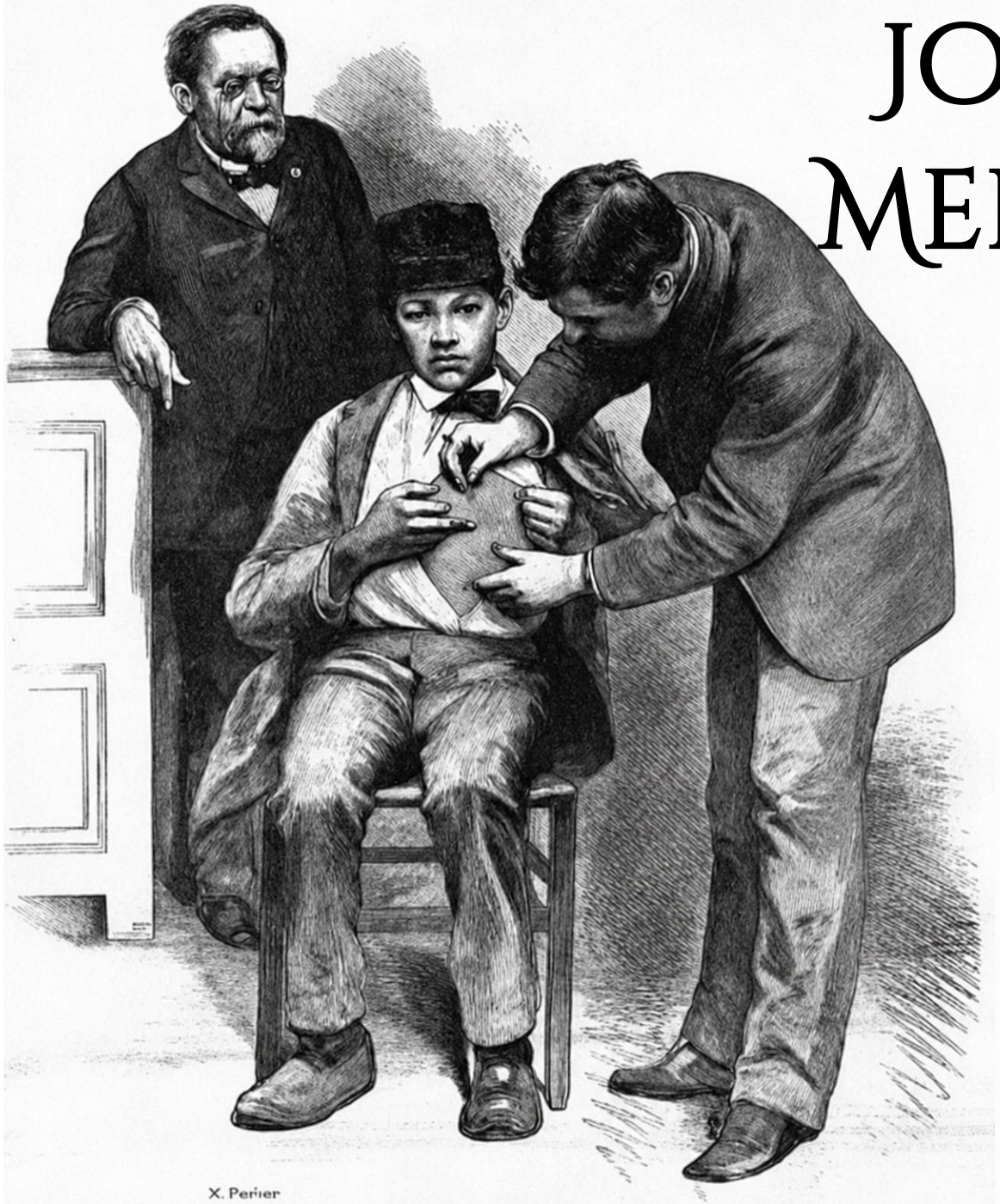

1798
1885
1888


LOUIS
PASTEUR

1796

1885

1888



JOSEPH MEISTER

1796

1885

1888

JOSEPH MEISTER

04/07/1885 A 06/07/1885

MORDIDO 14 VEZES POR UM CÃO COM RAIVA, TORNA-SE O PRIMEIRO HUMANO SALVO PELA VACINA CONTRA A RAIVA.

1885

1888

1955

1888

1955

POLIOMYELITIS VACCINE
Code 127-78
Three doses
During manufacture, 100 units penicillin and 100 micrograms streptomycin added to each cc.
CAUTION: Contains no preservative. Do not use if cloudy. Use within 6 hours after opening bottle.
SEE ENCLOSED DIRECTIONS
CUTTER Laboratories
BERKELEY, CALIF., U.S.A.

História da vacinação no Brasil

ORDEM E PROGRESSO

1804

1834
1849

© Pictorial Parade/Archive Getty Images

INÍCIO DA VACINAÇÃO

1804

1834
1849

1903

1804

1834
1849

1903

RIO DE JANEIRO

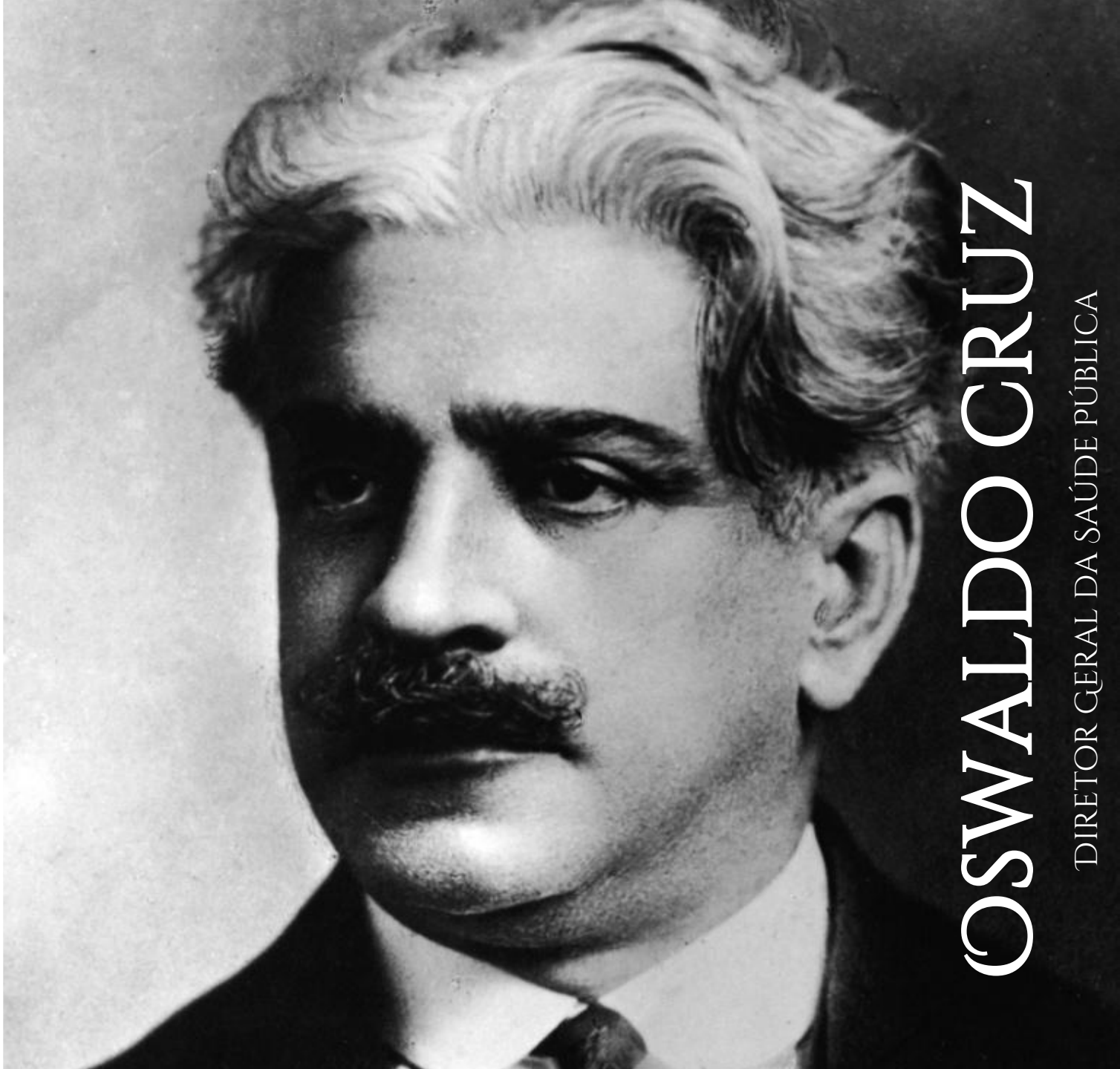
REGISTROS DAS EPIDEMIAS DE VARÍOLA E DE FEBRE AMARELA, RESPECTIVAMENTE.

Fonte: BIBLIOTECA NACIONAL, 1876

1834
1849

1903

1904



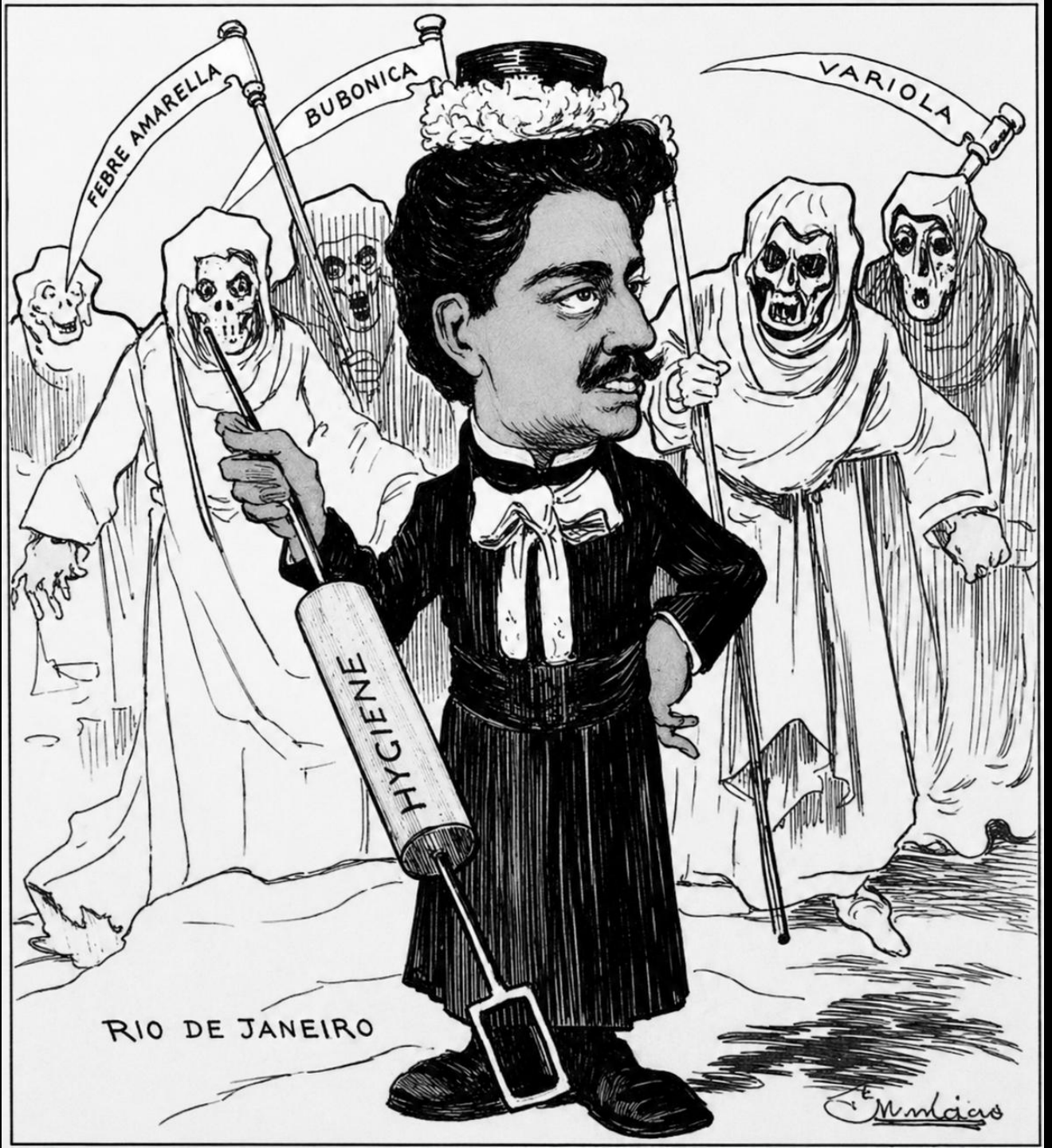
OSWALDO CRUZ

DIRETOR GERAL DA SAÚDE PÚBLICA

1834
1849

1903

1904



1834
1849

1903

1904

RIO DE JANEIRO

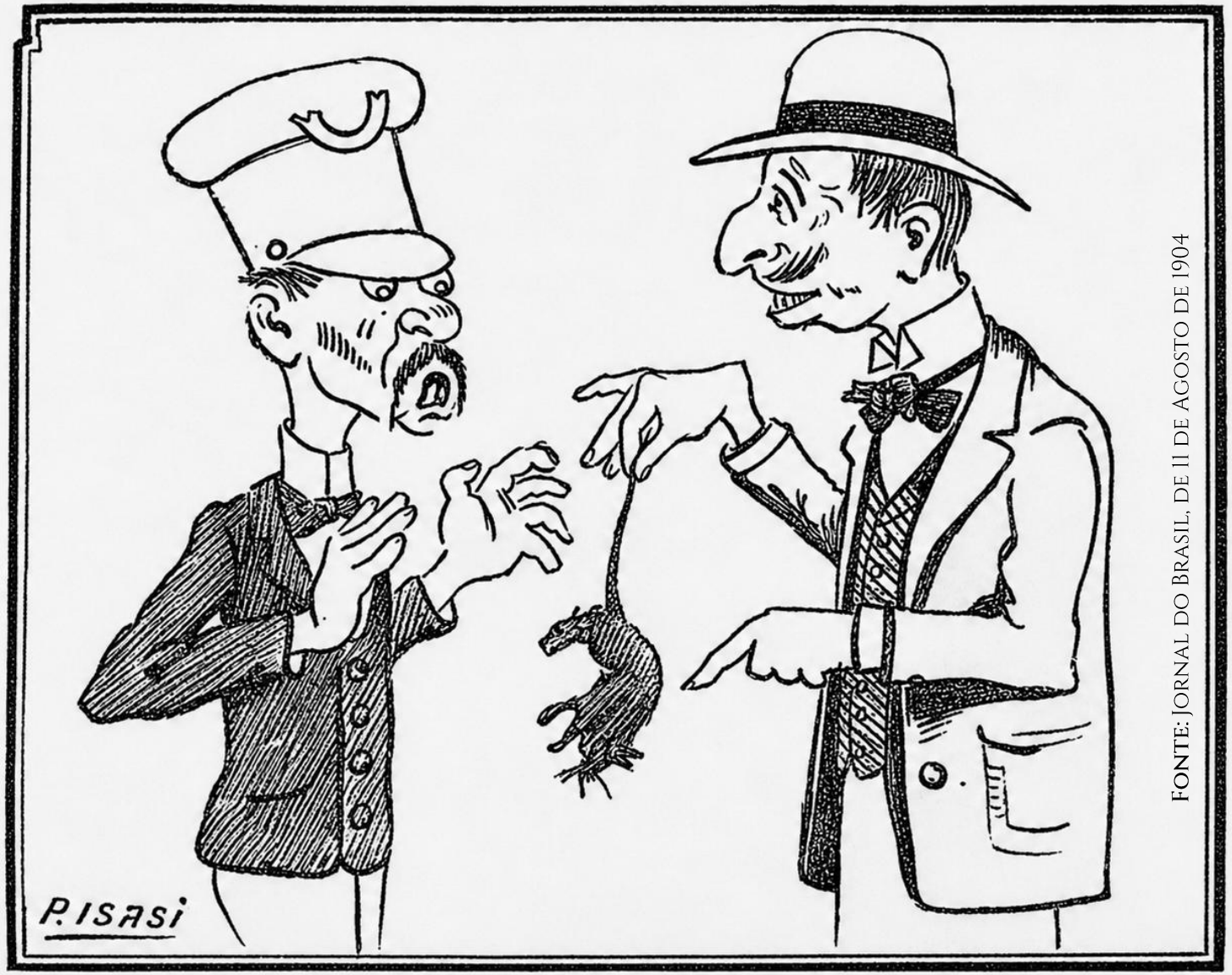
SURTO DE FEBRE AMARELA, PESTE BUBÔNICA E VARÍOLA.

1834
1849

1903

1904

A AUTHENTICIDADE DOS RATOS



FONTE: JORNAL DO BRASIL, DE 11 DE AGOSTO DE 1904

— Prove que o rato nasceu, viveu e morreu nesta capital e te 300 rs. por elle.
— ?
— Por meio de uma justificação com tres testemunhas de vista.

1834
1849

1903

1904

RIO DE JANEIRO

CASOS DE CORRUPÇÃO.

A AUTHENTICIDADE DOS RATOS

— Prove que o rato nasceu, viveu e morreu nesta capital e te 300 rs. por elle.
— ?
— Por meio de uma justificação com tres testemunhas de vista.

1903

1904

1971

– O Sr. sabe em que fase em que fase dos Srs. deputados...
Só póde atacar as cazas dos particulares, e não os povos; o a creeper já's houve no povo miúdo.

– XXX –

FONTE: TAGARELA, DE 21 ABRIL DE 1904

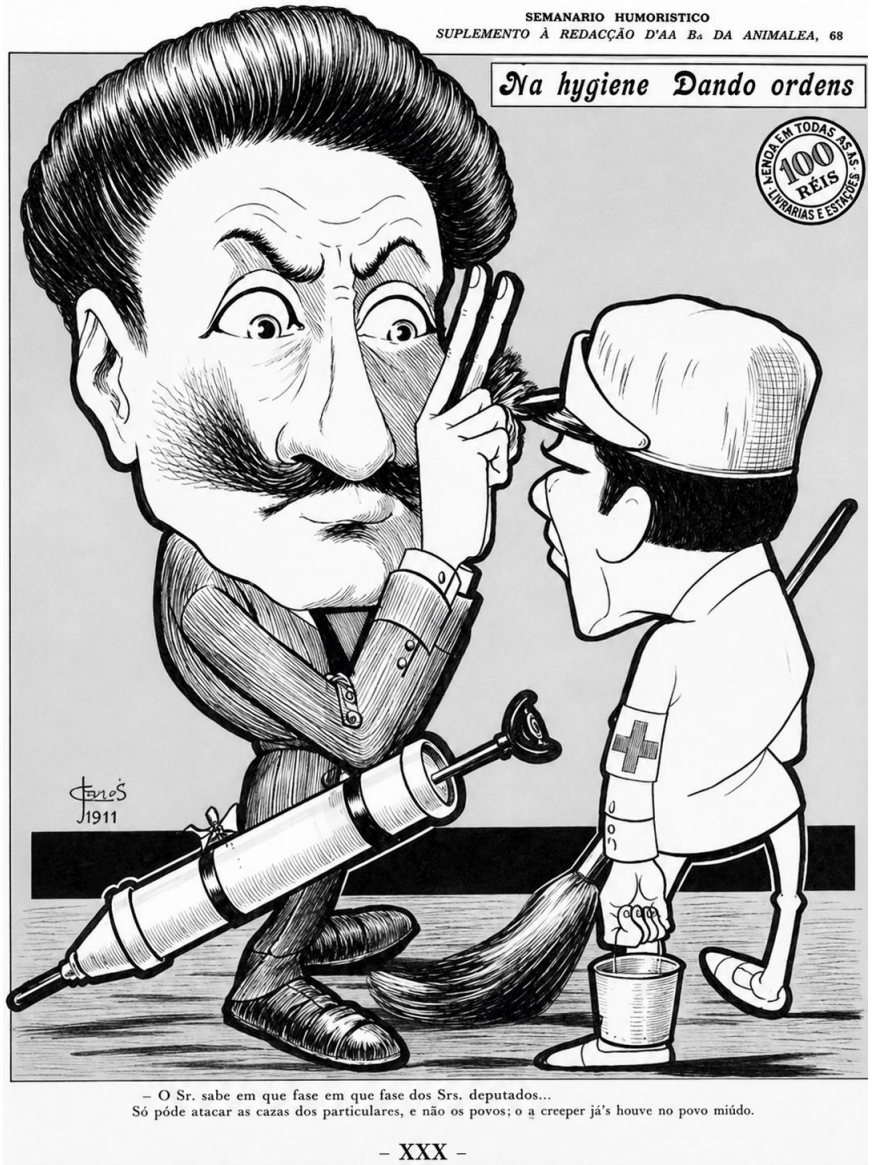
1903

1904

1971

RIO DE JANEIRO

CASOS DE DESINFORMAÇÃO.



FONTE: TAGARELA, DE 21 ABRIL DE 1904

1903

1904

1971

FONTE: REVISTA O MALHO, DE 29 DE OUTUBRO DE 1904

1903

1904

1971

O Cruzado da saúde

As Campanhas Sanitárias e a Imprensa

Nas duas primeiras décadas do século XX, Oswaldo Cruz foi certamente um dos homens públicos mais retratados pela imprensa brasileira.

Especialmente o período que abrange as grandes campanhas sanitárias na capital da Republica, e a inspeção aos portos do litoral, mas também as viagens à Amazonia, o sanitarista se tornou tema permanente dos desenhos de humor e das sátiras de jornais e revistas como *Fon-Fon!*, *Tagarela, Careta, O Malho.*

Dos traços físicos à personalidade, passando por sua atuação à frente da Diretoria Geral de Saúde Pública, tudo o que dizia respeito a Oswaldo Cruz foi alvo de uma crítica severa, muitas vezes bem-humorada, mas frequentemente impiedosa.

Entretanto, à medida que o êxito das campanhas começa a demonstrar o acerto de suas ações, as críticas vão aos poucos cedendo lugar ao aplauso, ao reconhecimento do trabalho do sanitarista. Sob o traço agora favorável dos caricaturistas, Oswaldo Cruz se torna o "saneador do Rio de Janeiro", o reformador da higiene e da saúde pública no país.

próxima página

1903

1904

1971

31/10/1904
REFORMA OSWALDO CRUZ

LEI QUE TORNOU A VACINA OBRIGATÓRIA CONTRA A VARÍOLA NO PAÍS.

OSWALDO

O Cruzado da saúde

AS CAMPANHAS SANITÁRIAS E A IMPRENSA

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próxima página

FONTE: HTTPS://OSWALDOCRUZ.FIOCRUZ.BR/JORNAL/INDEX.HTML

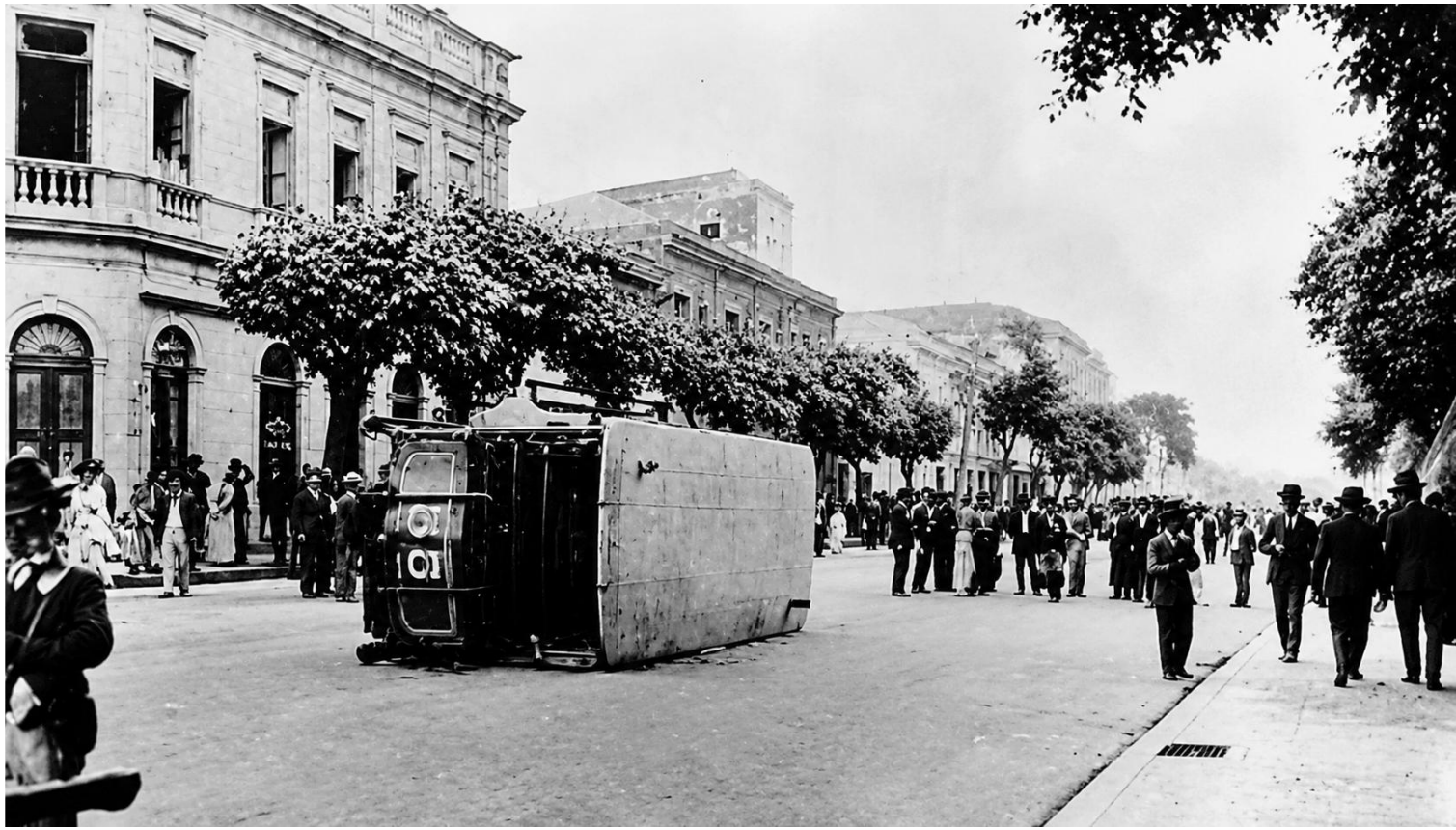
1903

1904

1971

DIA 10 A 16 DE NOVEMBRO REVOLTA DAS VACINAS NO RJ

DESCONTENTAMENTO POPULAR EXPLODIU NAS RUAS EM OPOSIÇÃO AOS PROGRAMAS DE HIGIENIZAÇÃO DO ESPAÇO URBANO.



FONTE: HTTPS://PORTAL.FIOCRUZ.BR/

1903

1904

1971

DIA 10 A 16 DE NOVEMBRO REVOLTA DAS VACINAS NO RJ

RESULTADO: 30 MORTOS E 945 PRESOS.

FONTE: HTTPS://PORTAL.FIOCRUZ.BR/

1904

1971

1973

RIO DE JANEIRO

REGISTRO DOS ÚLTIMOS 19 CASOS DE VARÍOLA NO PAÍS.

1971

1973

1980

CRIAÇÃO DO

1971

1973

1980

COMPETÊNCIAS DO PNI

IMPLANTAR E IMPLEMENTAR AÇÕES RELACIONADAS COM AS VACINAÇÕES DE CARÁTER OBRIGATÓRIO.

1971

1973

1980

COMPETÊNCIAS DO PNI

CENTRALIZAR, ANALISAR E DIVULGAR AS INFORMAÇÕES REFERENTES AO PNI.

1971

1973

1980

COMPETÊNCIAS DO PNI

ESTABELECER CRITÉRIOS E PRESTAR APOIO TÉCNICO E FINANCEIRO.

1971

1973

1980

COMPETÊNCIAS DO PNI

Normas básicas para a execução de vacinações.

1971

1973

1980

COMPETÊNCIAS DO PNI

SUPERVISIONAR E AVALIAR A EXECUÇÃO DAS VACINAÇÕES EM TERRITÓRIO NACIONAL.

1971

1973

1980

COMPROMISSOS DO PNI

INTENSIFICAÇÃO DAS AÇÕES DE PREVENÇÃO E CONTROLE DA FEBRE AMARELA.

1971

1973

1980

COMPROMISSOS DO PNI

MANUTENÇÃO DA ERRADICAÇÃO DA POLIOMIELITE NO BRASIL.

1971

1973

1980

COMPROMISSOS DO PNI

VIGILÂNCIA E CONTROLE DAS HEPATITES VIRAIS A E B.

1971

1973

1980

COMPROMISSOS DO PNI

ERRADICAÇÃO DO SARAMPO E ELIMINAÇÃO DA RUBÉOLA E TÉTANO NEONATAL.

1971

1973

1980

CRIAÇÃO DO PROGRAMA NACIONAL DE IMUNIZAÇÕES (PNI)

- CRIANÇA
- ADOLESCENTE
- ADULTO
- IDOSO
- GESTANTE
- POVOS INDÍGENAS

FONTE: HTTP://WWW.SAUDE.GOV.BR/SAUDE-DE-A-Z/VACINACAO/VACINE-SE

INÍCIO DAS CAMPANHAS NACIONAIS DE VACINAÇÃO PARA A POLIOMIELITE

1973

1980

1984

1980

1984

1986

Programa de Autossuficiência Nacional em Imunobiológicos

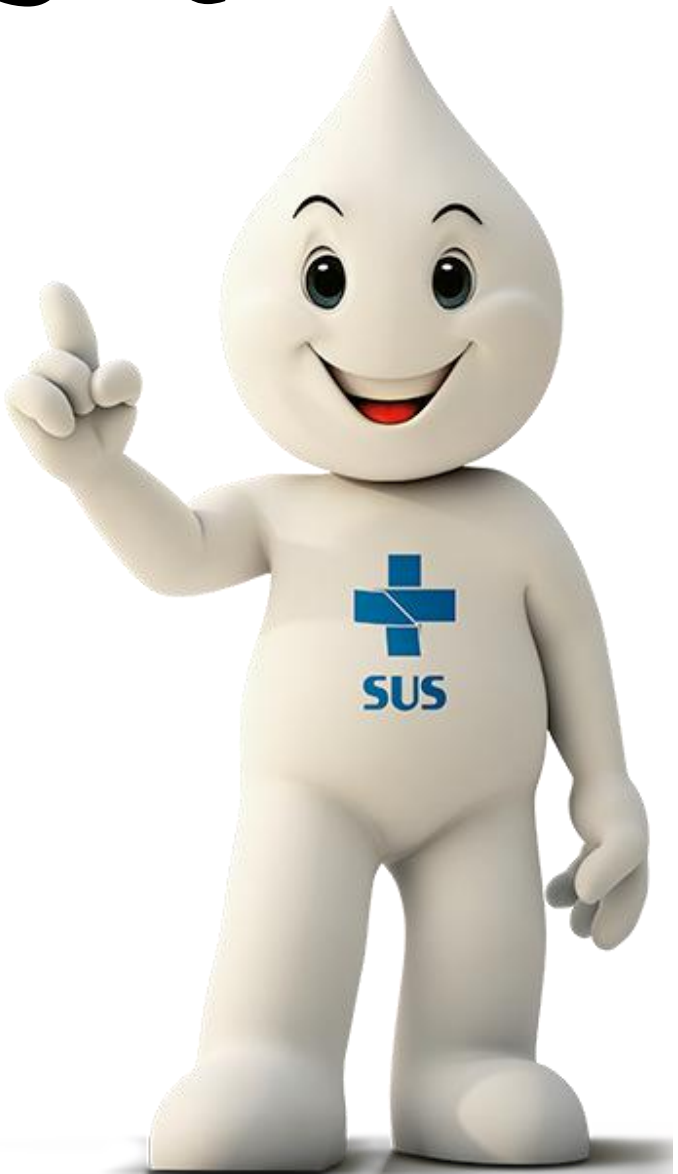
Atender à demanda nacional e tentar eliminar a necessidade de importação.

1984

1986

1989

NASCE O PERSONAGEM ZÉ GOTINHA.



FONTE: HTTP://WWW.BLOG.SAUDE.GOV.BR

1986

1989

2022

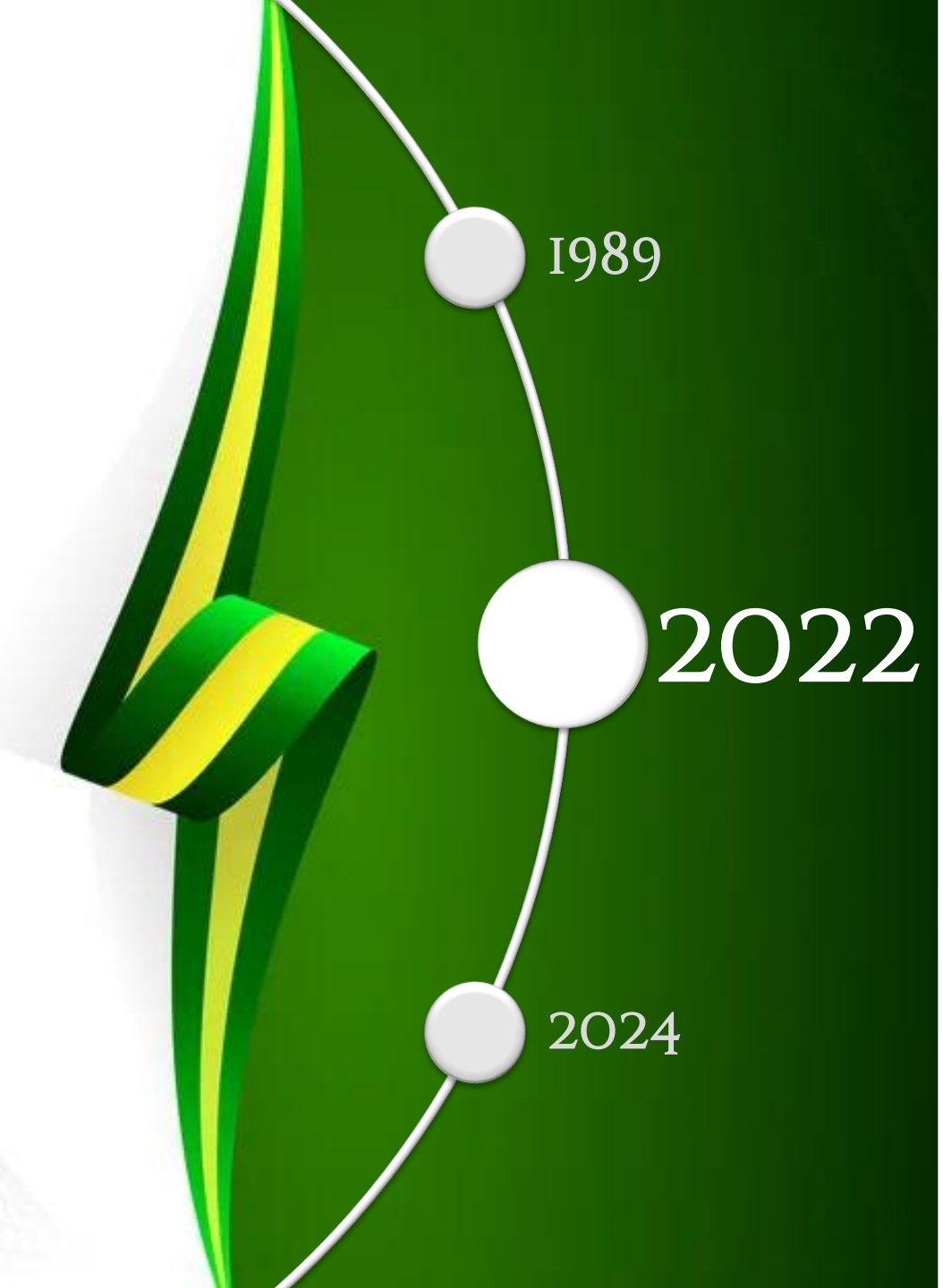
ÚLTIMO CASO DE POLIOMIELITE NO PAÍS

'Pretendo ser o último caso de pólio no Brasil', diz paciente diagnosticado com a doença no país em 1989

Em entrevista ao GLOBO, Deivson Rodrigues, de 34 anos, conta que teve paralisia severa decorrente da doença ainda criança e atribui a boa recuperação à vacina

Por Deivson Rodrigues*

29/09/2022 04h31 · Atualizado há 5 meses

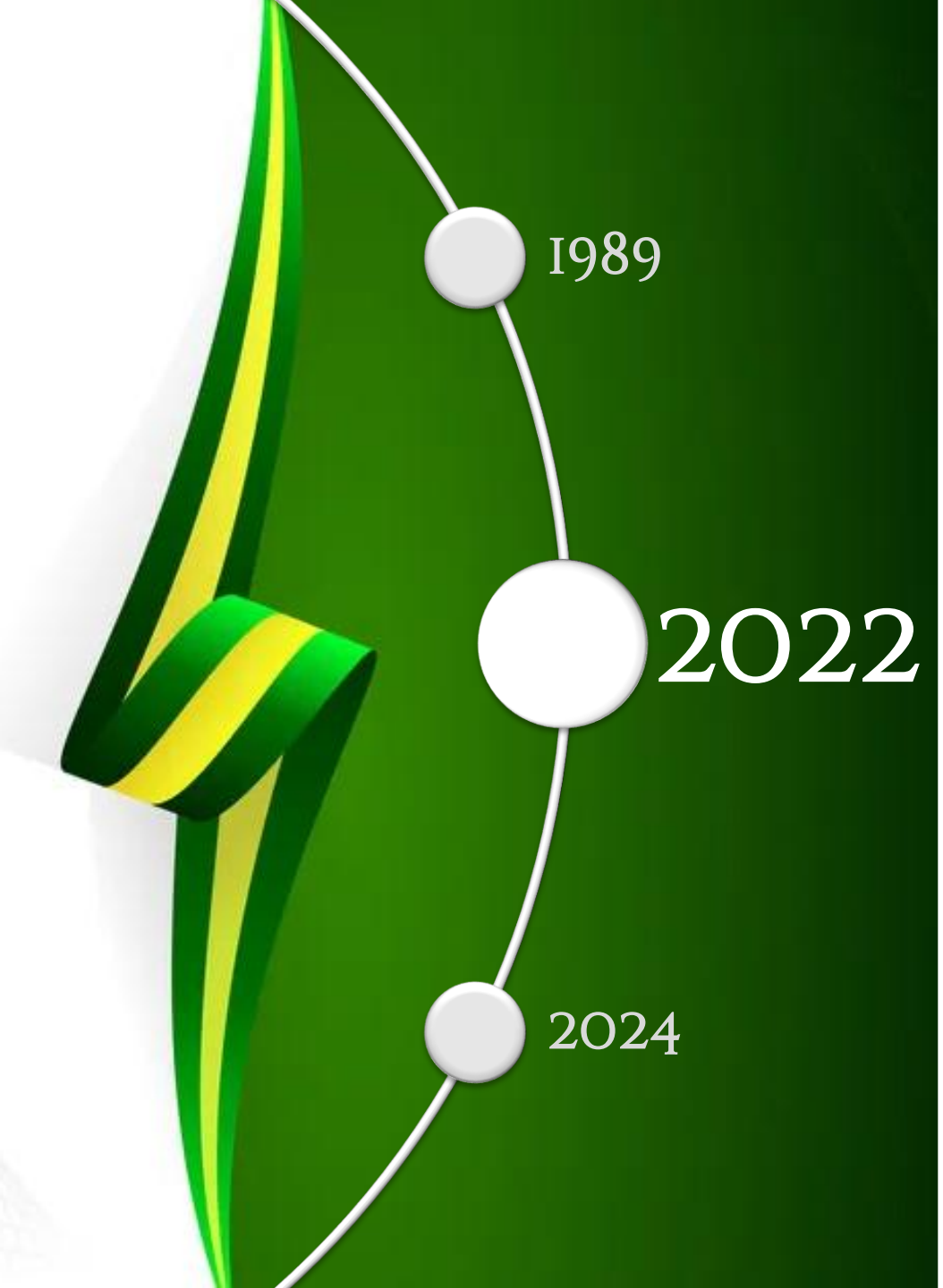


MINISTÉRIO DA SAÚDE MOVIMENTO NACIONAL PELA VACINAÇÃO

MOBILIZAR TODA POPULAÇÃO PARA QUE O BRASIL VOLTE A SER REFERÊNCIA EM ALTAS COBERTURAS VACINAIS;

ÍNDICES DE IMUNIZAÇÃO CAÍRAM NOS ÚLTIMOS ANOS, O QUE AUMENTA O RISCO DE DOENÇAS.

MOVIMENTO VACINA BRASIL



MINISTÉRIO DA SAÚDE MOVIMENTO NACIONAL PELA VACINAÇÃO

gov.br Órgãos do Governo Acesso à Informação Legislação Acessibilidade PT Entrar com o gov.br

Ministério da Saúde

O que você procura?

Assuntos > Notícias > 2022 > Setembro > Brasil atinge 52% de cobertura vacinal contra a poliomielite; entenda a importância da vacinação

PREVENÇÃO

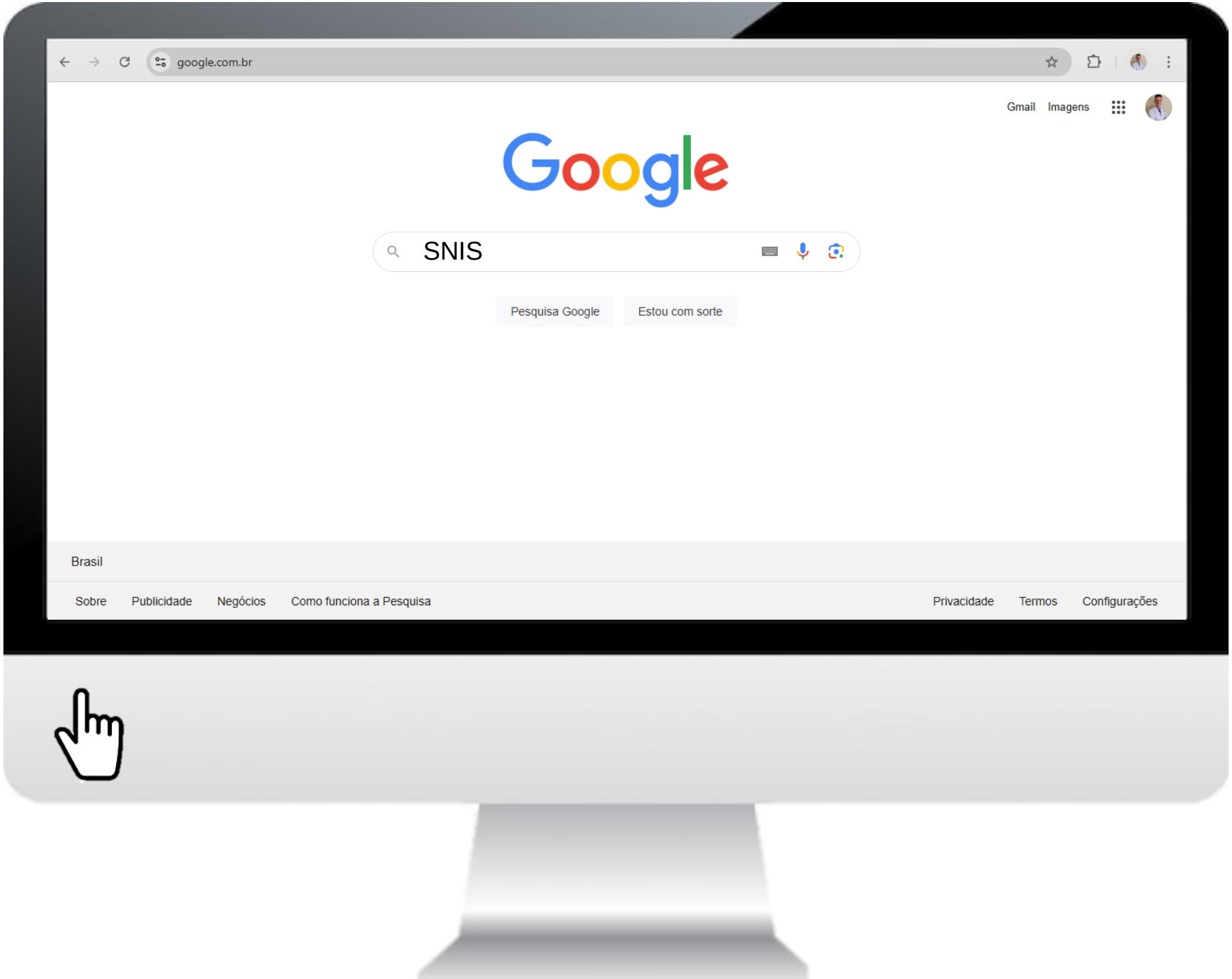
Brasil atinge 52% de cobertura vacinal contra a poliomielite; entenda a importância da vacinação

Toda a população com menos de cinco anos precisa ser vacinada para evitar a reintrodução do vírus que causa a paralisia infantil

Publicado em 26/09/2022 18h22 | Atualizado em 03/11/2022 12h02 Compartilhe:

Das 14,3 milhões de crianças menores de cinco anos que fazem parte do público-alvo da Campanha Nacional de Vacinação contra a Poliomielite e Multivacinação, 11,5 milhões são crianças entre um e menores de cinco anos. Desse público, mais de 52% estão imunizadas. Até agora, cerca de 6 milhões de doses foram aplicadas no Brasil. A campanha foi prorrogada pelo Ministério da Saúde até o dia 30 de setembro (Ministério da Saúde prorroga Campanha Nacional de Vacinação contra Poliomielite e Multivacinação até 30 de setembro).

FONTE: HTTPS://WWW.GOV.BR/SAUDE/PT-BR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Gmail
Imagens
Google
SNIS
Pesquisa Google
Estou com sorte
Brasil
Sobre
Publicidade
Negócios
Como funciona a Pesquisa
Privacidade
Termos
Configurações

www.gov.br/cidades/pt-br/acesso-a-informacao/acoes-e-programas/saneamento/snis
gov.br
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Ministério das Cidades
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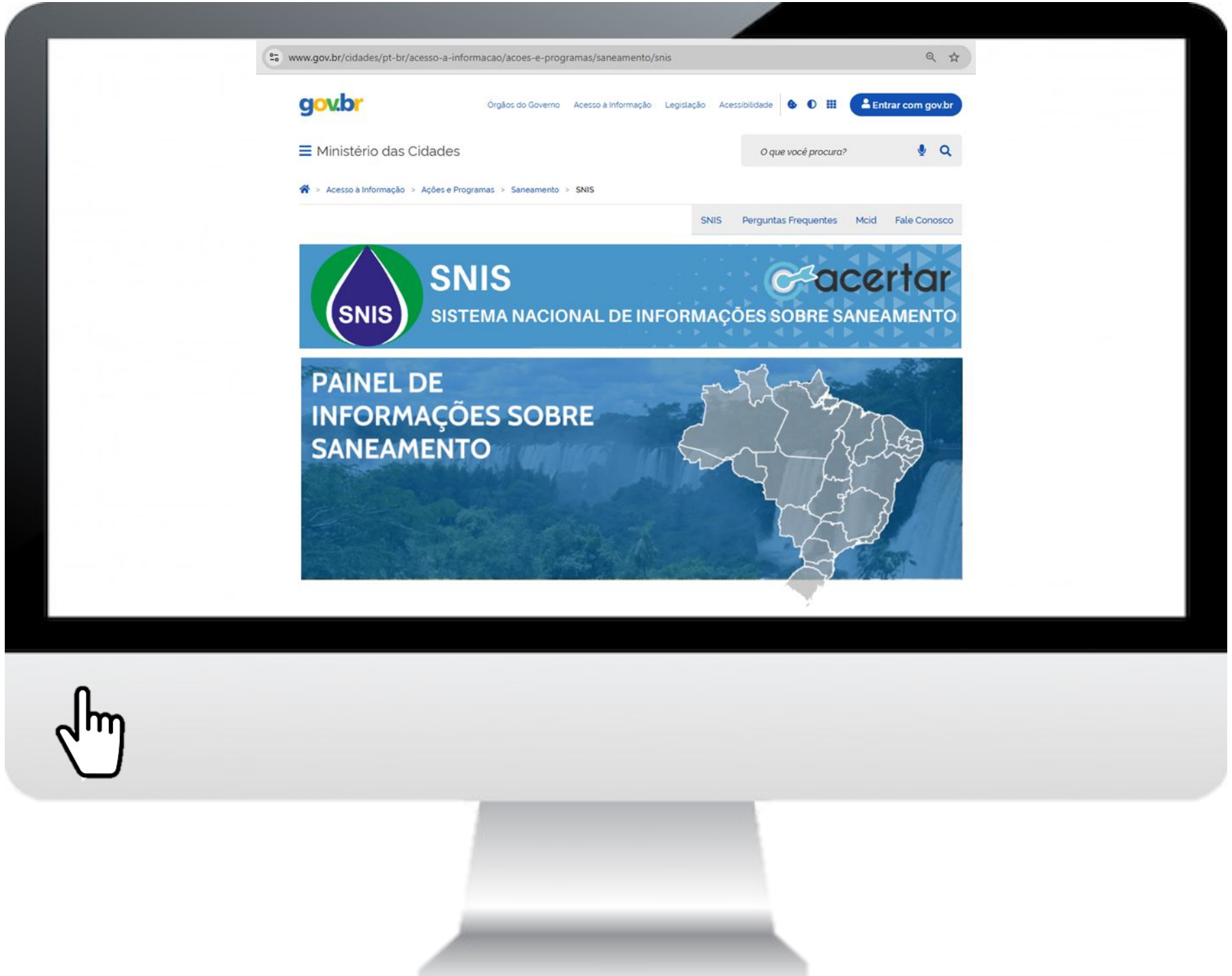
SNIS
Perguntas Frequentes
Mcid
Fale Conosco
SNIS
SNIS
SISTEMA NACIONAL DE INFORMAÇÕES SOBRE SANEAMENTO
acertar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SNIS
SISTEMA NACIONAL DE INFORMAÇÕES SOBRE SANEAMENTO
acertar
PAINEL DE INFORMAÇÕES SOBRE SANEAMENTO

Pesquisa nas **100 maiores cidades do país**, que concentram aproximadamente 40% da população brasileira.

- ✓ **Cobertura de Água Tratada:** 94,4%;
- ✓ Mais de **35 milhões não tem acesso à água tratada** em todo o país;
- ✓ **Exemplo:** Porto Velho com apenas 32,9%.

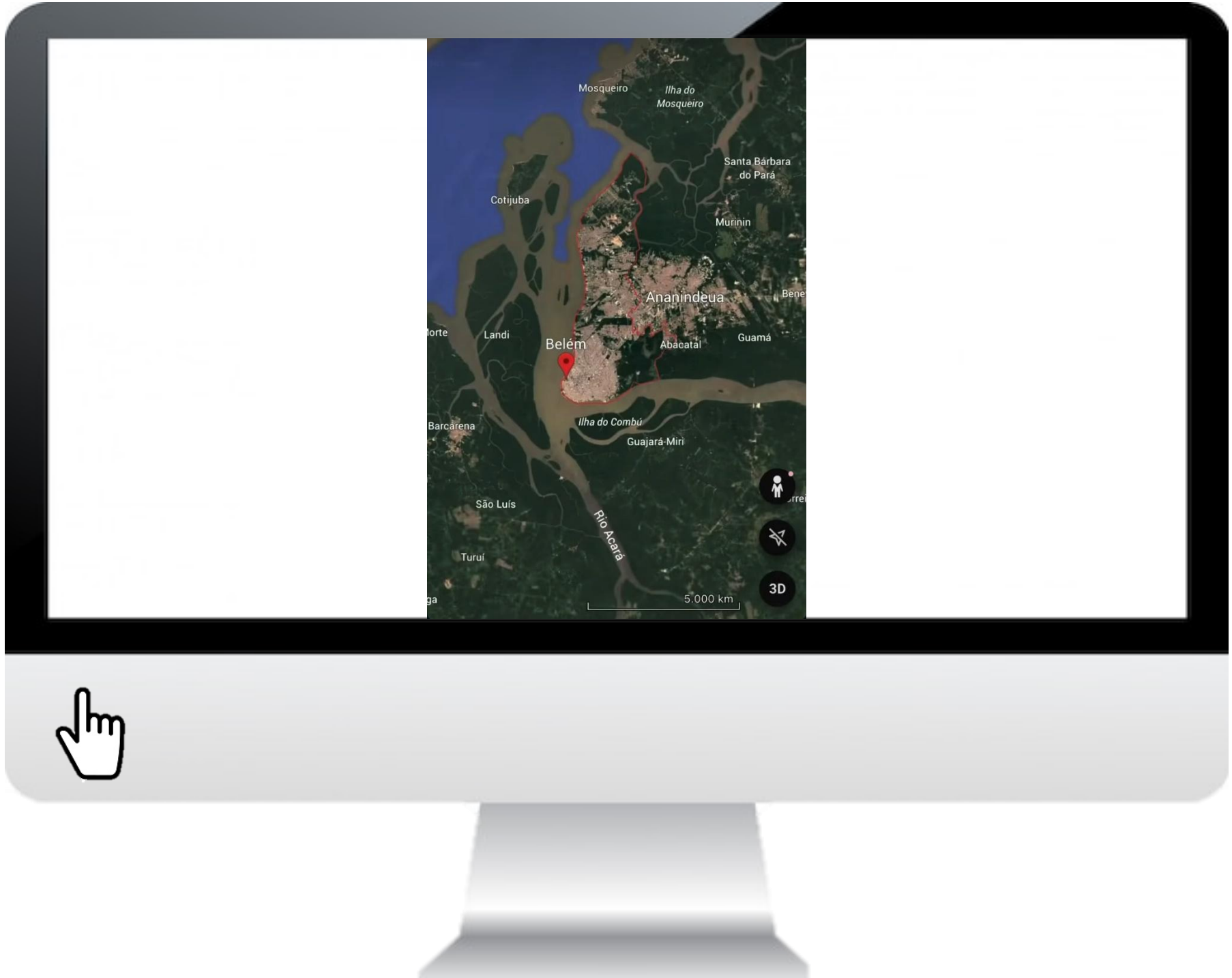
- ✓ **Esgoto Tratado:** 64,1%;
- ✓ **Desperdício de Água:** 36,3%;
- ✓ **Acesso a Coleta de Esgoto:** 75,7%.

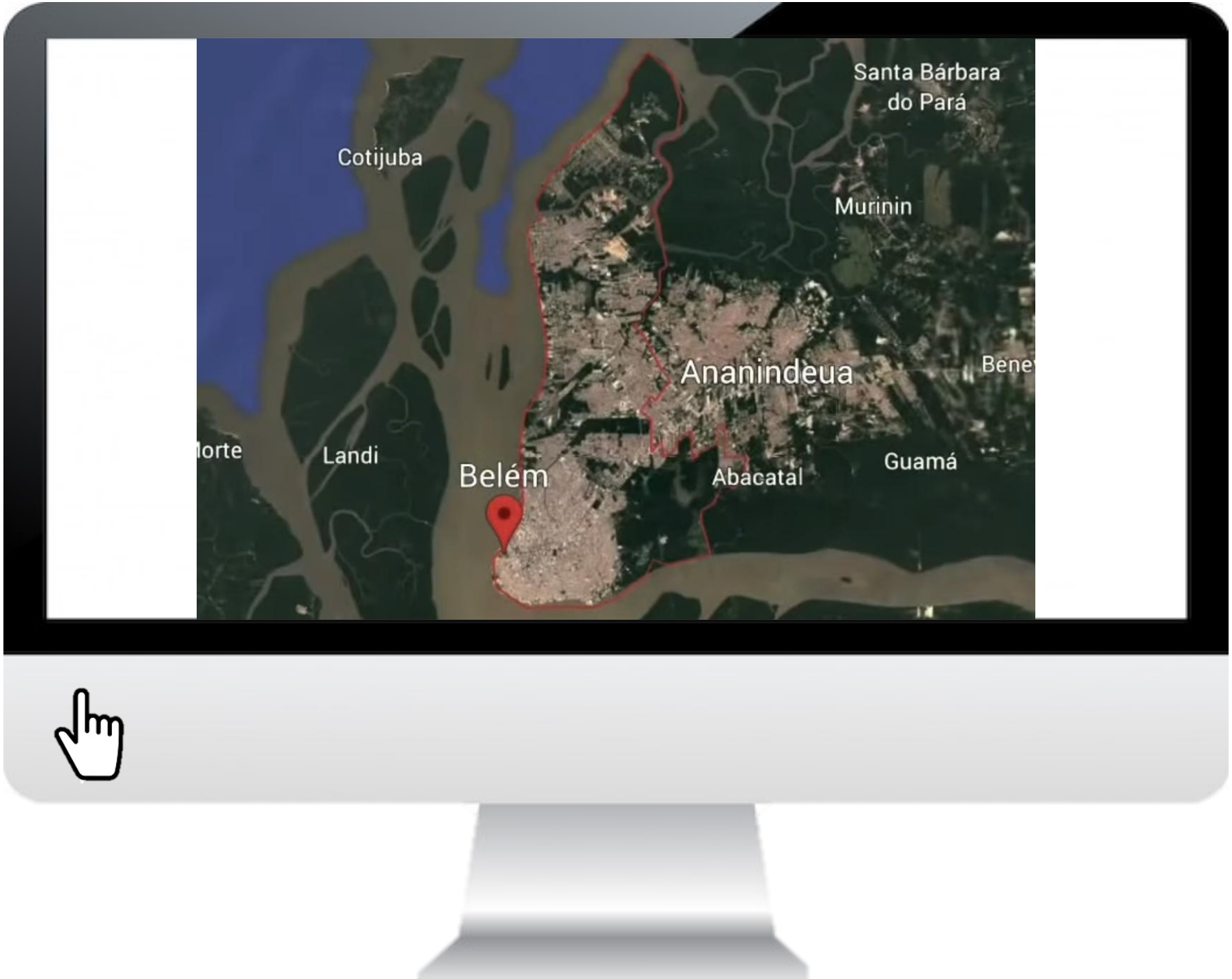
Pesquisa nas **100 maiores cidades do país**, que concentram aproximadamente 40% da população brasileira.

- ✓ **100 milhões NÃO tem coleta de esgoto:**
  - ➢ Apenas 50% do esgoto é tratado no Brasil.

NEGO
QUÆ SERA
AMEN

✓ Ananindeua: 4,1%;
✓ Piracicaba e Bauru: 100%


Mosqueiro
Ilha do Mosqueiro
Santa Bárbara do Pará
Cotijuba
Murinin
Ananindeua
Landi
Belém
Abaetetal
Guamá
Barcarena
Ilha do Combú
Guajará-Miri
São Luís
Rio Acará
Turuí
3D
5.000 km


Santa Bárbara
do Pará
Cotijuba
Murinin
Ananindeua
Bene
lorte
Landi
Belém
Abacatal
Guamá

Historicamente, as diferenças regionais e políticas públicas

Historicamente, as diferenças regionais e políticas públicas

Estados nas Melhores Posições:

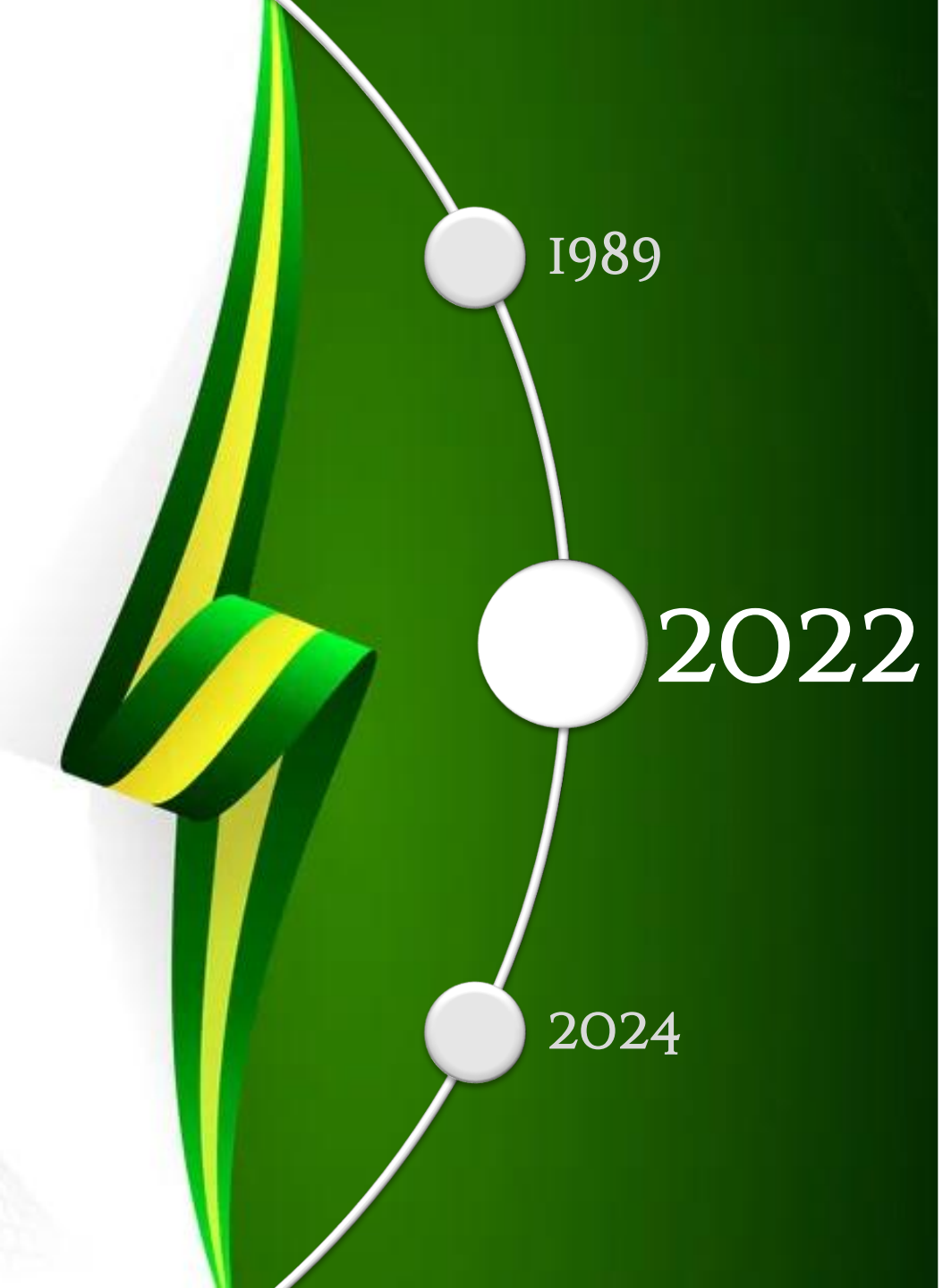
✓ **Curitiba:** 12ª posição.

Historicamente, as diferenças regionais e políticas públicas

Piores Municípios:

✓ Cidades nas regiões Norte, Nordeste e Rio de Janeiro.

Relacionada ao **estabelecimento de políticas públicas** que <u>incentivem o saneamento</u>.



MINISTÉRIO DA SAÚDE MOVIMENTO NACIONAL PELA VACINAÇÃO

gov.br Órgãos do Governo Acesso à Informação Legislação Acessibilidade PT Entrar com o gov.br

Ministério da Saúde

O que você procura?

Assuntos > Notícias > 2022 > Setembro > Brasil atinge 52% de cobertura vacinal contra a poliomielite; entenda a importância da vacinação

PREVENÇÃO

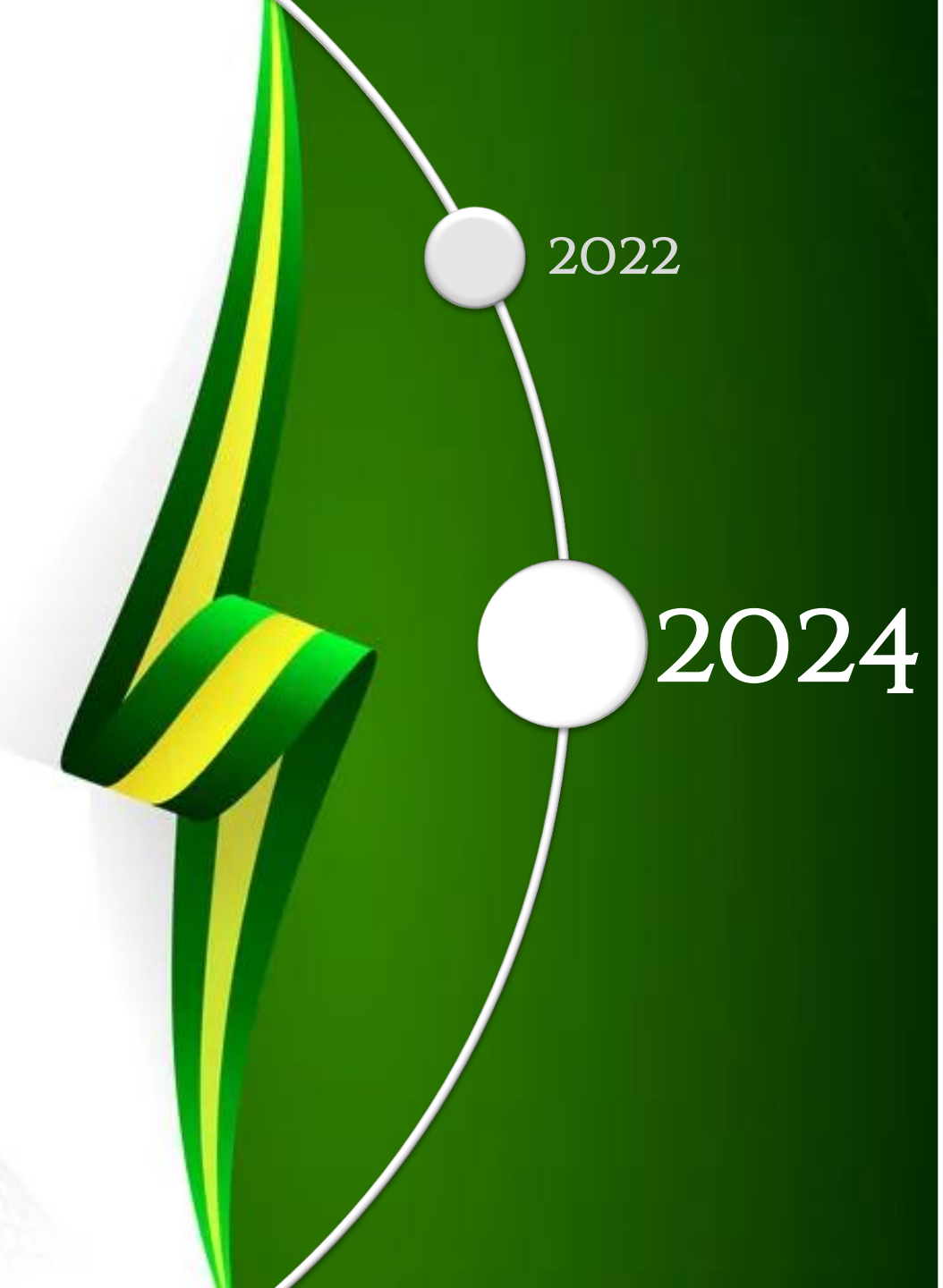
Brasil atinge 52% de cobertura vacinal contra a poliomielite; entenda a importância da vacinação

Toda a população com menos de cinco anos precisa ser vacinada para evitar a reintrodução do vírus que causa a paralisia infantil

Publicado em 26/09/2022 18h22 | Atualizado em 03/11/2022 12h02 Compartilhe:

Das 14,3 milhões de crianças menores de cinco anos que fazem parte do público-alvo da Campanha Nacional de Vacinação contra a Poliomielite e Multivacinação, 11,5 milhões são crianças entre um e menores de cinco anos. Desse público, mais de 52% estão imunizadas. Até agora, cerca de 6 milhões de doses foram aplicadas no Brasil. A campanha foi prorrogada pelo Ministério da Saúde até o dia 30 de setembro (Ministério da Saúde prorroga Campanha Nacional de Vacinação contra Poliomielite e Multivacinação até 30 de setembro).

FONTE: HTTPS://WWW.GOV.BR/SAUDE/PT-BR



MINISTÉRIO DA SAÚDE MOVIMENTO NACIONAL PELA VACINAÇÃO

gov.br Órgãos do Governo Acesso à Informação Legislação Acessibilidade PT Entrar com o gov.br

Ministério da Saúde

O que você procura?

Assuntos > Notícias para os estados > Distrito Federal > 2024 > Maio > Distrito Federal: Ministério da Saúde lança Campanha Nacional de Vacinação contra Poliomielite

IMUNIZAÇÃO

Distrito Federal: Ministério da Saúde lança Campanha Nacional de Vacinação contra Poliomielite

Ação quer reduzir o risco de reintrodução do poliovírus no Brasil, além de reforçar medidas para a erradicação da doença

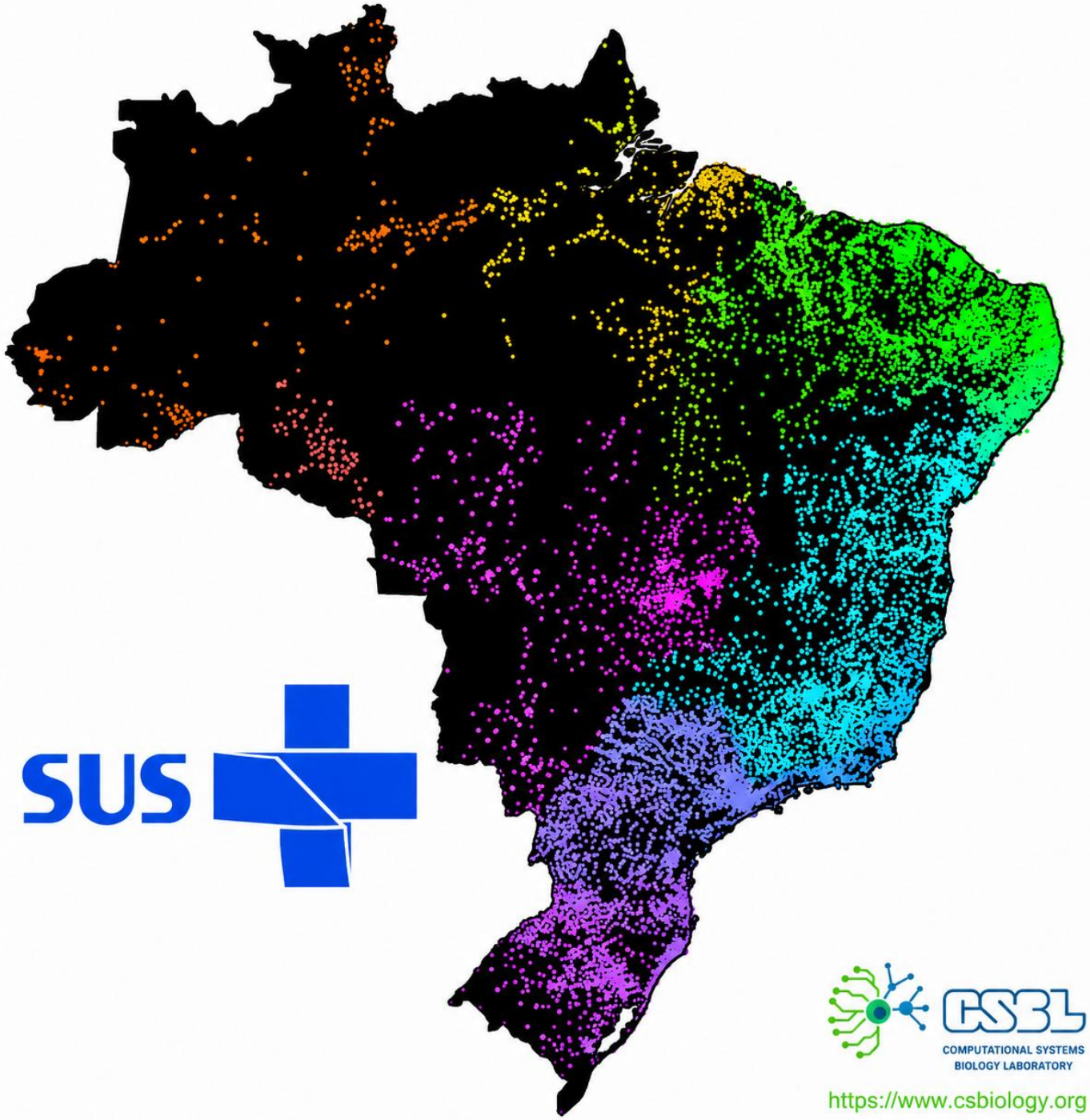
Publicado em 28/05/2024 18h21 Compartilhe:

O Ministério da Saúde lançou nesta segunda-feira (27) a Campanha Nacional de Vacinação contra a Poliomielite, que vai acontecer até o dia 14 de junho. A meta é vacinar, no mínimo, 95% do público-alvo, que abrange cerca de 13 milhões de crianças menores de cinco anos. A expectativa da campanha é reduzir o número de crianças não vacinadas e o risco de reintrodução do poliovírus no Brasil, além de reforçar medidas para a erradicação da doença. Em 2023, a pasta registrou aumento na cobertura vacinal dos principais imunizantes do calendário infantil no Distrito Federal. A aplicação da poliomielite injetável avançou 6,23 pontos percentuais, passando de 78,33% em 2022 para 84,56% no ano passado.

FONTE: HTTPS://WWW.GOV.BR/SAUDE/PT-BR

2022

2024



SUS

COMPUTATIONAL SYSTEMS BIOLOGY LABORATORY

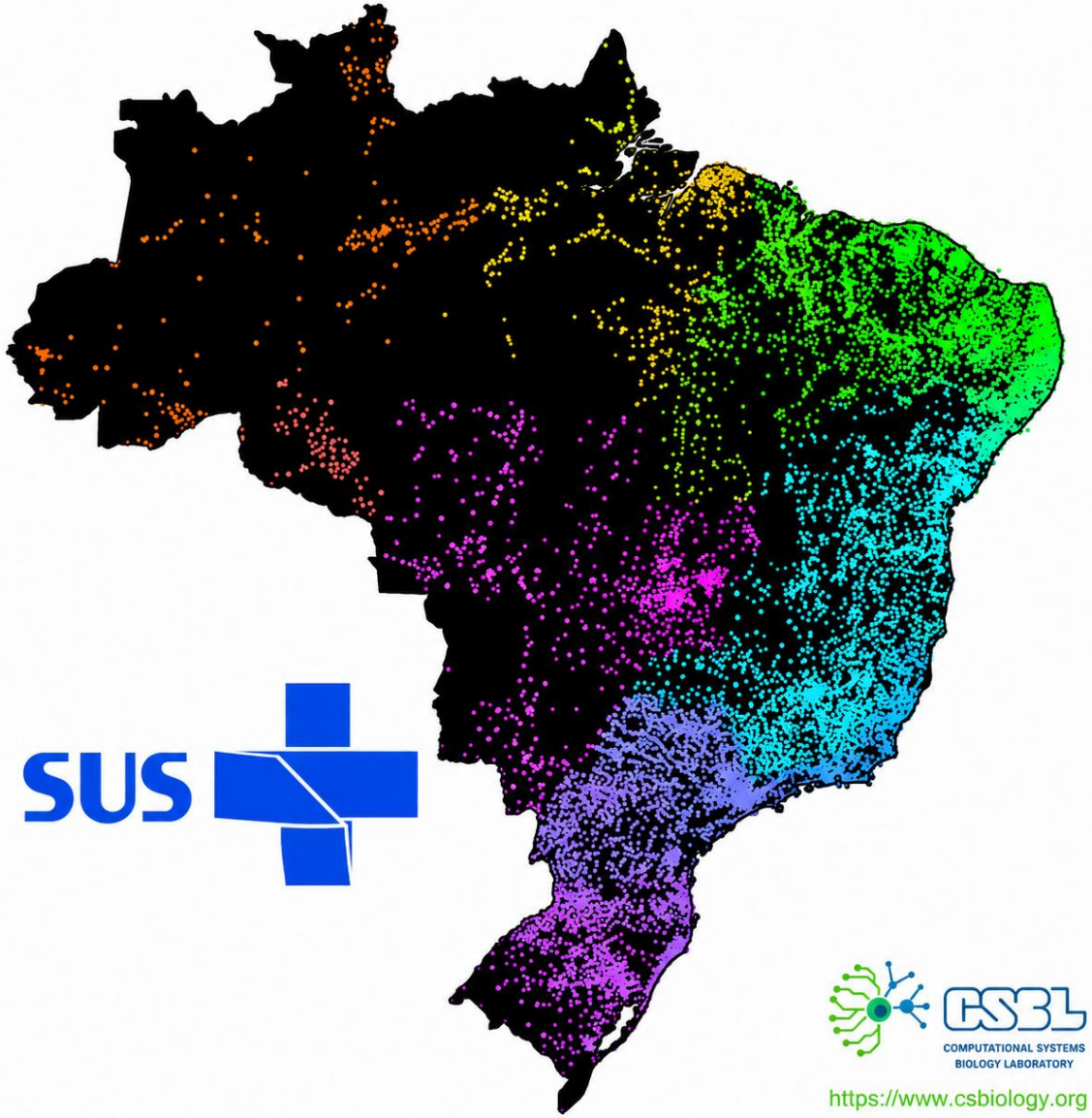
https://www.csbiology.org

FONTE: HTTP://WWW.BLOG.SAUDE.GOV.BR

2022

2024

42 MIL POSTOS DE SAÚDE ESPALHADOS PELO BRASIL



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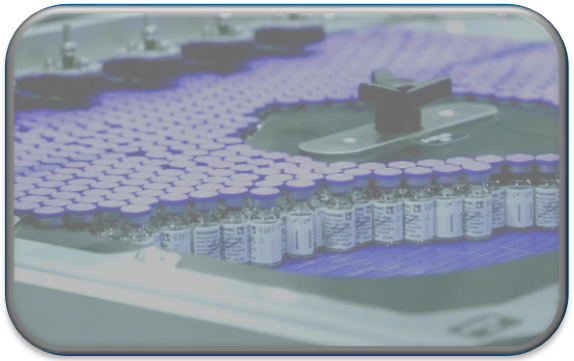
ETAPAS DA AULA

01

HISTÓRIA DA VACINAÇÃO

02

IMUNO BIOLÓGICOS



03

P&D FASES

04

DESAFIOS E DEBATES ATUAIS

ETAPAS DA AULA

01

HISTÓRIA DA VACINAÇÃO

02

IMUNO BIOLÓGICOS

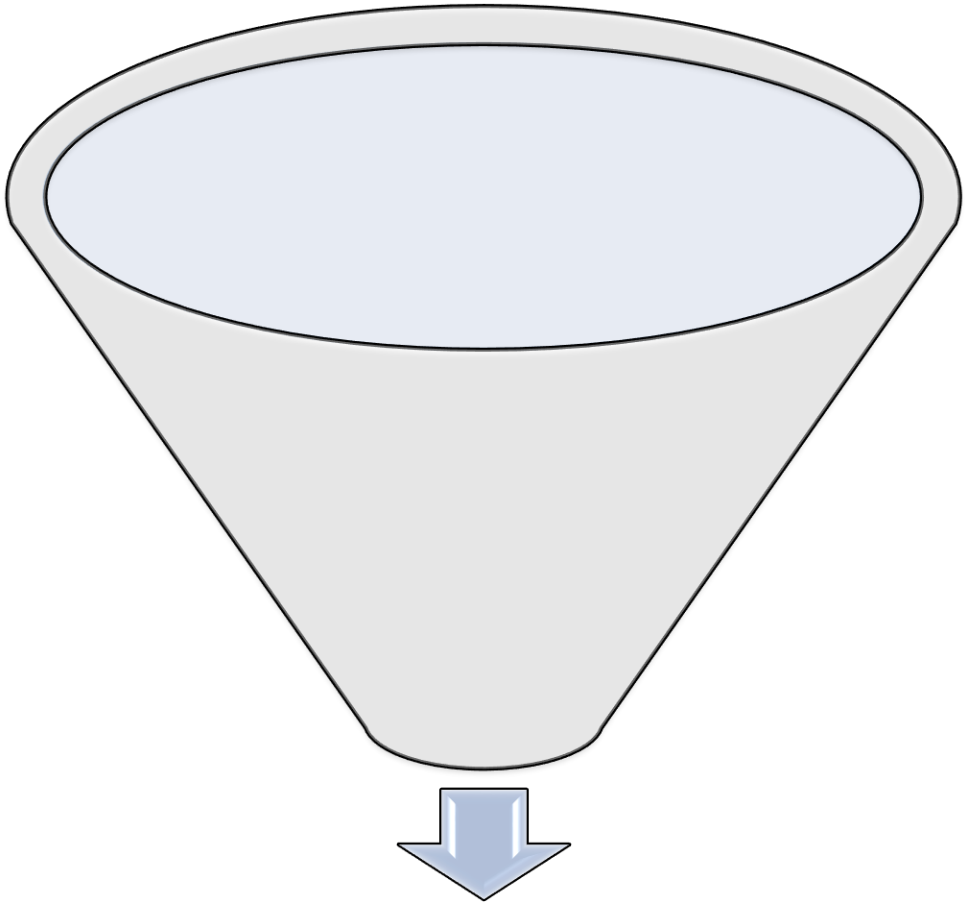
03

P&D FASES

04

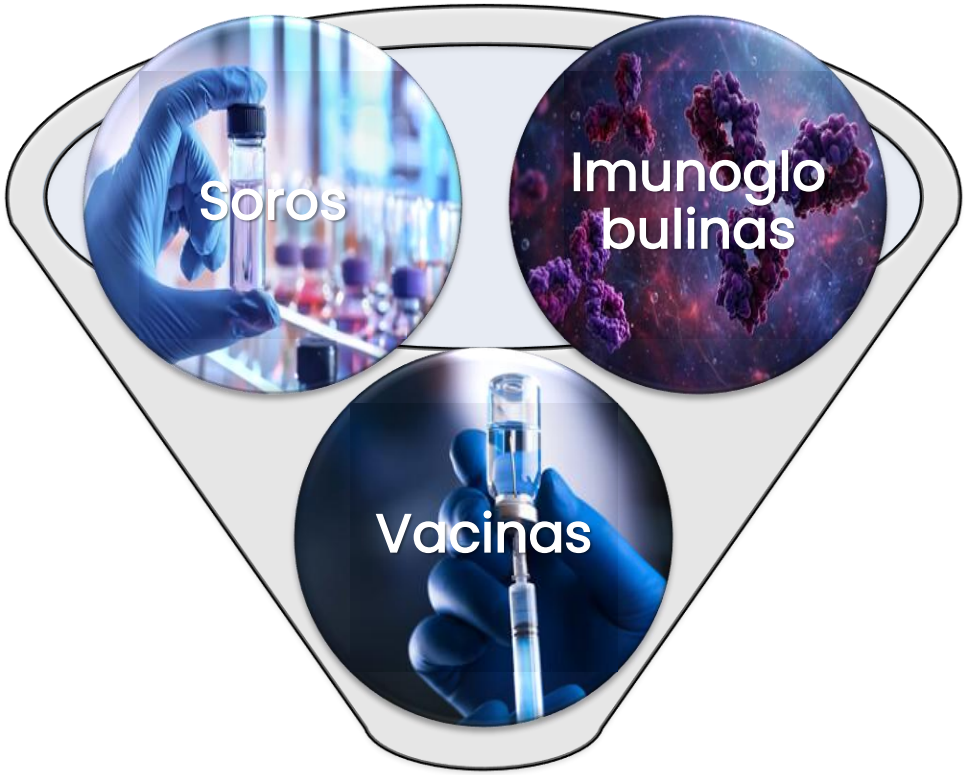
DESAFIOS E DEBATES ATUAIS

Preventiva



Curativas

Fonte: Adaptado pelo o Autor



Preventiva

Curativas

IMUNOBIOLÓGICOS

Toda substância de origem biológica capaz de induzir uma resposta imunológica no ser humano, usados na prevenção e tratamento de doenças.

Fonte: Adaptado pelo o Autor

RESPOSTA IMUNOLÓGICA

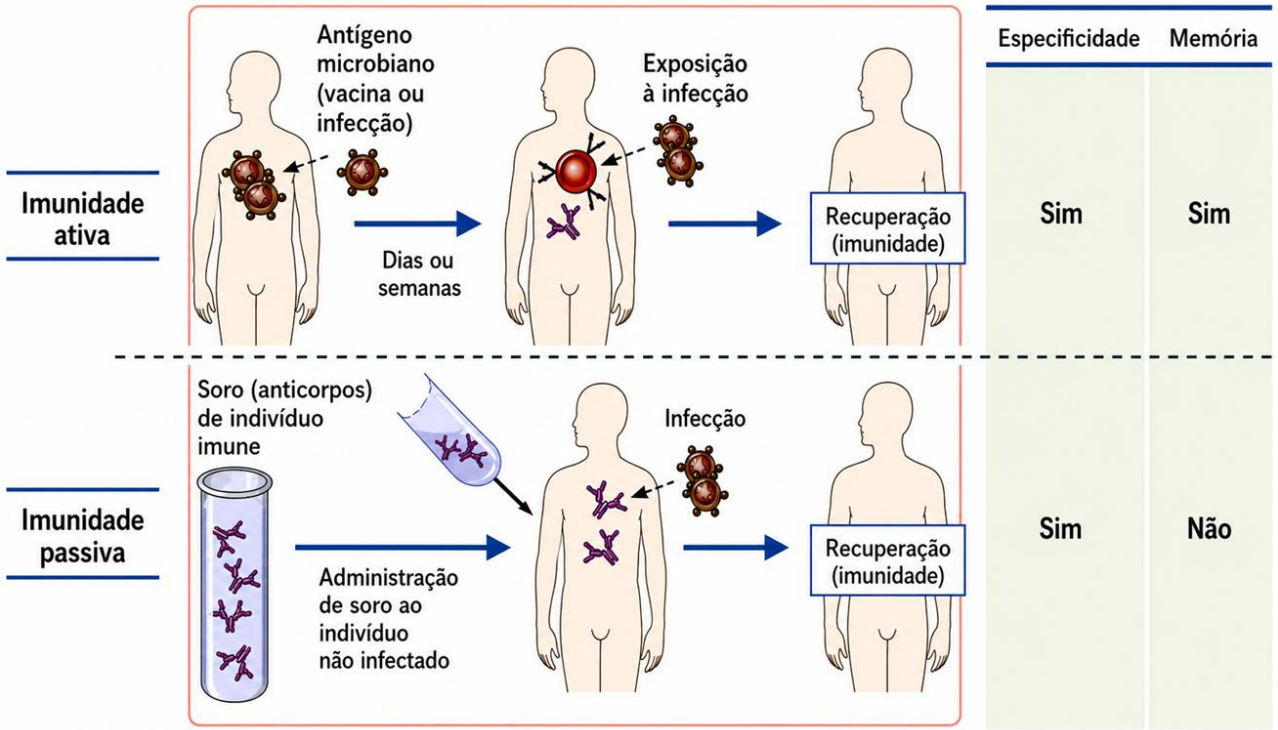
Classificação da Resposta Imune Quanto ao Modo de Aquisição:

Resposta Ativa

✓ Naturalmente Adquirida;
✓ Artificialmente Adquirida.

Resposta Passiva

✓ Naturalmente Adquirida;
✓ Artificialmente Adquirida.



	Especificidade	Memória
Imunidade ativa	Sim	Sim
Imunidade passiva	Sim	Não

RESPOSTA IMUNOLÓGICA

	Cenários			
Ig	1	2	3	4
IgM	+	−	+	−
IgG	−	+	+	−

+ Reagente
− Não Reagente

Fonte: Adaptado pelo o Autor

Cenário 1

Indica infecção de fase ativa.

Cenário 2

Indica infecção passada e/ou vacina.

Cenário 3

Indica infecção de fase ativa e/ou transitória.

Cenário 4

Indica ausência de infecção e de indicação para vacinação.

Imunoglobulinas

IgM → Momento

No MOMENTO o organismo está em combate

→ Fase Ativa

IgG → Ganhou

Organismo ja ganhou o combate, indica um contato com o agente anteriormente

→ Fase crônica/ imunização

IgM x IgG:
entenda de forma simples!

INDICA:

- Infecção recente
- Resposta inicial do organismo

INDICA:

- Infecção passada (ou imunidade)
- Permanece por mais tempo

INÍCIO DA INFECÇÃO → **APÓS ALGUM TEMPO**

IgM aparece primeiro

@bioodescomplica

IgG aparece depois e persiste

MACETE:

M = Momento (AGORA, recente)

G = Guardado (MEMÓRIA, passado)

MUITO COBRADO EM PROVAS E NA PRÁTICA CLÍNICA!

Interpretar sorologia é entender o tempo da infecção.

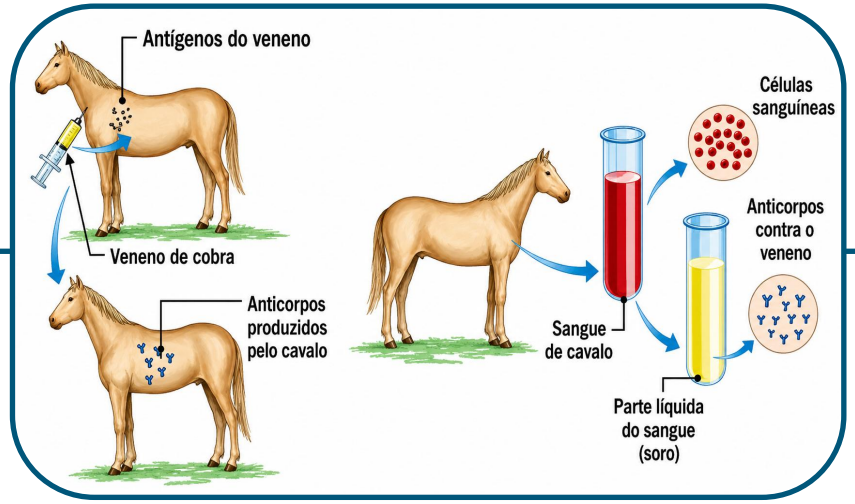
SOROS IMUNOLÓGICOS

- ✓ Utilizados para tratar:
  - ➢ Intoxicações provocadas pelo:
    - ❖ Veneno de animais peçonhentos;
    - ❖ Toxinas de agentes infecciosos.

Instituto Butantan:

- ✓ Responsável por produzir cerca de 80% dos imunobiológicos utilizados no Brasil:
  - ➢ Enviados ao Ministério da Saúde:
    - ❖ Redistribuídos às secretarias de Saúde dos Estados.

SOROS HETERÓLOGOS



Produção:

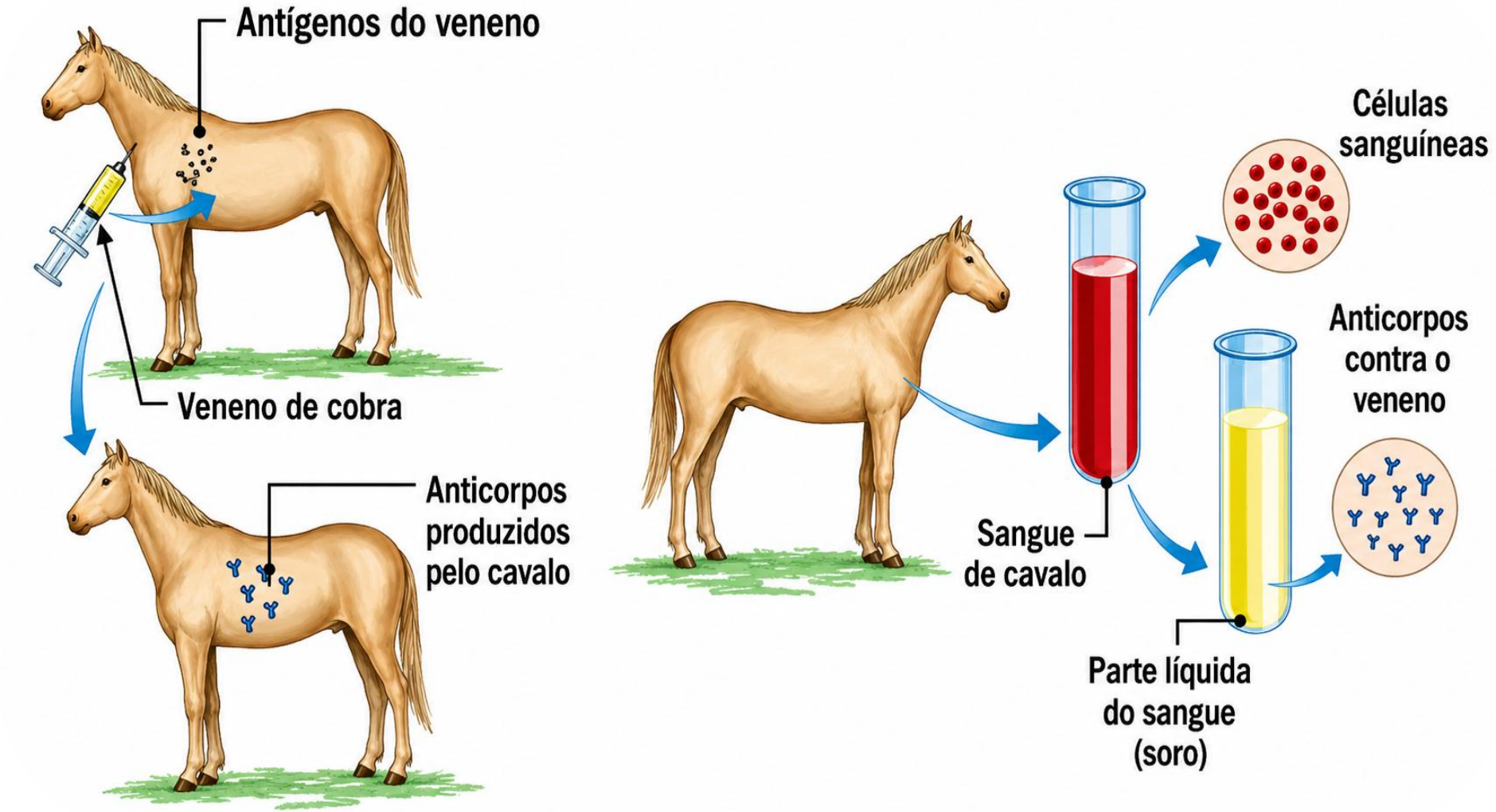
✓ Em indivíduos de espécies diferentes daquela que irá receber o soro;

✓ Dificilmente transmite doenças para humanos, porém, existe o risco de reações de hipersensibilidade.

Exemplo:

✓ Hiperimunizar cavalos com os venenos e depois extrair destes animais os anticorpos específicos para estes antígenos.

SOROS HETERÓLOGOS



PRODUÇÃO DE SOROS

Extração do Veneno:

- ✓ Também chamado peçonha:
  - ➢ É submetida a um processo de liofilizacão:
    - ❖ Desidrata e cristaliza o veneno.

Exemplos:

- ✓ Serpentes;
- ✓ Escorpiões;
- ✓ Aranhas;
- ✓ Taturanas.

PRODUÇÃO DE SOROS

Hiperimunização:

- ✓ O veneno liofilizado (antígeno) é diluído e injetado no cavalo, em doses adequadas:
  - ➢ Esse processo leva 40 dias.

Sangria Exploratória:

- ✓ Após a hiperimunizacão, retira-se uma amostra de sangue:
  - ➢ Medir o teor de anticorpos produzidos em resposta às injeções do antígeno.

PRODUÇÃO DE SOROS

Sangria Final:

- ✓ É realizada quando o teor de anticorpos atinge o nível desejado:
  - ➢ Retirar cerca de quinze litros de sangue de um cavalo de 500 Kg em três etapas, num intervalo de 48 horas.

Plasma e Hemácias, respectivamente:

- ✓ O soro é obtido a partir da purificação e concentração desse plasma;
- ✓ São devolvidas ao animal, através de uma técnica desenvolvida no Instituto Butantan – Plasmaferese:
  - ➢ Reduz os efeitos colaterais provocados pela sangria do animal.

PRODUÇÃO DE SOROS

Controle de Qualidade:

- ✓ No final do processo, <u>o soro obtido é submetido à testes</u>:
  - ➢ **Atividade Biológica:**
    - ❖ Verificação da quantidade de anticorpos produzidos.
  - ➢ **Esterilidade:**
    - ❖ Detecção de eventuais contaminações durante a produção.
  - ➢ **Inocuidade:**
    - ❖ Teste de segurança para o uso humano.
  - ➢ **Pirogênio:**
    - ❖ Detectar a presença dessa substância, que provoca alterações de temperatura nos pacientes.
  - ➢ **Testes Físico-Químicos**.

SOROS HOMÓLOGOS

Produção:

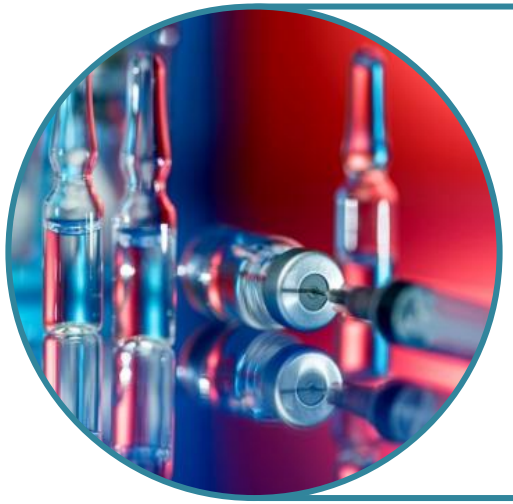
✓ A partir de anticorpos obtidos de seres humanos – menor risco de reação de hipersensibilidade;

✓ Geralmente do plasma de doadores previamente imunizados ou que se recuperaram de uma determinada infecção;

Exemplos:

✓ Imunoglobulinas Humanas Antitetânica, Antirrábica e Anti-D.

VACINAS



- ✓ É uma das medidas **mais importantes de prevenção de doenças** :
  - ➢ Todo e qualquer imunógeno empregado para ativar uma resposta imune específica :
    - ❖ A um determinado agente infeccioso e/ou toxina produzida.

Objetivos:

- ✓ Segura;
- ✓ Eficaz;
- ✓ Mimetizar uma resposta imune primária:
  - ➢ Desenvolvimento de memória imunológica protetora e de longo termo.

CARACTERÍSTICAS DAS VACINAS

Boa Imunogenicidade:

- ✓ Imunizar grande proporção dos inoculados da exposição ao patógeno vivo;
- ✓ Dose única ou poucas doses;
- ✓ Imunidade de longa duração.

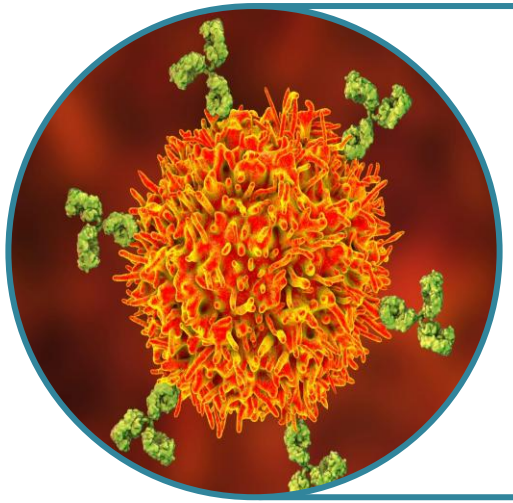
Pouca Reatogenicidade:

- ✓ Poucos efeitos adversos imediatos ou tardios;
- ✓ Não causar a doença ou morte.

CARACTERÍSTICAS DAS VACINAS

Facilidade:

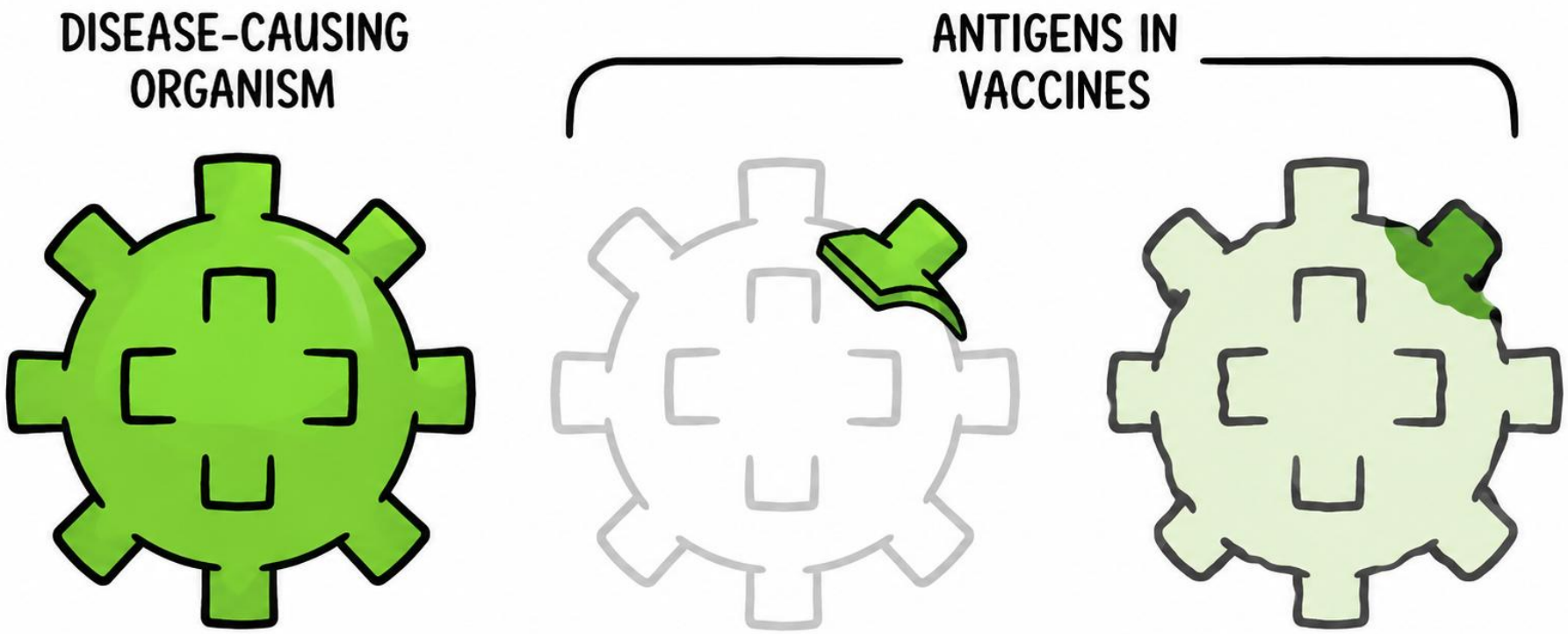
- ✓ Obtenção, com o menor custo possível;
- ✓ Distribuição;
- ✓ Redes de frio – armazenamento.



Induzir:

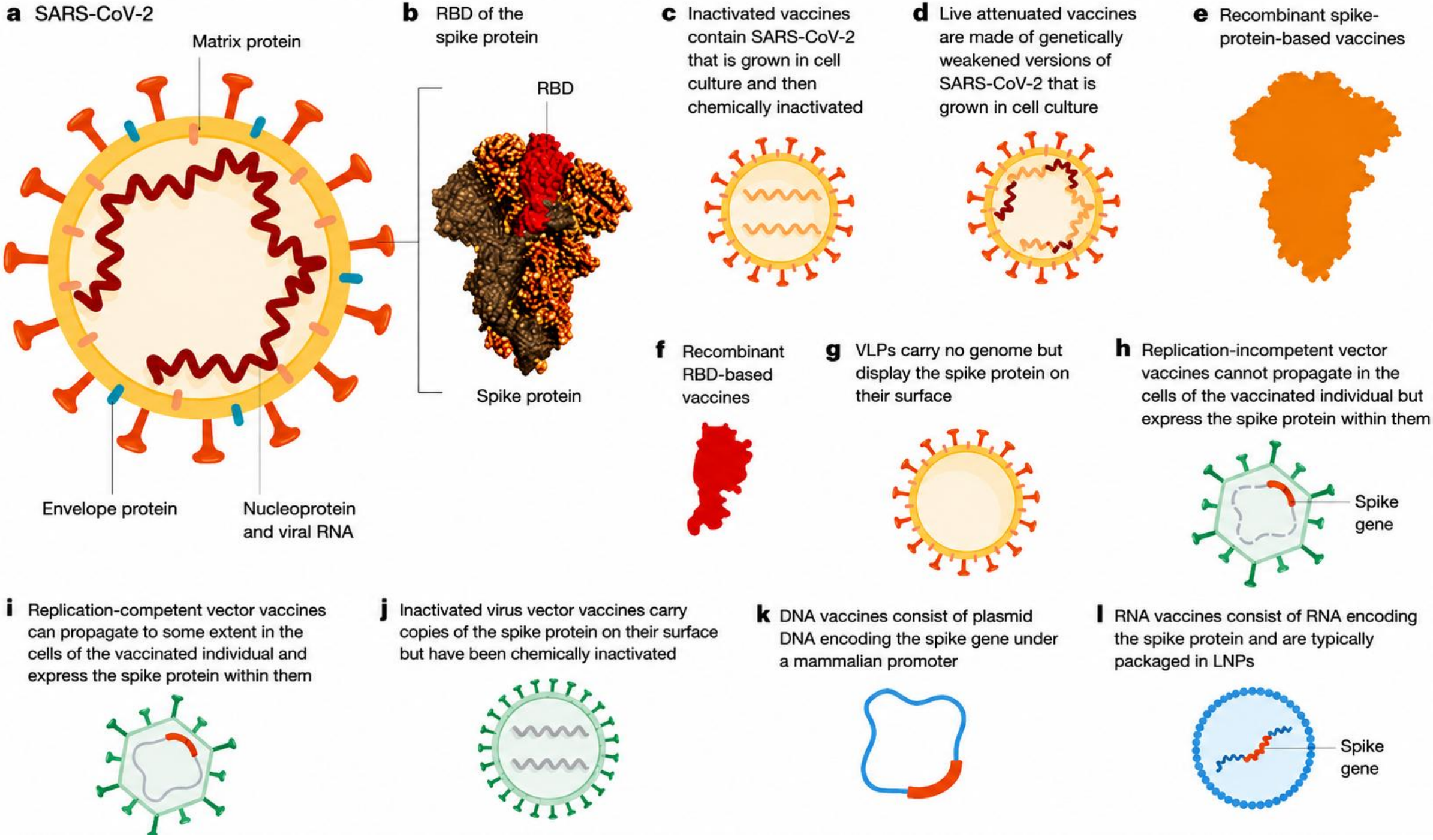
- ✓ Resposta Humoral: Produção de Imunoglobulinas;
- ✓ Resposta Celular: Linfócitos T;
- ✓ Desenvolvimento de memória imunológica.

TIPOS DE VACINAS

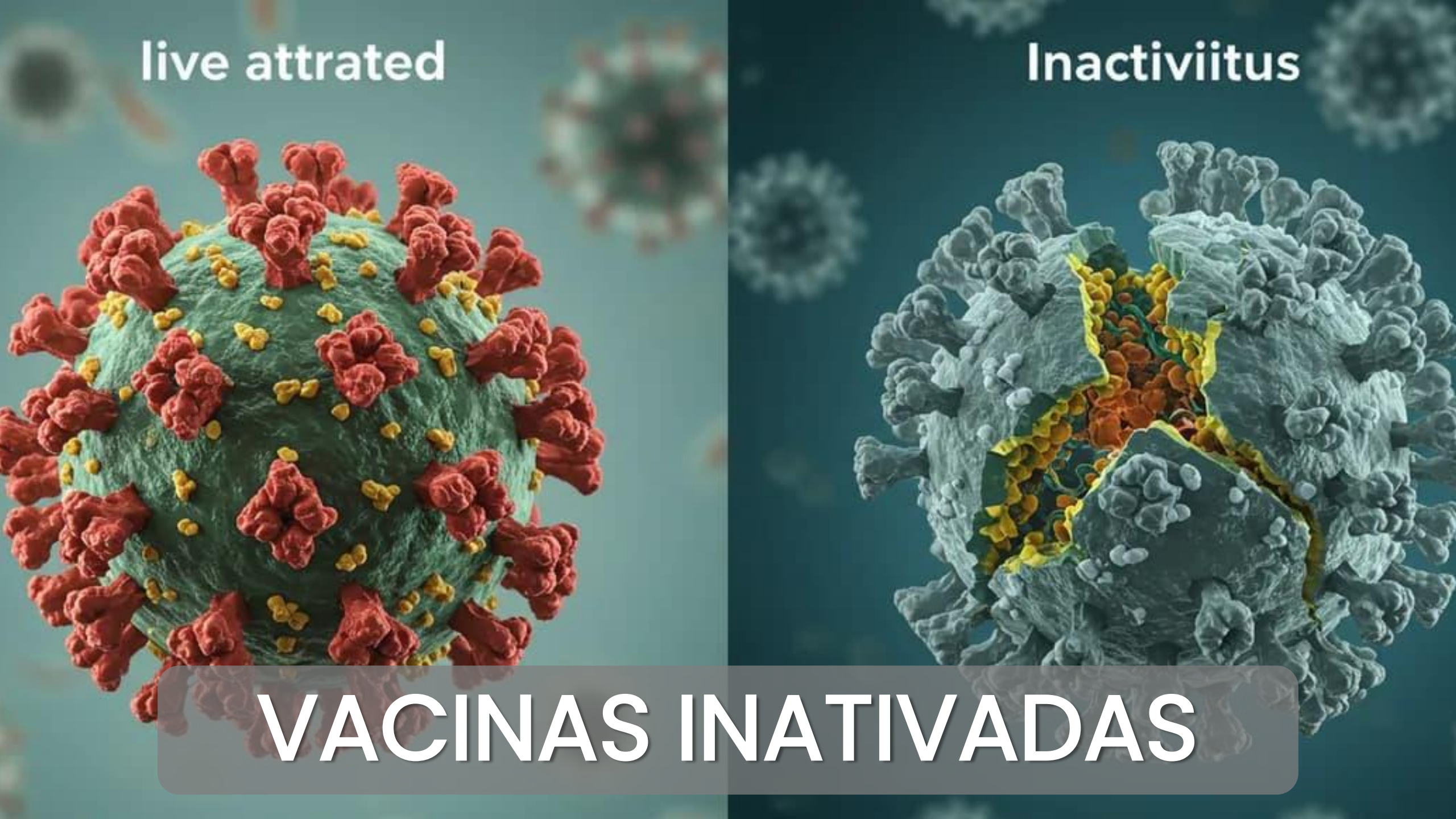


The key ingredient in a vaccine is the antigen. It's either a tiny part of the disease-causing organism, or a weakened, non-dangerous version, so your body can learn the specific way to fight it without getting sick.

Source: https://www.who.int/pt/news-room/feature-stories/detail/how-are-vaccines-developed

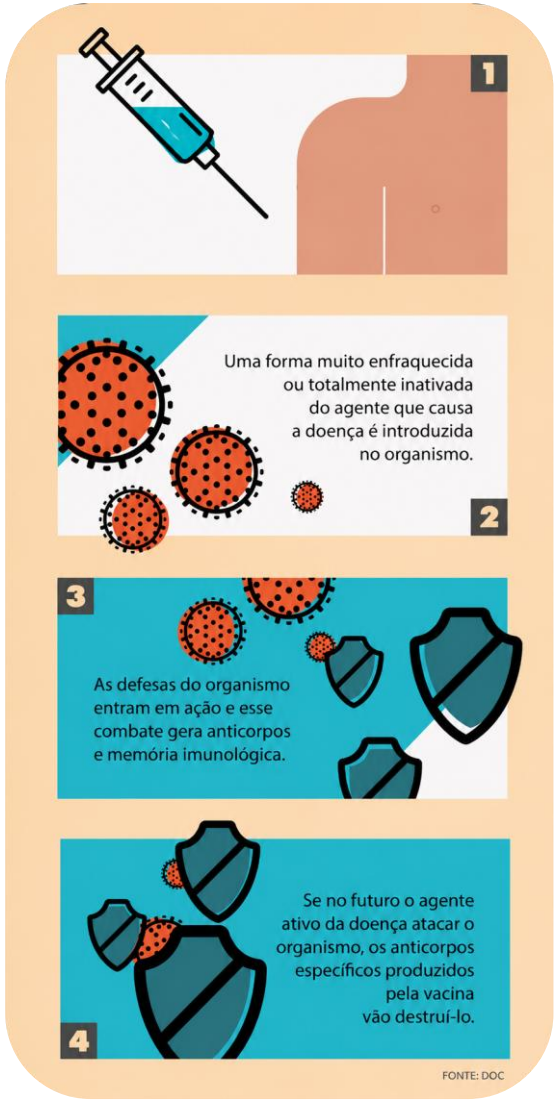


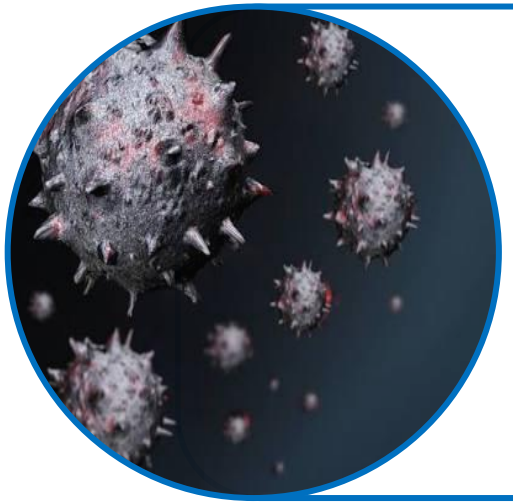
Fonte: https://www.nature.com/articles/s41586-020-2798-3



VACINAS INATIVADAS

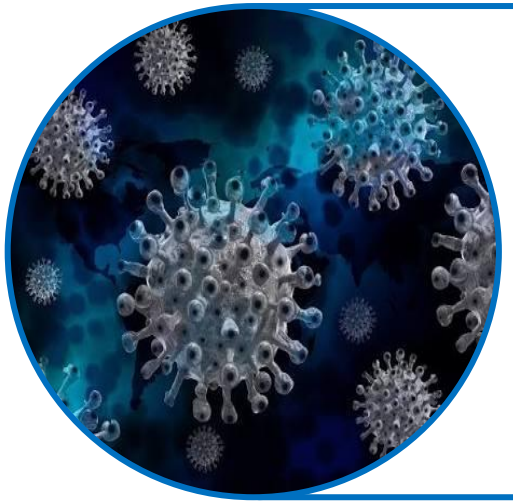
VACINAS INATIVADAS





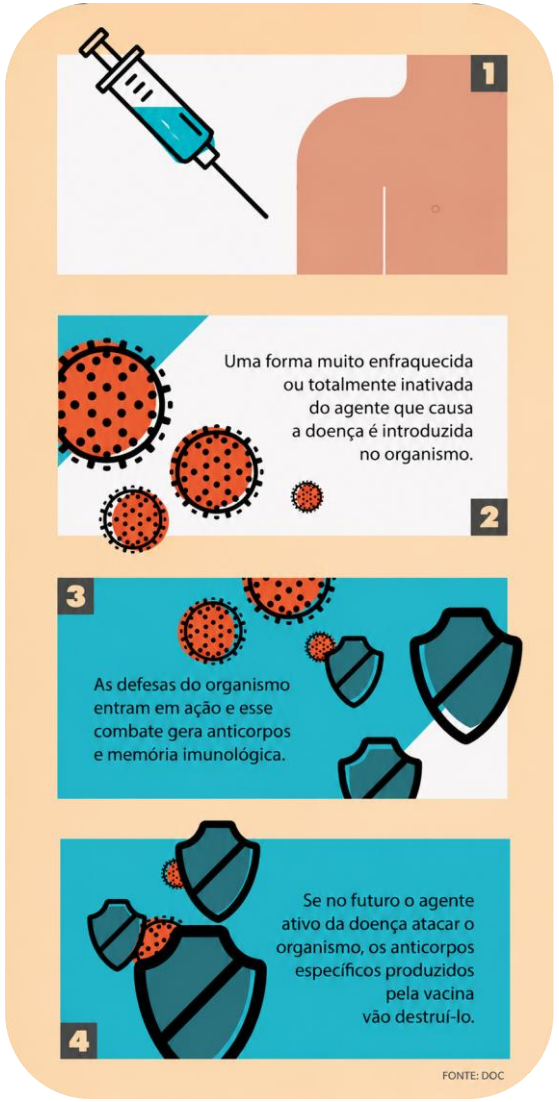
Produzidas:

- ✓ Com <u>microrganismos totalmente inativos</u>:
  - ➢ Quanto à sua capacidade de se reproduzir e portanto, causar uma infecção.



- ✓ Deverá **conter grande quantidade do antígeno** para ser eficaz:
  - ➢ Geralmente, <u>necessitam de mais de uma aplicação</u>, para conseguir gerar uma resposta duradoura.

VACINAS INATIVADAS

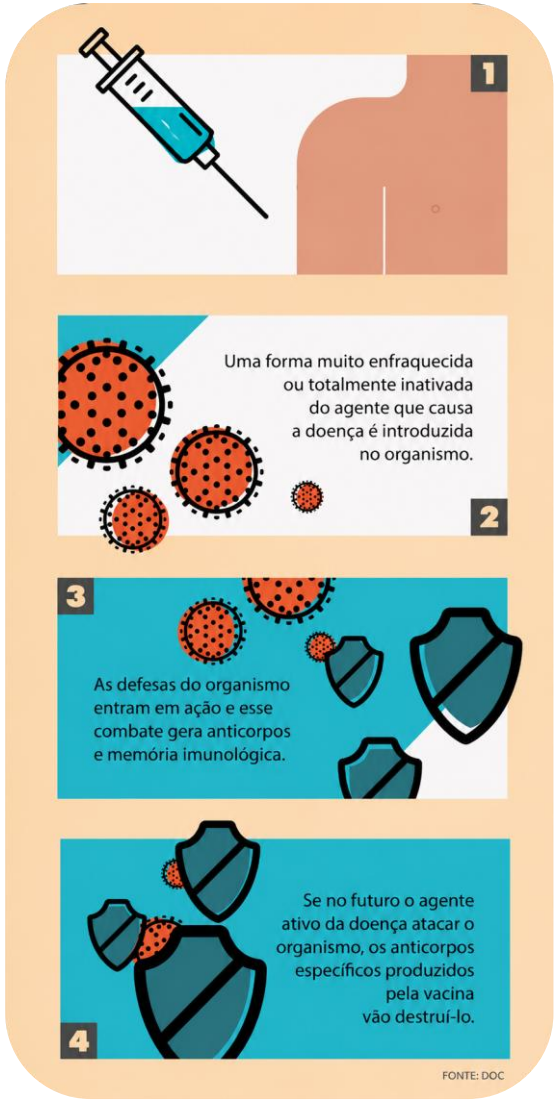


- ✓ Podem ser usadas em:
  - ➢ Gestantes;
  - ➢ Imunocomprometidas.

Exemplos:

- ✓ Hepatite A;
- ✓ Poliomielite;
- ✓ *Influenza* Inativada;
- ✓ Raiva;
- ✓ SARS-CoV-2.

VACINAS INATIVADAS



Eventos Adversos:

- ✓ Em geral, são precoces, aparecendo 24 a 48 horas após a vacinação:
  - ➢ Relacionados com a <u>resposta inflamatória no local da aplicação</u>:
    - ❖ **Sinais cardinais – "4 ORs".**

FASES DE TESTES NO DESENVOLVIMENTO DE VACINAS

Fase	Pré-clínica	Fase 1	Fase 2	Fase 3	Fase 4
Objetivo	Testagem com modelos animais	Teste de segurança[1]	Teste de eficácia[2] e de efeitos colaterais A Prova de conceito[3] B Dose ótima[4]	Teste de eficácia, efetividade[5] e segurança	Resultados após aplicação massiva na população
Número de participantes	—	Dezenas de pessoas	Centenas de pessoas	Milhares a dezenas de milhares de pessoas	Centenas de milhares a milhões de pessoas

Coordenação	País	Plataforma	Fase de desenvolvimento
Sinovac (CoronaVac)	China	Vírus inativado	**Fase 3** Iniciou ensaio clínico no Brasil em 21/7 em parceria com Instituto Butantan (SP), e em agosto na Indonésia; no Brasil a Anvisa autorizou a importação de 6 milhões de doses pelo governo paulista, e de insumos para a produção de outros 40 milhões de doses pelo Butantan; em 9/11 os testes foram suspensos para investigação de morte de participante, mas logo retomados por não ter relação com a vacinação; em 3/12 um carregamento a São Paulo e o governo paulista anunciou o plano de iniciar a vacinação em janeiro de 2021
BioNTech, Fosun Pharma, Pfizer	EUA	RNA	**Fase 2/3** Ensaio em fases combinadas nos Estados Unidos, na Argentina e no Brasil, e em fase 3 na África do Sul, na Alemanha e na Turquia; em 18/11 a empresa anunciou resultados finais que indicam uma eficácia de 95% - e 94% em idosos; o desafio de produção e distribuição é grande, já que as doses precisam ser mantidas a -70°C; em 2/12 o Reino Unido obteve autorização para vacinação emergencial
Moderna e Inst. Nacional de Alergias e Doenças Infecciosas (Niaid)	EUA	RNA	**Fase 3** Ensaio clínico fase 3 iniciado em 27/7 nos Estados Unidos; no dia 22/10 atingiu a meta de recrutar 30 mil voluntários; em 16/11 anunciou uma eficácia de 94,5% entre 95 infectados entre os voluntários; em 30/11, submeteu pedido de autorização para uso emergencial
Universidade de Oxford e AstraZeneca	Reino Unido	Vetor viral não replicante	**Fase 2b/3** Iniciados em 17/08, ensaios clínicos em fase 2/3 na Inglaterra e na Índia, e fase 3 no Brasil, na África do Sul e nos Estados Unidos foram interrompidos para investigação de efeitos adversos e depois retomados – nos EUA por último, em 23/10; dados anunciados em 23/11 indicam uma eficácia de 90% caso seja aplicada uma meia dose, seguida por uma dose completa – em contraposição a 60% com duas doses completas; a empresa tem enfrentado questionamentos sobre os procedimentos de testes
Inst. Produtos Biológicos de Wuhan/Beijing e Sinopharm	China	Vírus inativado	**Fase 3** Ensaio clínico fase 3 iniciado em julho nos Emirados Árabes Unidos, onde em 14/9 foi autorizado o uso emergencial para profissionais da saúde, em agosto no Peru e em Marrocos, e em setembro na Argentina; em 14/10 foram noticiados planos de oferecer a vacina gratuitamente para chineses que pretendessem fazer estudos superiores no exterior
Companhia Farmacêutica Janssen	EUA	Vetor viral não replicante	**Fase 3** Início de ensaio clínico fase 3 em curso nos Estados Unidos e no Reino Unido; em 12/10 ensaio foi paralisado para investigação de efeito adverso e retomado
Novavax	EUA	Subunidade proteica	**Fase 3** Iniciou fase 2b de ensaios clínicos em agosto na África do Sul e fase 3 em setembro no Reino Unido; início da fase 3 nos Estados Unidos está previsto para o início de novembro
CanSino e Inst. Biotecnológico de Beijing	China	Vetor viral não replicante	**Fase 3** Ensaio clínico fase 3 iniciado em 11/9 na Rússia, 15/9 no Paquistão e em 30/10 no México
Instituto Gamaleia, Moscou (Sputnik V)	Rússia	Vetor viral não replicante	**Fase 3** Iniciou fase 3 de ensaio clínico em setembro na Rússia e na Belorrússia; em 2/12 governo russo autorizou a vacinação em massa, mesmo antes de se finalizarem testes
Bharat Biotech (Covaxin)	Índia	Vírus inativado	**Fase 2/3** Iniciou ensaios clínicos em julho na Índia; início fase 3 em 11/11
Instituto Vector (EpiVacCorona)	Rússia	Peptídeo	**Fase 1/2** Ensaio clínico em fases 1 e 2 conjugadas foi registrado em agosto para teste em 100 participantes; em 14 de outubro o governo russo anunciou a aprovação da vacina antes mesmo de iniciada um fase 3 de testes
Fiocruz (RJ) e Instituto Butantan (SP)	Brasil	Vetor viral replicante	**Pré-clínica**
Universidade de São Paulo (USP)	Brasil	Partícula semelhante a vírus (VLP)	**Pré-clínica**

Fonte: https://revistapesquisa.fapesp.br/vacinas-em-desenvolvimento-contra-covid-19

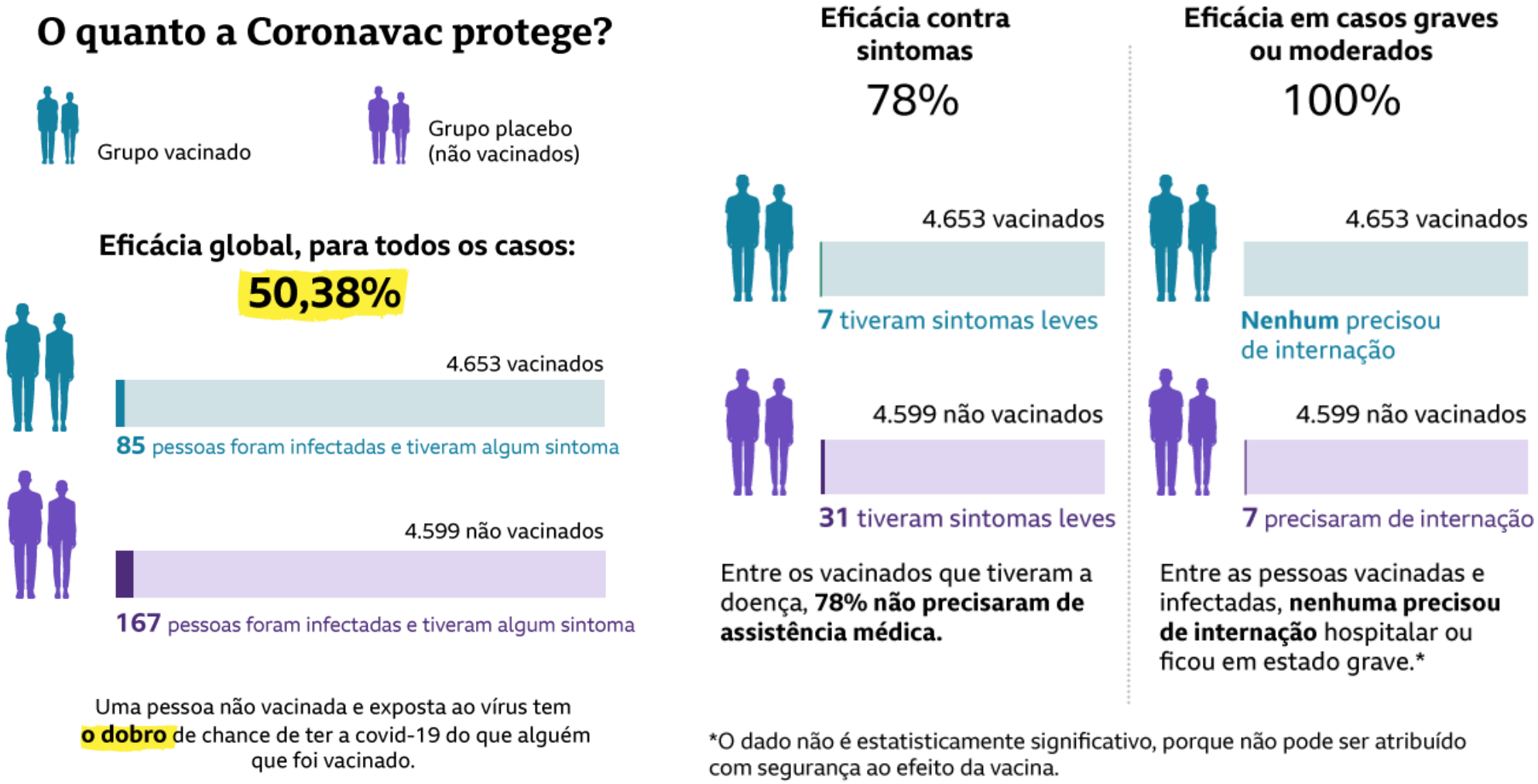
CORONAVAC

- ✓ Produzidas pelo Instituto Butantan em parceria com a Sinovac;

Informações Técnicas:

- ✓ **Eficácia:** 50,38%;
- ✓ **Testada:** Profissionais da saúde;
- ✓ **Duas doses:** com intervalo de 21 dias;
- ✓ **Variantes:** Britânica, brasileira e sul-africana.

CORONAVAC



Fonte: Adaptado de https://www.bbc.com/portuguese/brasil-55655588

Article

Title Efficacy and safety of a COVID-19 inactivated vaccine in healthcare professionals in Brazil: The PROFISCOV study

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Roberta de Oliveira Piorelli, MD [1]

Luiz Carlos Pereira Júnior, MD, PhD [2]

Sonia Mara Raboni, MD, PhD [3]

Fabiano Ramos, MD, PhD [4]

Gustavo Adolfo Sierra Romero, MD, PhD [5]

Fábio Eudes Leal, MD, PhD [6]

Luis Fernando Aranha Camargo, MD, PhD [7]

Francisco Hideo Aoki, MD, PhD [8]

Eduardo Barbosa Coelho, MD, PhD [9]

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André Machado de Siqueira, MD, PhD [14]

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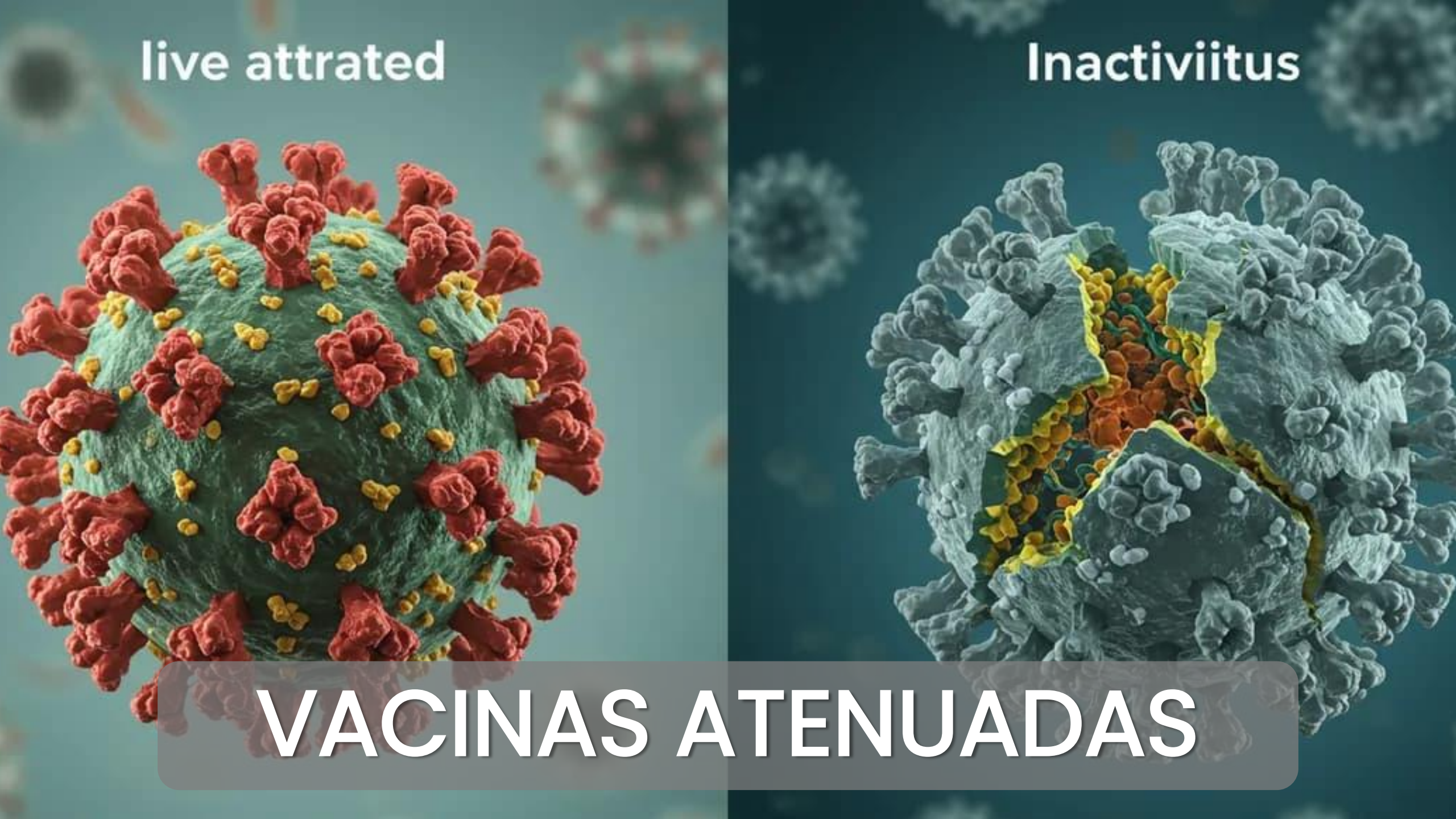
Preprint not peer reviewed

Fonte: https://static.poder360.com.br/2021/04/estudo-preliminar-coronavac-11.abr_.2021.pdf



VACINAS ATENUADAS

VACINAS ATENUADAS

Produzidas:

- ✓ Com <u>microrganismos de laboratório vivos enfraquecidos</u>:
  - ➢ Perderam a capacidade de causar doenças;
  - ➢ Porém, sem contudo a capacidade de se reproduzirem.

- ✓ **Não necessitam de grande quantidade do antígeno** para um amplo efeito:
  - ➢ Os antígenos se <u>multiplicarão em contato com o organismo do receptor</u>, conferindo uma ótima e prolongada proteção

VACINAS ATENUADAS

- ✓ **Contraindicadas para:**
  - ➢ Gestantes;
  - ➢ Imunocomprometidas.

Efeitos Adversos:

- ✓ Maior risco de causar efeitos similares com a doença selvagem, apesar de serem mais brandos:
  - ➢ 05 a 20 dias após a aplicação.

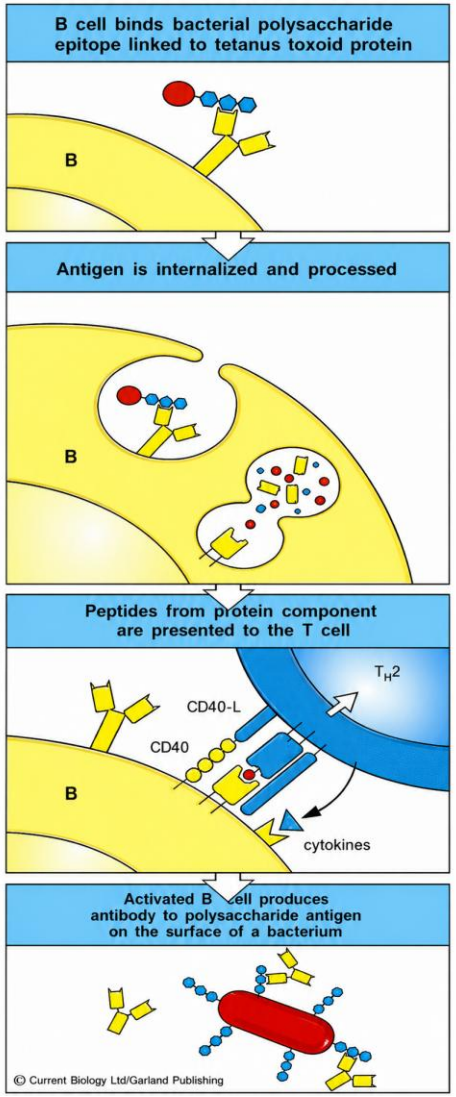
VACINAS ATENUADAS

Exemplos:

- ✓ Tuberculose – BCG;
- ✓ Febre Amarela;
- ✓ Tríplice Viral:
  - ➢ Sarampo;
  - ➢ Caxumba;
  - ➢ Rubéola.
- ✓ Tetraviral:
  - ➢ Sarampo;
  - ➢ Caxumba;
  - ➢ Rubéola;
  - ➢ Varicela.

VACINAS TOXÓIDES

VACINAS TOXÓIDES



- ✓ <u>Utilizam toxinas manipuladas que são inativadas, perdendo sua toxicidade</u>:
  - ➢ Preservando-se porém a estrutura antigênica.

- ✓ Quando administradas induzem à produção de anticorpos neutralizantes:
  - ➢ Sem que sejam capazes de produzir nenhum efeito tóxico no receptor.

VACINAS TOXÓIDES

Doses de Reforço:

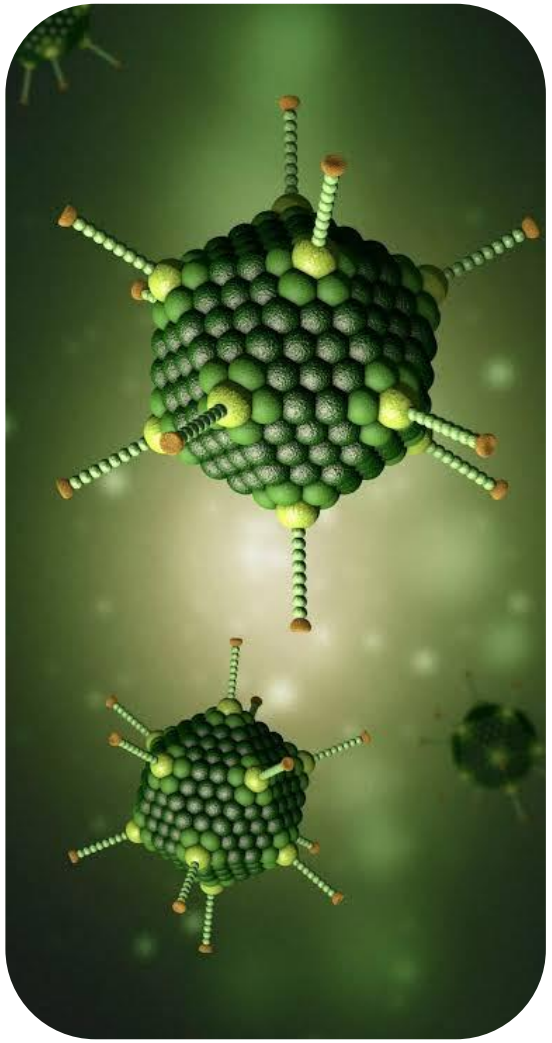
- ✓ Os níveis de anticorpos diminuem ao longo dos anos, tornando necessárias doses – a cada 10 anos – para manter a proteção.

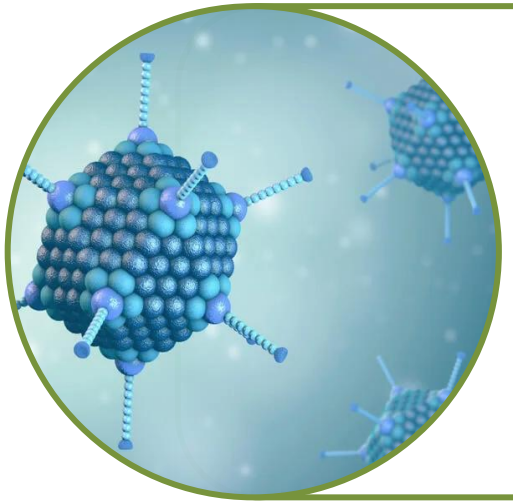
Exemplos:

- ✓ Tetânica – *Clostridium tetani* ;
- ✓ Difteria – *Corynebacterium diphtheriae* ;
- ✓ Coqueluche – *Bordetella Pertussis* .

VACINAS VETORIAIS NÃO REPLICANTES

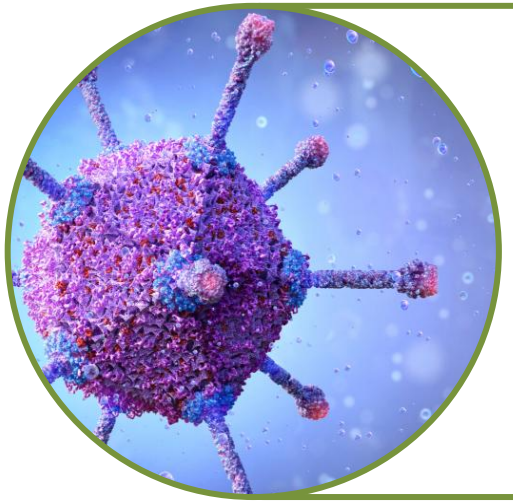
VACINAS VETORIAIS VIRAIS NÃO REPLICANTES





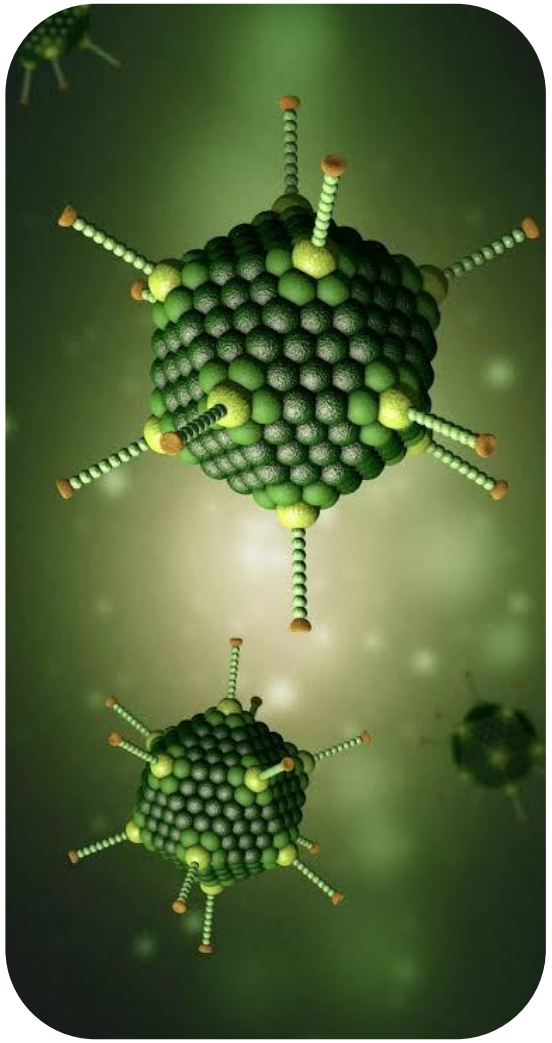
Vetor – Vírus Recombinante:

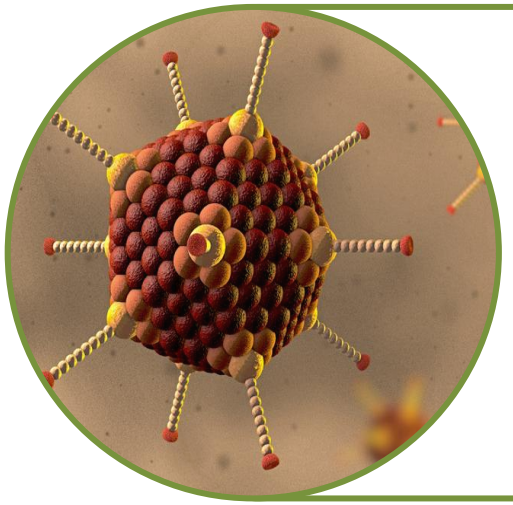
- ✓ Normalmente é um <u>Adenovírus modificado inócuo</u>:
  - ➢ É adicionado o gene que codifica o antígeno específico:
    - ❖ Proteína *spike* do SARS-CoV-2.



- ✓ **Não se reproduz:**
  - ➢ Penetra nas células humanas, onde ativa o gene do antígeno que produz a *spike*.

VACINAS VETORIAIS VIRAIS NÃO REPLICANTES

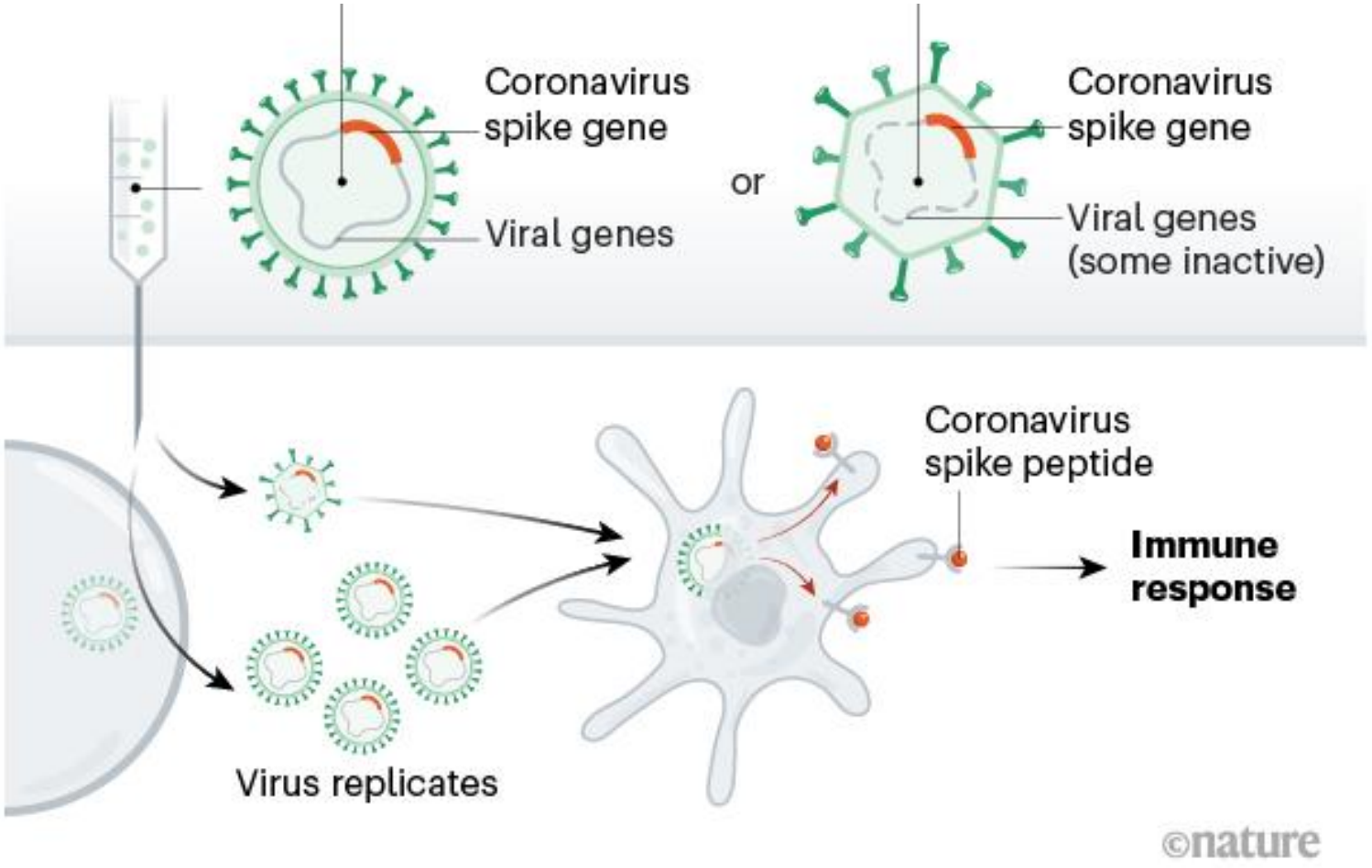




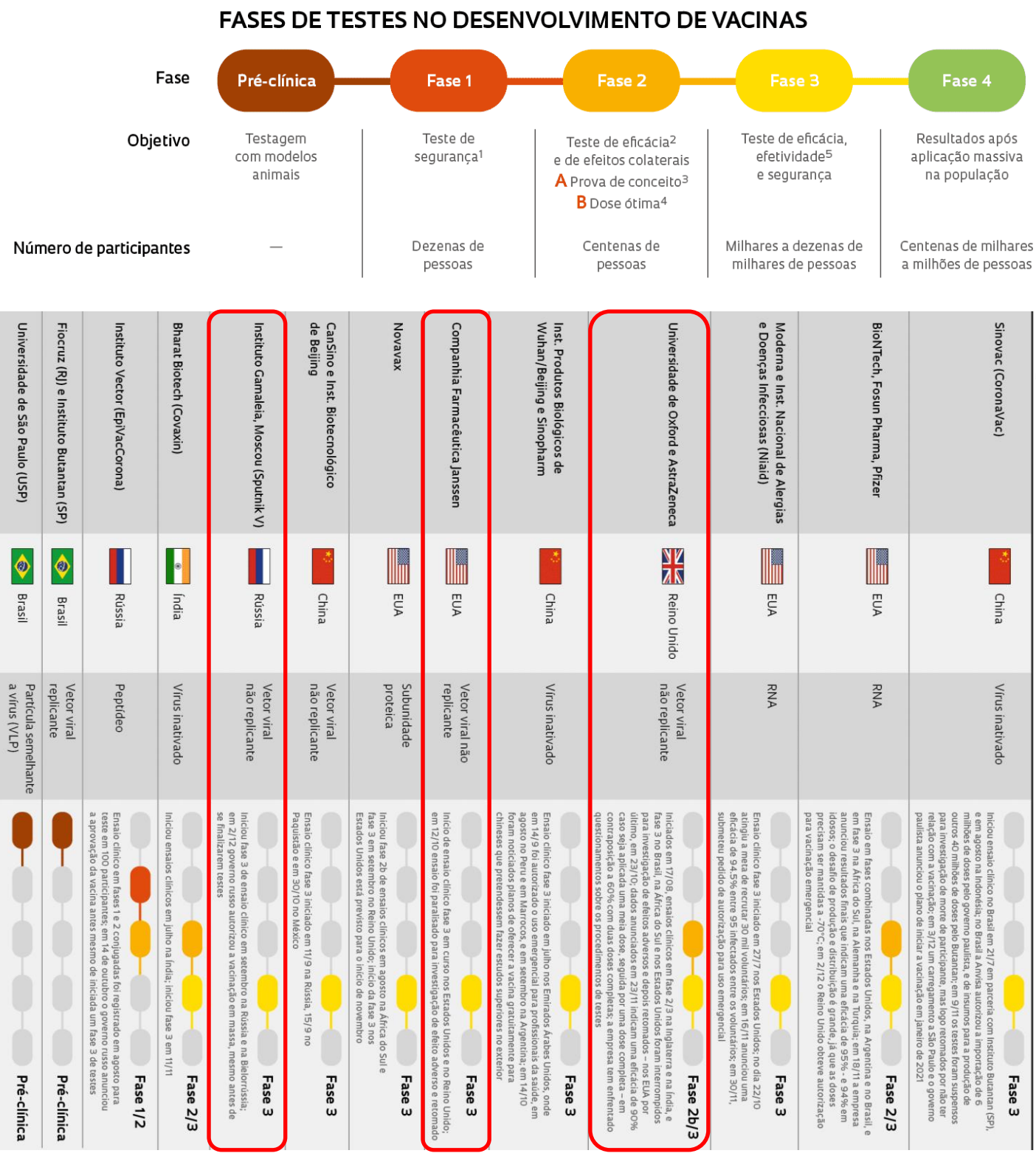
Exemplos:

- ✓ COVID-19;
- ✓ Ebola.

VACINAS VETORIAIS VIRAIS NÃO REPLICANTES



Vacina de vetores virais replicantes (à esquerda) e não replicantes (à direita).



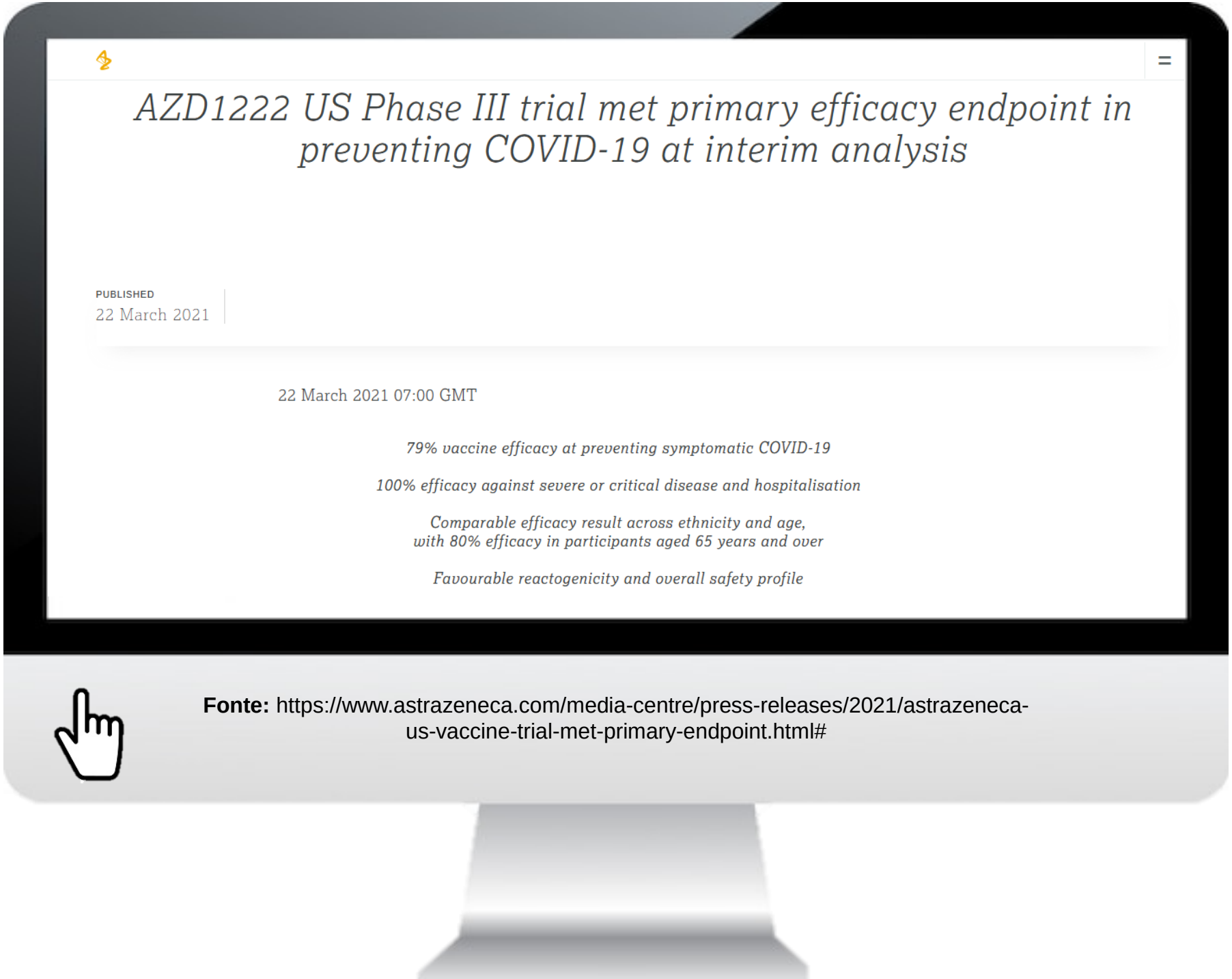
Fonte: https://revistapesquisa.fapesp.br/vacinas-em-desenvolvimento-contra-covid-19

OSFORD | ASTRAZENECA

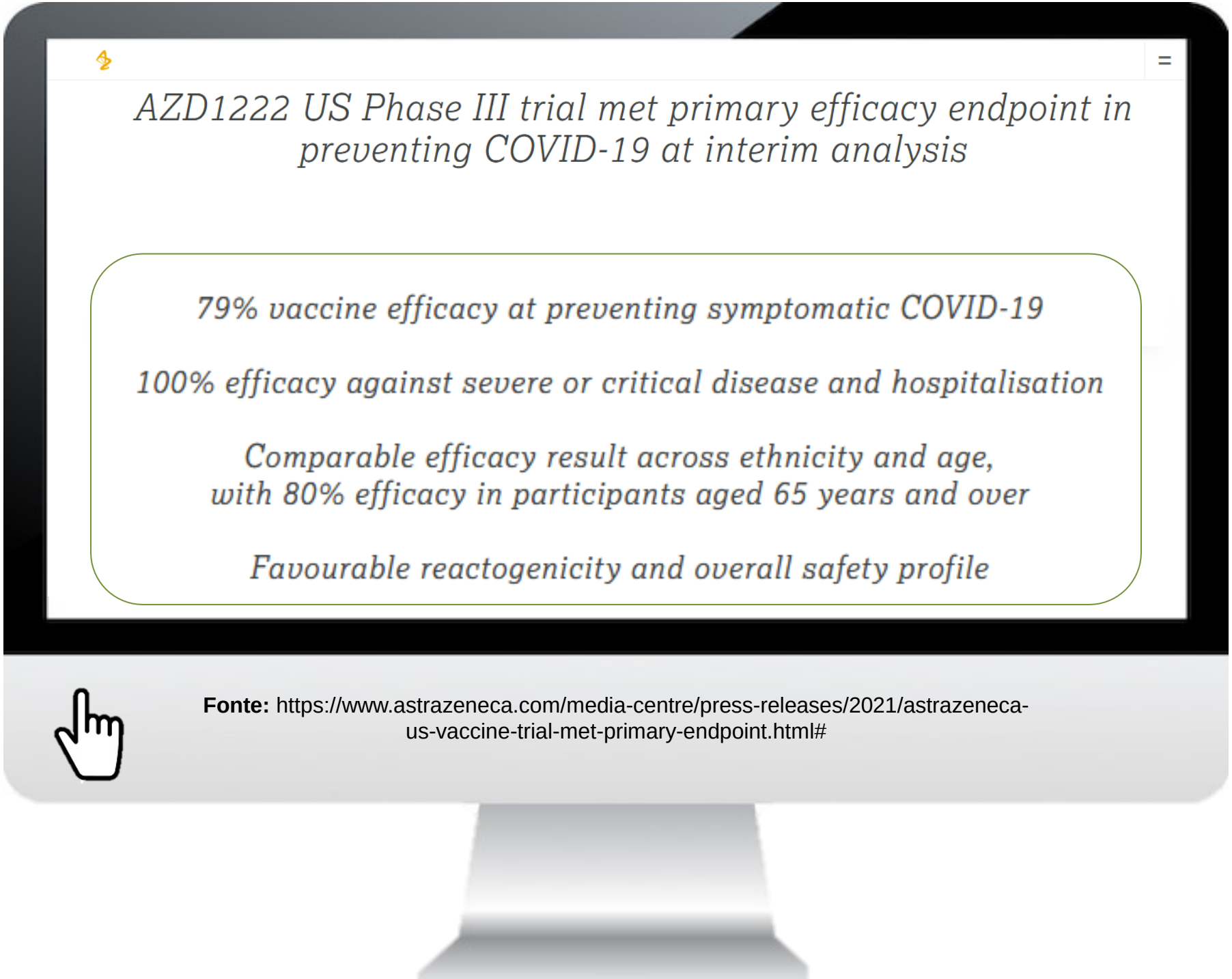
- ✓ Produzidas pelo Instituto Oswaldo Cruz (FIOCRUZ) em parceria com a Oxford / Astrazeneca;

Informações Técnicas:

- ✓ **Eficácias:** 70,4% e 82,4%;
- ✓ **Duas doses:** com intervalo de 14 e 90 dias, respectivamente;
- ✓ **Transmissão:** Redução de até 67,6%.



Fonte: https://www.astrazeneca.com/media-centre/press-releases/2021/astrazeneca-us-vaccine-trial-met-primary-endpoint.html#



Fonte: https://www.astrazeneca.com/media-centre/press-releases/2021/astrazeneca-us-vaccine-trial-met-primary-endpoint.html#

OSFORD | ASTRAZENECA

Dados Importantes:

- ✓ 35 milhões de doses aplicadas na Europa:
  - ➢ **222 casos de coágulos:**
    - ❖ 1 caso para <u>cada 175.000 vacinados</u>.

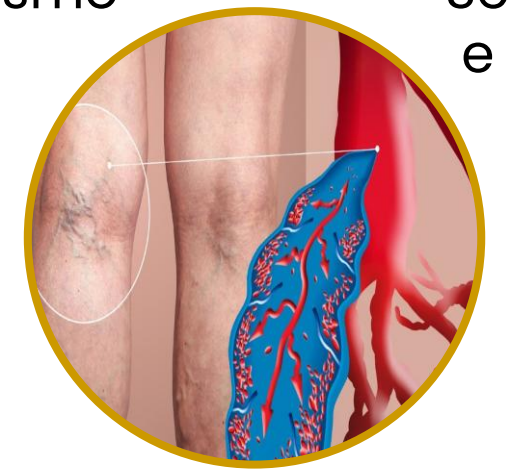
FATORES DE RISCO – TROMBOSE

Tabagismo

Sedentarismo
e Obesidade

Gravidez

Idade
(60+ Anos)

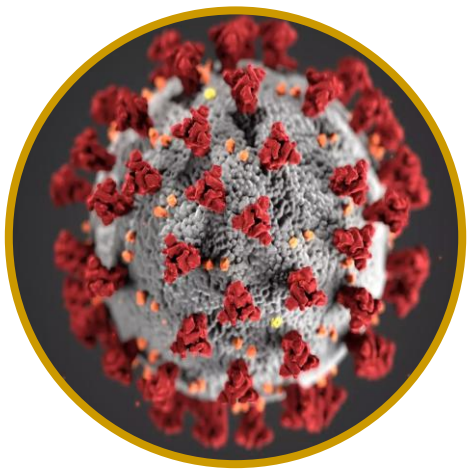


Varizes

Uso de Anticoncepcionais

Hereditariedade

RISCO DE TROMBOSE



COVID-19
09 – 20%

Gravidez
0,14%

Tabagismo
0,18%

Uso de Anticoncepcionais
0,05 – 0,12%

Vacinas
0,0004%

OSFORD | ASTRAZENECA

Dados Importantes:

- ✓ 35 milhões de doses aplicadas na Europa:
  - ➢ **222 casos de coágulos:**
    - ❖ 1 caso para <u>cada 175.000 vacinados</u>.

Dados Importantes:

- ✓ Aproximadamente <u>uma em cada 1.000 mulheres que tomam anticoncepcionais sofrem um coágulo deste tipo</u>:
  - ➢ **Melhor vacinar ou melhor arriscar?.**

Safety and immunogenicity of ChAdOx1 nCoV-19 vaccine administered in a prime-boost regimen in young and old adults (COV002): a single-blind, randomised, controlled, phase 2/3 trial

Maheshi N Ramasamy*, Angela M Minassian*, Katie J Ewer*, Amy L Flaxman*, Pedro M Folegatti*, Daniel R Owens*, Merryn Voysey*, Parvinder K Aley, Brian Angus, Gavin Babbage, Sandra Belij-Rammerstorfer, Lisa Berry, Sagida Bibi, Mustapha Bittaye, Katrina Cathie, Harry Chappell, Sue Charlton, Paola Cicconi, Elizabeth A Clutterbuck, Rachel Colin-Jones, Christina Dold, Katherine R W Emary, Sofiyah Fedosyuk, Michelle Fuskova, Diane Gbesemete, Catherine Green, Bassam Hallis, Mimi M Hou, Daniel Jenkin, Carina C D Joe, Elizabeth J Kelly, Simon Kerridge, Alison M Lawrie, Alice Lelliott, May N Lwin, Rebecca Makinson, Natalie G Marchevsky, Yama Mujadidi, Alexdair P S Munro, Mihaela Pacurar, Emma Plested, Jade Rand, Thomas Rawlinson, Sarah Rhead, Hannah Robinson, Adam J Ritchie, Amy L Ross-Russell, Stephen Saich, Nisha Singh, Catherine C Smith, Matthew D Snape, Rinn Song, Richard Tarrant, Yrene Themistocleous, Kelly M Thomas, Tonya L Villafana, Sarah C Warren, Marion E E Watson, Alexander D Douglas*, Adrian V S Hill*, Teresa Lambe*, Sarah C Gilbert*, Saul N Faust*, Andrew J Pollard*, and the Oxford COVID Vaccine Trial Group

Summary

Background Older adults (aged ≥70 years) are at increased risk of severe disease and death if they develop COVID-19 and are therefore a priority for immunisation should an efficacious vaccine be developed. Immunogenicity of vaccines is often worse in older adults as a result of immunosenescence. We have reported the immunogenicity of a novel chimpanzee adenovirus-vectored vaccine, ChAdOx1 nCoV-19, in young adults, and now describe the safety and immunogenicity of this vaccine in a wider range of participants, including adults aged 70 years and older.

Methods In this report of the phase 2 component of a single-blind, randomised, controlled, phase 2/3 trial (COV002), healthy adults aged 18 years and older were enrolled at two UK clinical research facilities, in an age-escalation manner, into 18–55 years, 56–69 years, and 70 years and older immunogenicity subgroups. Participants were eligible if they did not have severe or uncontrolled medical comorbidities or a high frailty score (if aged ≥65 years). First, participants were recruited to a low-dose cohort, and within each age group, participants were randomly assigned to receive either intramuscular ChAdOx1 nCoV-19 ($2·2 \times 10^{10}$ virus particles) or a control vaccine, MenACWY, using block randomisation and stratified by age and dose group and study site, using the following ratios: in the 18–55 years group, 1:1 to either two doses of ChAdOx1 nCoV-19 or two doses of MenACWY; in the 56–69 years group, 3:1:3:1 to one dose of ChAdOx1 nCoV-19, one dose of MenACWY, two doses of ChAdOx1 nCoV-19, or two doses of MenACWY; and in the 70 years and older, 5:1:5:1 to one dose of ChAdOx1 nCoV-19, one dose of MenACWY, two doses of ChAdOx1 nCoV-19, or two doses of MenACWY. Prime-booster regimens were given 28 days apart. Participants were then recruited to the standard-dose cohort ($3·5–6·5 \times 10^{10}$ virus particles of ChAdOx1 nCoV-19) and the same randomisation procedures were followed, except the 18–55 years group was assigned in a 5:1 ratio to two doses of ChAdOx1 nCoV-19 or two doses of MenACWY. Participants and investigators, but not staff administering the vaccine, were masked to vaccine allocation. The specific objectives of this report were to assess the safety and humoral and cellular immunogenicity of a single-dose and two-dose schedule in adults older than 55 years. Humoral responses at baseline and after each vaccination until 1 year after the booster were assessed using an in-house standardised ELISA, a multiplex immunoassay, and a live severe acute respiratory syndrome coronavirus 2 (SARS-CoV-2) microneutralisation assay (MNA_{80}). Cellular responses were assessed using an ex-vivo IFN-γ enzyme-linked immunospot assay. The coprimary outcomes of the trial were efficacy, as measured by the number of cases of symptomatic, virologically confirmed COVID-19, and safety, as measured by the occurrence of serious adverse events. Analyses were by group allocation in participants who received the vaccine. Here, we report the preliminary findings on safety, reactogenicity, and cellular and humoral immune responses. This study is ongoing and is registered with ClinicalTrials.gov, NCT04400838, and ISRCTN, 15281137.

Findings Between May 30 and Aug 8, 2020, 560 participants were enrolled: 160 aged 18–55 years (100 assigned to ChAdOx1 nCoV-19, 60 assigned to MenACWY), 160 aged 56–69 years (120 assigned to ChAdOx1 nCoV-19: 40 assigned to MenACWY), and 240 aged 70 years and older (200 assigned to ChAdOx1 nCoV-19: 40 assigned to MenACWY). Seven participants did not receive the boost dose of their assigned two-dose regimen, one participant received the incorrect vaccine, and three were excluded from immunogenicity analyses due to incorrectly labelled samples. 280 (50%) of 552 analysable participants were female. Local and systemic reactions were more common in participants given ChAdOx1 nCoV-19 than in those given the control vaccine, and similar in nature to those previously reported

Published Online November 19, 2020 https://doi.org/10.1016/S0140-6736(20)32466-1

See Online/Comment https://doi.org/10.1016/S0140-6736(20)32481-8

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Fonte: https://www.thelancet.com/action/showPdf?pii=S0140-6736%2820%2932466-1

SPUTNIK V

- ✓ Produzidas pela *N. F. Gamaleya National Center of Epidemiology and Microbiology*,

Informações Técnicas:

- ✓ **Eficácias:** 91,6%;
- ✓ **Duas doses:** com Adenovírus diferentes em cada dose, em um intervalo de 21 dias.

SPUTNIK V

INFORMAÇÃO IMPORTANTE:

- ✓ **VETADA PELA ANVISA:**
  - ➢ Um dos Adenovírus se demonstrou ser replicante;
  - ➢ Nunca foi incorporada ao PNI.

Sputnik Light:

- ✓ **Eficácia:** 79,4%;
- ✓ **Dose única.**

Safety and efficacy of an rAd26 and rAd5 vector-based heterologous prime-boost COVID-19 vaccine: an interim analysis of a randomised controlled phase 3 trial in Russia

Denis Y Logunov*, Inna V Dolzhikova*, Dmitry V Shcheblyakov, Amir I Tukhvatulin, Olga V Zubkova, Alina S Dzharullaeva, Anna V Kovyrshina, Nadezhda L Lubenets, Daria M Grousova, Alina S Erokhova, Andrei G Botikov, Fatima M Izhaeva, Olga Popova, Tatiana A Ozharovskaya, Ilias B Esmagambetov, Irina A Favorskaya, Denis I Zrelkin, Daria V Voronina, Dmitry N Shcherbinin, Alexander S Semikhin, Yana V Simakova, Elizaveta A Tokarskaya, Daria A Egorova, Maksim M Shmarov, Natalia A Nikitenko, Vladimir A Gushchin, Elena A Smolyarchuk, Sergey K Zyryanov, Sergei V Borisevich, Boris S Naroditsky, Alexander L Gintsburg, and the Gam-COVID-Vac Vaccine Trial Group†

Summary

Background A heterologous recombinant adenovirus (rAd)-based vaccine, Gam-COVID-Vac (Sputnik V), showed a good safety profile and induced strong humoral and cellular immune responses in participants in phase 1/2 clinical trials. Here, we report preliminary results on the efficacy and safety of Gam-COVID-Vac from the interim analysis of this phase 3 trial.

Methods We did a randomised, double-blind, placebo-controlled, phase 3 trial at 25 hospitals and polyclinics in Moscow, Russia. We included participants aged at least 18 years, with negative SARS-CoV-2 PCR and IgG and IgM tests, no infectious diseases in the 14 days before enrolment, and no other vaccinations in the 30 days before enrolment. Participants were randomly assigned (3:1) to receive vaccine or placebo, with stratification by age group. Investigators, participants, and all study staff were masked to group assignment. The vaccine was administered (0·5 mL/dose) intramuscularly in a prime-boost regimen: a 21-day interval between the first dose (rAd26) and the second dose (rAd5), both vectors carrying the gene for the full-length SARS-CoV-2 glycoprotein S. The primary outcome was the proportion of participants with PCR-confirmed COVID-19 from day 21 after receiving the first dose. All analyses excluded participants with protocol violations: the primary outcome was assessed in participants who had received two doses of vaccine or placebo, serious adverse events were assessed in all participants who had received at least one dose at the time of database lock, and rare adverse events were assessed in all participants who had received two doses and for whom all available data were verified in the case report form at the time of database lock. The trial is registered at ClinicalTrials.gov (NCT04530396).

Findings Between Sept 7 and Nov 24, 2020, 21 977 adults were randomly assigned to the vaccine group (n=16 501) or the placebo group (n=5476). 19 866 received two doses of vaccine or placebo and were included in the primary outcome analysis. From 21 days after the first dose of vaccine (the day of dose 2), 16 (0·1%) of 14 964 participants in the vaccine group and 62 (1·3%) of 4902 in the placebo group were confirmed to have COVID-19; vaccine efficacy was 91·6% (95% CI 85·6–95·2). Most reported adverse events were grade 1 (7485 [94·0%] of 7966 total events). 45 (0·3%) of 16 427 participants in the vaccine group and 23 (0·4%) of 5435 participants in the placebo group had serious adverse events; none were considered associated with vaccination, with confirmation from the independent data monitoring committee. Four deaths were reported during the study (three [<0·1%] of 16 427 participants in the vaccine group and one [<0·1%] of 5435 participants in the placebo group), none of which were considered related to the vaccine.

Interpretation This interim analysis of the phase 3 trial of Gam-COVID-Vac showed 91·6% efficacy against COVID-19 and was well tolerated in a large cohort.

Funding Moscow City Health Department, Russian Direct Investment Fund, Sberbank, and RUSAL.

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Introduction

The COVID-19 pandemic has led to more than 98 million confirmed cases and more than 2 million deaths (at the time of publication). There are a few provisionally licensed vaccines against COVID-19, and global efforts are focusing on developing safe and efficacious vaccines for COVID-19 prevention. According to the WHO draft landscape of COVID-19 candidate vaccines,[1] 64 candidates are in clinical assessment (including 13 at phase 3) and 173 are in preclinical analyses. The phase 3 vaccine candidates include a variety of vaccine platforms: vector vaccines (Gamaleya National Research Centre for Epidemiology and Microbiology [NRCEM; this study], University of Oxford/AstraZeneca,[2] CanSino Biological Inc/Beijing Institute of Biotechnology, and Janssen Pharmaceutical Companies), mRNA-based vaccines

Published Online February 2, 2021 https://doi.org/10.1016/S0140-6736(21)00234-8

See Online/Comment https://doi.org/10.1016/S0140-6736(21)00191-4

* Contributed equally

† Trial group members are listed in the appendix

Federal State Budget Institution "National Research Centre for Epidemiology and Microbiology named after Honorary Academician N F Gamaleya" of the Ministry of Health of the Russian Federation, Moscow, Russia (D Y Logunov DSc, I V Dolzhikova PhD, D V Shcheblyakov PhD, A I Tukhvatulin PhD, O V Zubkova PhD, A S Dzharullaeva MSc, A V Kovyrshina MSc, N L Lubenets MSc, D M Grousova MSc, A S Erokhova MSc, A G Botikov MSc, F M Izhaeva MSc, O Popova M Sc, T A Ozharovskaya M Sc, I B Esmagambetov PhD, I A Favorskaya PhD, D I Zrelkin MSc, D V Voronina M Sc, D N Shcherbinin PhD, A S Semikhin PhD, Y V Simakova MSc, E A Tokarskaya PhD, D A Egorova PhD, M M Shmarov DSc, N A Nikitenko PhD, V A Gushchin PhD, Prof B S Naroditsky DSc, Prof A L Gintsburg DSc); Federal State Autonomous Educational Institution of Higher Education I M Sechenov First Moscow State Medical University of the Ministry of Health of the Russian Federation (Sechenov University), Moscow, Russia (E A Smolyarchuk PhD,

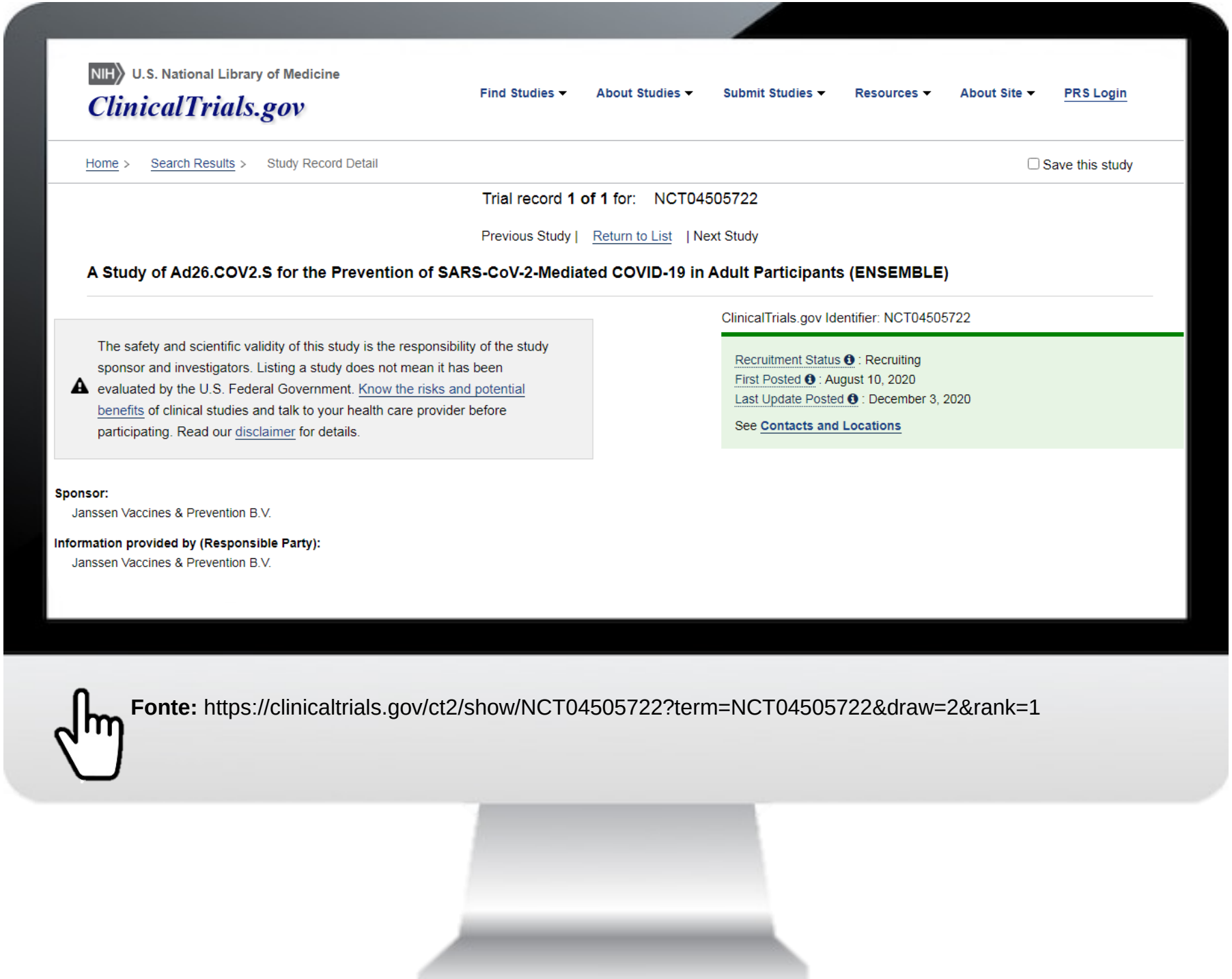
Fonte:https://www.thelancet.com/action/showPdf?pii=S0140-6736%2821%2900234-8

J&J | JANSSEN

- ✓ Produzidas pela farmacêutica Johnson & Johnson;

Informações Técnicas:

- ✓ **Eficácias:** 66%;
- ✓ **Dose Única:** na época, estrategicamente falando, era uma excelente candidata para o Brasil.



Fonte: https://clinicaltrials.gov/ct2/show/NCT04505722?term=NCT04505722&draw=2&rank=1

J&J | JANSSEN

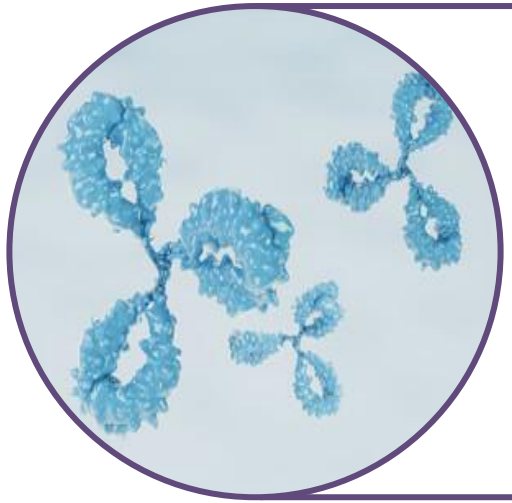
Dados Importantes:

- ✓ 08 milhões de doses aplicadas:
  - ➢ **15 casos de coágulos:**
    - ❖ Representam 0,0001875% dos vacinados;
    - ❖ Melhor vacinar ou melhor arriscar?.

VACINAS DE RNAm
(RNA mensageiro)

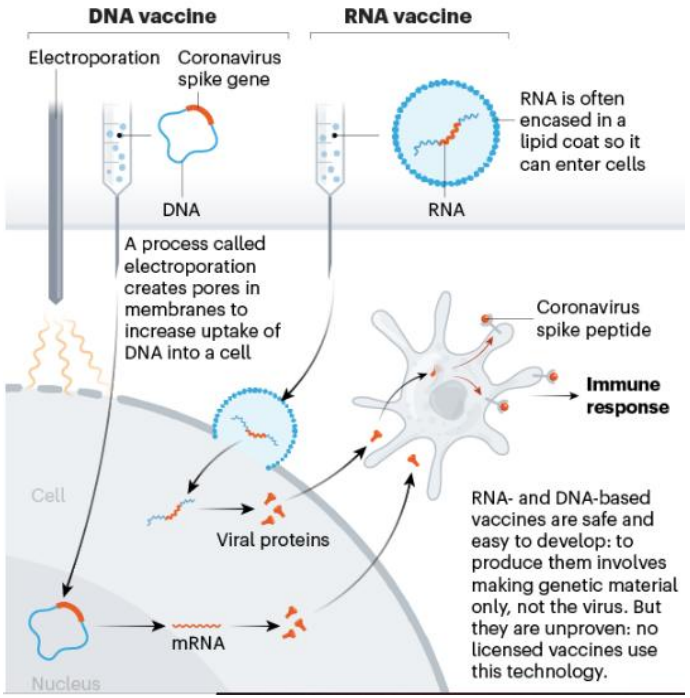
ENGENHARIA GENÉTICA

- ✓ Manipulação genética de microrganismos:
  - ➢ **Deleção** ou **inativação dos genes que codificam os fatores de virulência**.



- ✓ Sejam capazes de **biossintetizar os antígenos** que <u>desencadeiam a produção de anticorpos protetores</u>:
  - ➢ Induzem memória de longo prazo.

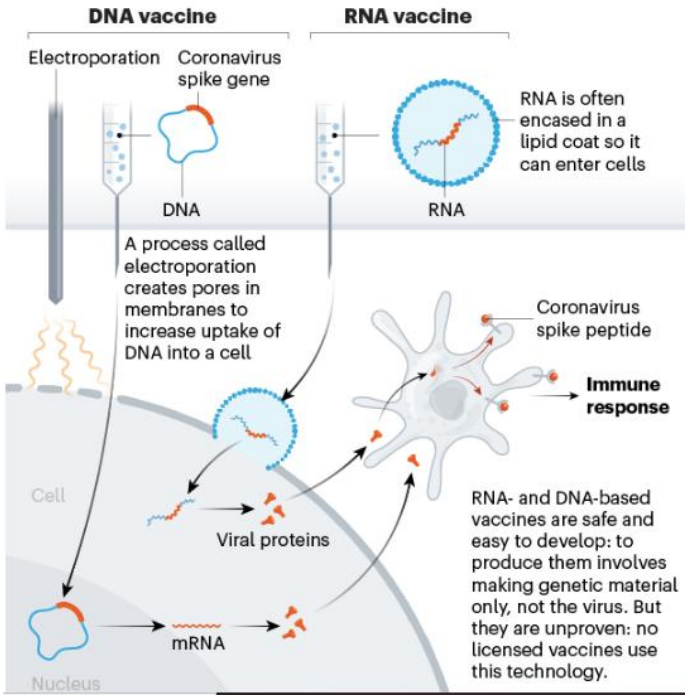
VACINAS DE RNAm

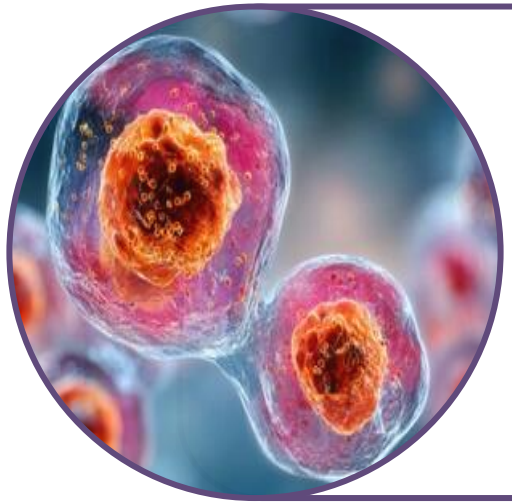


- ✓ Introduzir no receptor do imunizante a própria molécula do RNAm :
  - ➢ A <u>"receita química"</u> usada no processo celular para estimular defesa contra o patógeno.

- ✓ **Mais rápidas e fáceis de serem produzidas:**
  - ➢ O RNA é sintetizado em laboratório e no organismo é transportado pelos lipídios.

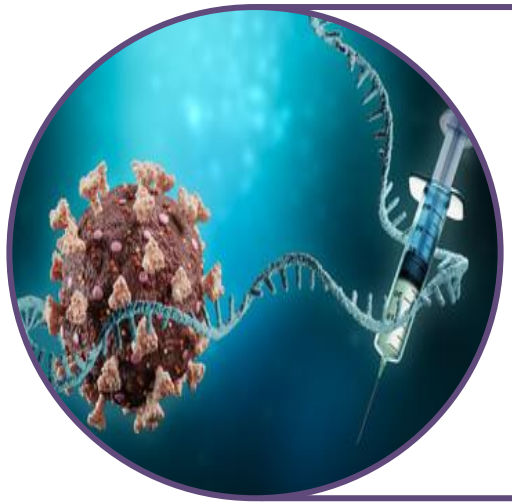
VACINAS DE RNAm





Lipossolubilidade:

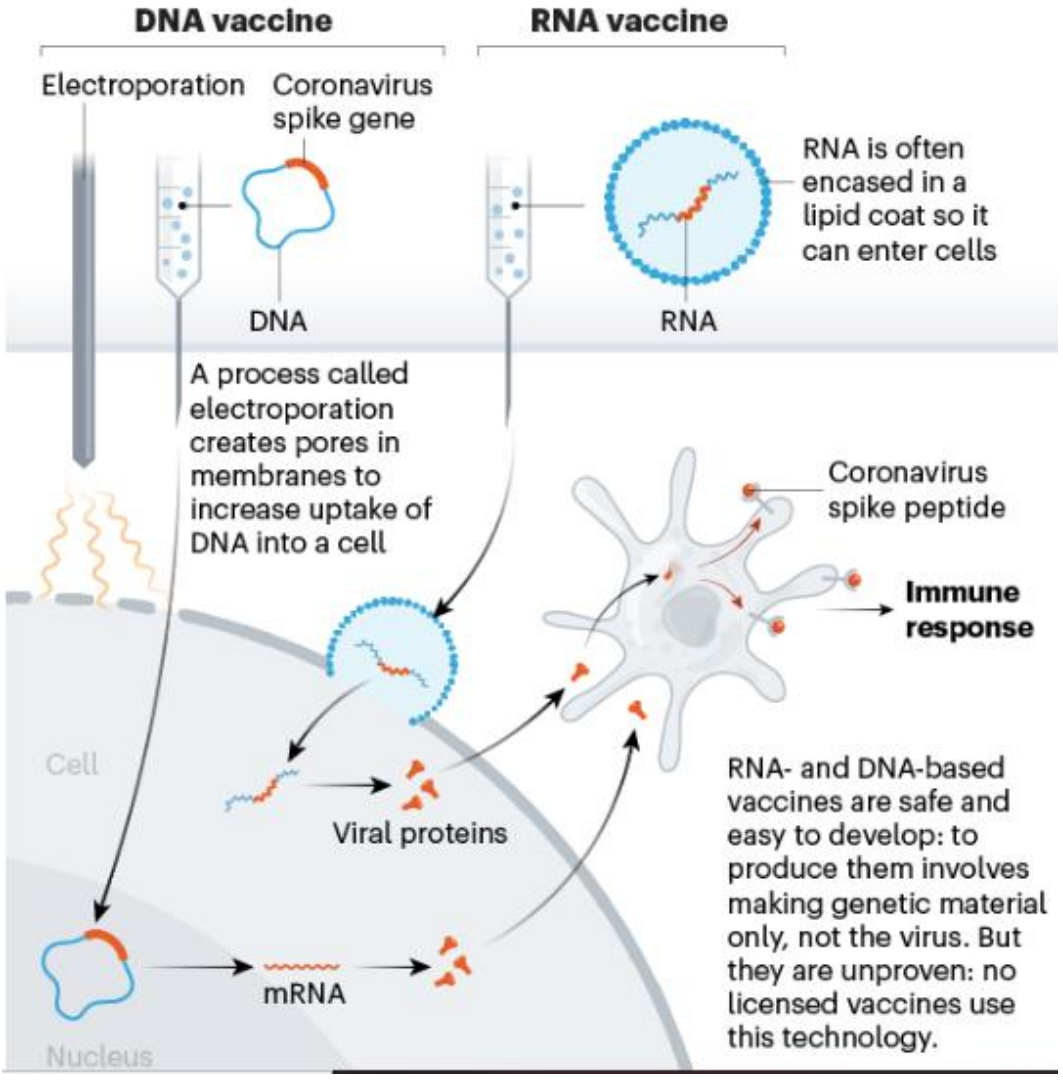
- ✓ Atravessam facilmente as células humanas:
  - ✓ O RNA vai ficar no citoplasma e produz a proteína de interesse.



COVID-19:

- ✓ A proteína *spike*:
  - ➢ NÃO ENTRA NO NÚCLEO!

VACINAS DE RNAm



FASES DE TESTES NO DESENVOLVIMENTO DE VACINAS

Fase	Objetivo	Número de participantes
Pré-clínica	Testagem com modelos animais	—
Fase 1	Teste de segurança[1]	Dezenas de pessoas
Fase 2	Teste de eficácia[2] e de efeitos colaterais **A** Prova de conceito[3] **B** Dose ótima[4]	Centenas de pessoas
Fase 3	Teste de eficácia, efetividade[5] e segurança	Milhares a dezenas de milhares de pessoas
Fase 4	Resultados após aplicação massiva na população	Centenas de milhares a milhões de pessoas

Coordenação	País	Plataforma	Fase de desenvolvimento
Sinovac (CoronaVac)	China	Vírus inativado	**Fase 3** Iniciou ensaio clínico no Brasil em 21/7 em parceria com Instituto Butantan (SP), e em agosto na Indonésia; no Brasil a Anvisa autorizou a importação de 6 milhões de doses pelo governo paulista, e de insumos para a produção de outros 40 milhões de doses pelo Butantan; em 9/11 os testes foram suspensos para investigação de morte de participante, mas logo retomados por não ter relação com a vacinação; em 3/12 um carregamento a São Paulo e o governo paulista anunciou o plano de iniciar a vacinação em janeiro de 2021
BioNTech, Fosun Pharma, Pfizer	EUA	RNA	**Fase 2/3** Ensaio em fases combinadas nos Estados Unidos, na Argentina e no Brasil, e em fase 3 na África do Sul, na Alemanha e na Turquia; em 18/11 a empresa anunciou resultados finais que indicam uma eficácia de 95% - e 94% em idosos; o desafio de produção e distribuição é grande, já que as doses precisam ser mantidas a -70°C; em 2/12 o Reino Unido obteve autorização para vacinação emergencial
Moderna e Inst. Nacional de Alergias e Doenças Infecciosas (Niaid)	EUA	RNA	**Fase 3** Ensaio clínico fase 3 iniciado em 27/7 nos Estados Unidos; no dia 22/10 atingiu a meta de recrutar 30 mil voluntários; em 16/11 anunciou uma eficácia de 94.5% entre 95 infectados entre os voluntários; em 30/11, submeteu pedido de autorização para uso emergencial
Universidade de Oxford e AstraZeneca	Reino Unido	Vetor viral não replicante	**Fase 2b/3** Iniciados em 17/08, ensaios clínicos em fase 2/3 na Inglaterra e na Índia, e fase 3 no Brasil, na África do Sul e nos Estados Unidos foram interrompidos para investigação de efeitos adversos e depois retomados – nos EUA por último, em 23/10; dados anunciados em 23/11 indicam uma eficácia de 90% caso seja aplicada uma meia dose, seguida por uma dose completa – em contraposição a 60% com duas doses completas; a empresa tem enfrentado questionamentos sobre os procedimentos de testes
Inst. Produtos Biológicos de Wuhan/Beijing e Sinopharm	China	Vírus inativado	**Fase 3** Ensaio clínico fase 3 iniciado em julho nos Emirados Árabes Unidos, onde em 14/9 foi autorizado o uso emergencial para profissionais da saúde, em agosto no Peru e em Marrocos, e em setembro na Argentina; em 14/10 foram noticiados planos de oferecer a vacina gratuitamente para chineses que pretendessem fazer estudos superiores no exterior
Companhia Farmacêutica Janssen	EUA	Vetor viral não replicante	**Fase 3** Início de ensaio clínico fase 3 em curso nos Estados Unidos e no Reino Unido; em 12/10 ensaio foi paralisado para investigação de efeito adverso e retomado
Novavax	EUA	Subunidade proteica	**Fase 3** Iniciou fase 2b de ensaios clínicos em agosto na África do Sul e fase 3 em setembro no Reino Unido; início da fase 3 nos Estados Unidos está previsto para o início de novembro
CanSino e Inst. Biotecnológico de Beijing	China	Vetor viral não replicante	**Fase 3** Ensaio clínico fase 3 iniciado em 11/9 na Rússia, 15/9 no Paquistão e em 30/10 no México
Instituto Gamaleia, Moscou (Sputnik V)	Rússia	Vetor viral não replicante	**Fase 3** Iniciou fase 3 de ensaio clínico em setembro na Rússia e na Bielorrússia; em 2/12 governo russo autorizou a vacinação em massa, mesmo antes de se finalizarem testes
Bharat Biotech (Covaxin)	Índia	Vírus inativado	**Fase 2/3** Iniciou ensaios clínicos em julho na Índia; iniciou fase 3 em 11/11
Instituto Vector (EpiVacCorona)	Rússia	Peptídeo	**Fase 1/2** Ensaio clínico em fases 1 e 2 conjugadas foi registrado em agosto para teste em 100 participantes; em 14 de outubro o governo russo anunciou a aprovação da vacina antes mesmo de iniciada um fase 3 de testes
Fiocruz (RJ) e Instituto Butantan (SP)	Brasil	Vetor viral replicante	**Pré-clínica**
Universidade de São Paulo (USP)	Brasil	Partícula semelhante a vírus (VLP)	**Pré-clínica**

Fonte: https://revistapesquisa.fapesp.br/vacinas-em-desenvolvimento-contra-covid-19

PFIZER | BIONTECH

Transmissão:

- ✓ Demonstrou redução de 75% menos de um mês após a aplicação da primeira dose.

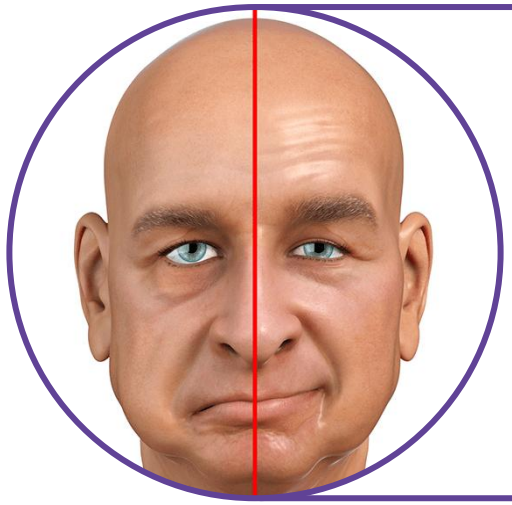
Informações Técnicas:

- ✓ **Eficácia:** 95%;
- ✓ **Duas doses:** com intervalo de 28 dias;
- ✓ **Armazenamento:** -70°C e -25°C.

PFIZER | BIONTECH

Crianças e Adolescentes:

- ✓ **100% de Eficácia:**
  - ➢ 2.259 Adolescentes: 12 e 15 anos.
- ✓ **90,7% de Eficácia:**
  - ➢ 2.268 Crianças: 05 e 11 anos.



Dados Importantes:

- ✓ Milhões de doses aplicadas:
  - ➢ 66 casos de *Bells Palsy* e foram controlados;
  - ➢ Melhor vacinar ou melhor arriscar?.

The NEW ENGLAND JOURNAL of MEDICINE

RESEARCH SUMMARY

Safety and Efficacy of the BNT162b2 mRNA Covid-19 Vaccine

F.P. Polack, et al. DOI: 10.1056/NEJMoa2034577

CLINICAL PROBLEM

Safe and effective vaccines to prevent severe acute respiratory syndrome coronavirus 2 (SARS-CoV-2) infection and Covid-19 are urgently needed. No vaccines that protect against betacoronaviruses are currently available, and mRNA-based vaccines have not been widely tested.

CLINICAL TRIAL

A randomized, double-blind study of an mRNA vaccine encoding the SARS-CoV-2 spike protein.

43,548 participants ≥16 years old were assigned to receive the vaccine or placebo by intramuscular injection on day 0 and day 21. Participants were followed for safety and for the development of symptomatic Covid-19 for a median of 2 months.

RESULTS

Safety:
Vaccine recipients had local reactions (pain, erythema, swelling) and systemic reactions (e.g., fever, headache, myalgias) at higher rates than placebo recipients, with more reactions following the second dose. Most were mild to moderate and resolved rapidly.

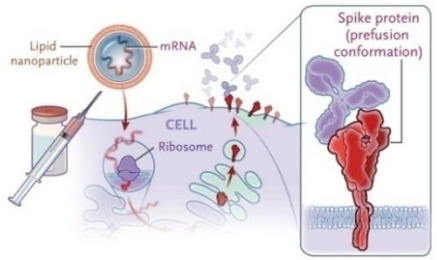
Efficacy:
The vaccine showed some early protection 12 days after the first dose; 7 days after the second dose, 95% efficacy was observed.

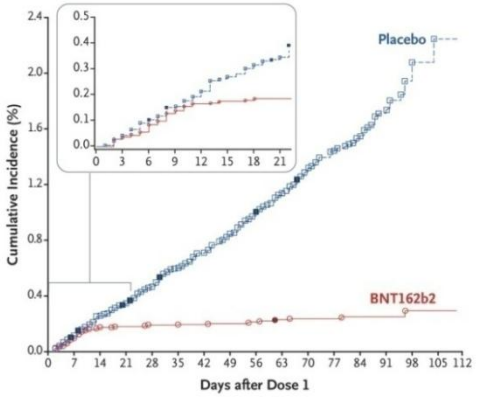
LIMITATIONS AND REMAINING QUESTIONS

Further study is required to understand the following:

- Safety and efficacy beyond 2 months and in groups not included in this trial (e.g., children, pregnant women, and immunocompromised persons).
- Whether the vaccine protects against asymptomatic infection and transmission to unvaccinated persons.
- How to deal with those who miss the second vaccine dose.

Links: Full article | NEJM QuickTake | Editorial





	BNT162b2 Vaccine	Placebo
Symptomatic Covid-19	8	162
	N=18198	N=18325
Severe Covid-19	1	9
	N=21669	N=21686

Vaccine efficacy of 95% (95% credible interval, 90.3–97.6%)

CONCLUSIONS

Two doses of an mRNA-based vaccine were safe over a median of two months and provided 95% protection against symptomatic Covid-19 in persons 16 years of age or older.

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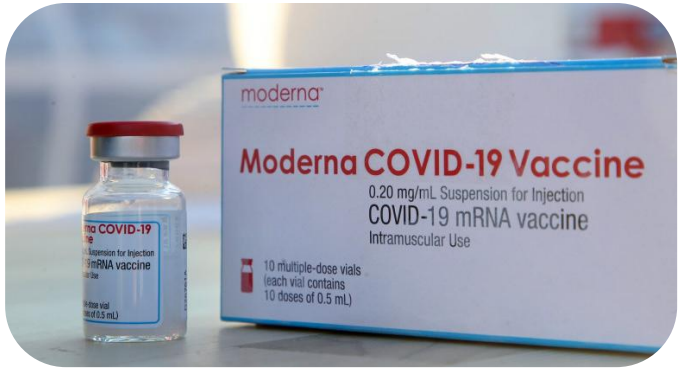
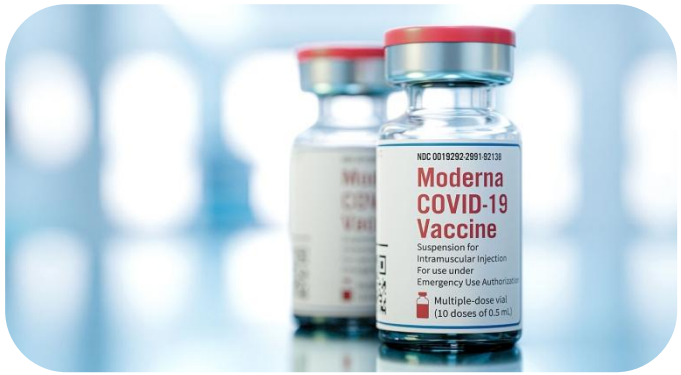
Fonte: https://www.nejm.org/doi/full/10.1056/NEJMoa2034577

MODERNA





Informações Técnicas:

- ✓ **Eficácia:** 94,1%;
- ✓ **Duas doses:** com intervalo de 28 dias;
- ✓ **Armazenamento:** -20°C.

The NEW ENGLAND JOURNAL of MEDICINE

RESEARCH SUMMARY

Efficacy and Safety of mRNA-1273 SARS-CoV-2 Vaccine

L.R. Baden, et al. DOI: 10.1056/NEJMoa2035389

CLINICAL PROBLEM

The Covid-19 pandemic continues and expands. Additional data regarding vaccines to prevent symptomatic severe acute respiratory syndrome coronavirus 2 (SARS-CoV-2) infection are needed. The mRNA-1273 vaccine is a lipid-encapsulated mRNA vaccine encoding the prefusion stabilized spike protein of SARS-CoV-2.

CLINICAL TRIAL

A randomized, double-blind trial to evaluate the efficacy and safety of mRNA-1273.

30,420 participants ≥18 years old were assigned to receive either the vaccine or placebo in two intramuscular injections 28 days apart. Participants were followed for safety and the development of laboratory-confirmed, symptomatic Covid-19 over a median of 2 months after the second dose.

RESULTS

Safety:
Vaccine recipients had higher rates of local reactions (e.g., pain, erythema, swelling) and systemic reactions (e.g., headache, fatigue, myalgia) than placebo recipients. Most reactions were mild to moderate and resolved over 1–3 days.

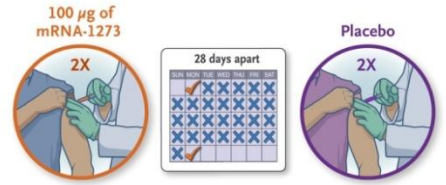
Efficacy:
The incidence of Covid-19 was lower among vaccine recipients than among placebo recipients as early as 14 days after the first dose. Protection in the vaccine group persisted for the period of follow-up.

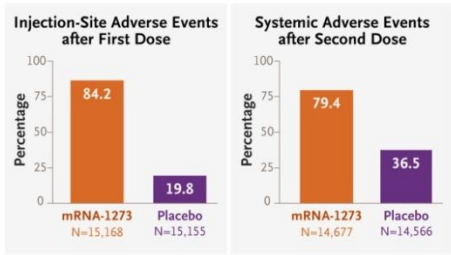
LIMITATIONS AND REMAINING QUESTIONS

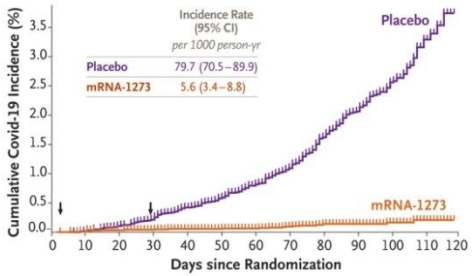
Further study is required to understand the following:

- Safety and efficacy over a longer period of time, in a larger population, and in pregnant women and children.
- Whether the vaccine protects against asymptomatic infection and transmission to unvaccinated persons.
- How to care for those who miss the second vaccine dose.

Links: Full article | NEJM Quick Take | Editorial







	mRNA-1273 Vaccine N=14,550	Placebo N=14,598
Symptomatic Covid-19	11	185
Severe Covid-19	0	30

Vaccine efficacy of 94.1% (95% CI, 89.3–96.8%; P<0.001)

CONCLUSIONS

Two doses of a SARS-CoV-2 mRNA-based vaccine were safe and provided 94% efficacy against symptomatic Covid-19 in persons 18 or older.

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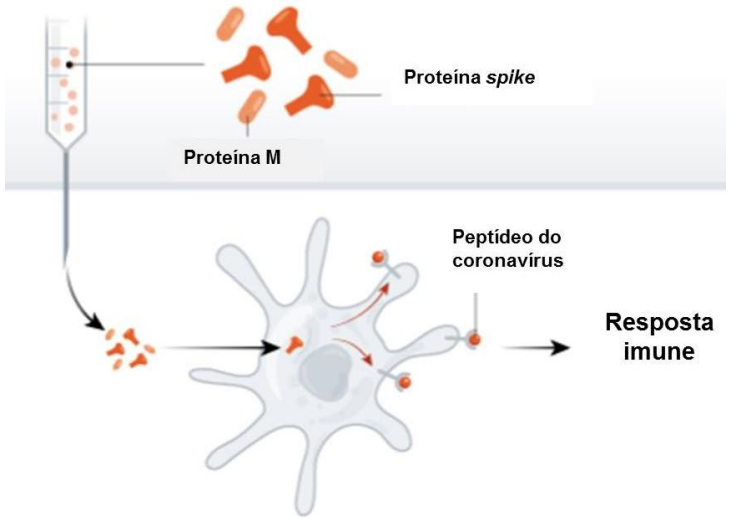
Fonte: https://www.nejm.org/doi/full/10.1056/NEJMoa2035389

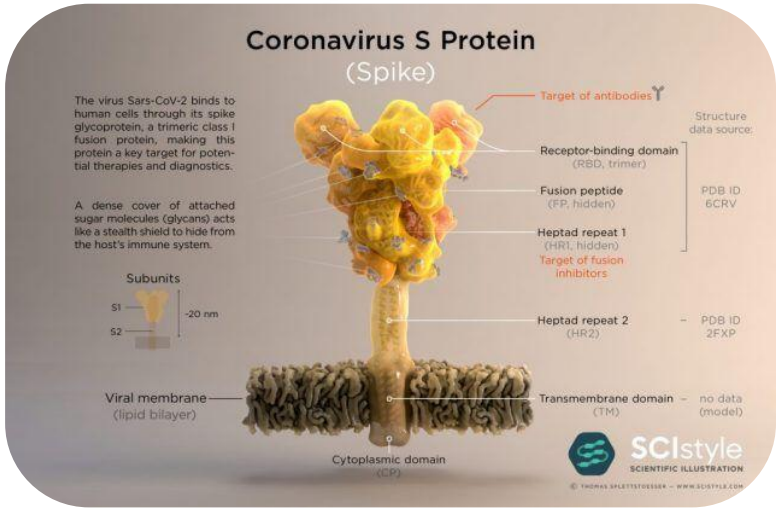
Fonte: https://www.medrxiv.org/content/10.1101/2021.05.06.21256403v1

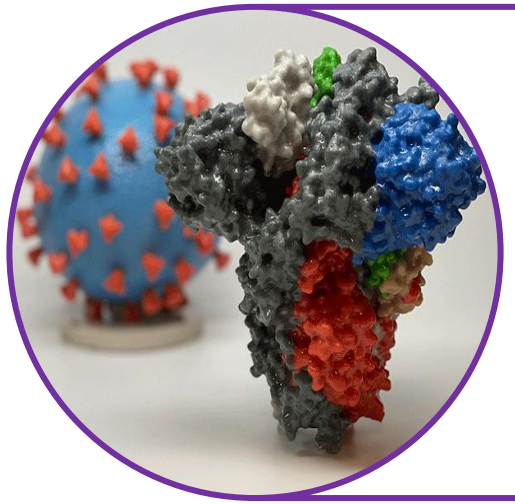
VACINAS DE SUB-UNIDADES PROTEICAS

VACINAS DE SUB-UNIDADES PROTEICAS



- ✓ Não utilizam o patógeno inteiro, mas <u>apenas seus antígenos</u>:
  - ➢ Partes de sua estrutura reconhecidas pelo organismo:
    - ❖ Suficientes para provocar uma resposta imunológica.

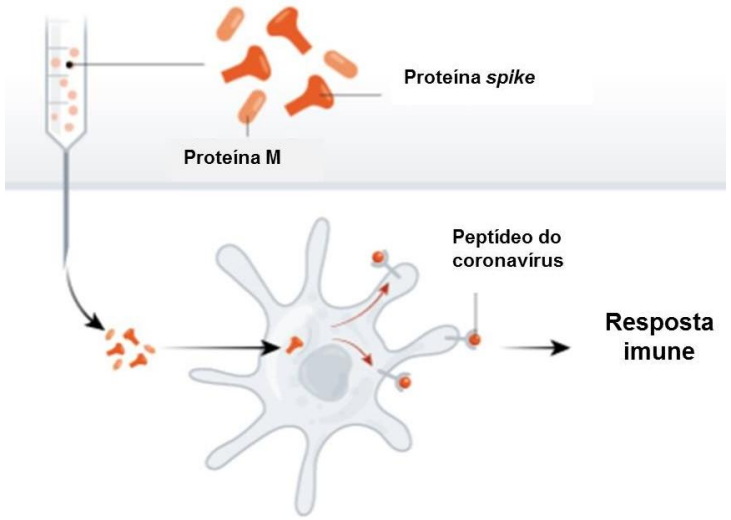


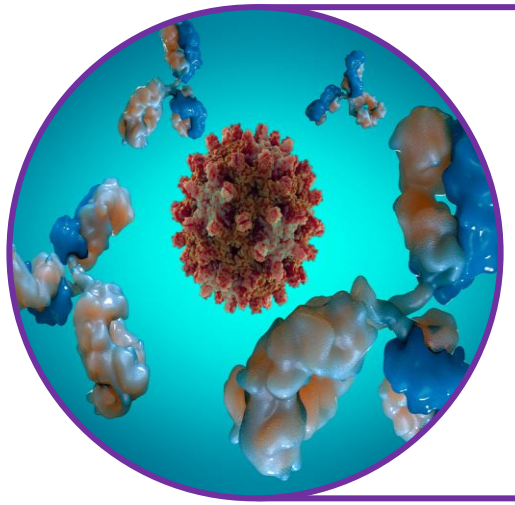


VLP:

- ✓ Partículas similares ao vírus, como se fosse uma “casca”, contendo a estrutura externa, sem o conteúdo interior:
  - ➢ Não possuem o material genético do vírus.

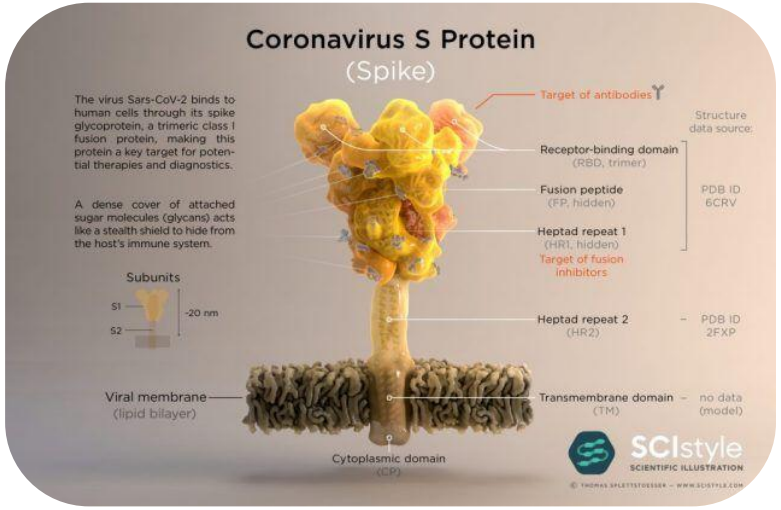
VACINAS DE SUB-UNIDADES PROTEICAS





SARS-CoV-2:

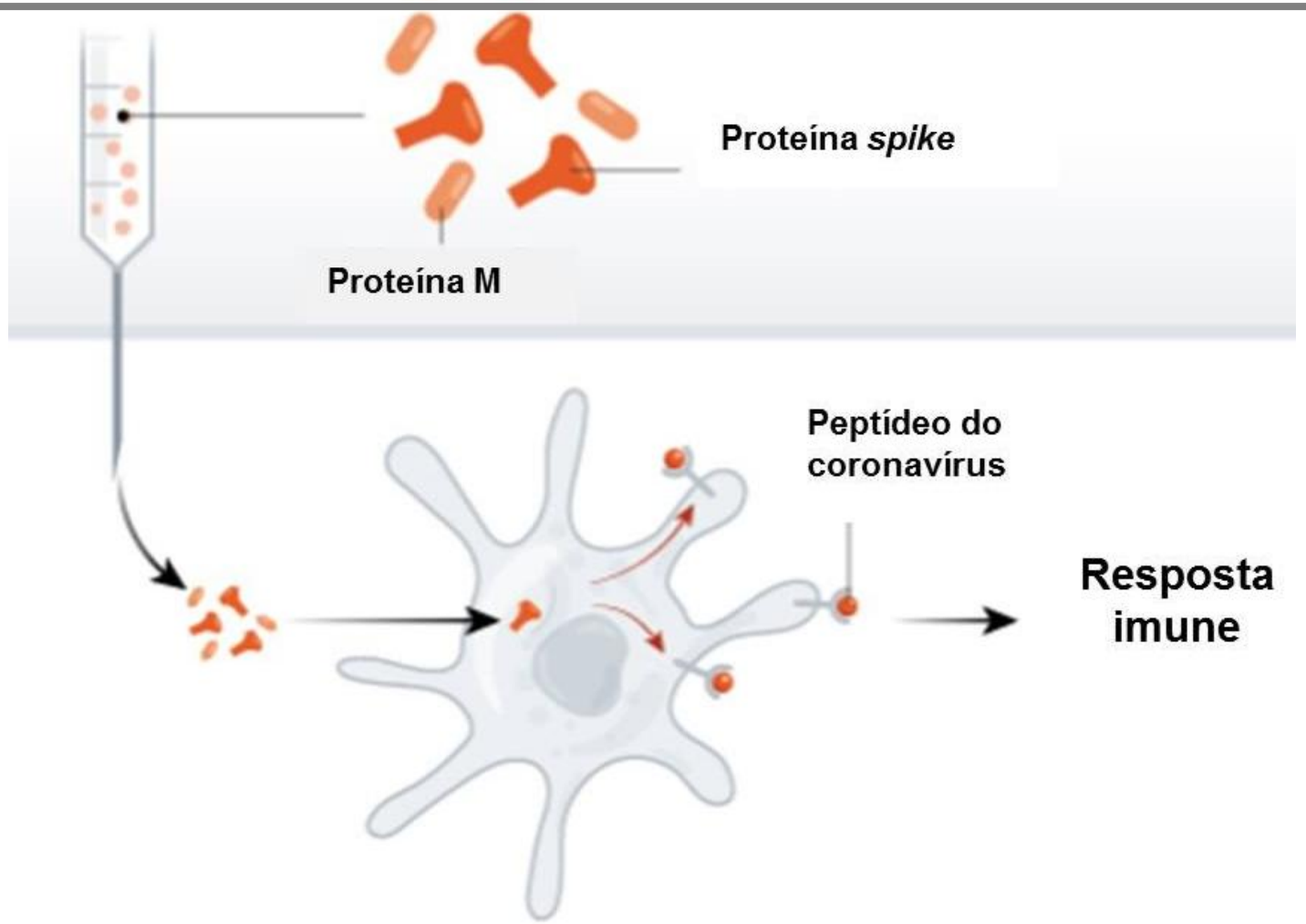
- ✓ Espículas responsáveis por ajudar o vírus a penetrar nas células humanas – *spike*:
  - ➢ Adicionado ao VLP com intuito de estimular o sistema imune.



Exemplos:

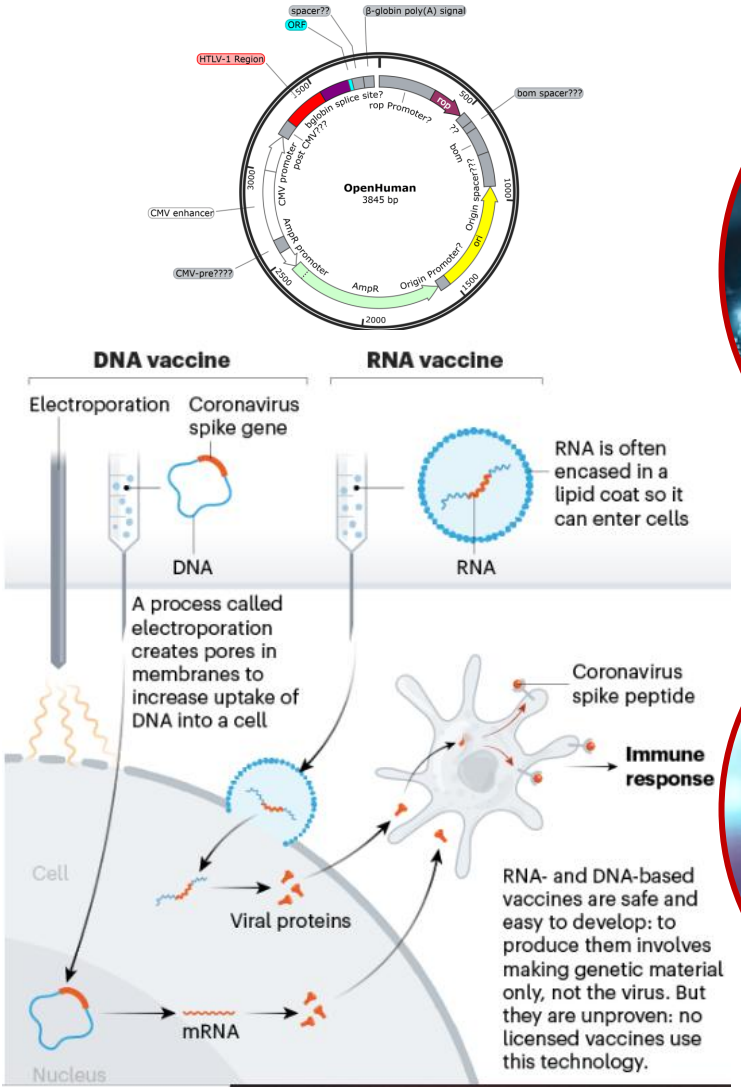
- ✓ HPV;
- ✓ Hepatite B;
- ✓ Herpes-Zóster;
- ✓ COVID-19 (Novavax).

VACINAS DE SUB-UNIDADES PROTEICAS



VACINAS DE DNA

VACINAS DE DNA



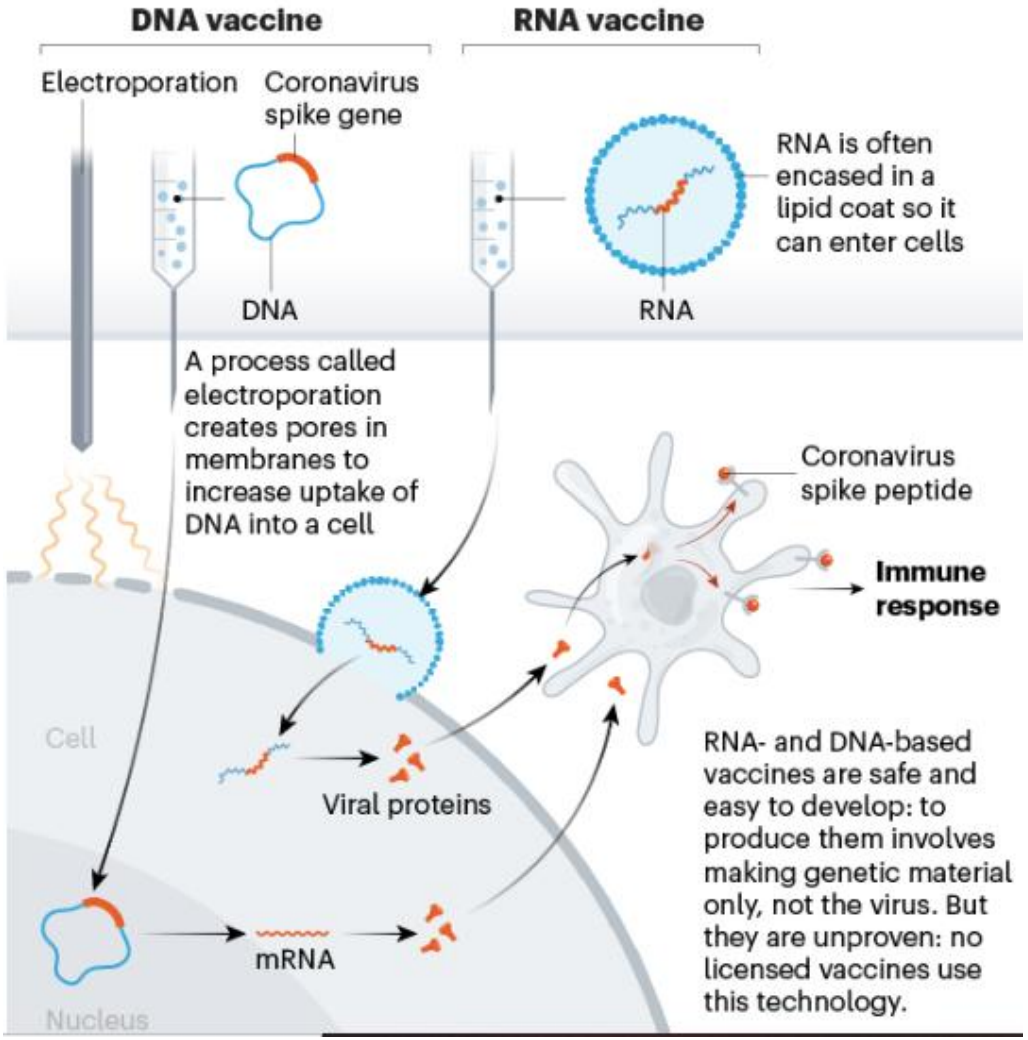
Ainda experimentais:

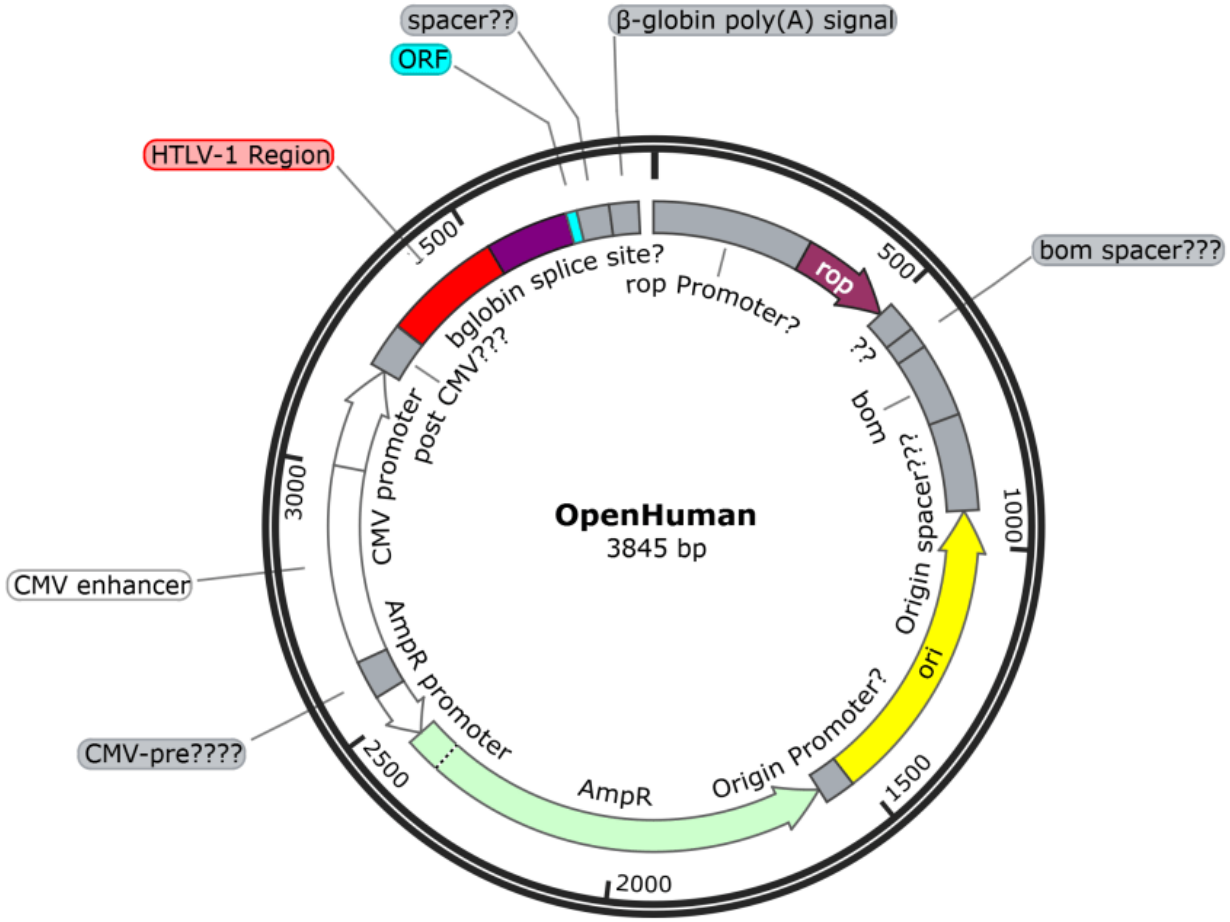
✓ Utilizam <u>plasmídeos contendo DNA</u> que codifica o antígeno ou a sequência genômica desejada.

Fáceis de desenvolver:

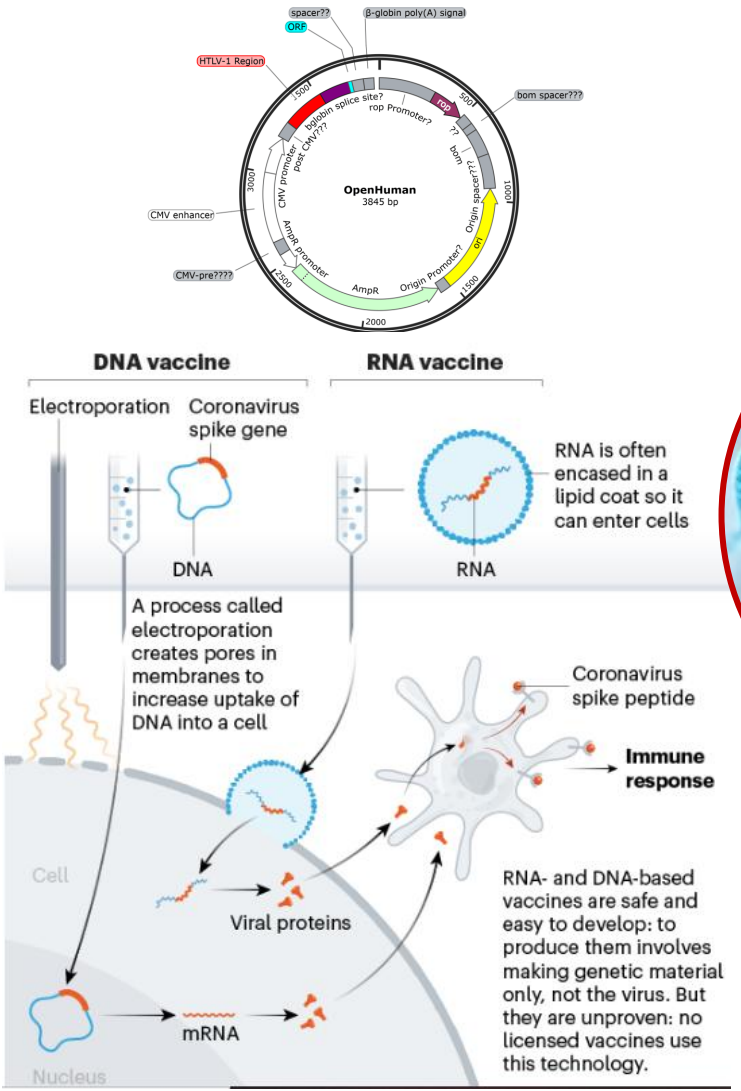
✓ Envolvem apenas o material genético, e não o vírus.

VACINAS DE DNA





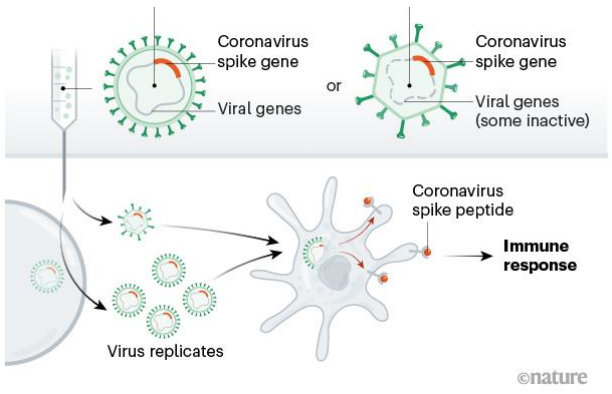
VACINAS DE DNA



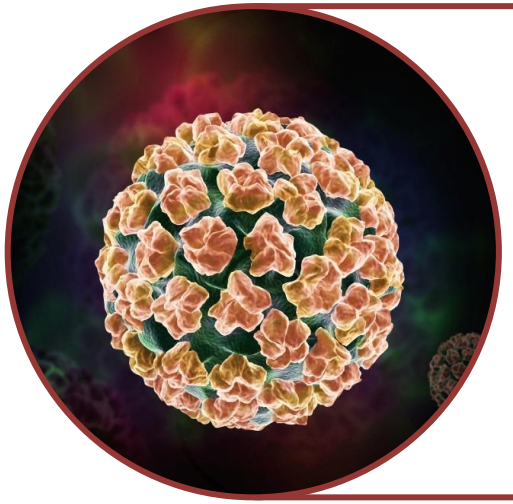
- ✓ As células do organismo, <u>leem a sequência e copiam na forma de uma molécula mais simples (RNAm)</u>:
  - ➢ E produzem o antígeno.

VACINAS VETORIAIS VIRAIS REPLICANTES

VACINAS VETORIAIS VIRAIS REPLICANTES

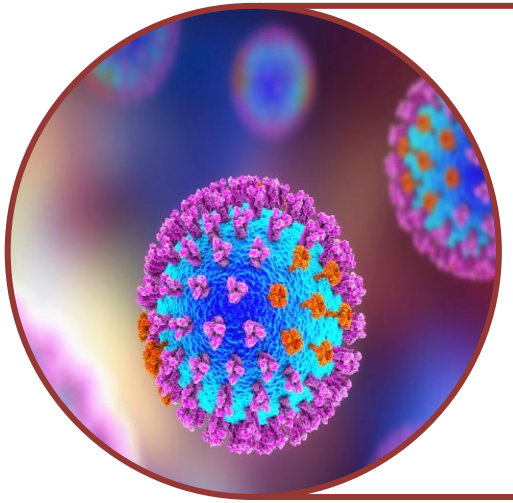


Vacina de vetores virais replicantes (à esquerda) e não replicantes (à direita).



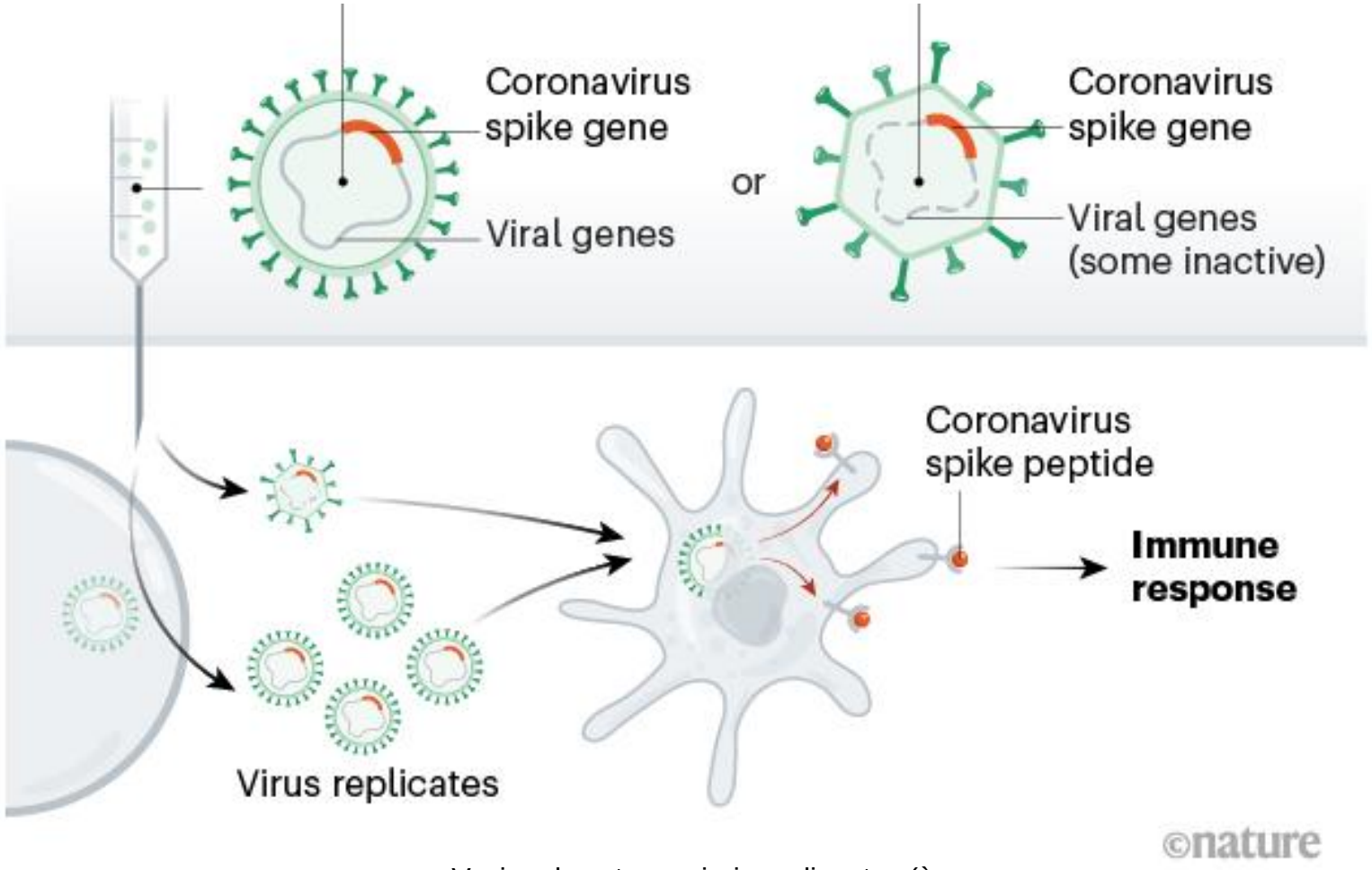
Vetor – Vírus Recombinante:

- ✓ Normalmente é um <u>Sarampo, Influenza, Poliomielite modificado inócuo</u>:
  - ➢ É adicionado o gene que codifica o antígeno específico:
    - ❖ Proteína *spike* do SARS-CoV-2.



- ✓ **O vírus recombinante se reproduz:**
  - ➢ Nas células humanas e Pode não produzir a imunidade desejada se a pessoa tiver anticorpos contra o vetor viral.

VACINAS VETORIAIS VIRAIS REPLICANTES



Vacina de vetores virais replicantes (à esquerda) e não replicantes (à direita).

EFICÁCIA DAS VACINAS

***United States of America* (USA):**

- *About 328 million (328,000,000) people live in the U.S.*

Group	*Group Size*	*Number Infected*	*Infection Risk*	*If in U.S. Population*
Placebo	21,830	162	$\frac{162}{21{,}830} = 0.74\%$	2,427,200 (*about 2.5 million*)
Vaccine	21,830	8	$\frac{8}{21{,}830} = 0.04\%$	131,200 (*131 thousand*)

Difference between the risks in both groups.

- 0.74% - 0.04% = 0.7%

$$\frac{0.7\%}{0.74\%} \cong 95\%$$

Fonte: Adaptado de https://www.nytimes.com/2020/12/13/learning/what-does-95-effective-mean-teaching-the-math-of-vaccine-efficacy.html

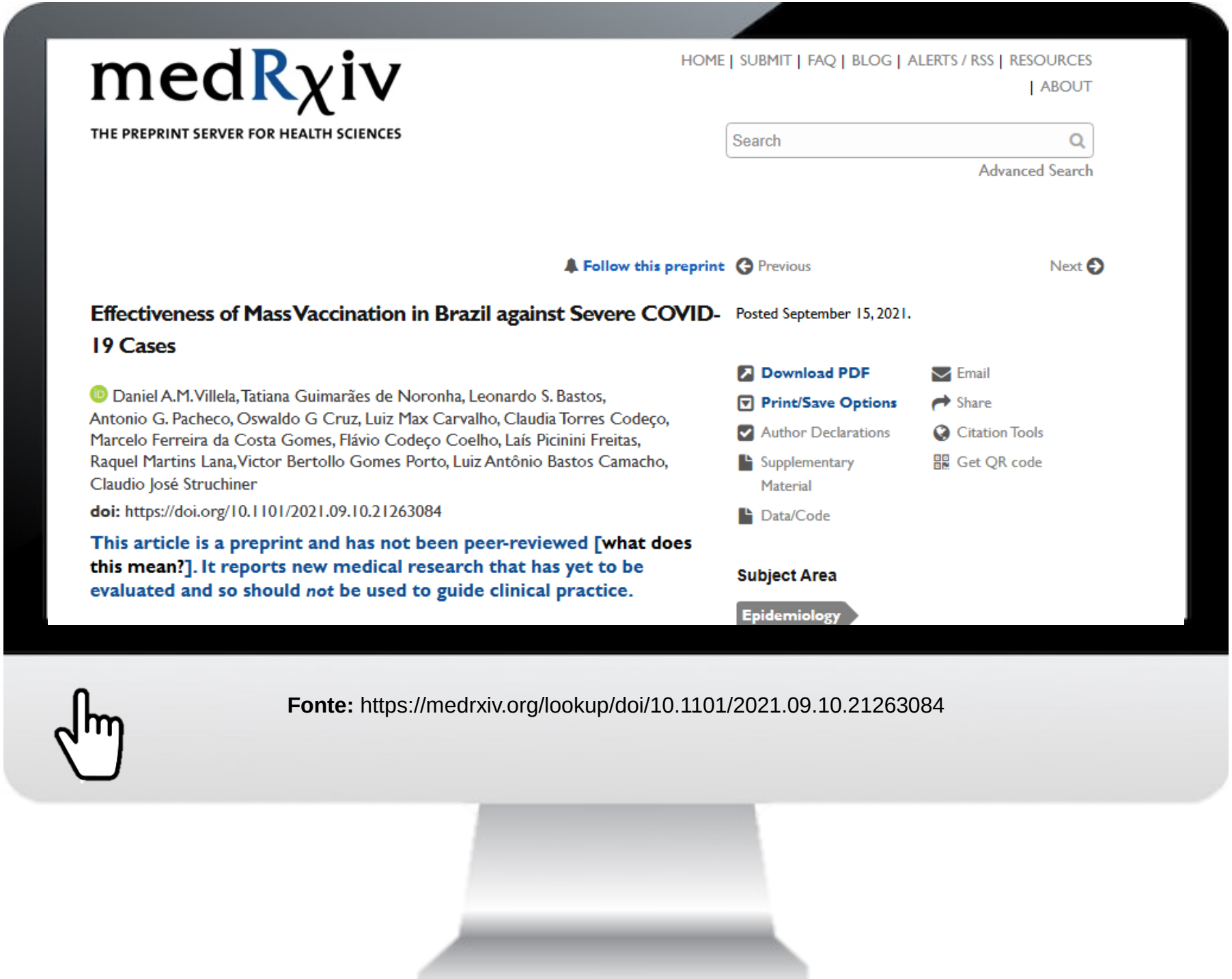
EFICÁCIA DAS VACINAS

Pesquisas de Vacinas:

- Poucas testaram infectividade;
- Tiveram poucos voluntários com desfechos graves.

Vaccine	*Doses*	*Total Sample*	*Total Endpoints*	*All Endpoints*	*Severe Infections*
CoronaVac	2	> 12,000	252	***Vaccine:*** 85 ***Placebo:*** 167	***Vaccine:*** 0 ***Placebo:*** 7
Sputinik V	2	21,977	76	***Vaccine:*** 16 ***Placebo:*** 62	***Vaccine:*** 0 ***Placebo:*** 20
J&J	1	43,783	464	***Vaccine:*** 116 ***Placebo:*** 348	***Vaccine:*** 5 ***Placebo:*** 34
Pfizer	2	43,355	170	***Vaccine:*** 8 ***Placebo:*** 162	***Vaccine:*** 1 ***Placebo:*** 9
Moderna	2	28,207	236	***Vaccine:*** 11 ***Placebo:*** 225	***Vaccine:*** 0 ***Placebo:*** 30
Novavax	2	> 15,000	62	***Vaccine:*** 6 ***Placebo:*** 56	***Vaccine:*** 0 ***Placebo:*** 1

Fonte: Adaptado de Takeda



Fonte: https://medrxiv.org/lookup/doi/10.1101/2021.09.10.21263084

Pesquisa da FIOCRUZ – Efetividade das Vacinas no Mundo Real:

✓ Acompanharam 66 milhões de vacinados contra casos graves ou mortes.

Fonte: https://medrxiv.org/lookup/doi/10.1101/2021.09.10.21263084

CoronaVac:

- ✓ **20 a 39 anos:** 58,4%;
- ✓ **60 a 79 anos:** 60,4%;
- ✓ **+ 80 anos:** 29,6%.

Fonte: https://medrxiv.org/lookup/doi/10.1101/2021.09.10.21263084

Oxford/AstraZeneca:

- ✓ **20 a 39 anos:** 97,9%;
- ✓ **40 a 59 anos:** 82,7%;
- ✓ **60 a 79 anos:** 79,6%;
- ✓ **+ 80 anos:** 66,7%.

Fonte: https://medrxiv.org/lookup/doi/10.1101/2021.09.10.21263084

Pfizer (uma dose):

✓ **40 a 59 anos:** 89,9%.

Fonte: https://medrxiv.org/lookup/doi/10.1101/2021.09.10.21263084

ANVISA

- ✓ Autarquia sob regime especial:
  - ➢ Vinculada ao Ministério da Saúde, criada pela Lei nº 9.782/1999.

Objetivo:

- ✓ Proteger e promover a saúde da população:
  - ➢ Por meio do controle sanitário da produção e consumo de produtos e serviços sujeitos à vigilância sanitária.

ANVISA

17/01/2021 – Apresentação e Liberação das Vacinas:

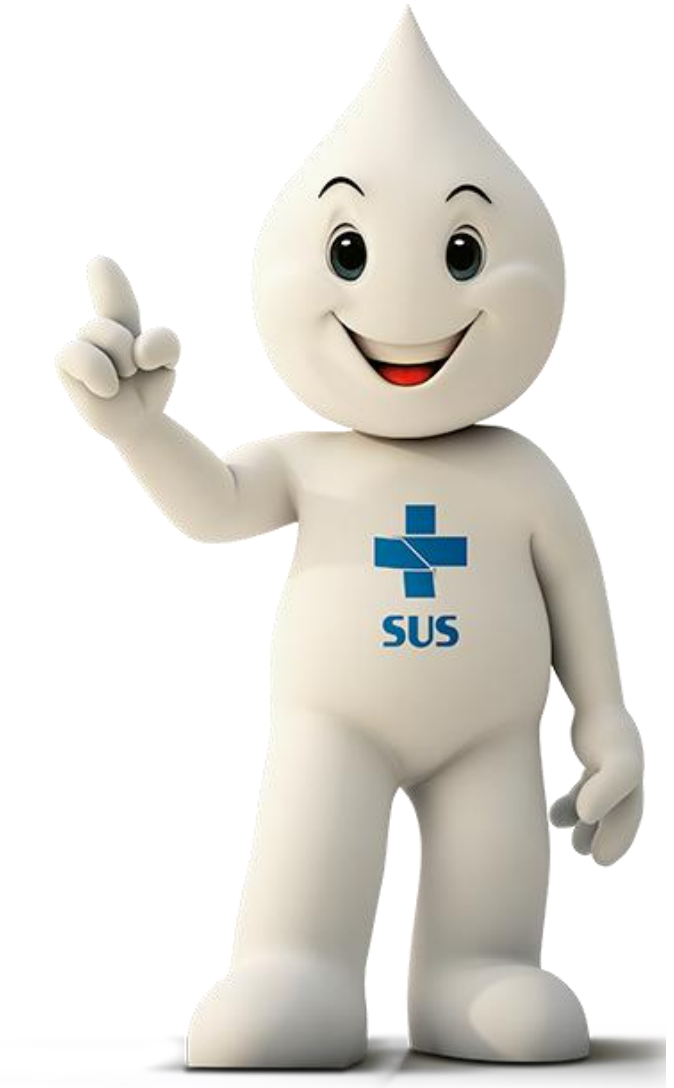
- ✓ Linguagem clara e técnica;
- ✓ Reinterpretaram o que foi entregue pelas empresas;
- ✓ Apontaram limitações.

Levantaram perguntas a serem respondidas (à época):

- ✓ Quanto tempo dura a imunidade?;
- ✓ Qual o melhor regime de dose?;
- ✓ Próximos grupos de vacinação: crianças, gestantes e lactantes?

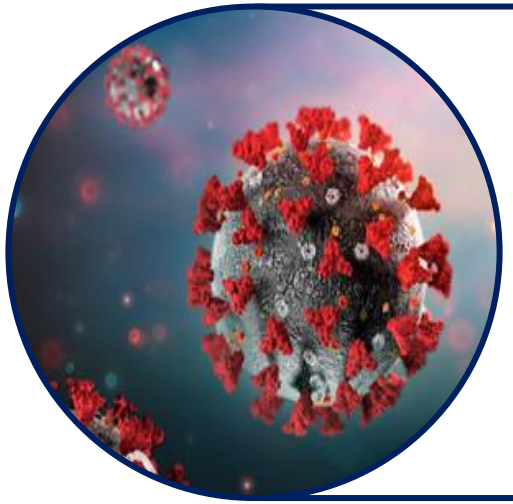
VACINAÇÃO NO BRASIL

- O Brasil teve assegurado para 2021:
  - 545,5 milhões de doses:
    - ✓ **Sinovac/Butantan:**
      - ➢ 100 milhões de doses.
    - ✓ **Oxford/AstraZeneca:**
      - ➢ 100 milhões de doses até julho;
      - ➢ 110 milhões de doses produzidos no Brasil no segundo semestre.
    - ✓ **Pfizer:**
      - ➢ 200 milhões doses até o final do ano.
    - ✓ **Janssen:**
      - ➢ 38 milhões de doses.
    - ✓ **Sputink V:**
      - ➢ 37 milhões de doses.
    - ✓ **Consórcio Covax:**
      - ➢ 33,3 milhões de doses.
    - ✓ **Covaxin:**
      - ➢ 20 milhões de doses.



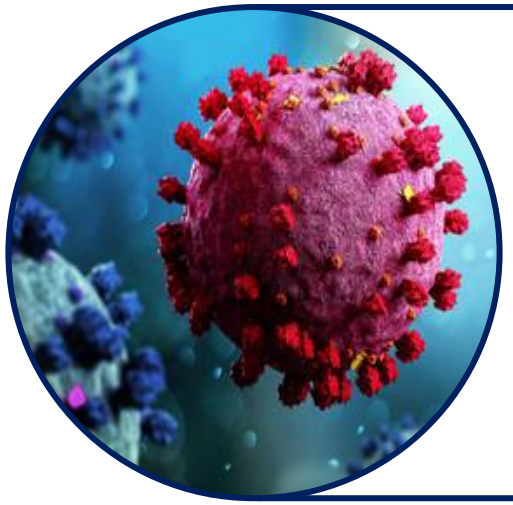
Fonte: http://www.blog.saude.gov.br

VACINAÇÃO | COVID-19



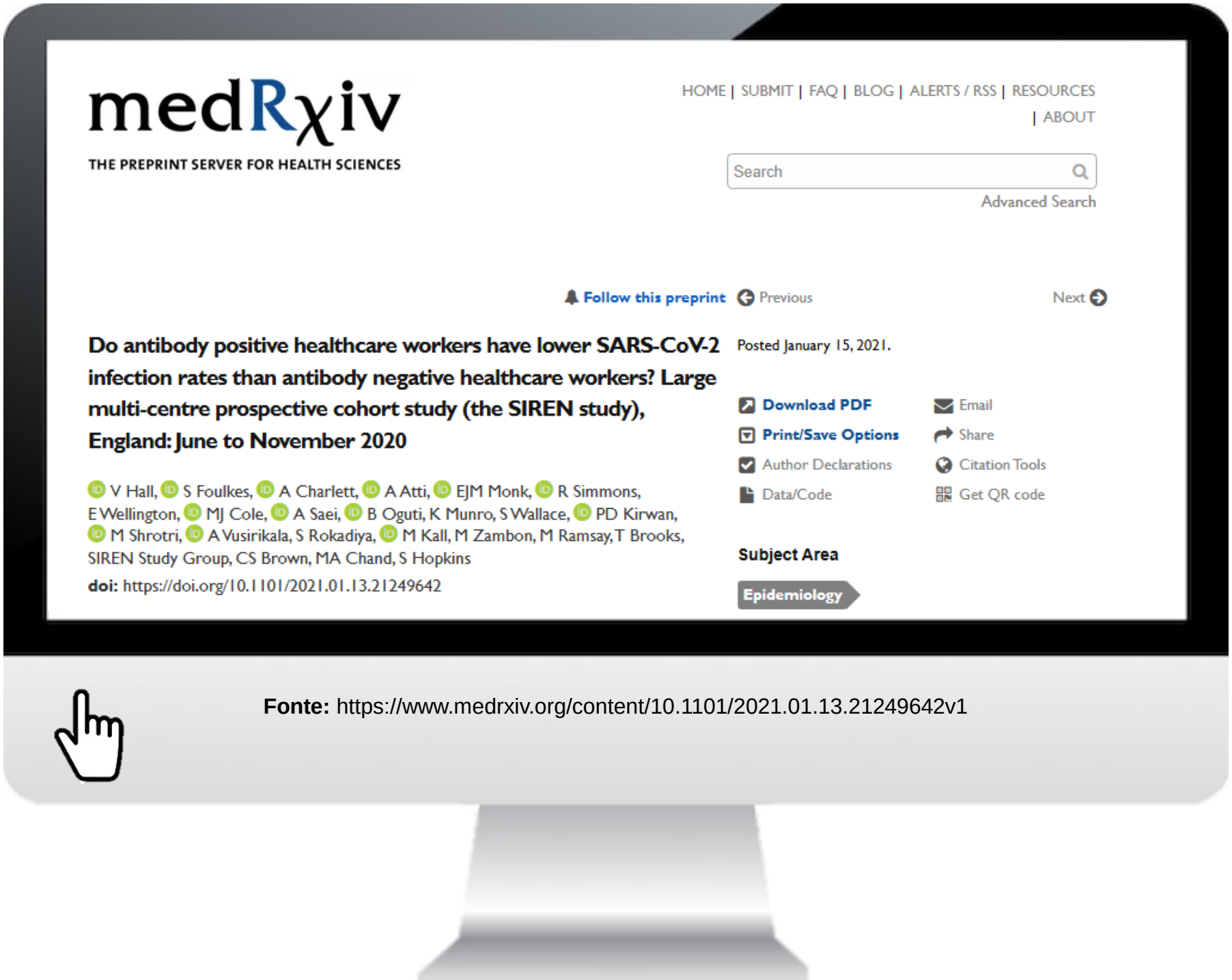
Estudos demonstraram:

- ✓ A imunidade conferida pela infecção natural tem durado no **mínimo três meses**.

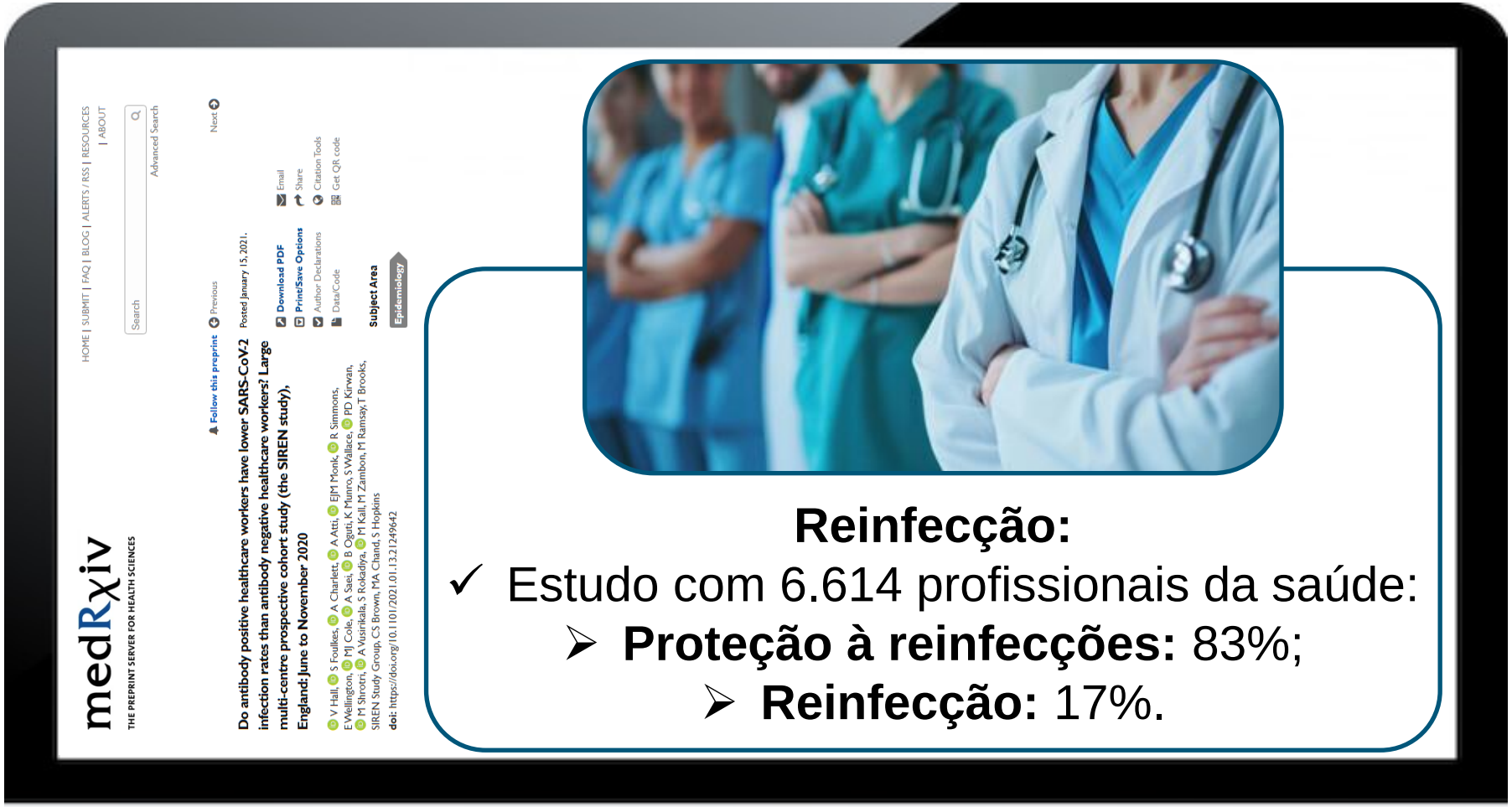


Quem teve infecção natural pela COVID-19:

- ✓ Devia esperar 04 semanas após o período de isolamento;
- ✓ Principalmente os profissionais da saúde, os quais estavam muito expostos:
  - ➢ DEVEM SE VACINAR!!!



Fonte: https://www.medrxiv.org/content/10.1101/2021.01.13.21249642v1

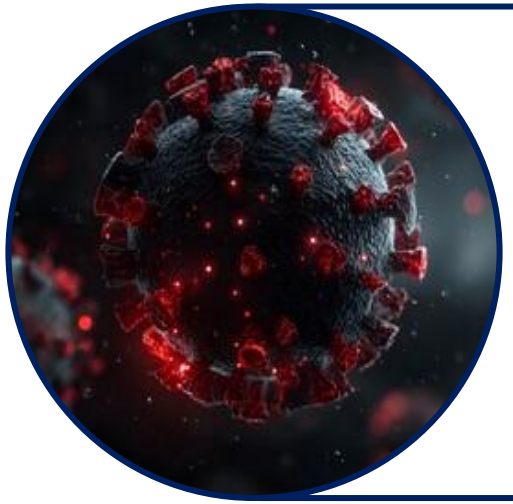


Reinfecção:

- ✓ Estudo com 6.614 profissionais da saúde:
  - ➢ **Proteção à reinfecções:** 83%;
  - ➢ **Reinfecção:** 17%.

Fonte: https://www.medrxiv.org/content/10.1101/2021.01.13.21249642v1

VACINAÇÃO | COVID-19



Imunologia Básica:

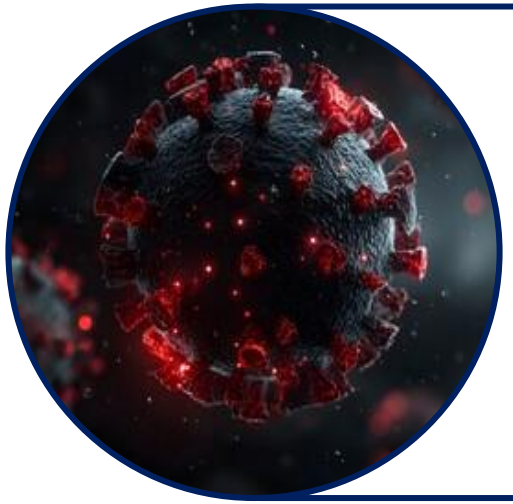
✓ A infecção é **MENOS PREVISÍVEL** do que a vacinação!.

A Vacinação DEMONSTROU:

✓ Proteção por células T contra variantes do SARS-CoV-2:
- ➢ Foi melhor quando vacinado do que quando previamente infectado.

✓ Justamente, em pessoas previamente infectadas:
- ➢ A vacina atua ainda melhor, pois aumenta o número de linfócitos maduros e imunoglobulinas.

VACINAÇÃO | COVID-19



Imunologia Básica:

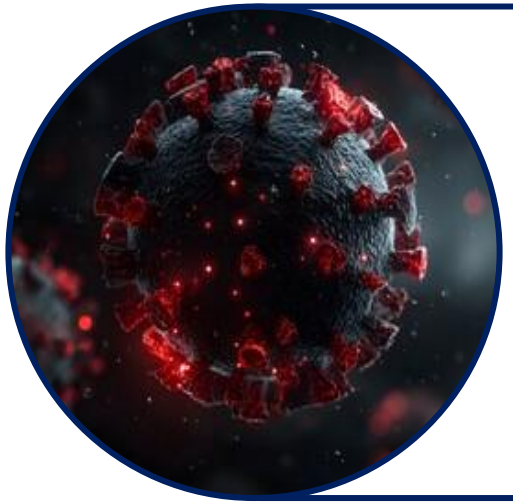
✓ A infecção é **MENOS PREVISÍVEL** do que a vacinação!.

A Vacinação DEMONSTROU:

✓ <u>A proteção contra a doença grave</u>:

- Se reduz mais lentamente com a vacina do que com infecção.

VACINAÇÃO | COVID-19



Imunologia Básica:

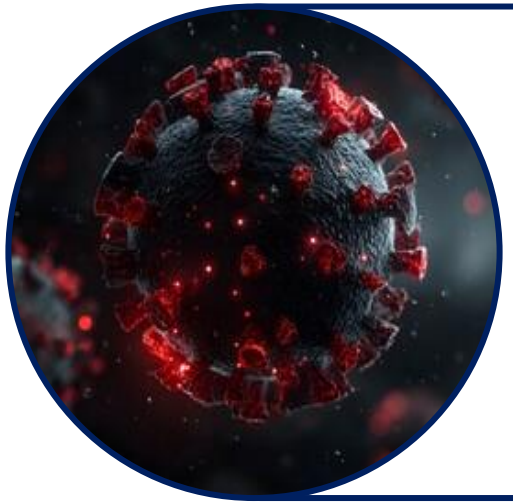
✓ A infecção é **MENOS PREVISÍVEL** do que a vacinação!.

A Vacinação DEMONSTROU:

✓ <u>Populações com elevadas coberturas vacinais</u>:

- ➢ Oferecem menores oportunidades ao vírus para acumular mutações;
- ➢ **Além de EVITAR o surgimento de novas variantes.**

VACINAÇÃO | COVID-19



Imunologia Básica:

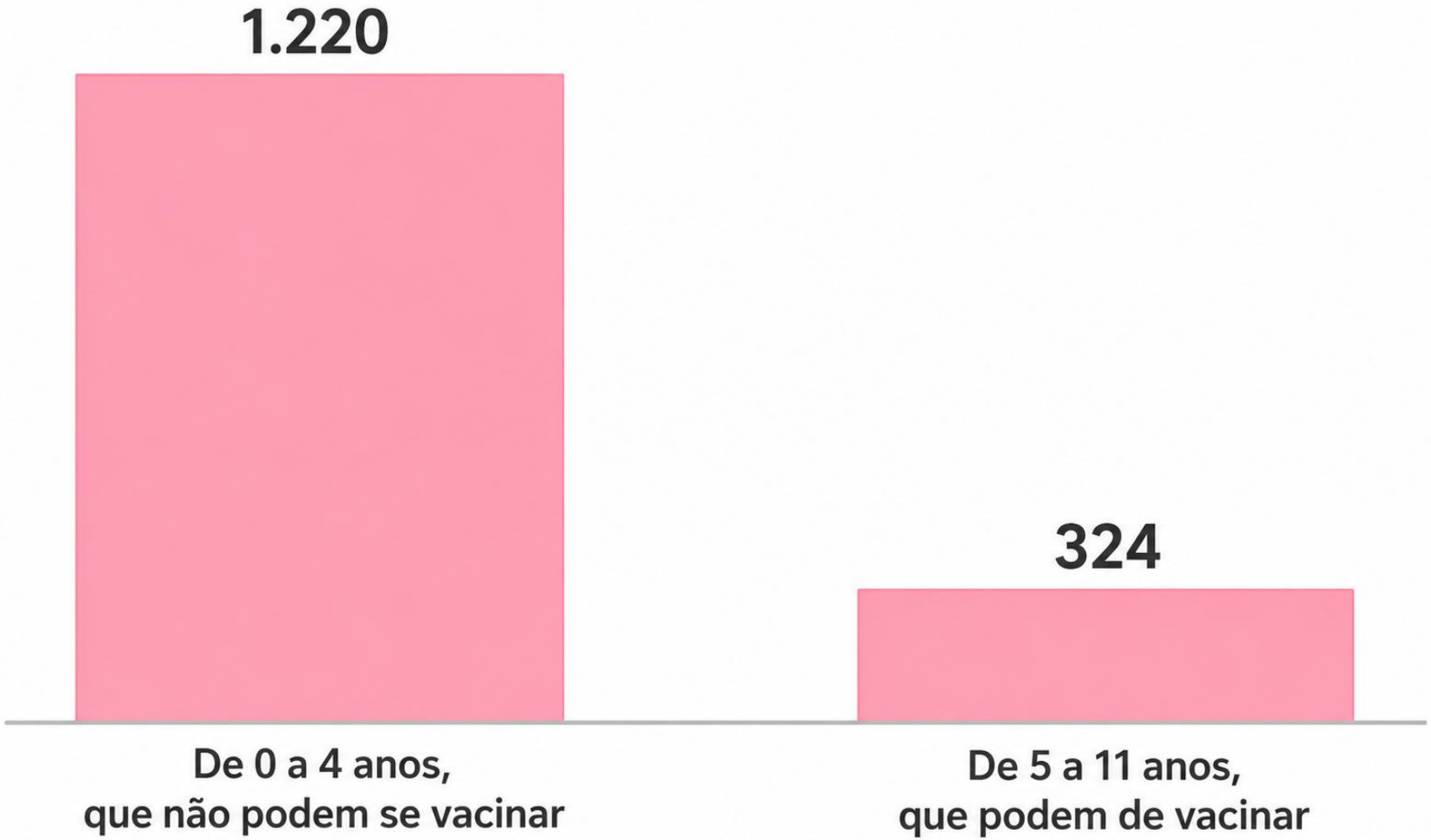
✓ A infecção é **MENOS PREVISÍVEL** do que a vacinação!.

A Vacinação DEMONSTROU:

✓ A eficácia da vacinação com <u>três doses em previamente infectados é ainda maior.</u>

VACINAÇÃO INFANTIL

- **Mortes por Síndrome Respiratória Aguda Grave (SRAG) por COVID-19 Confirmadas:**



Fonte: Dados do Sivep-Gripe e *Vital Strategies*. Dados de Janeiro de 2020 até 10 de Dezembro de 2021.

VACINAÇÃO INFANTIL

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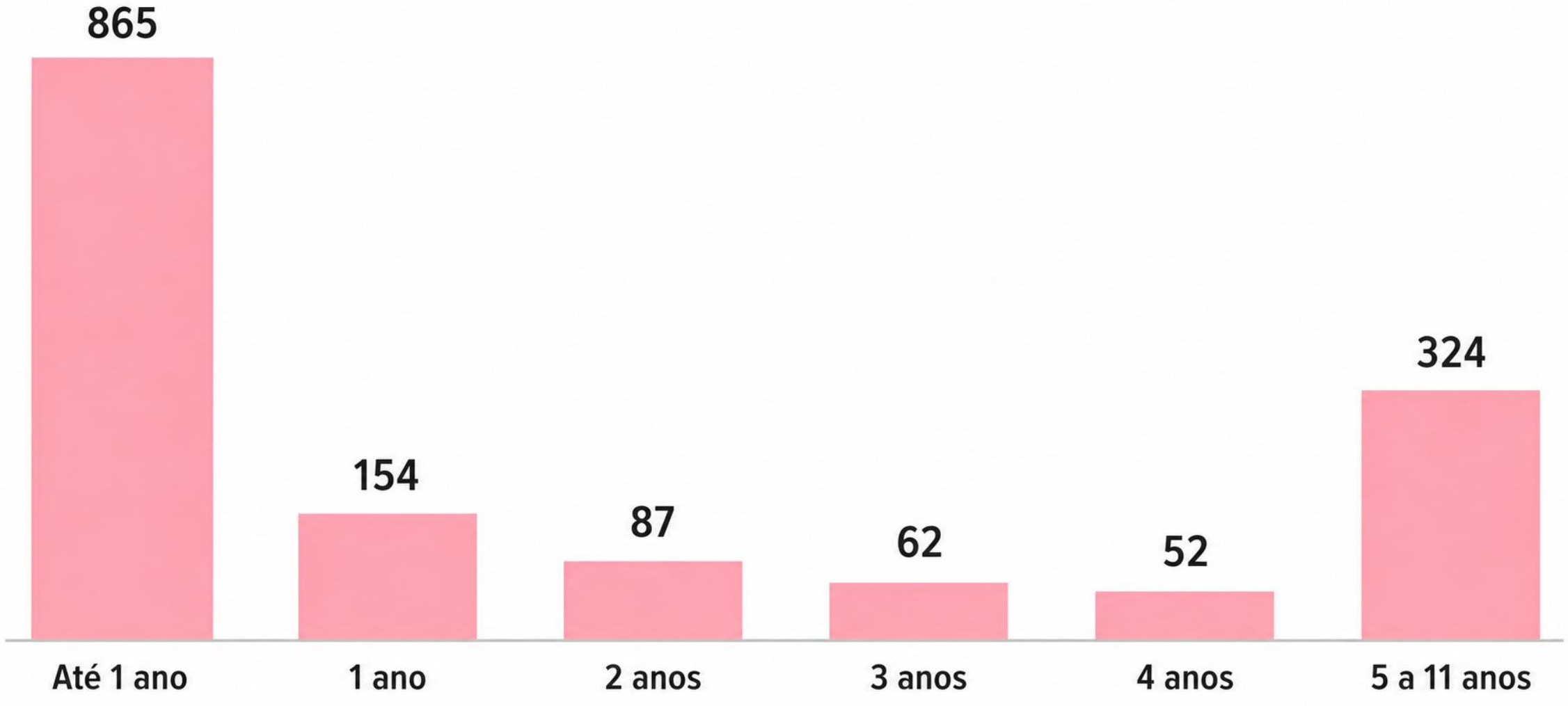

Fonte: Dados do Sivep-Gripe e *Vital Strategies*. Dados de Janeiro de 2020 até 10 de Dezembro de 2021.

VACINAÇÃO INFANTIL

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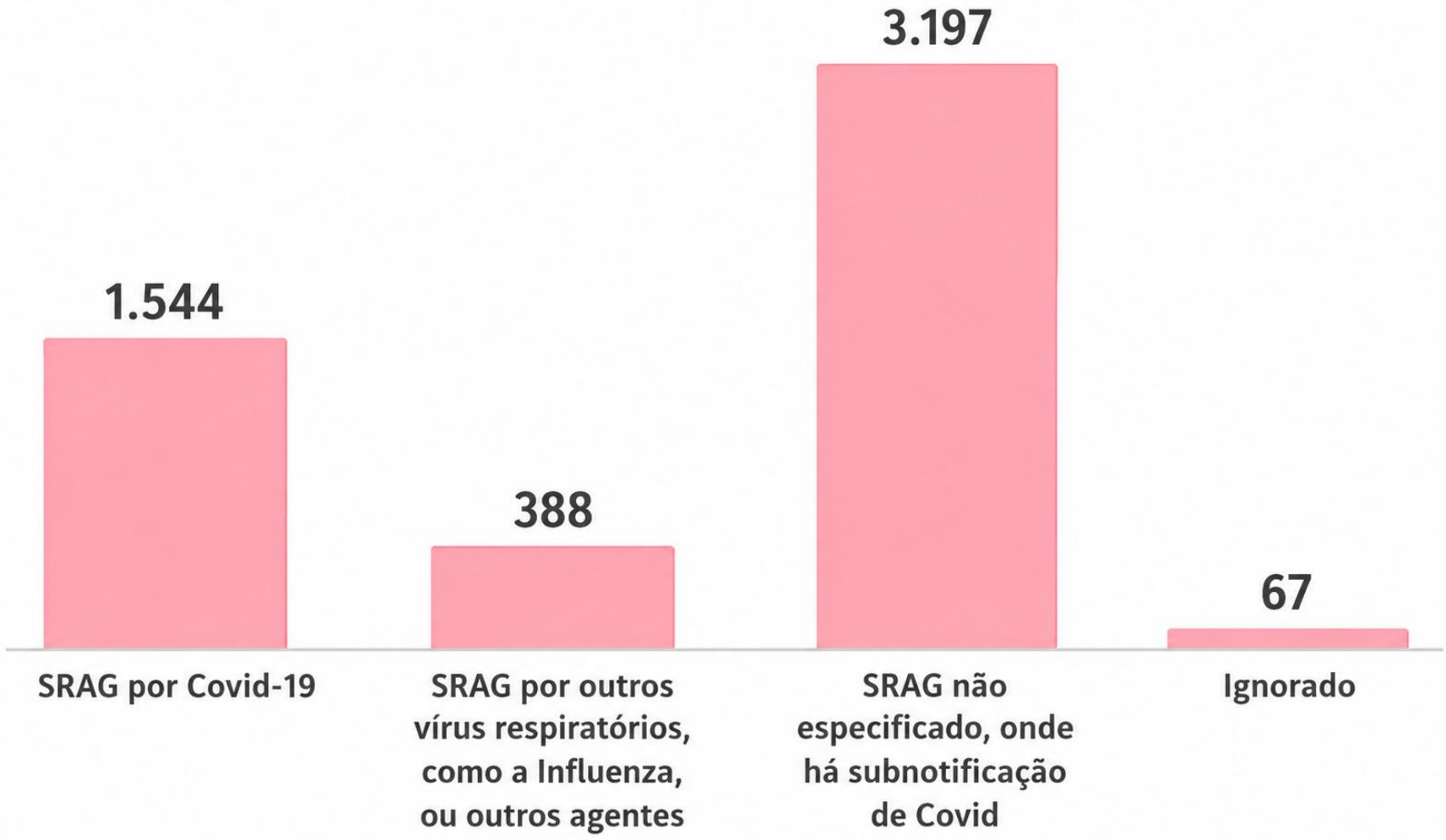

Fonte: Dados do Sivep-Gripe e *Vital Strategies*. Dados de Janeiro de 2020 até 10 de Dezembro de 2021.

ETAPAS DA AULA

01

HISTÓRIA DA VACINAÇÃO

02

IMUNO BIOLÓGICOS

03

P&D FASES

04

DESAFIOS E DEBATES ATUAIS

ETAPAS DA AULA

01

HISTÓRIA DA VACINAÇÃO

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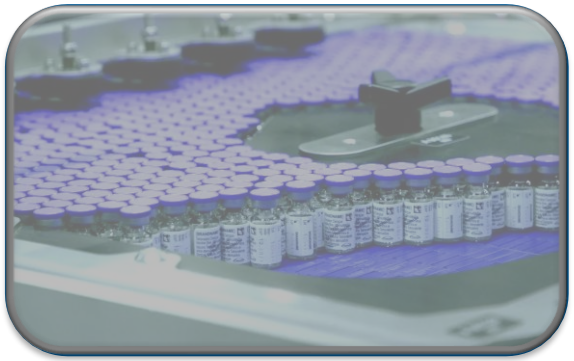


03

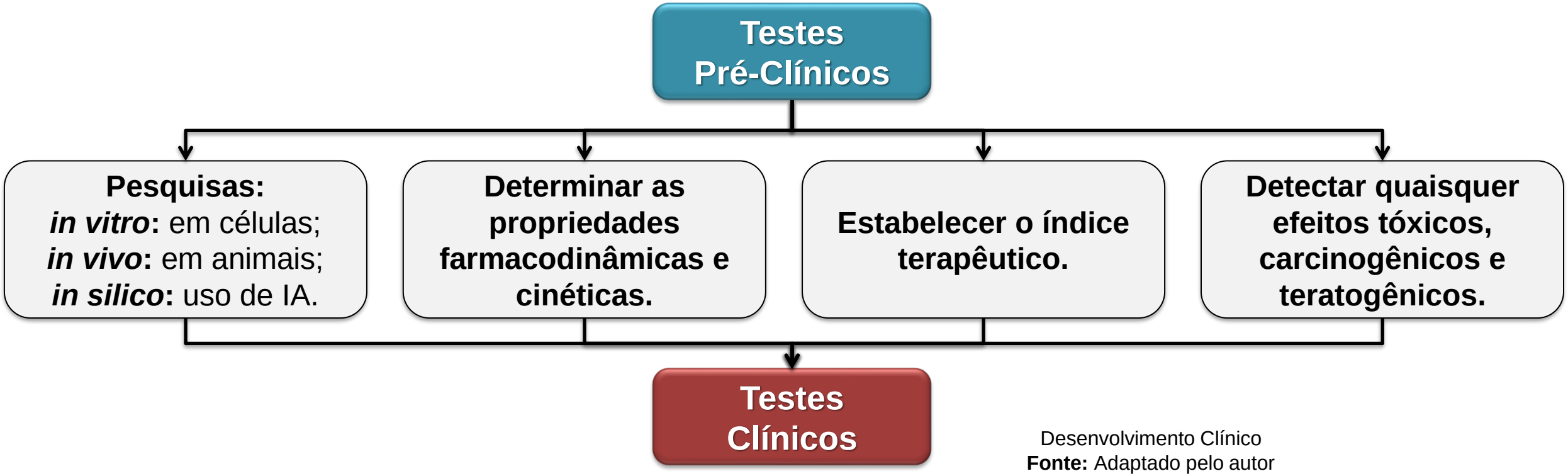
P&D FASES

04

DESAFIOS E DEBATES ATUAIS

DESENVOLVIMENTO CLÍNICO

- ✓ Durante o desenvolvimento e a fase de testes de novos fármacos e/ou vacinas:
  - ➢ A segurança e a utilidade das substâncias, devem ser determinados das seguintes maneiras:



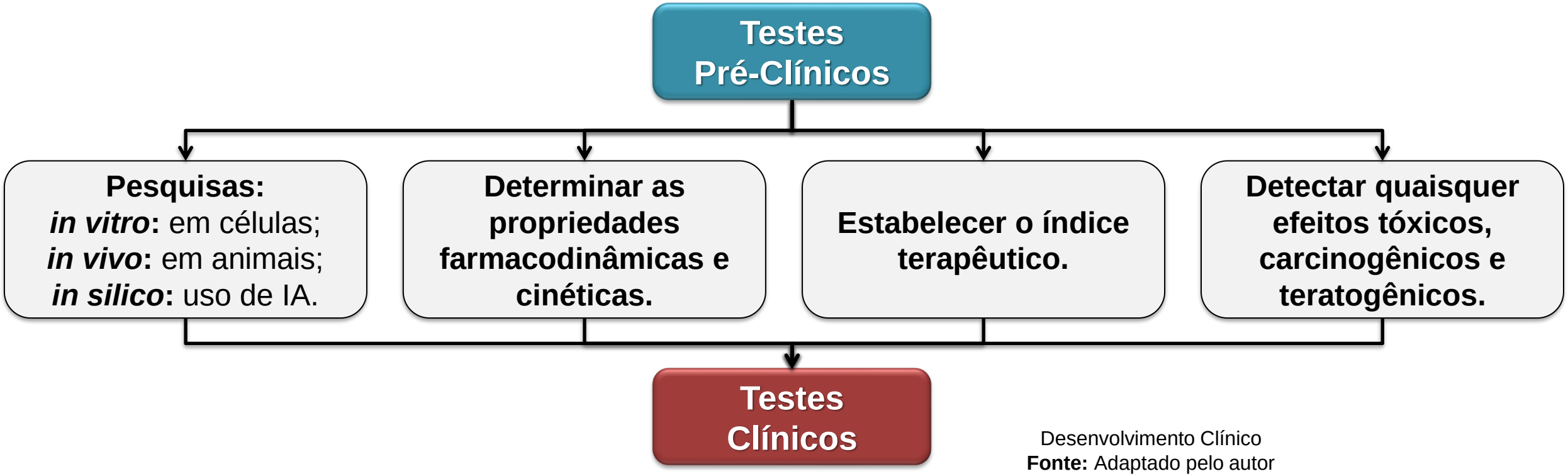
Desenvolvimento Clínico
Fonte: Adaptado pelo autor

DESENVOLVIMENTO CLÍNICO

Exemplo:

✓ De cada 10.000 moléculas estudadas:

➢ Apenas 1 (UMA) chega aos pacientes!!!



Desenvolvimento Clínico
Fonte: Adaptado pelo autor

DESENVOLVIMENTO CLÍNICO

Todas as fases devem ser cuidadosamente:

- ✓ Planejados;
- ✓ Bem projetados;
- ✓ E controlados.

Testes Pré-Clínicos

- **Pesquisas:**
  ***in vitro*:** em células;
  ***in vivo*:** em animais;
  ***in silico*:** uso de IA.
- **Determinar as propriedades farmacodinâmicas e cinéticas.**
- **Estabelecer o índice terapêutico.**
- **Detectar quaisquer efeitos tóxicos, carcinogênicos e teratogênicos.**

Testes Clínicos

Desenvolvimento Clínico
Fonte: Adaptado pelo autor

PESQUISA CLÍNICA

- Processo Tradicional de Desenvolvimento de Compostos:

Fonte: Adaptado de https://www.nature.com/articles/s41586-020-2798-3

PESQUISA CLÍNICA

- **Processo Tradicional de Desenvolvimento de Compostos:**

Fonte: Adaptado de https://www.nature.com/articles/s41586-020-2798-3

BBC NEWS BRASIL

Notícias Brasil Internacional Economia Saúde Ciência Tecnologia Vídeos Podcasts

As doenças antes incuráveis que estão ganhando tratamentos graças à IA

A inteligência artificial está inventando novos remédios contra Parkinson, superbactérias resistentes a antibióticos e muitas doenças raras

Fonte: https://www.bbc.com/portuguese/articles/cz0e7m77en1o

RSFLIX

RSFLIX

Quem está assistindo?

Candidatos Biomedicina

Candidatos Enfermagem

Candidatos Farmácia

Candidatos Fisioterapia

Rafael Saraiva

Adicionar perfil

GERENCIAR PERFIS

RSFLIX

A GUERRA DO AMANHÃ

★ ★ ★ ★ ★ Ação / Ficção Científica

Quando viajantes do tempo surgem com um alerta, o presente se torna a última esperança para salvar o futuro.

Top 2 filmes sobre pesquisa *in silico*

1

2

RSFLIX

PANTERA NEGRA: WAKANDA

★ ★ ★ ★ ★ Ação / Ficção Científica

A perda de T'Challa e a chegada de um novo adversário colocam o destino de Wakanda em jogo.

Top 2 filmes sobre pesquisa *in silico*

1

2

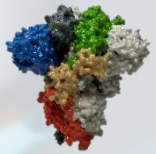
PESQUISA CLÍNICA

- **Processo Tradicional de Desenvolvimento de Compostos:**

Fonte: Adaptado de https://www.nature.com/articles/s41586-020-2798-3

PESQUISA CLÍNICA

- Processo Desenvolvimento de Vacinas Contra a COVID-19:



Fase Exploratória

Dados pré existentes de SARS e MERS

Fonte: Adaptado de https://www.nature.com/articles/s41586-020-2798-3

PESQUISA CLÍNICA

Environmental Sustainability (2021) 4:455–467
https://doi.org/10.1007/s42398-021-00165-x

REVIEW

Linkages between environmental issues and zoonotic diseases: with reference to COVID-19 pandemic

Jitendra Mishra[1] · Priya Mishra[1] · Naveen Kumar Arora[1]

Received: 17 November 2020 / Revised: 28 February 2021 / Accepted: 2 March 2021 / Published online: 29 March 2021
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Abstract
Coronavirus disease (COVID-19) caused by severe acute respiratory syndrome (SARS) coronavirus 2 (SARS-CoV-2) has taken mankind by surprise with various unprecedented impacts on human life around the globe. This zoonotic pandemic is proving to be the most destructive disease outbreaks since decades. The increasing human population and anthropogenic activities have impacted the environment and have direct linkages with the current and other recent outbreaks of zoonotic diseases. Despite having a difference in their origin, major reasons behind the emergence and spread of zoonotic pandemics are related to activities such as habitat fragmentation, deforestation, biodiversity loss, intensive agriculture and livestock farming, uncontrolled urbanization, pollution, climate change and bushmeat hunting and trading. It is important to focus on environmental and climatic factors that are involved in the emergence of such pandemics involving novel human pathogens and viruses in particular. Research and data analysis, particularly in relation to COVID-19, has shown that meteorological factors along with population density and living conditions (particularly in the urban and semi-urban areas) play a crucial role in the intensity, evolution and spread of SARS-CoV-2. This particular virus is novel but coronaviruses have a long history and are known to cause disease outbreaks earlier as well. COVID-19 pandemic provides learning for the future, in particular the importance of environmental sustainability for controlling such outbreaks. A strategic plan can be developed involving policy-makers, organizations, and governments to control the onset and spread of the novel pathogens. This review-based study recommends that prevention of COVID-19 like pandemics from re-occurring is through tackling the issues related to the environment by controlling anthropogenic activities. It will also be important to track the lineage and future evolution course of such human pathogens so as to determine the nexus of environmental and biological factors in the development and spread of novel strains.

Keywords COVID-19 · Pandemic · Zoonoses · Environment · Climate change · Anthropogenic

Introduction

Large-scale outbreaks of infectious diseases, also known as pandemics, have caused significant loss of human life along with economic and social issues. There have been several zoonotic pandemic outbreaks which have caused morbidity and mortality over a wide geographical region since recorded history (Fig. 1). Apart from direct impact on human life, these highly contagious zoonoses influence socio-economic activities such as education, employment, transportation, tourism, trade and health care system (Cascio et al. 2011; Torgerson and Macpherson 2011; Epp and Waldner 2012; Smith et al. 2019). Currently, the whole world is affected due to a novel zoonotic disease known as coronavirus disease (COVID-19). COVID-19 was first spotted in December 2019 in Wuhan, Hubei province of China, when dozens of people were hospitalized due to pneumonia-like symptoms with unknown causes (Zhu et al. 2020). Later, preliminary tests and epidemiological studies indicated that COVID -19 was caused by a new zoonotic coronavirus strain named as "severe acute respiratory syndrome (SARS) coronavirus 2" (SARS-CoV-2) by the International Committee on Taxonomy of Viruses (ICTV) (Wu et al. 2020a). SARS-CoV-2 showed genetic similarities with earlier reported highly pathogenic coronaviruses such as "severe acute respiratory syndrome coronavirus" (SARS-CoV) and the

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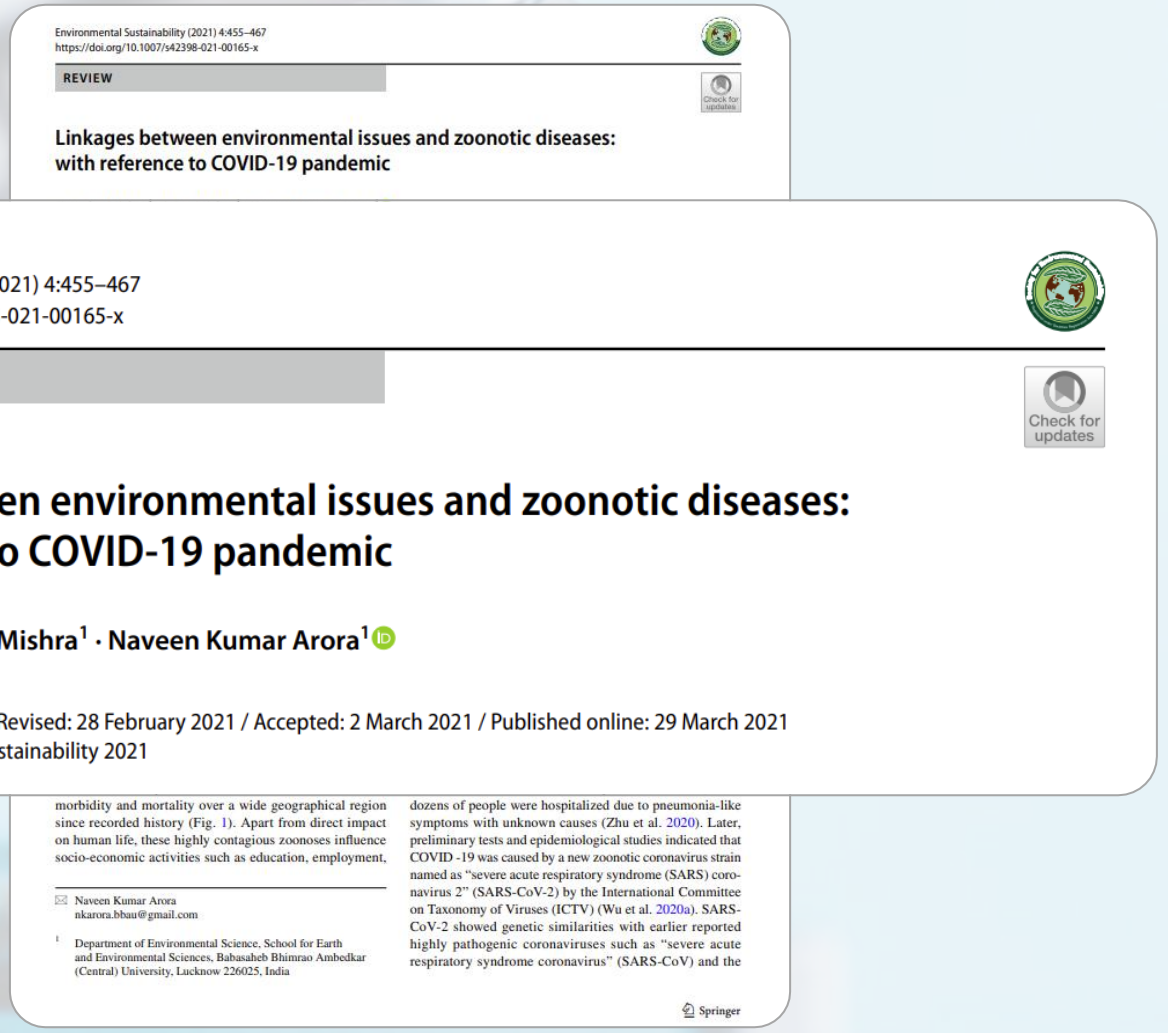
nkarora.bbau@gmail.com

[1] Department of Environmental Science, School for Earth and Environmental Sciences, Babasaheb Bhimrao Ambedkar (Central) University, Lucknow 226025, India

Fonte: https://www.ncbi.nlm.nih.gov/pmc/articles/PMC8005368/pdf/42398_2021_Article_165.pdf

PESQUISA CLÍNICA



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PESQUISA CLÍNICA

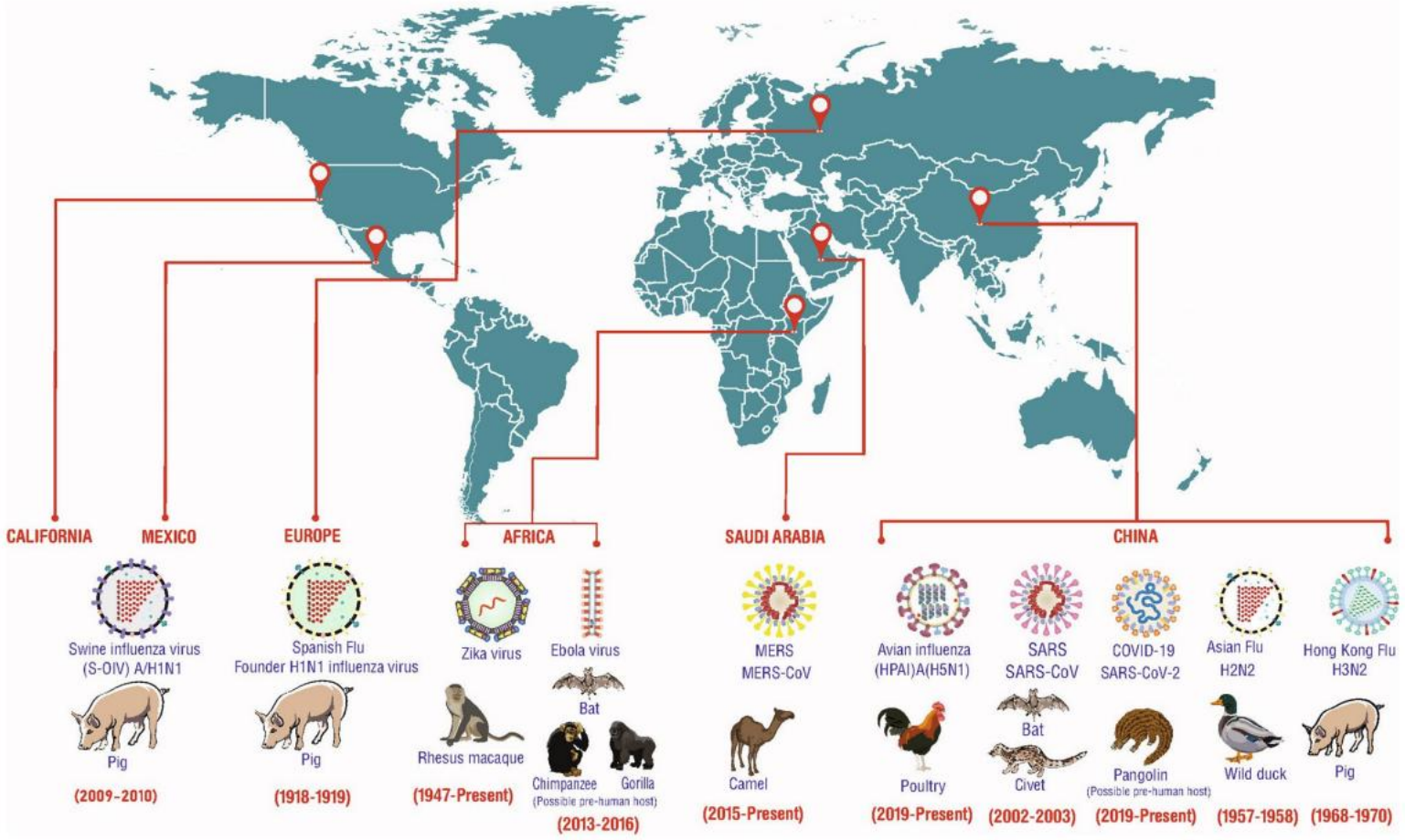
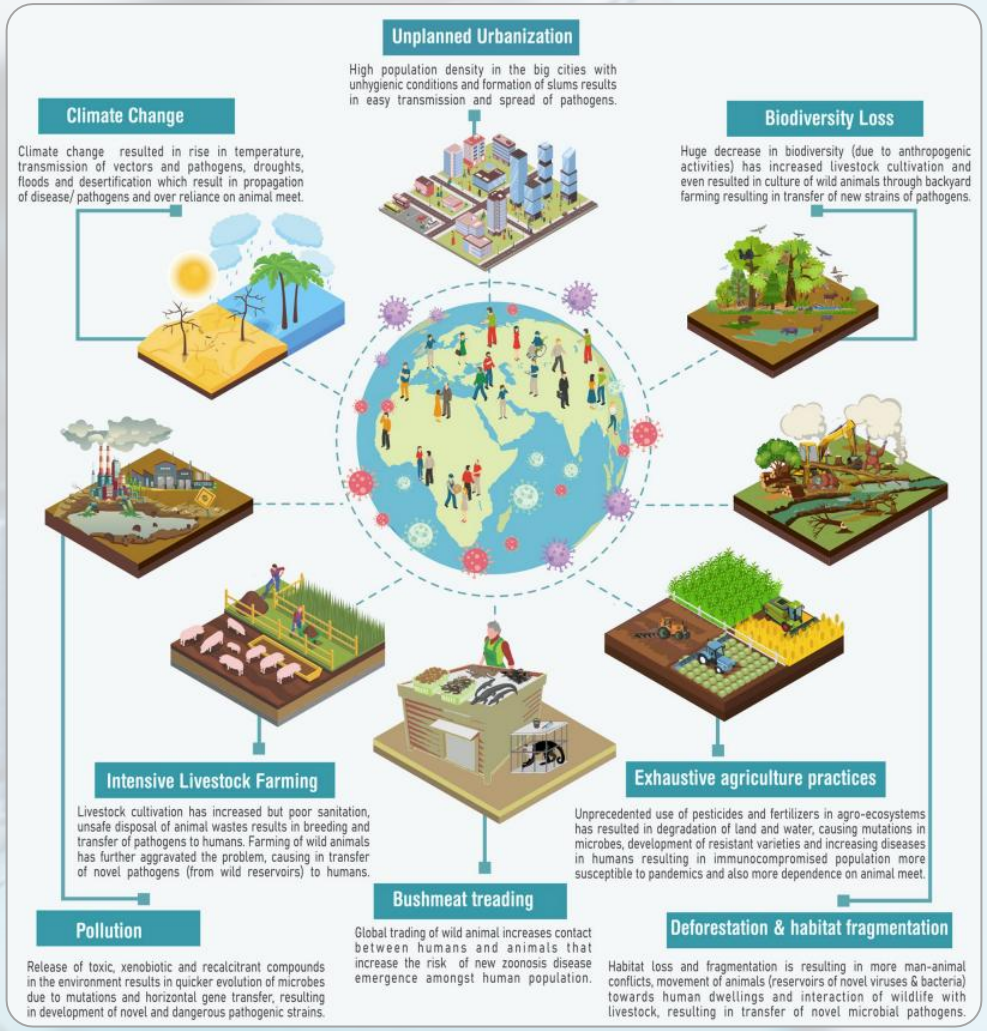


Fig. 1 Major zoonotic pandemic outbreaks in the recorded history

Fonte: https://www.ncbi.nlm.nih.gov/pmc/articles/PMC8005368/pdf/42398_2021_Article_165.pdf

PESQUISA CLÍNICA



Fonte: https://www.ncbi.nlm.nih.gov/pmc/articles/PMC8005368/pdf/42398_2021_Article_165.pdf

RSFLIX

OS SEM-FLORESTA

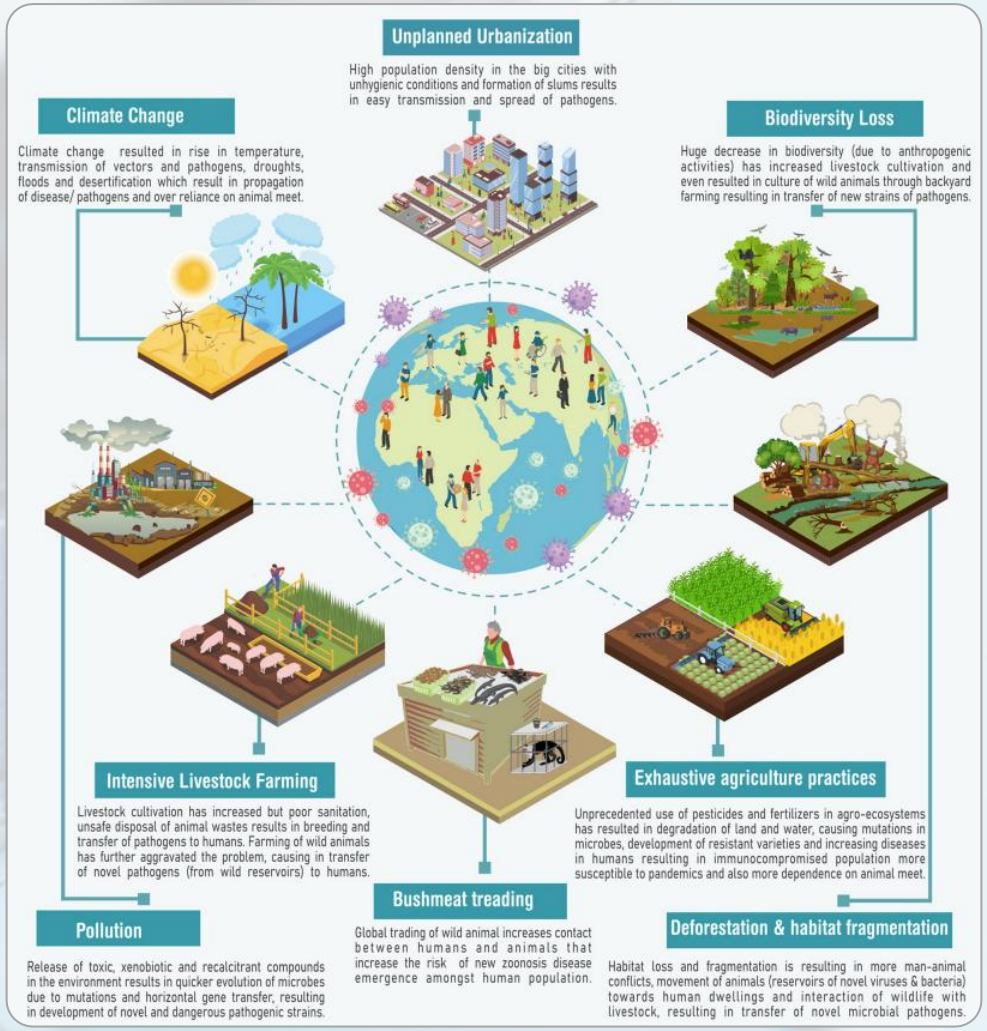
★ ★ ★ ★ ½ Infantil / Comédia

Com a cidade invadindo a floresta, a tartaruga Verne e seus amigos se desesperam. Mas o guaxinim RJ mostrará as vantagens de se ter humanos por perto.

Top 1 em filmes sobre o impacto humano na natureza

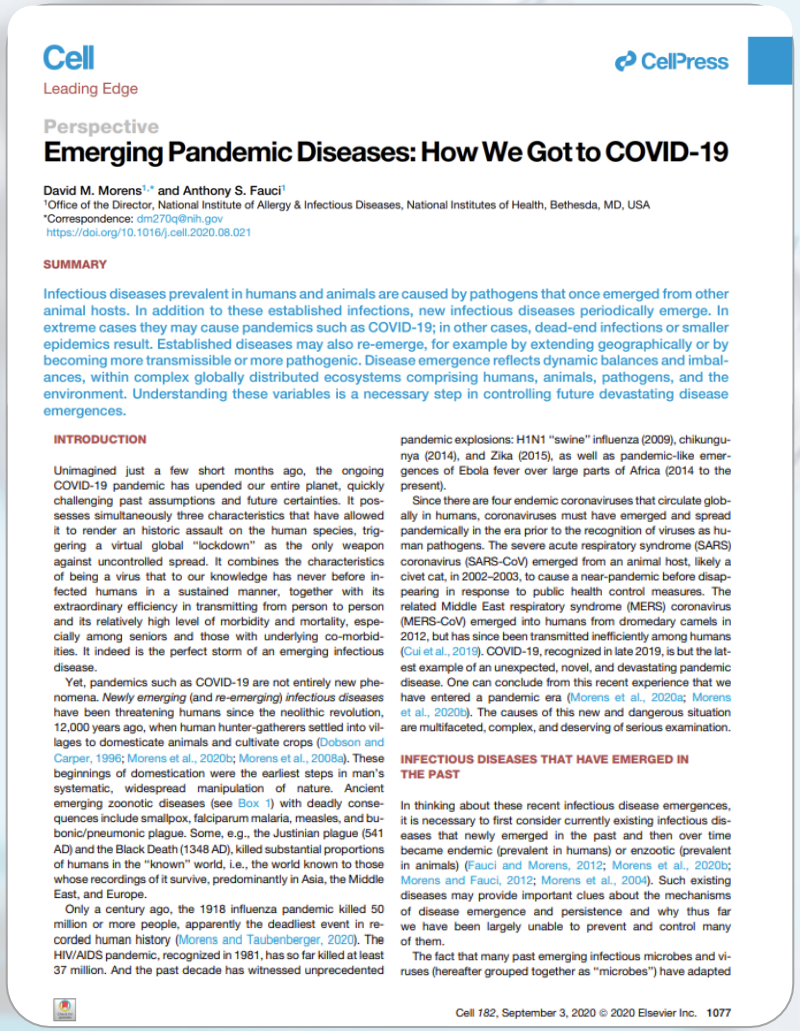
1

PESQUISA CLÍNICA



Fonte: https://www.ncbi.nlm.nih.gov/pmc/articles/PMC8005368/pdf/42398_2021_Article_165.pdf

PESQUISA CLÍNICA



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Emerging Pandemic Diseases: How We Got to COVID-19

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https://doi.org/10.1016/j.cell.2020.08.021

SUMMARY

Infectious diseases prevalent in humans and animals are caused by pathogens that once emerged from other animal hosts. In addition to these established infections, new infectious diseases periodically emerge. In extreme cases they may cause pandemics such as COVID-19; in other cases, dead-end infections or smaller epidemics result. Established diseases may also re-emerge, for example by extending geographically or by becoming more transmissible or more pathogenic. Disease emergence reflects dynamic balances and imbalances, within complex globally distributed ecosystems comprising humans, animals, pathogens, and the environment. Understanding these variables is a necessary step in controlling future devastating disease emergences.

INTRODUCTION

Unimagined just a few short months ago, the ongoing COVID-19 pandemic has upended our entire planet, quickly challenging past assumptions and future certainties. It possesses simultaneously three characteristics that have allowed it to render an historic assault on the human species, triggering a virtual global "lockdown" as the only weapon against uncontrolled spread. It combines the characteristics of being a virus that to our knowledge has never before infected humans in a sustained manner, together with its extraordinary efficiency in transmitting from person to person and its relatively high level of morbidity and mortality, especially among seniors and those with underlying co-morbidities. It indeed is the perfect storm of an emerging infectious disease.

Yet, pandemics such as COVID-19 are not entirely new phenomena. *Newly emerging* (and *re-emerging*) *infectious diseases* have been threatening humans since the neolithic revolution, 12,000 years ago, when human hunter-gatherers settled into villages to domesticate animals and cultivate crops (Dobson and Carper, 1996; Morens et al., 2020b; Morens et al., 2008a). These beginnings of domestication were the earliest steps in man's systematic, widespread manipulation of nature. Ancient emerging zoonotic diseases (see Box 1) with deadly consequences include smallpox, falciparum malaria, measles, and bubonic/pneumonic plague. Some, e.g., the Justinian plague (541 AD) and the Black Death (1348 AD), killed substantial proportions of humans in the "known" world, i.e., the world known to those whose recordings of it survive, predominantly in Asia, the Middle East, and Europe.

Only a century ago, the 1918 influenza pandemic killed 50 million or more people, apparently the deadliest event in recorded human history (Morens and Taubenberger, 2020). The HIV/AIDS pandemic, recognized in 1981, has so far killed at least 37 million. And the past decade has witnessed unprecedented pandemic explosions: H1N1 "swine" influenza (2009), chikungunya (2014), and Zika (2015), as well as pandemic-like emergences of Ebola fever over large parts of Africa (2014 to the present).

Since there are four endemic coronaviruses that circulate globally in humans, coronaviruses must have emerged and spread pandemically in the era prior to the recognition of viruses as human pathogens. The severe acute respiratory syndrome (SARS) coronavirus (SARS-CoV) emerged from an animal host, likely a civet cat, in 2002–2003, to cause a near-pandemic before disappearing in response to public health control measures. The related Middle East respiratory syndrome (MERS) coronavirus (MERS-CoV) emerged into humans from dromedary camels in 2012, but has since been transmitted inefficiently among humans (Cui et al., 2019). COVID-19, recognized in late 2019, is but the latest example of an unexpected, novel, and devastating pandemic disease. One can conclude from this recent experience that we have entered a pandemic era (Morens et al., 2020a; Morens et al., 2020b). The causes of this new and dangerous situation are multifaceted, complex, and deserving of serious examination.

INFECTIOUS DISEASES THAT HAVE EMERGED IN THE PAST

In thinking about these recent infectious disease emergences, it is necessary to first consider currently existing infectious diseases that newly emerged in the past and then over time became endemic (prevalent in humans) or enzootic (prevalent in animals) (Fauci and Morens, 2012; Morens et al., 2020b; Morens and Fauci, 2012; Morens et al., 2004). Such existing diseases may provide important clues about the mechanisms of disease emergence and persistence and why thus far we have been largely unable to prevent and control many of them.

The fact that many past emerging infectious microbes and viruses (hereafter grouped together as "microbes") have adapted

Fonte: https://www.sciencedirect.com/science/article/pii/S0092867420310126?via%3Dihub

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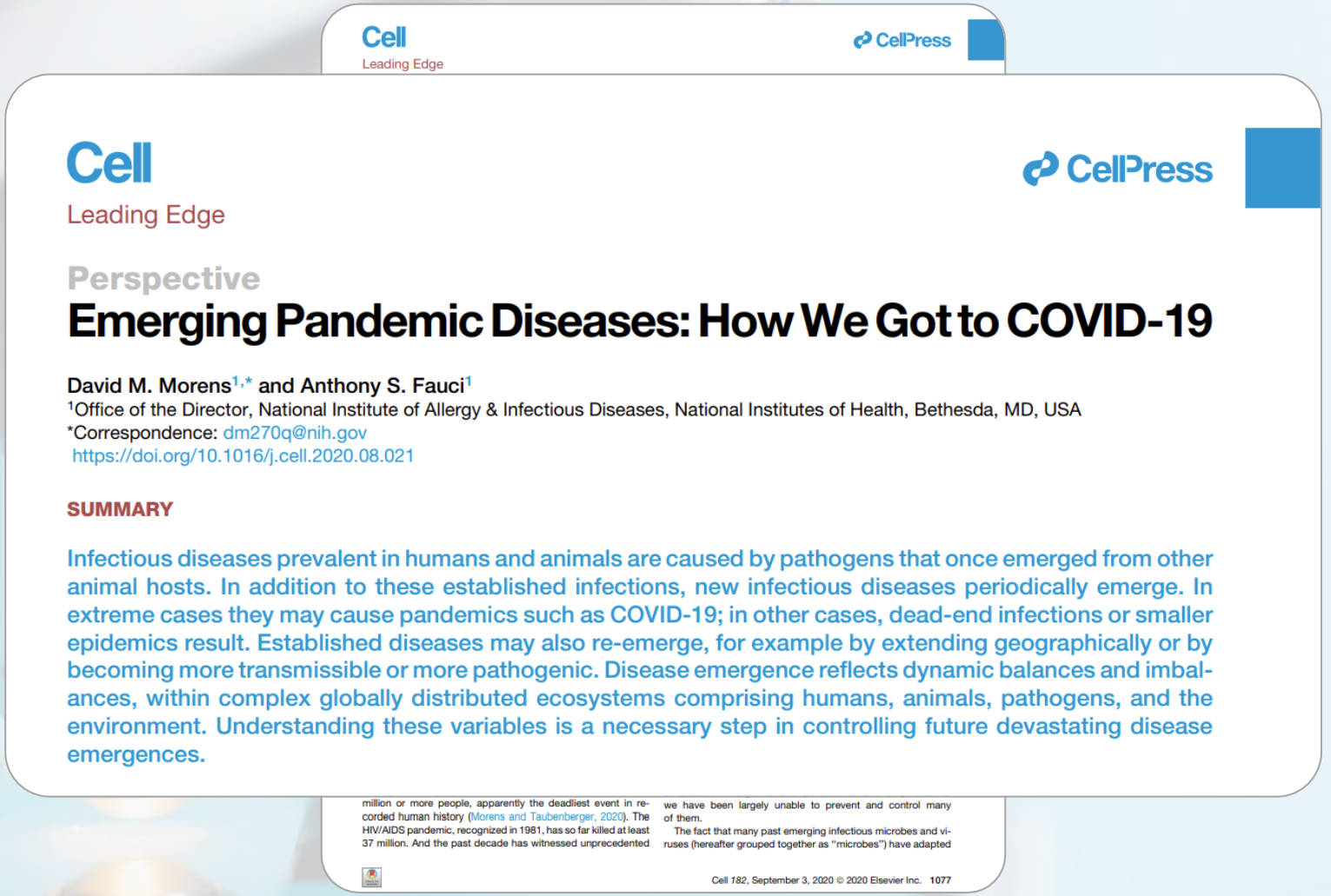

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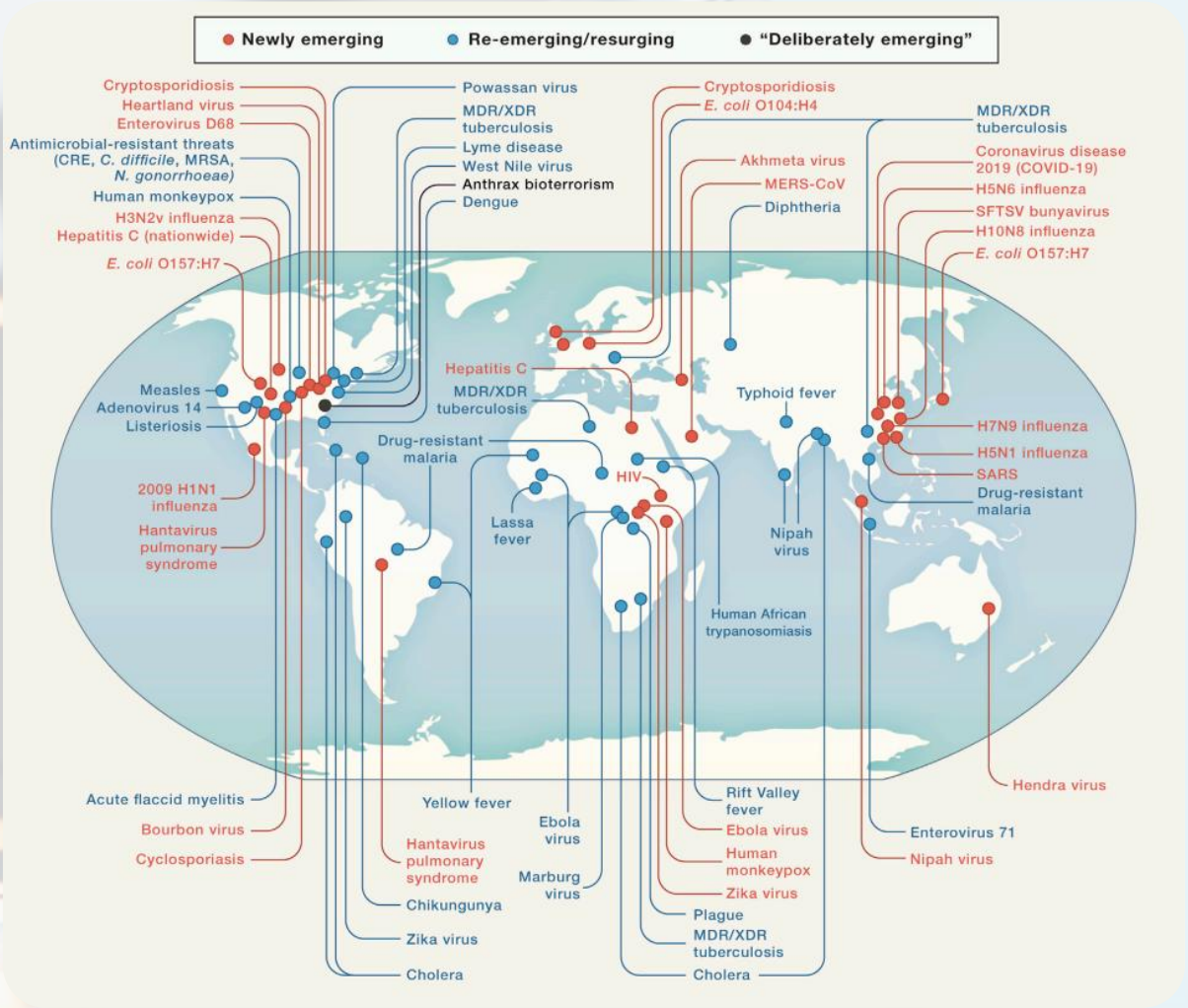

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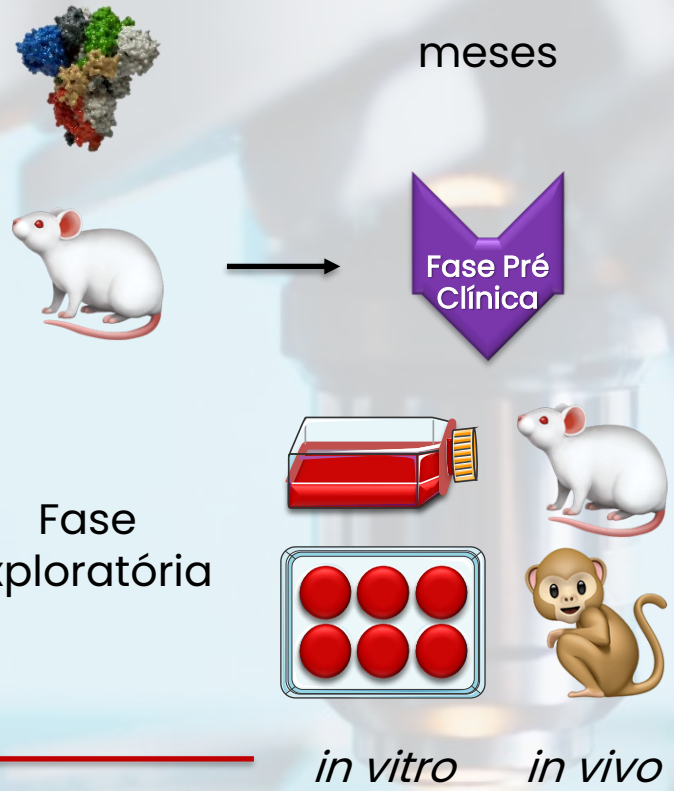

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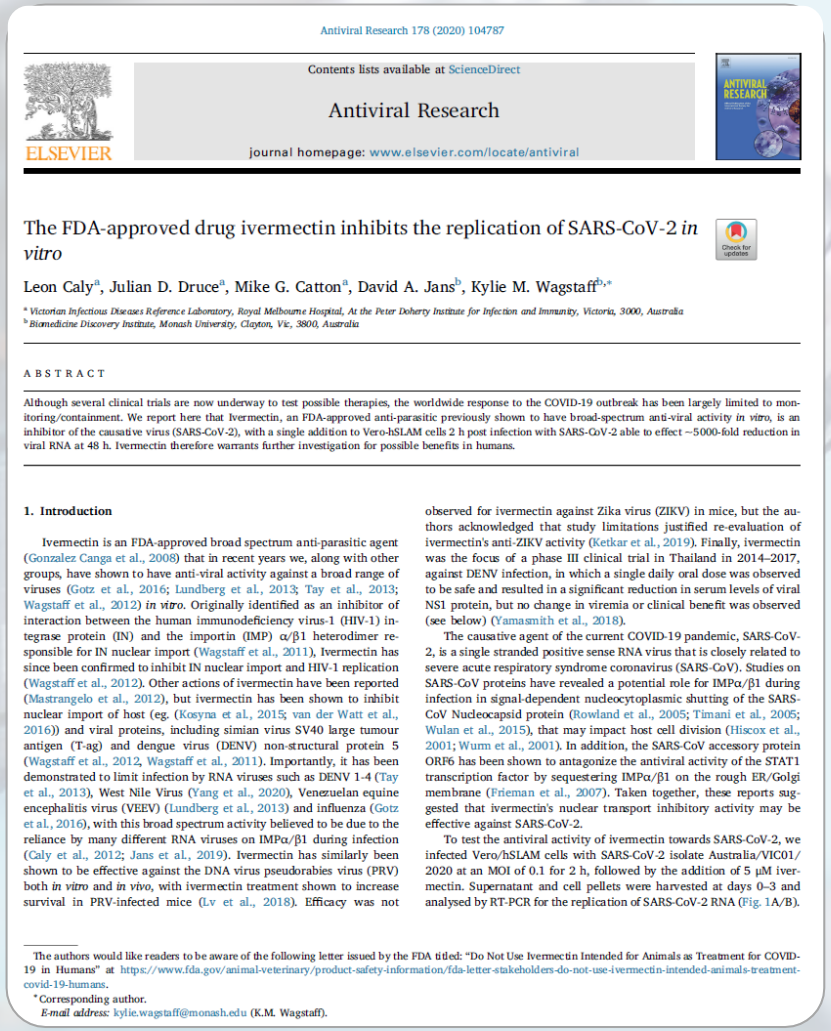

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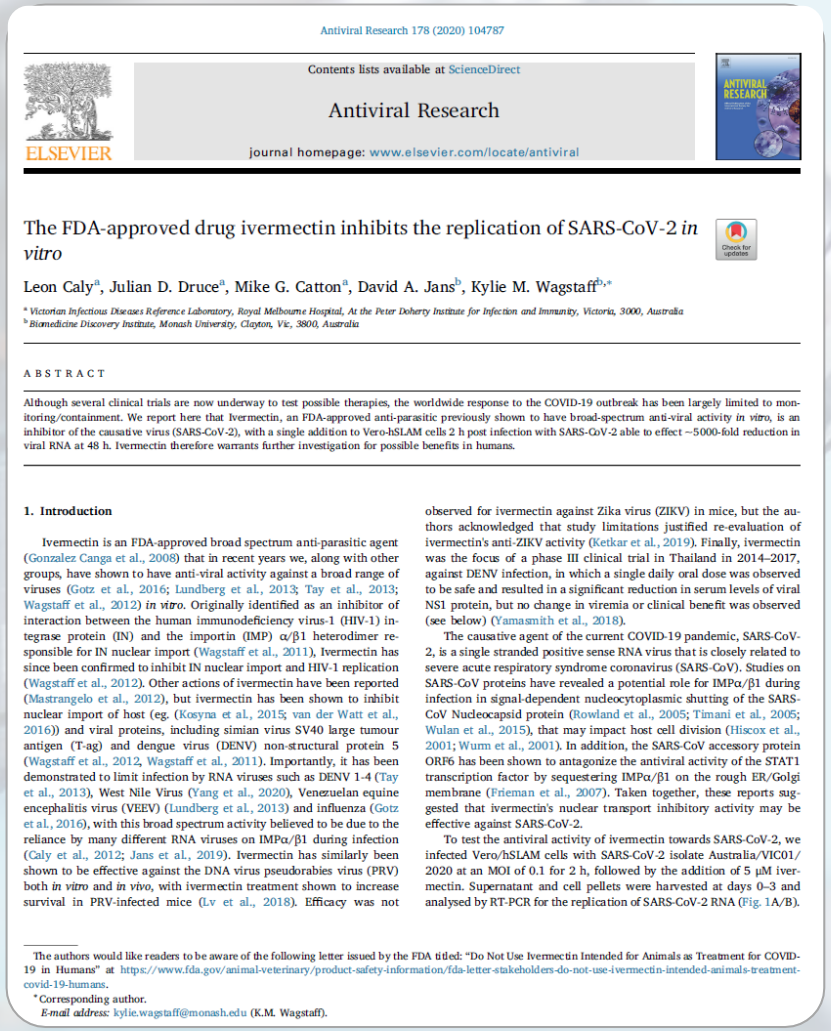

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Esse estudo demonstrou que:

✓ *in vitro*, este medicamento foi capaz de reduzir o RNA viral em aproximadamente 5.000 vezes em 48 horas.

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- ✓ Além de que, é **inviável alcançar no organismo humano a mesma concentração (5μM) de ivermectina** usada no estudo *in vitro*:
  - ➢ Ou seja, **seria necessária uma dose de 1.000 a 1.200 mg do medicamento**.

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Resumindo:

✓ A dose que reduziu a replicação do SARS-CoV-2, foi **pelo menos 100 vezes maior do que a quantidade dos comprimidos.**

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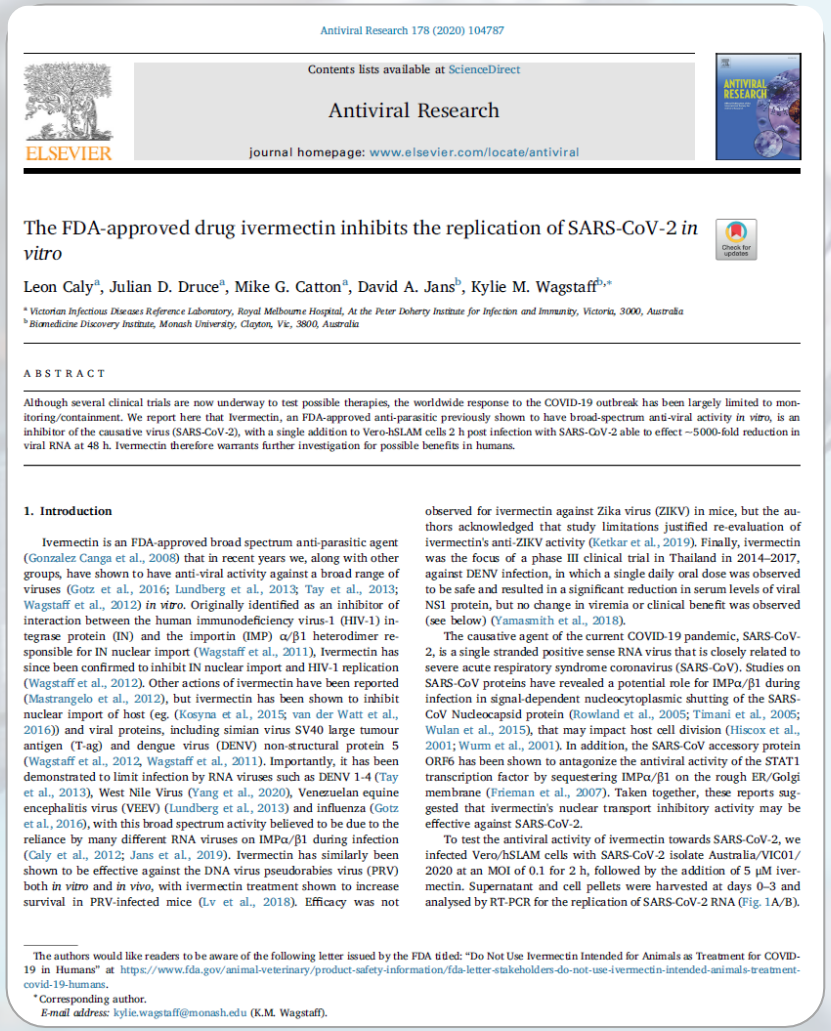

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RSFLIX

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Candidatos Biomedicina

Candidatos Enfermagem

Candidatos Farmácia

Candidatos Fisioterapia

Rafael Saraiva

Adicionar perfil

GERENCIAR PERFIS

RSFLIX

CONTÁGIO

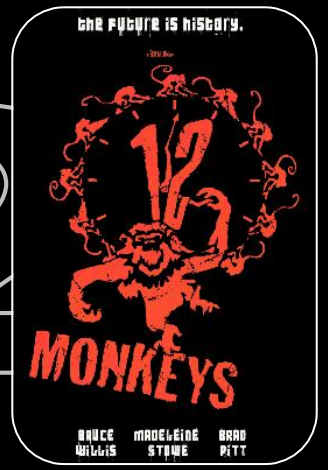
★ ★ ★ ★ ★ Thriller / Ficção Científica

Enquanto uma doença desconhecida se alastra, o mundo enfrenta o medo, o caos e a busca por uma cura.

Top 14 em filmes e séries sobre Pandemia

1

2



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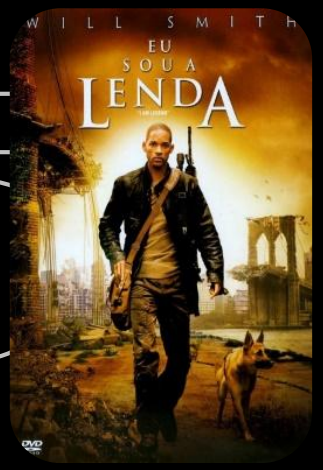

RSFLIX

OS 12 MACACOS

★ ★ ★ ★ ★ Ficção Científica

Enviado ao passado, um condenado tenta descobrir a origem de um vírus que quase extinguiu a população mundial.

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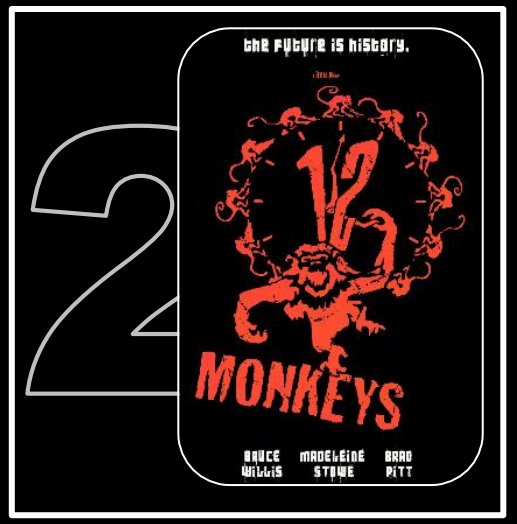

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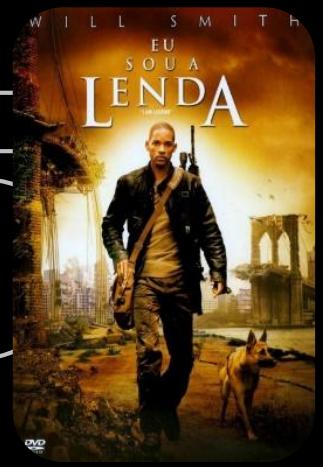

RSFLIX

GUERRA MUNDIAL Z

★ ★ ★ ★ ★ Terror / Ação

Com a humanidade à beira do colapso, um ex-agente da ONU busca a origem de uma epidemia devastadora.

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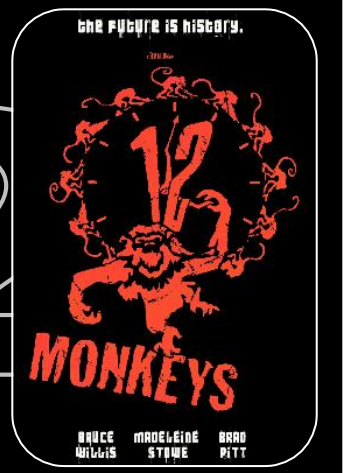

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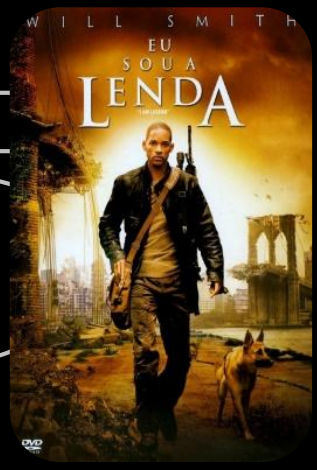


RSFLIX

PLANETA DOS MACACOS

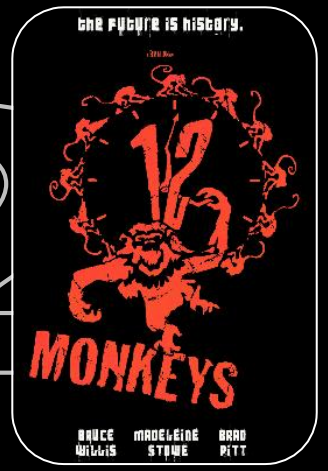
★ ★ ★ ★ ★ Ação / Ficção Científica

Um chimpanzé de laboratório desenvolve inteligência e emoções humanas e uma batalha épica surge para determinar qual espécie será dominante no planeta Terra.

Top 14 em filmes e séries sobre Pandemia

1

2



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4

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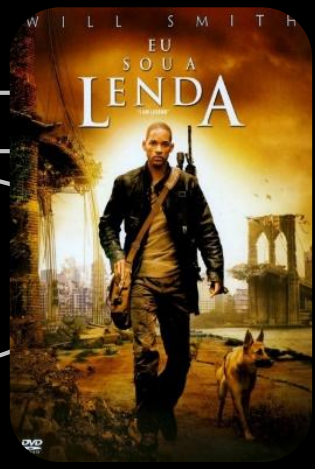


RSFLIX

EU SOU A LENDA

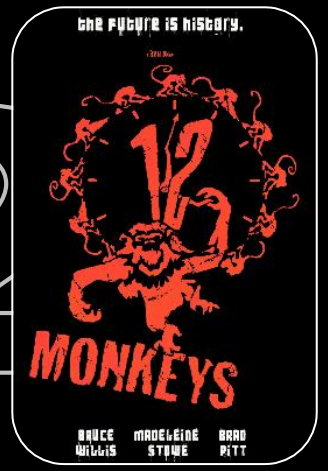
★ ★ ★ ★ ★ Ação / Terror

Imune a um vírus devastador, um cientista procura sobreviventes e uma cura para restaurar a humanidade.

Top 14 em filmes e séries sobre Pandemia

1

2



3

4

5

RSFLIX

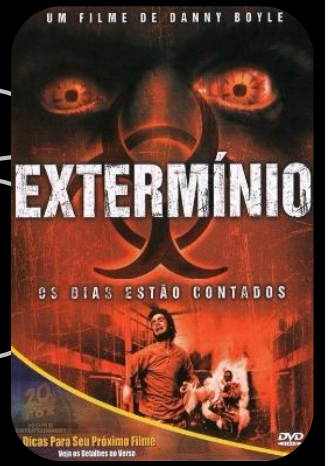
EXTERMÍNIO

★ ★ ★ ★ ★ Terror / Ficção Científica

Um grupo de sobreviventes enfrenta hordas de zumbis em uma jornada desesperada por um refúgio seguro.

Top 14 em filmes e séries sobre Pandemia

6



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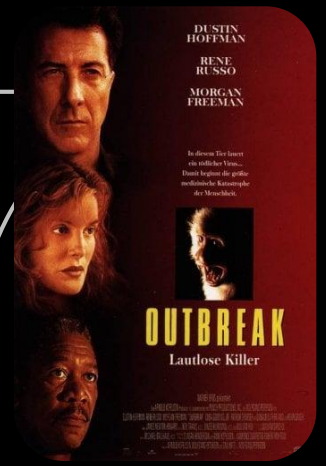

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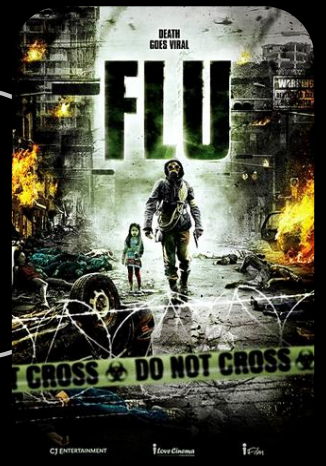


9

RSFLIX

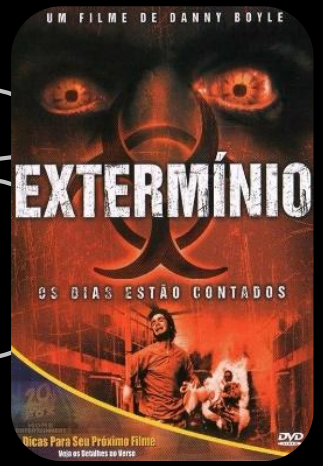
EPIDEMIA

★ ★ ★ ★ ★ Thriller / Ação

Após a chegada de um vírus letal, médicos e militares unem forças para conter uma epidemia devastadora.

Top 14 em filmes e séries sobre Pandemia

6



7

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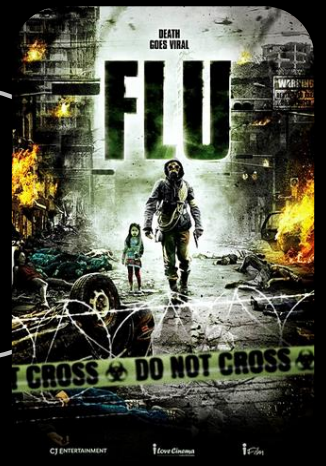


9

RSFLIX

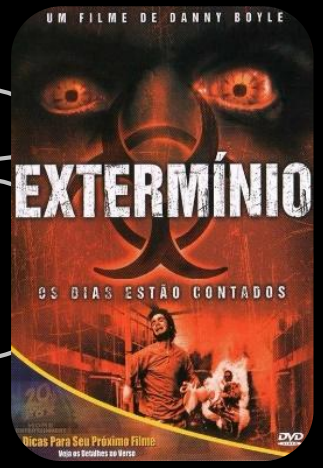
A GRIPE

★ ★ ★ ★ ★ Ficção Científica / Thriller

Uma doença desconhecida se espalha rapidamente, levando a população ao caos em busca de uma chance de sobreviver.

Top 14 em filmes e séries sobre Pandemia

6



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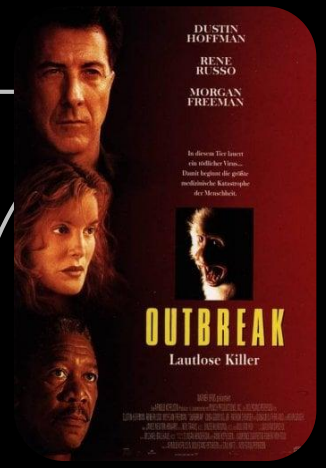

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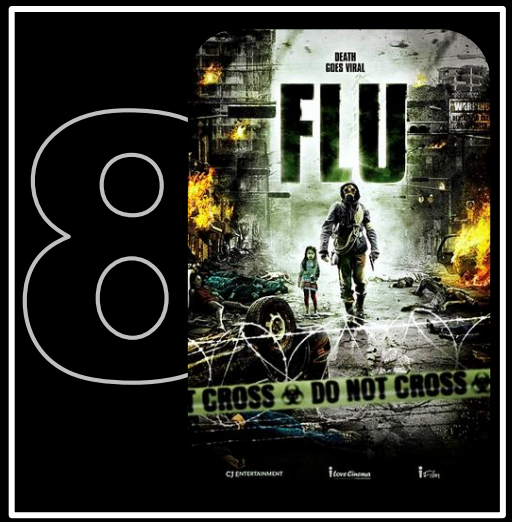


9

RSFLIX

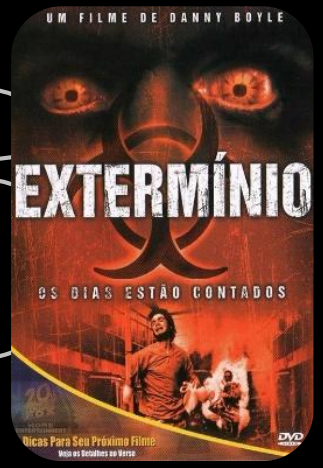
THE HOT ZONE

★ ★ ★ ★ ★ Drama / Suspense

O surgimento de um surto de ebola desencadeia uma crise global que desafia cientistas e governos.

Top 14 em filmes e séries sobre Pandemia

6



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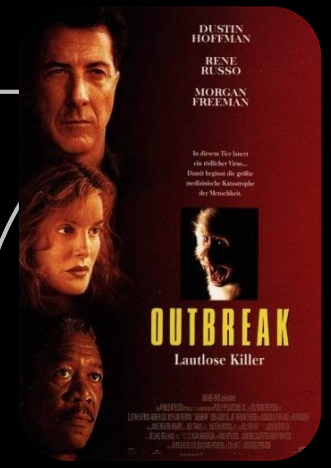

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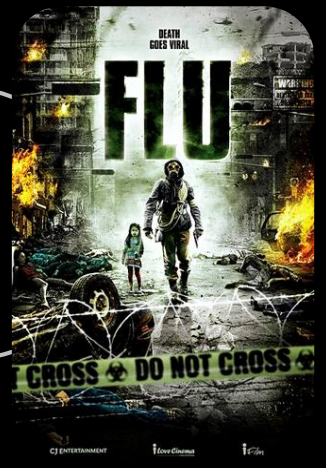


9

RSFLIX

THE LAST SHIP

★ ★ ★ ★ ★ Ação / Drama / Ficção

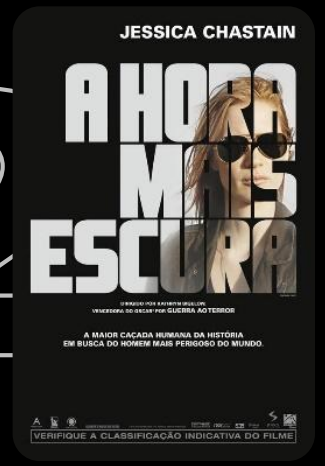
Após uma epidemia devastar o planeta, a tripulação de um navio de guerra busca uma forma de salvar a humanidade.

Top 14 em filmes e séries sobre Pandemia

10

11

12



RSFLIX

THE LAST OF US

★ ★ ★ ★ ★ Ação / Aventura / Drama / Terror

Em um mundo devastado por um fungo mortal, um homem escolta uma adolescente que pode salvar a humanidade.

Top 14 em filmes e séries sobre Pandemia

10

11

12

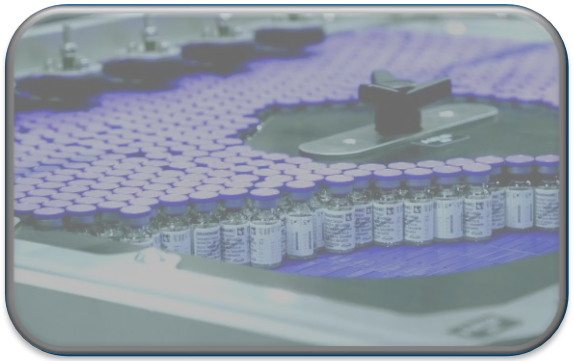
ETAPAS DA AULA

01

HISTÓRIA DA VACINAÇÃO

02

IMUNO BIOLÓGICOS



03

P&D FASES

04

DESAFIOS E DEBATES ATUAIS

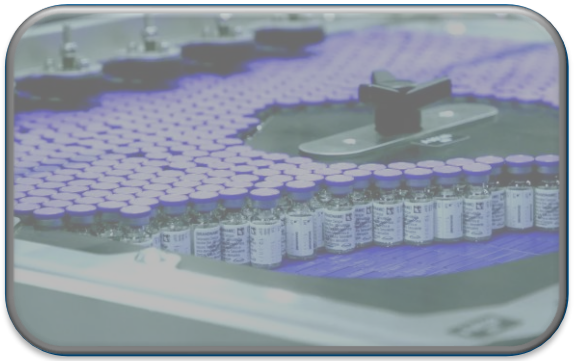
ETAPAS DA AULA

01

HISTÓRIA DA VACINAÇÃO

02

IMUNO BIOLÓGICOS



03

P&D FASES

04

DESAFIOS E DEBATES ATUAIS

DESAFIOS E DEBATES ATUAIS

FAKE NEWS & MOVIMENOS ANTI-VACINA

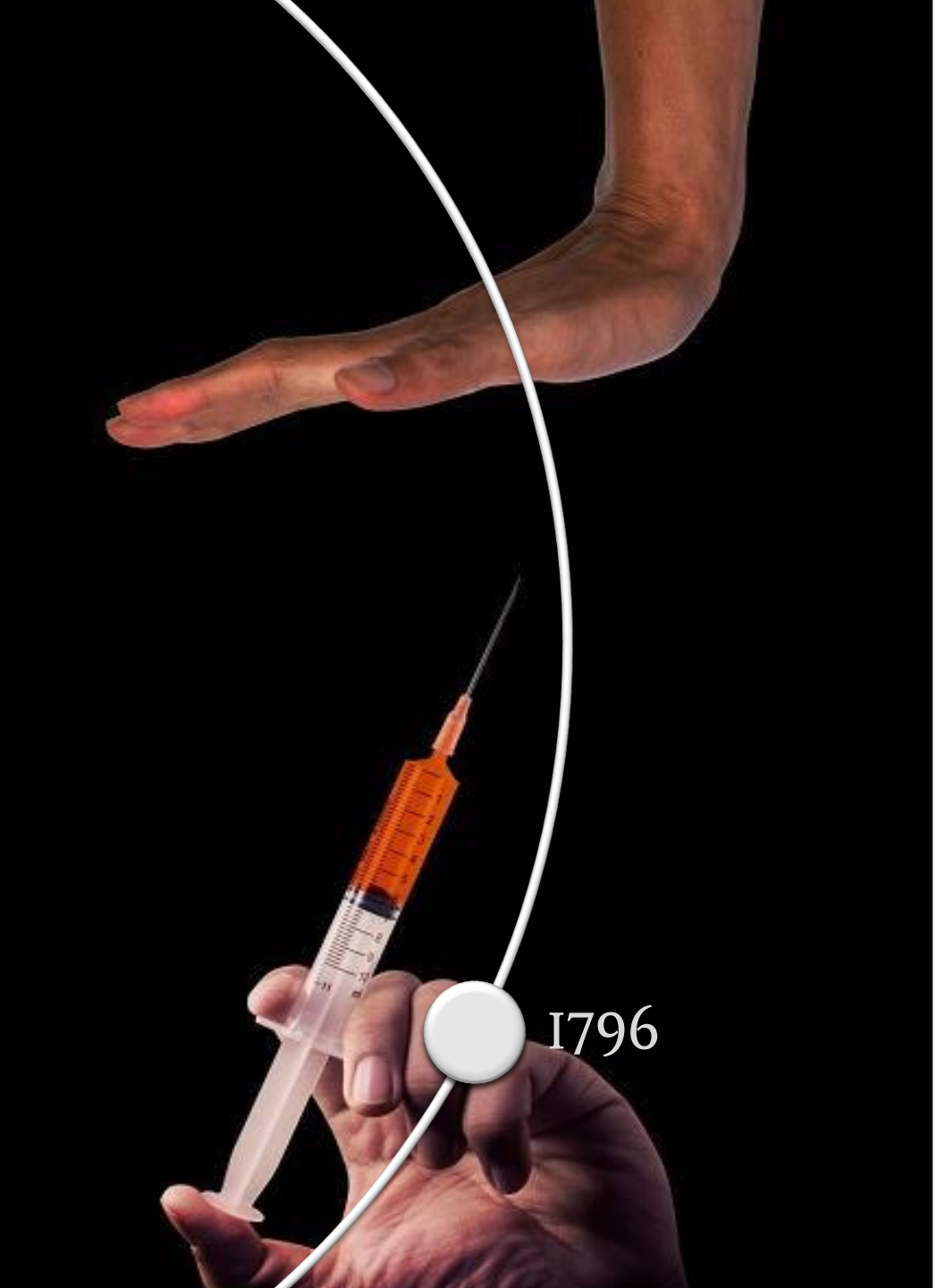
COMBATE AS *FAKE NEWS* E OS MOVIMENTOS ANTI-VACINA

SÃO TÃO ANTIGOS QUANTO
A PRÓPRIA VACINA;

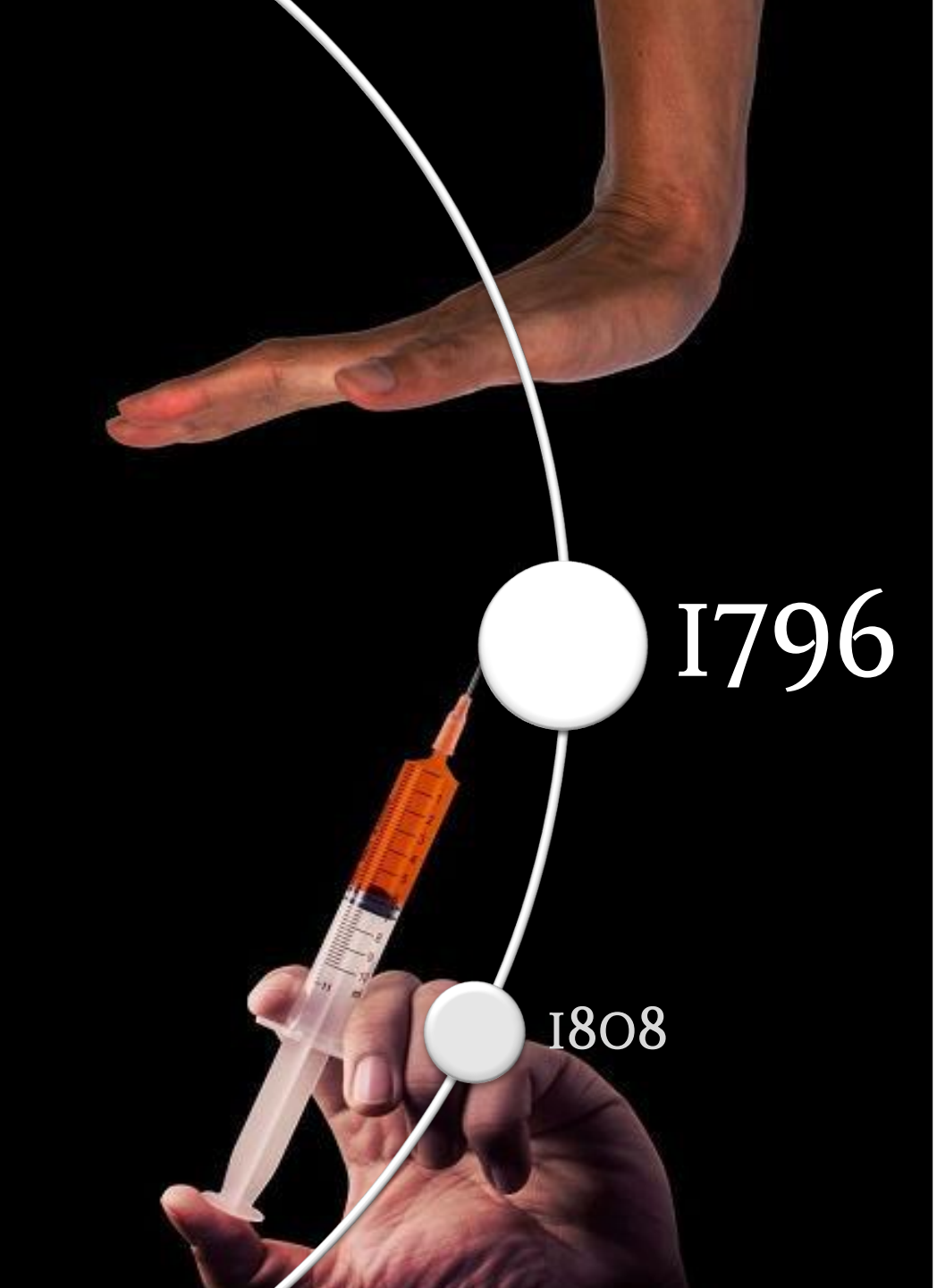
COMBATE AS *FAKE NEWS* E OS MOVIMENTOS ANTI-VACINA

COMBATE AS *FAKE NEWS* E OS MOVIMENTOS ANTI-VACINA

A DESINFORMAÇÃO NÃO É CATAPULTADA APENAS "POR QUEM, MÁ-FÉ OU IGNORÂNCIA", CRIAM TEORIAS CONTRAS ARMAS CRUZADAS CONTRA UM MÉTODO QUE PROTEGEU BILHÕES DE VIDAS DESDE SUA CRIAÇÃO.



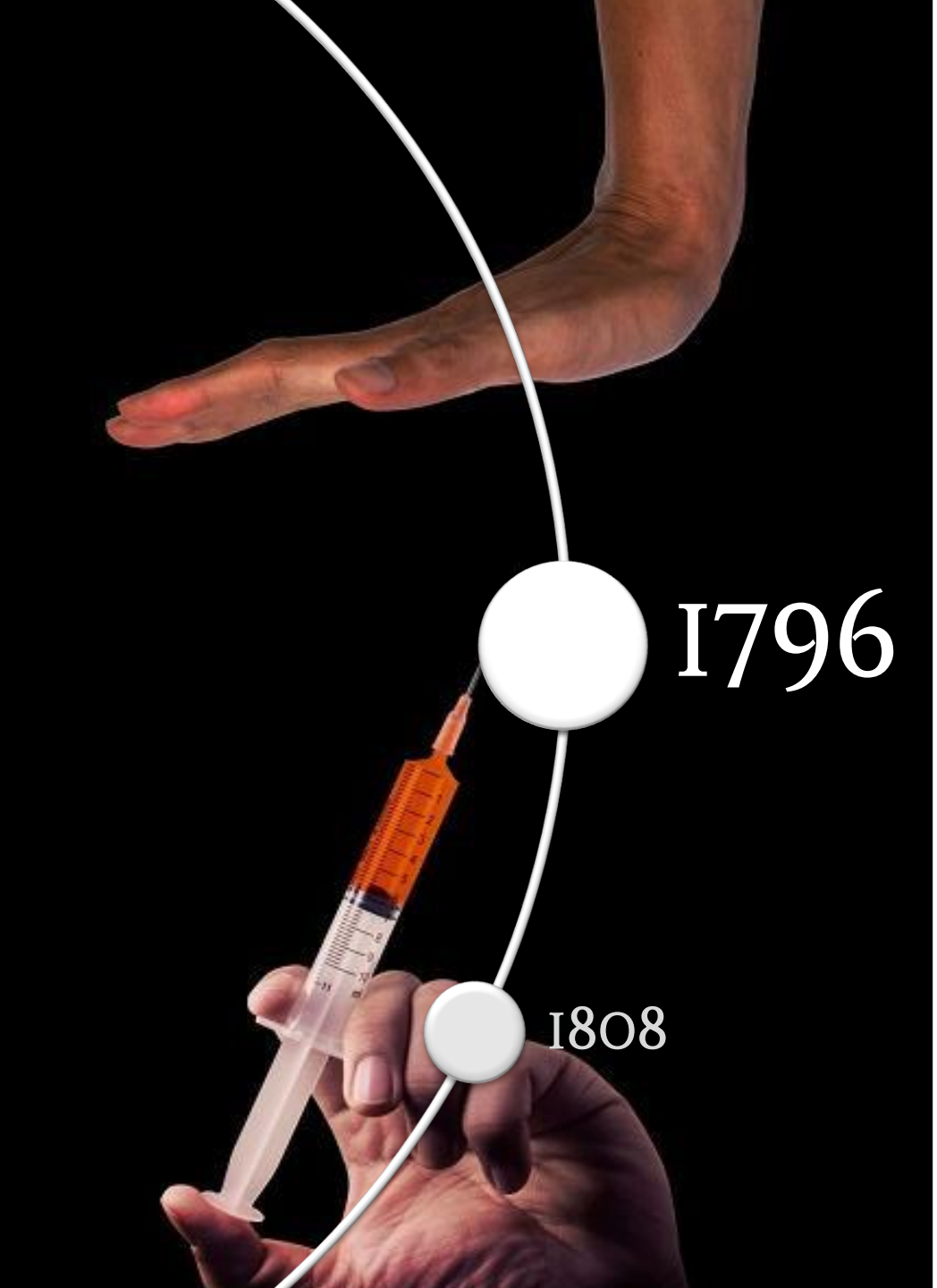
EDWARD JENNER



CHARGE DO MOVIMENTO ANTI-VACINA DA VARÍOLA

SATIRIZA O PROCEDIMENTO AO REPRESENTAR "OS MARAVILHOSOS EFEITOS DA NOVA INOCULAÇÃO".

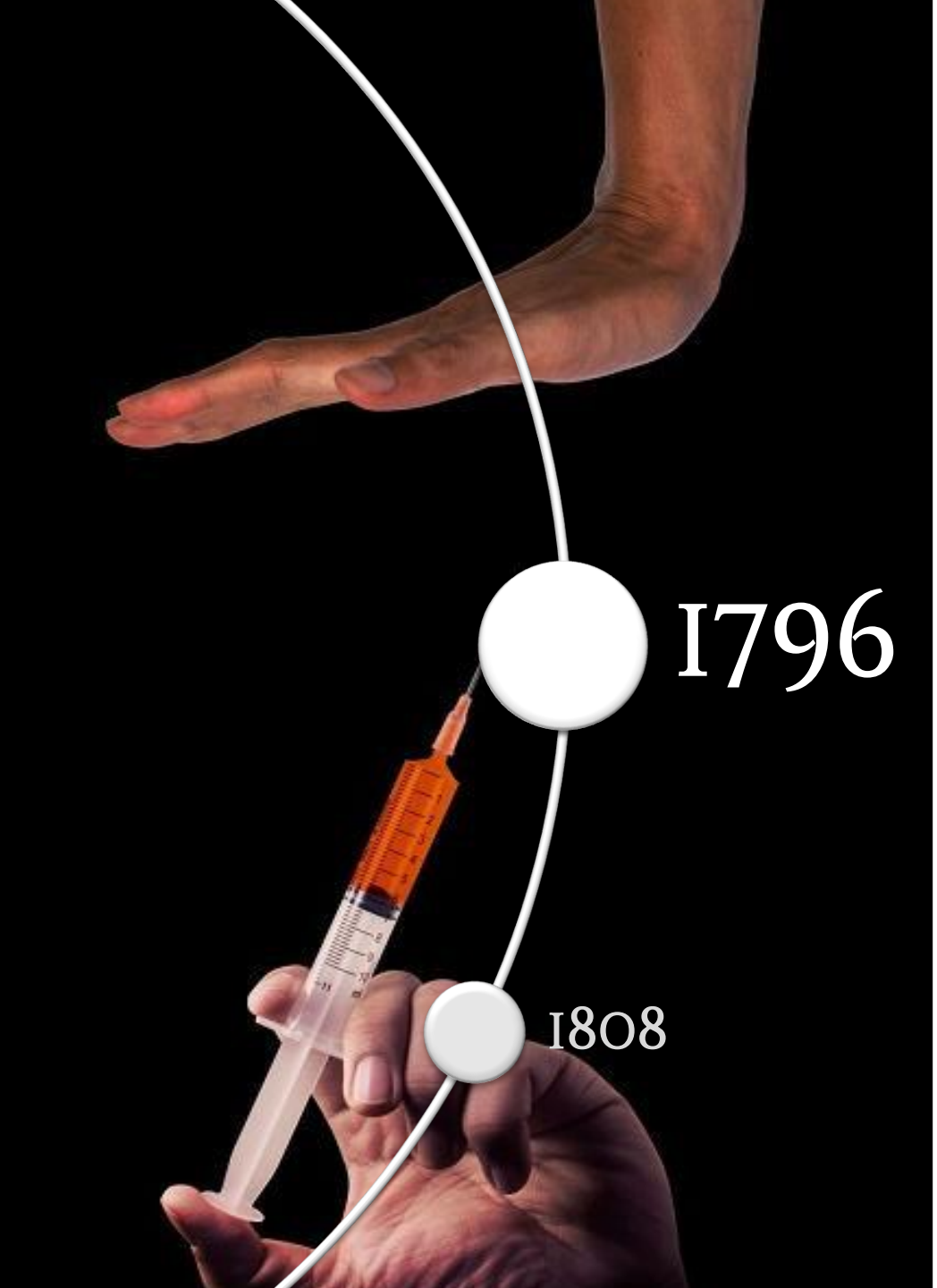
FONTE: HTTPS://WWW.BBC.COM/PORTUGUESE/VERT-FUT-54416626



Charge do Movimento Anti-Vacina da Varíola

Havia transferência de doenças que acometiam os animais para o homem.

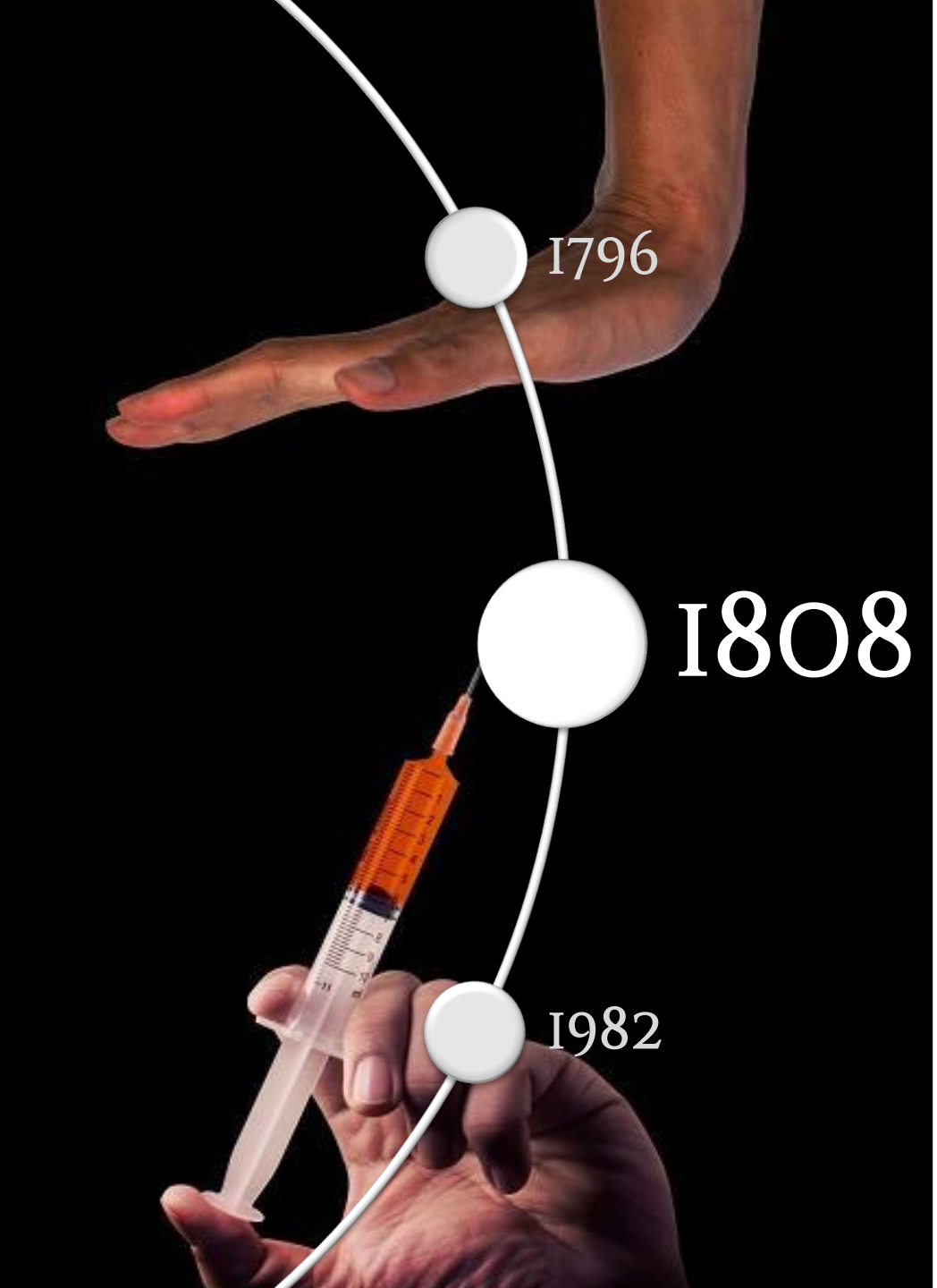
Fonte: https://www.bbc.com/portuguese/vert-fut-54416626



Charge do Movimento Anti-Vacina da Varíola

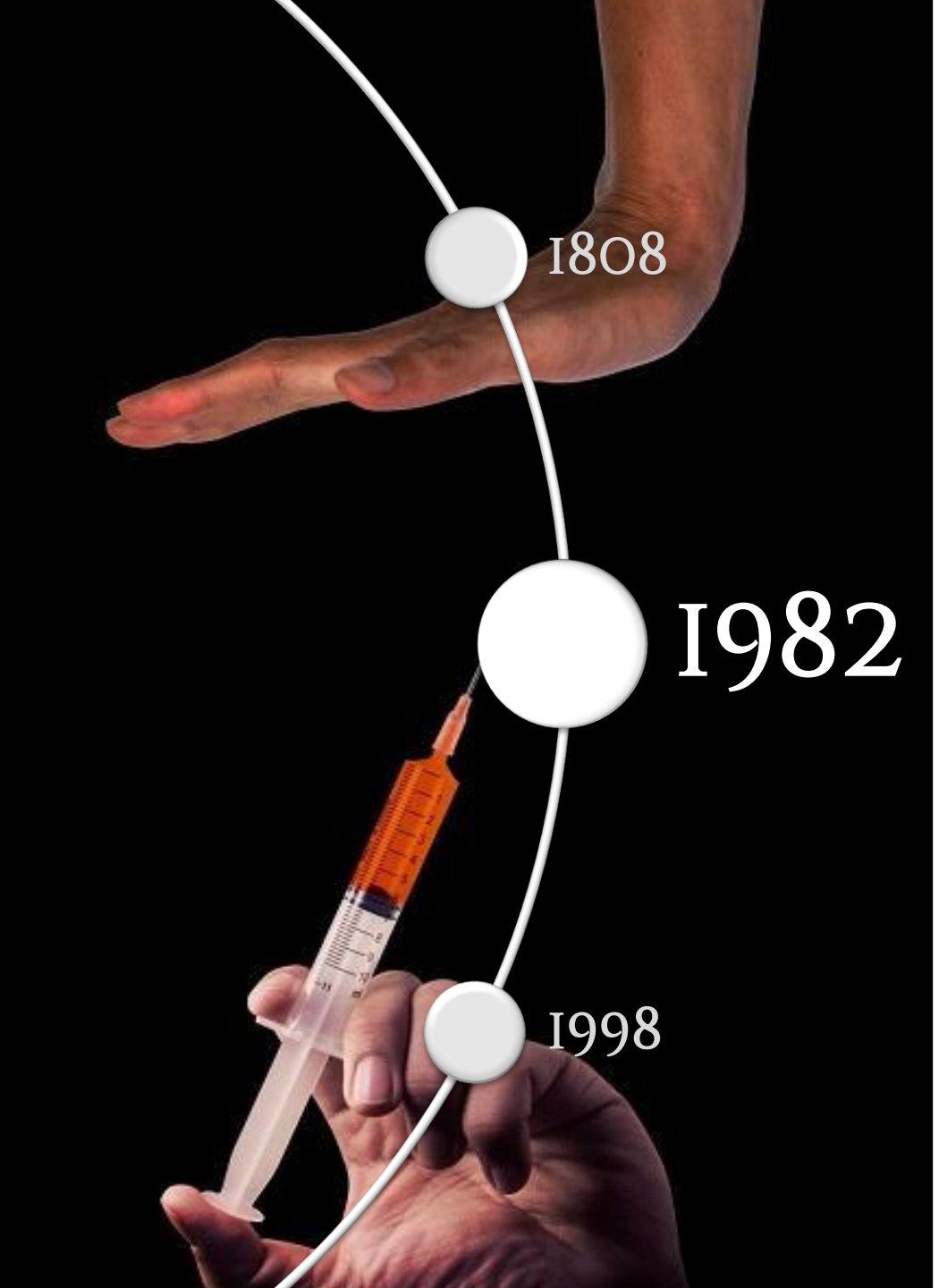
Além de "bestializar" os vacinados, dando fisionomia de vaca.

Fonte: https://www.bbc.com/portuguese/vert-fut-54416626



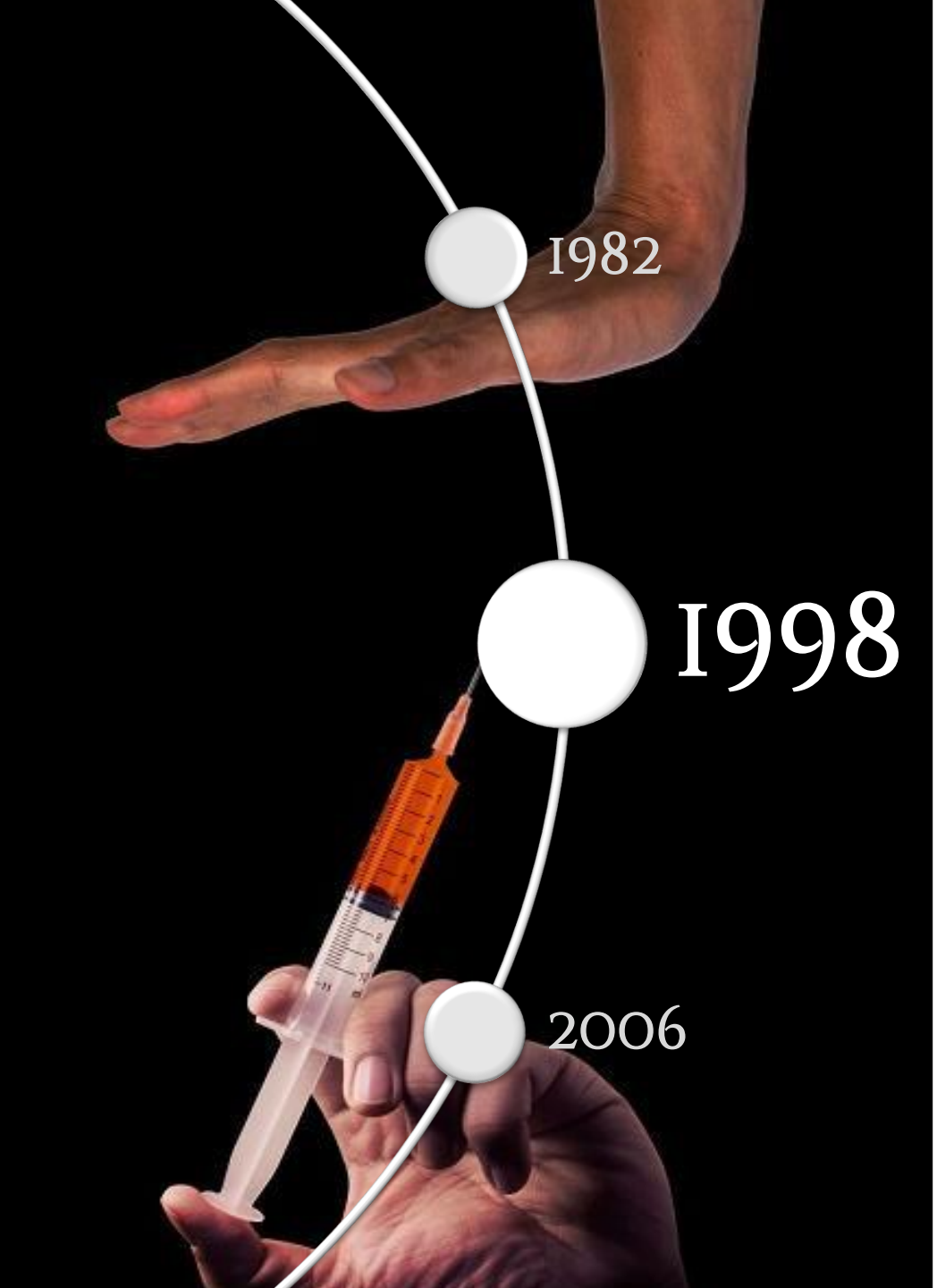
PUBLICAÇÃO LUSITANA

"E O CLERO PORTUGUÊS AFIRMAVA QUE OS VACINADOS RECEBIAM O PRÓPRIO DEMÔNIO NO CORPO, E SUAS ALMAS ERAM ROUBADAS".



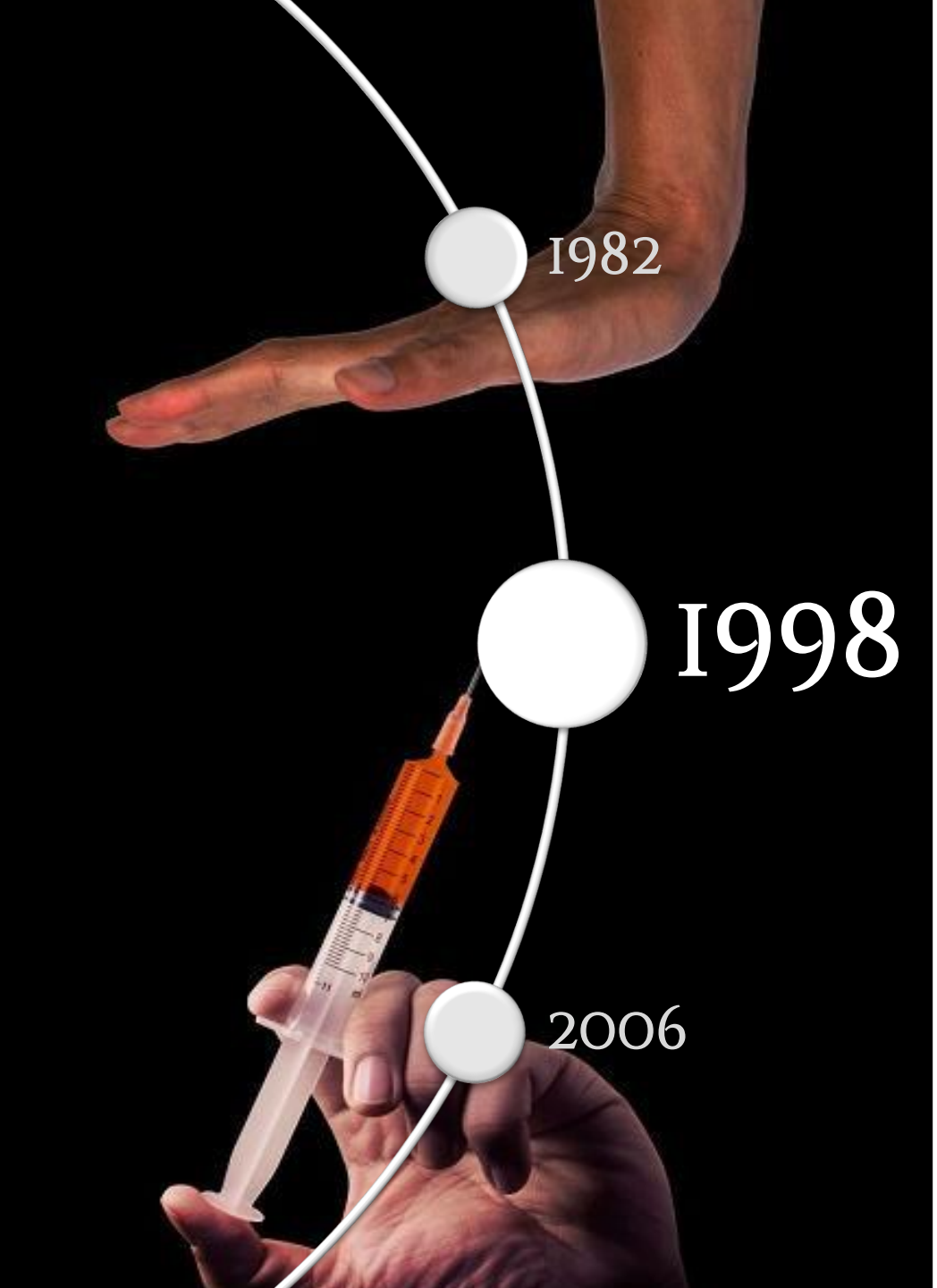
CANAL AMERICANO NBC
DPT: VACCINE ROULETTE

PRODUZIU UM ANTIVAX DE ESCALA GLOBAL, ASSOCIANDO A VACINA CONTRA "TÉTANO, DIFTERIA E COQUELUCHE", **A DANOS CEREBRAIS.**



Andrew Wakefield

Responsável pela atual onda anti-vacina



THE LANCET

The Lancet, Volume 351, Issue 9103, Pages 637 - 641, 28 February 1998
doi:10.1016/S0140-6736(97)11096-0

This article was retracted

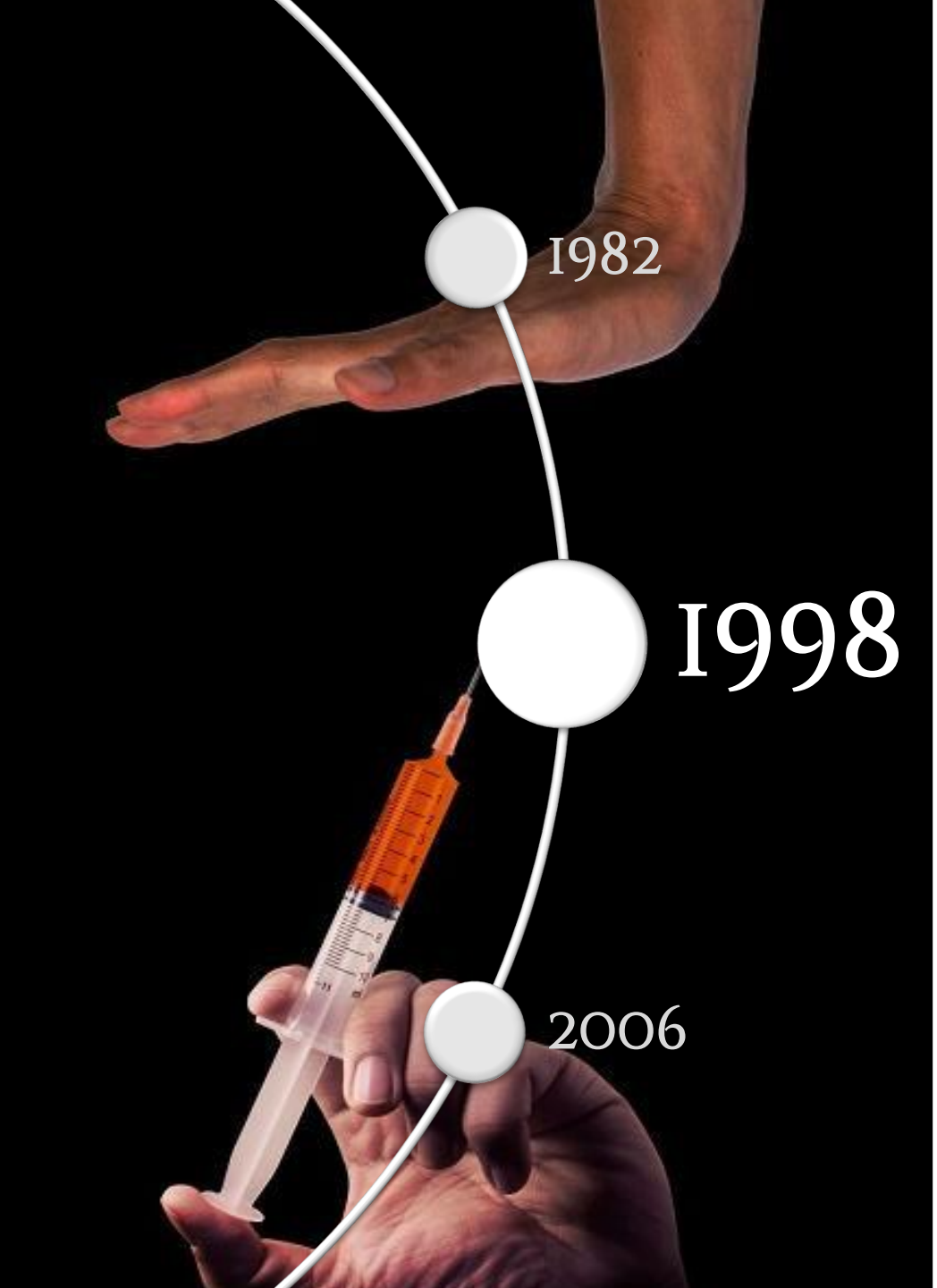
RETRACTED: Ileal-lymphoid-nodular hyperplasia, non-specific colitis, and pervasive developmental disorder in children

Dr AJ Wakefield FRCS a, SH Murch MB b, A Anthony MB a, J Linnell PhD a, DM Casson MRCP b, M Malik MRCP b, M Berelowitz FRCPsych c, AP Dhillon MRCPath a, MA Thomson FRCP b, P Harvey FRCP d, A Valentine FRCR e, SE Davies MRCPath a, JA Walker-Smith FRCP a

Summary

ARTIGO NA LANCET

VINCULOU A TRÍPLICE VIRAL, QUE PROTEGE CONTRA "SARAMPO, CAXUMBA E RUBÉOLA" AO AUTISMO.



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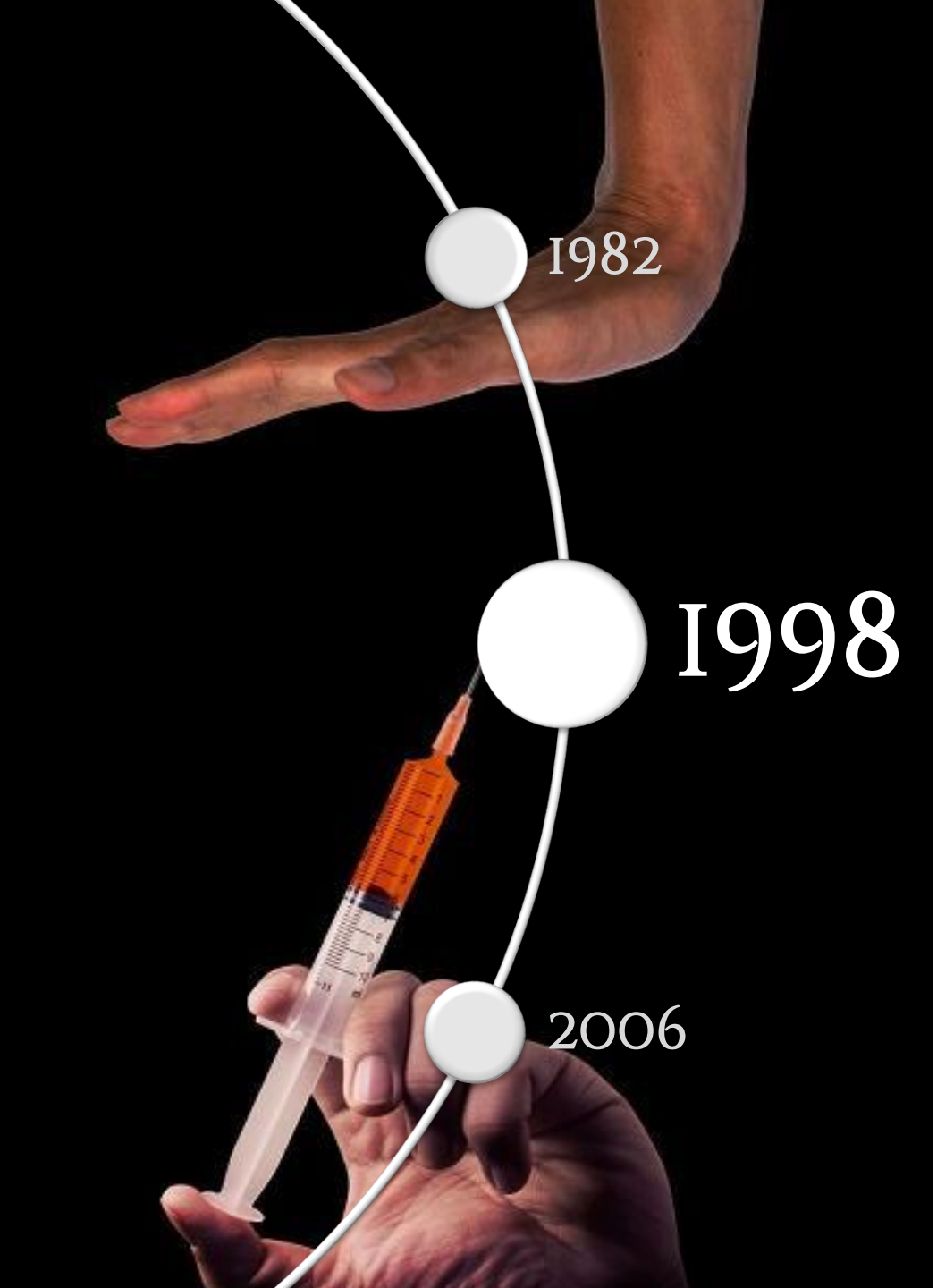
Summary

ARTIGO NA LANCET

DESCOBRIU-SE, QUE WAKEFIELD ESTAVA TENTANDO PATENTEAR UMA VACINA CONCORRENTE.



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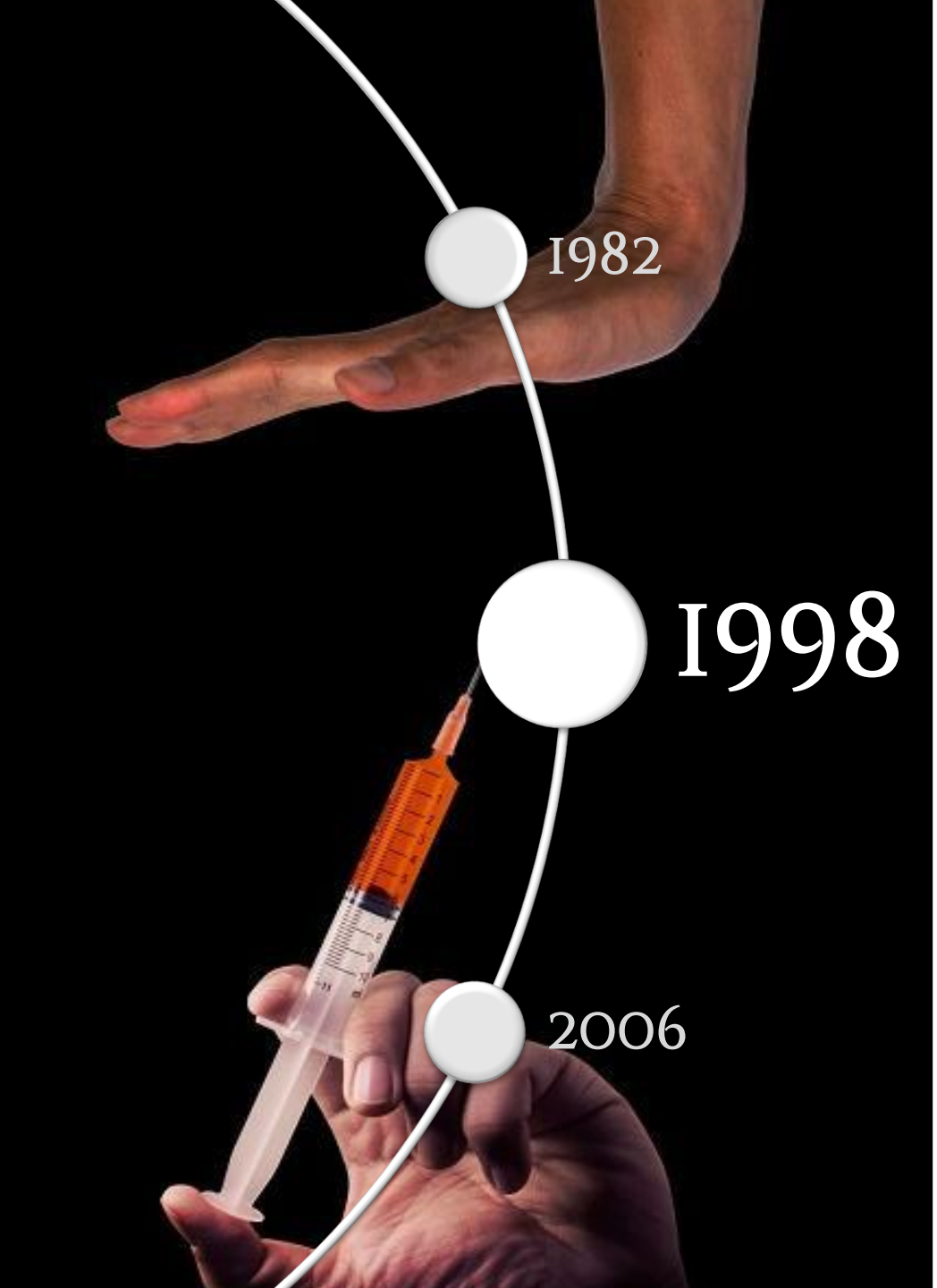
ARTIGO NA LANCET

PARA SEU "EXPERIMENTO": PAGOU CINCO LIBRAS A CRIANÇAS PARA COLETAR O SANGUE DELAS NA FESTA DE ANIVERSÁRIO DO FILHO.



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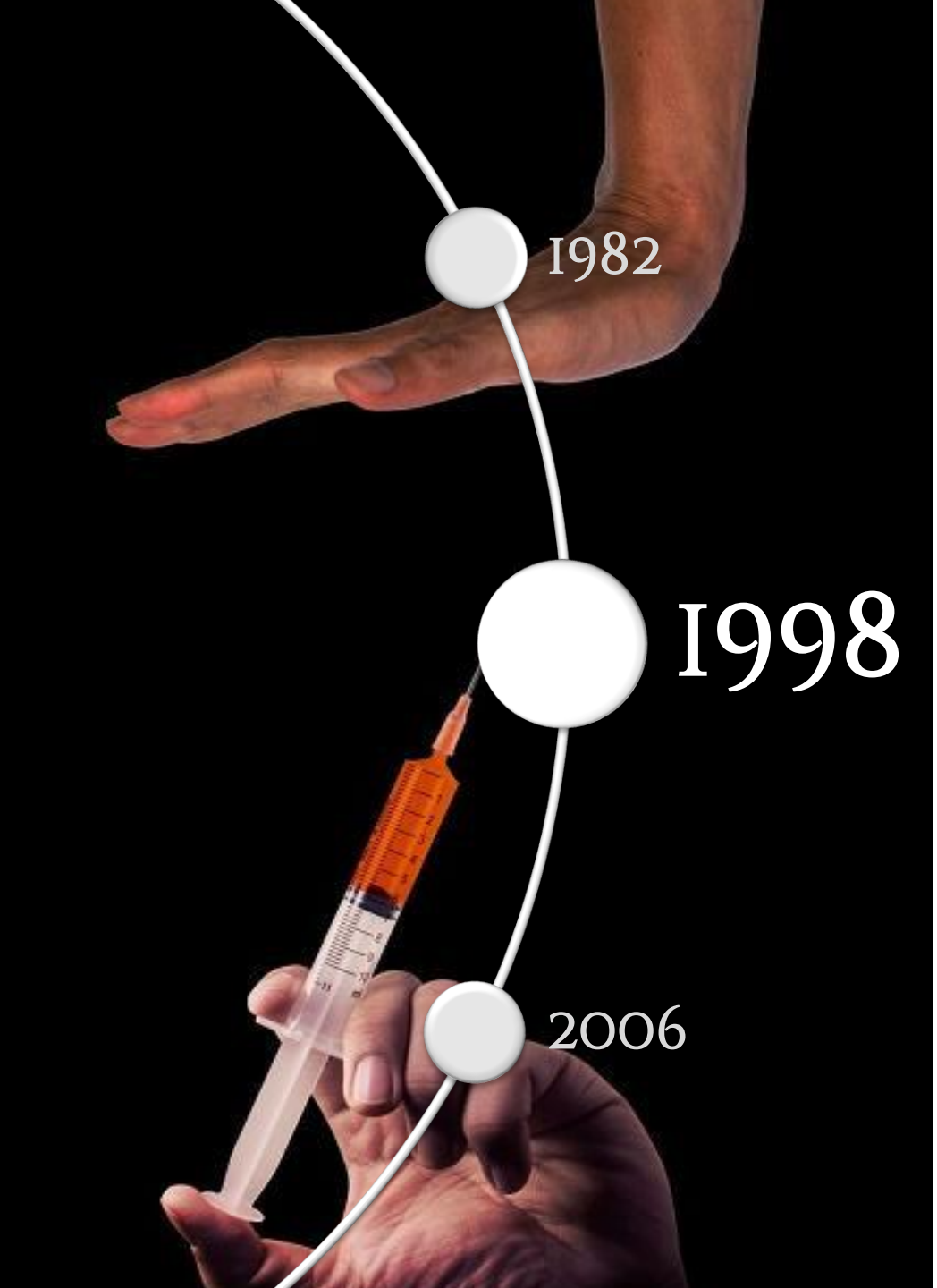

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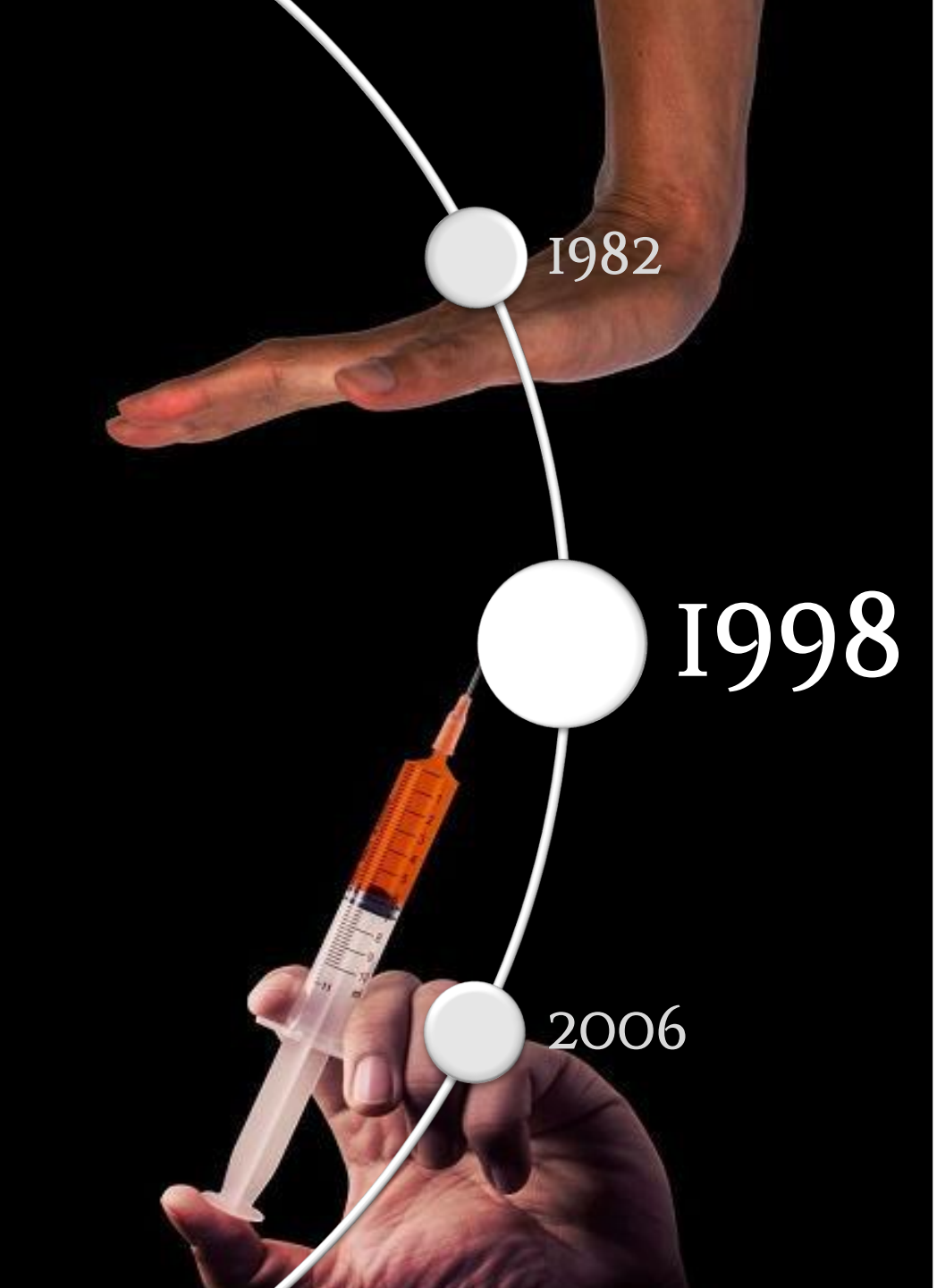
Summary

Background
We investigated a consecutive series of children with chronic enterocolitis and regressive developmental disorder.

Methods
12 children (mean age 6 years [range 3–10], 11 boys) were referred to a paediatric gastroenterology unit with a history of normal development followed by loss of acquired skills, including language, together with diarrhoea and abdominal pain. Children underwent gastroenterological,

RETRACTED

LANCET
SE RETRATOU!



MMR DOCS IN U-TURN

Dramatic retraction KOs autism link

Colleagues withdraw support for doctor who sparked MMR scare

Jab scare climbdown

MMR authors deny finding autism link

ANTI-MMR SCIENTISTS CHANGE THEIR MINDS

Major U-turn on triple jab risk

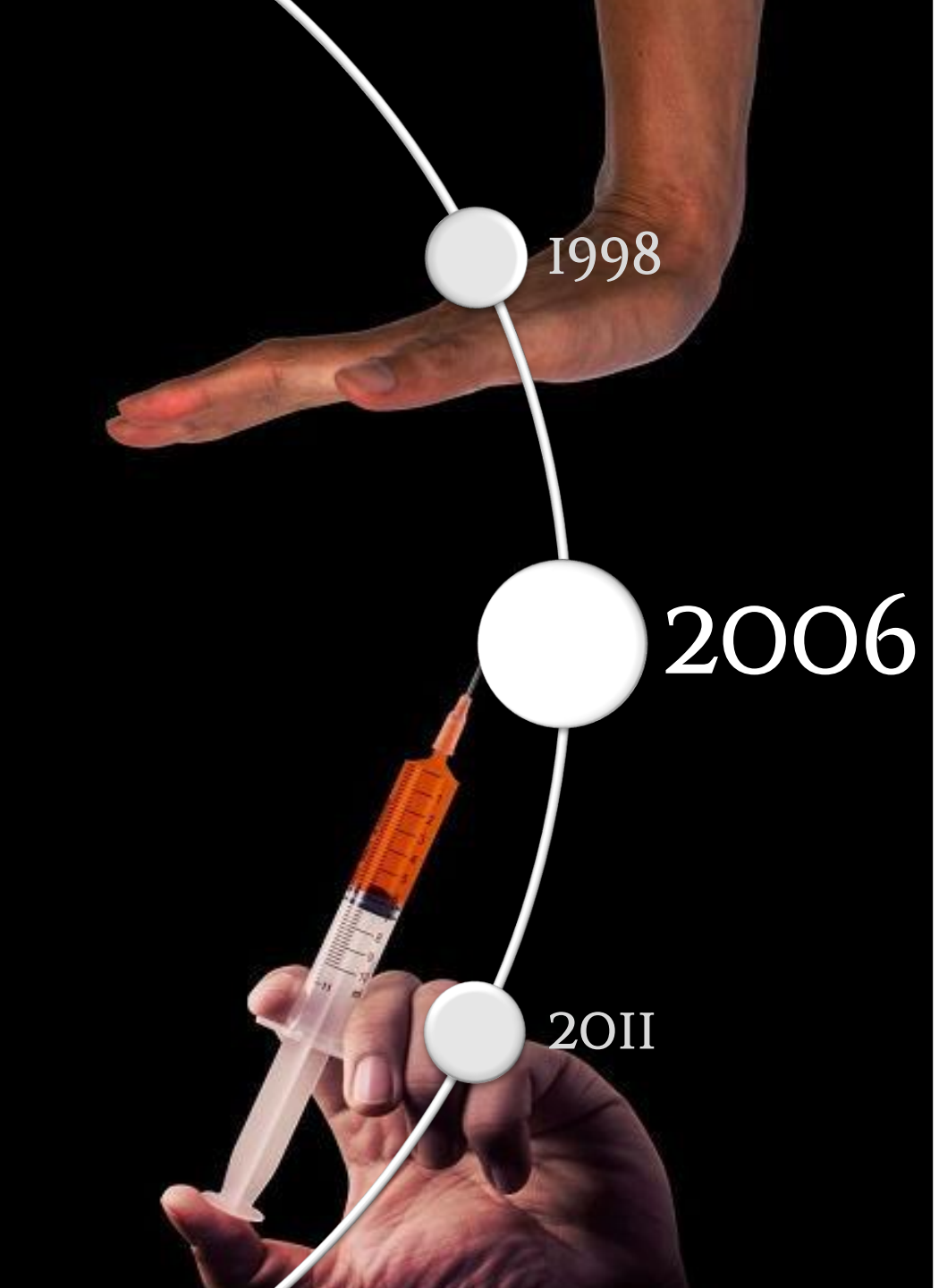
Retraction from MMR authors

MMR research scientists retract link with autism

MMR doctors disown Wakefield study

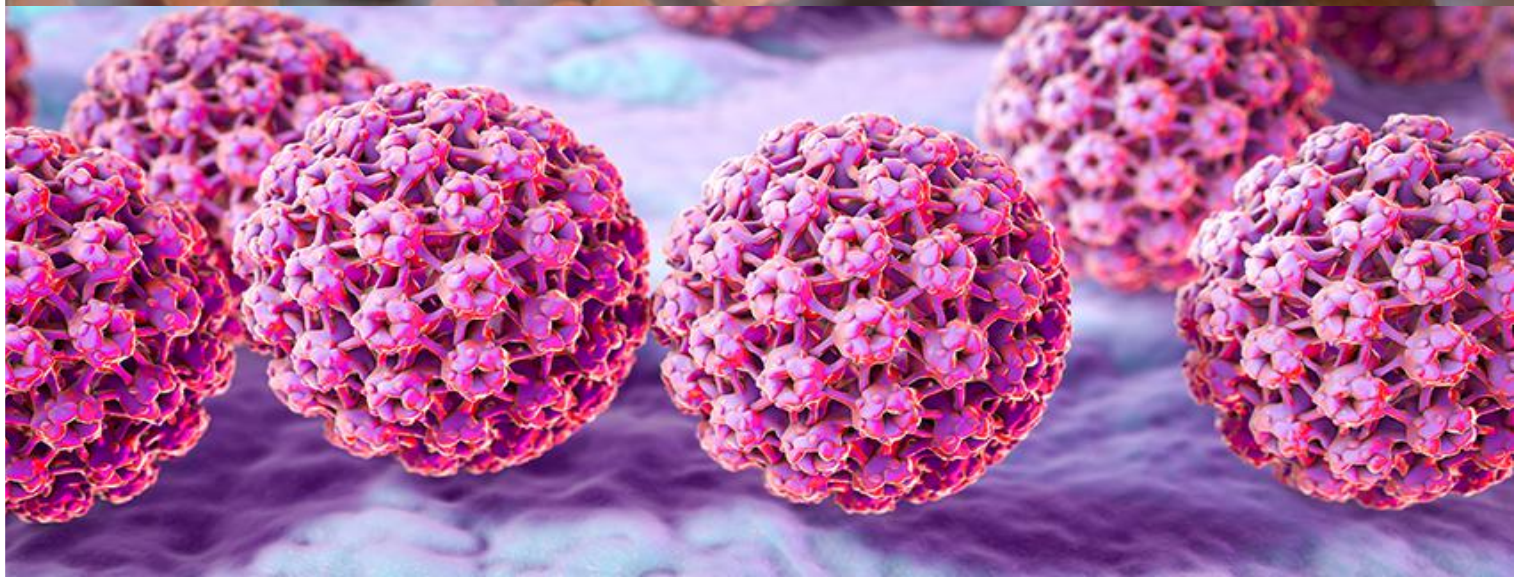
TEVE A LICENÇA MÉDICA CASSADA;

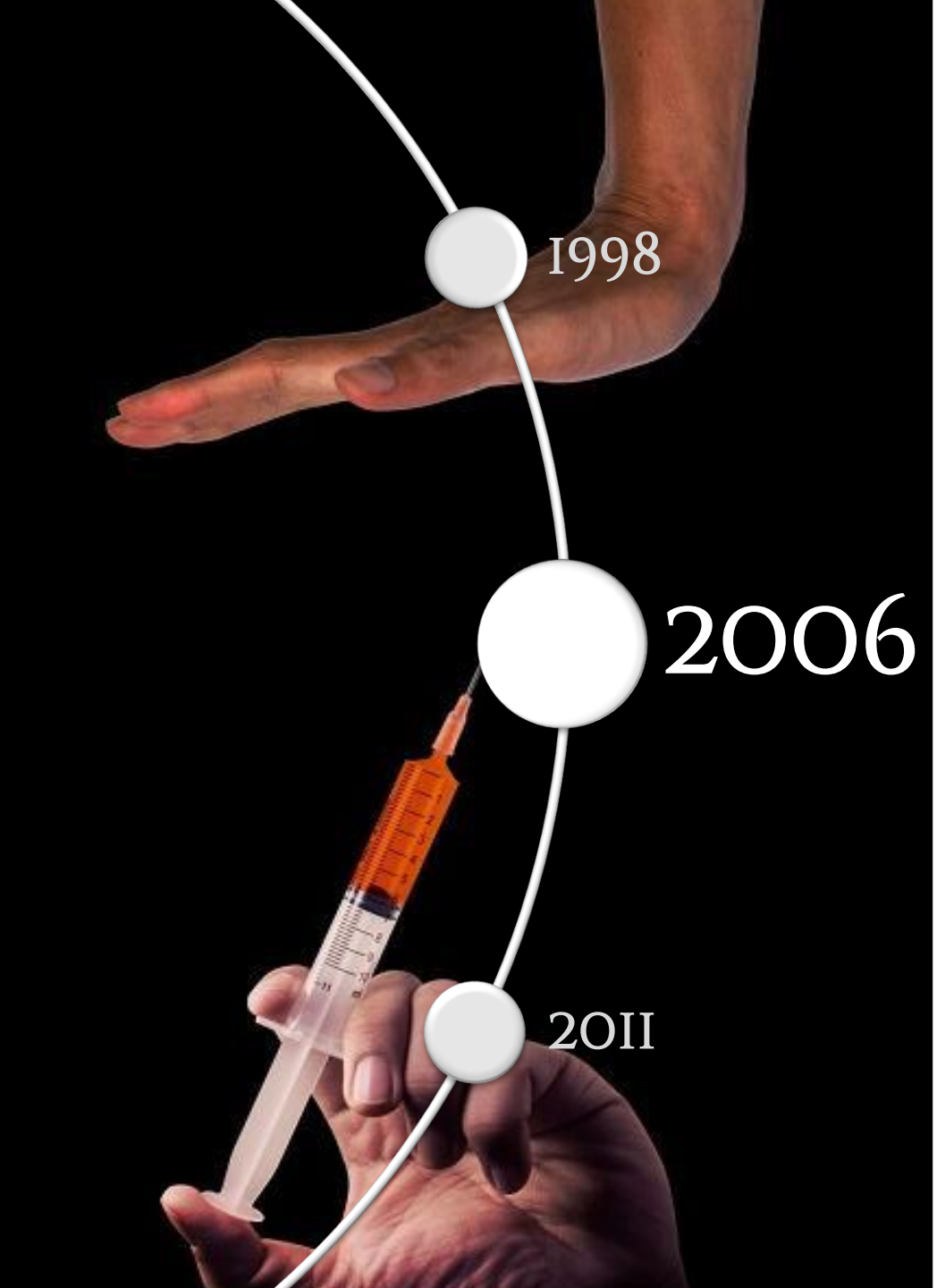
SUA PESQUISA NUNCA FOI RESPALDADA POR NOVOS ESTUDOS.



QUESTÕES RELIGIOSAS

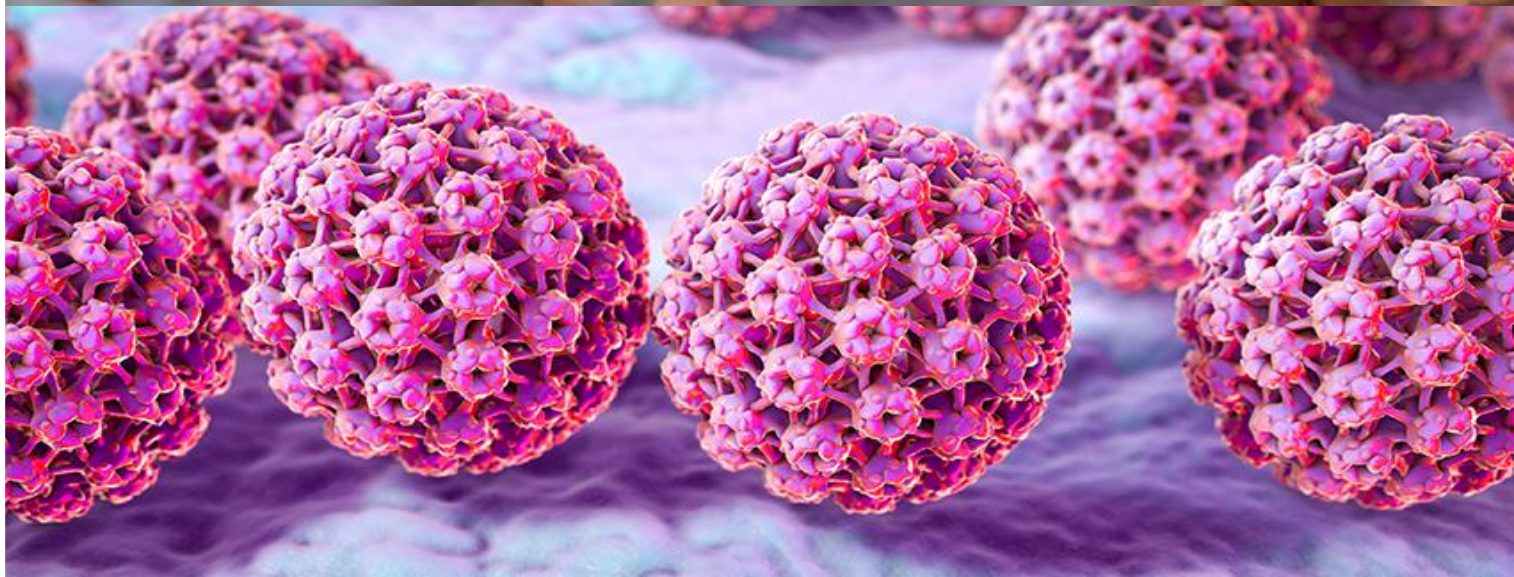
AS VACINAS CONTÊM FETOS ABORTADOS.

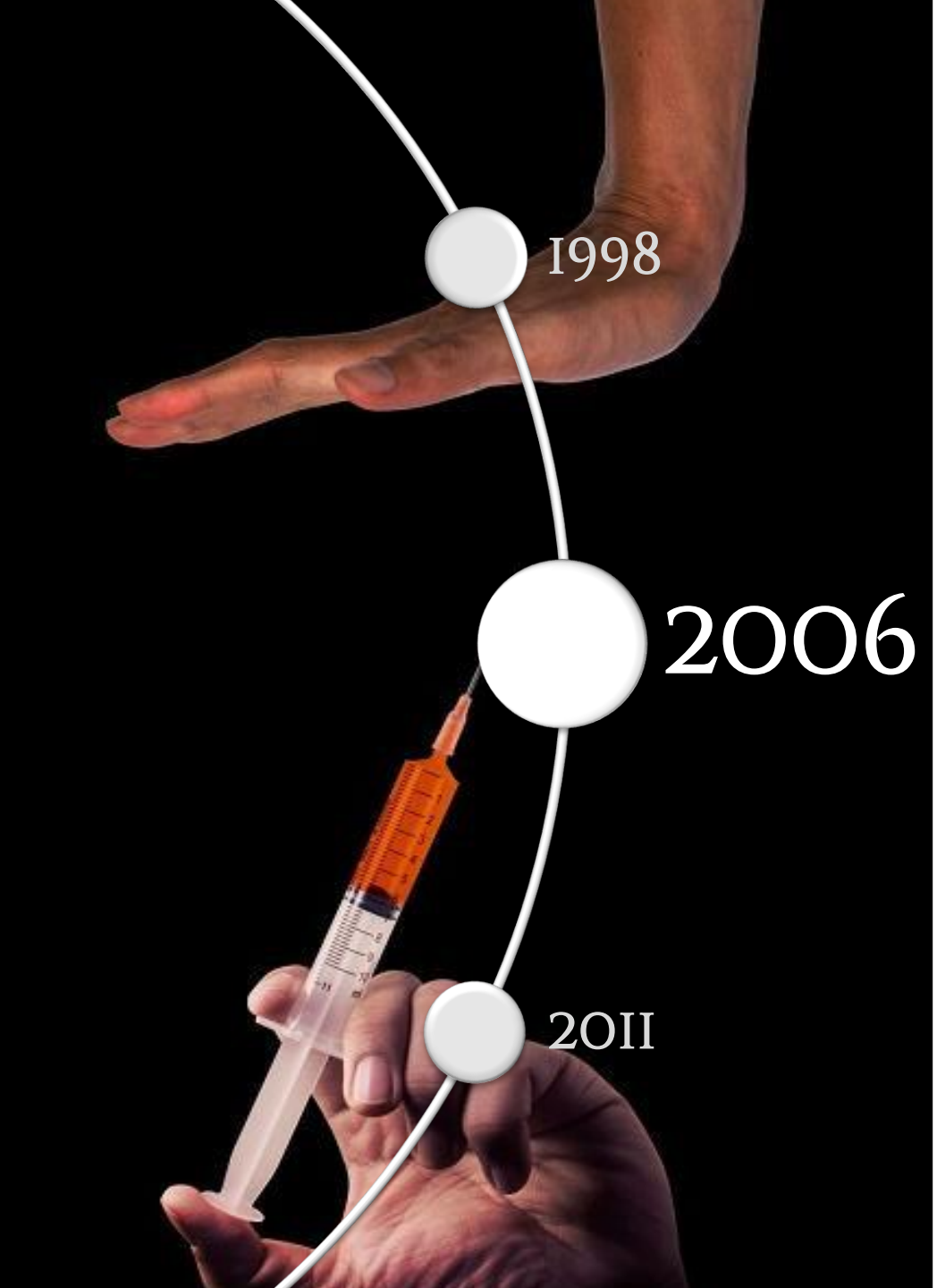




QUESTÕES RELIGIOSAS VACINA CONTRA O HPV

ASSOCIAR A VACINA COM PROMISCUIDADE, SENDO UM INCENTIVO A INICIAÇÃO PRECOCE DA VIDA SEXUAL.





VACINA CONTRA O HPV

The NEW ENGLAND JOURNAL of MEDICINE

ORIGINAL ARTICLE

HPV Vaccination and the Risk of Invasive Cervical Cancer

Jiayao Lei, Ph.D., Alexander Ploner, Ph.D., K. Miriam Elfström, Ph.D., Jiangrong Wang, Ph.D., Adam Roth, M.D., Ph.D., Fang Fang, M.D., Ph.D., Karin Sundström, M.D., Ph.D., Joakim Dillner, M.D., Ph.D., and Pär Sparén, Ph.D.

From the Departments of Medical Epidemiology and Biostatistics (J.L., A.P., P.S.) and Laboratory Medicine (K.M.E., J.W., K.S., J.D.) and the Institute of Environmental Medicine (F.F.), Karolinska Institutet, the Regional Cancer Center Stockholm Gotland (K.M.E.), and the Karolinska University Laboratory, Karolinska University Hospital (J.D.), Stockholm, the Department of Communicable Disease Control and Health Protection, Public Health Agency of Sweden, Solna (A.R.), and the Department of Translational Medicine, Lund University, Lund (A.R.) — all in Sweden. Address reprint requests to Dr. Lei at Nobels väg 12A, 171 65 Solna, Sweden, or at jiayao.lei@ki.se.

N Engl J Med 2020;383:1340-8.
DOI: 10.1056/NEJMoa1917338
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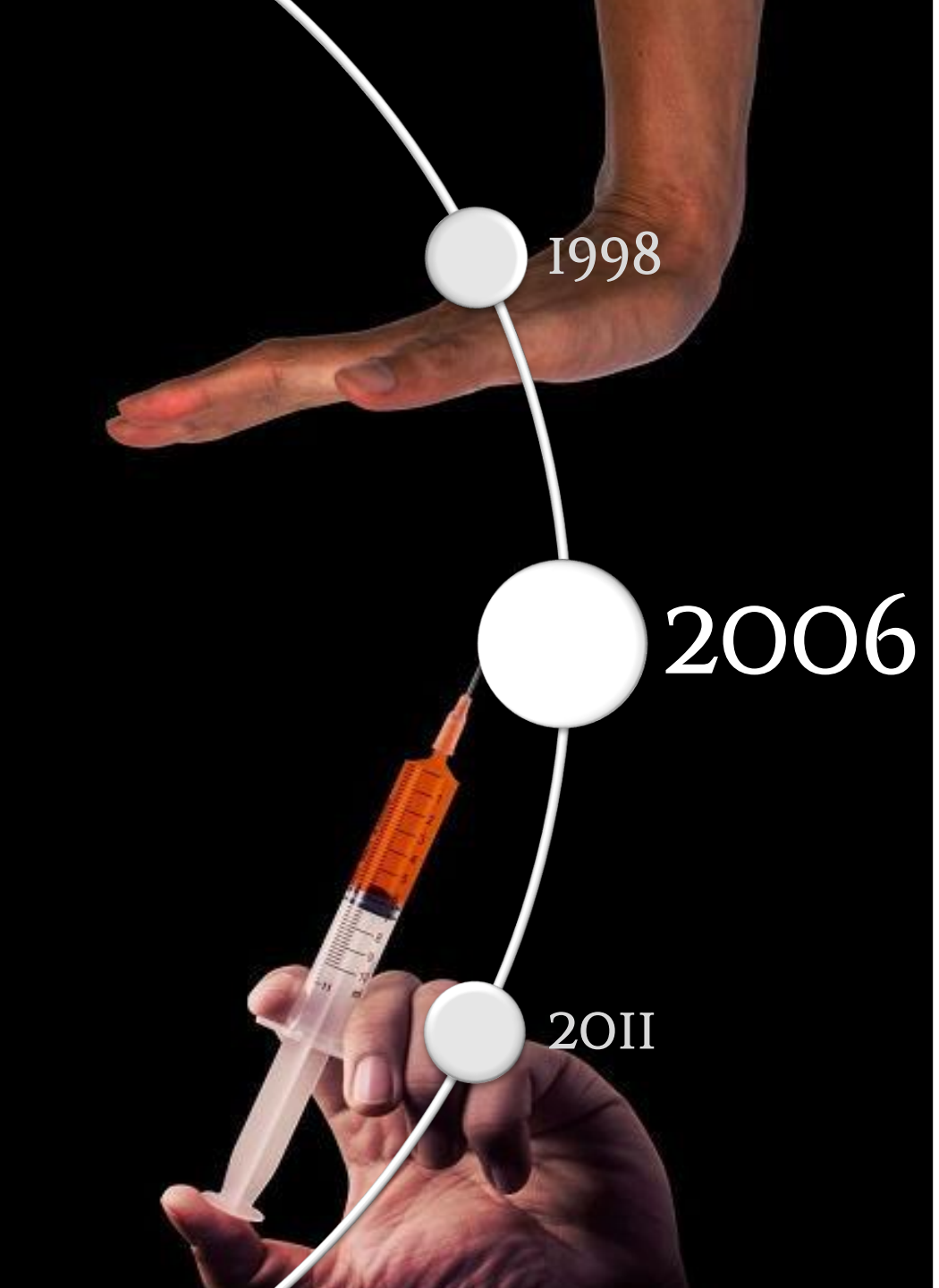
ABSTRACT

BACKGROUND
The efficacy and effectiveness of the quadrivalent human papillomavirus (HPV) vaccine in preventing high-grade cervical lesions have been shown. However, data to inform the relationship between quadrivalent HPV vaccination and the subsequent risk of invasive cervical cancer are lacking.

METHODS
We used nationwide Swedish demographic and health registers to follow an open population of 1,672,983 girls and women who were 10 to 30 years of age from 2006 through 2017. We assessed the association between HPV vaccination and the risk of invasive cervical cancer, controlling for age at follow-up, calendar year, county of residence, and parental characteristics, including education, household income, mother's country of birth, and maternal disease history.

RESULTS
During the study period, we evaluated girls and women for cervical cancer until their 31st birthday. Cervical cancer was diagnosed in 19 women who had received the quadrivalent HPV vaccine and in 538 women who had not received the vaccine. The cumulative incidence of cervical cancer was 47 cases per 100,000 persons among women who had been vaccinated and 94 cases per 100,000 persons among those who had not been vaccinated. After adjustment for age at follow-up, the incidence rate ratio for the comparison of the vaccinated population with the unvaccinated population was 0.51 (95% confidence interval [CI], 0.32 to 0.82). After additional adjustment for other covariates, the incidence rate ratio was 0.37 (95% CI, 0.21 to 0.57). After adjustment for all covariates, the incidence rate ratio was 0.12 (95% CI, 0.00 to 0.34) among women who had been vaccinated before the age of 17 years and 0.47 (95% CI, 0.27 to 0.75) among women who had been vaccinated at the age of 17 to 30 years.

CONCLUSIONS
Among Swedish girls and women 10 to 30 years old, quadrivalent HPV vaccination was associated with a substantially reduced risk of invasive cervical cancer at the population level. (Funded by the Swedish Foundation for Strategic Research and others.)



VACINA CONTRA O HPV

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ORIGINAL ARTICLE

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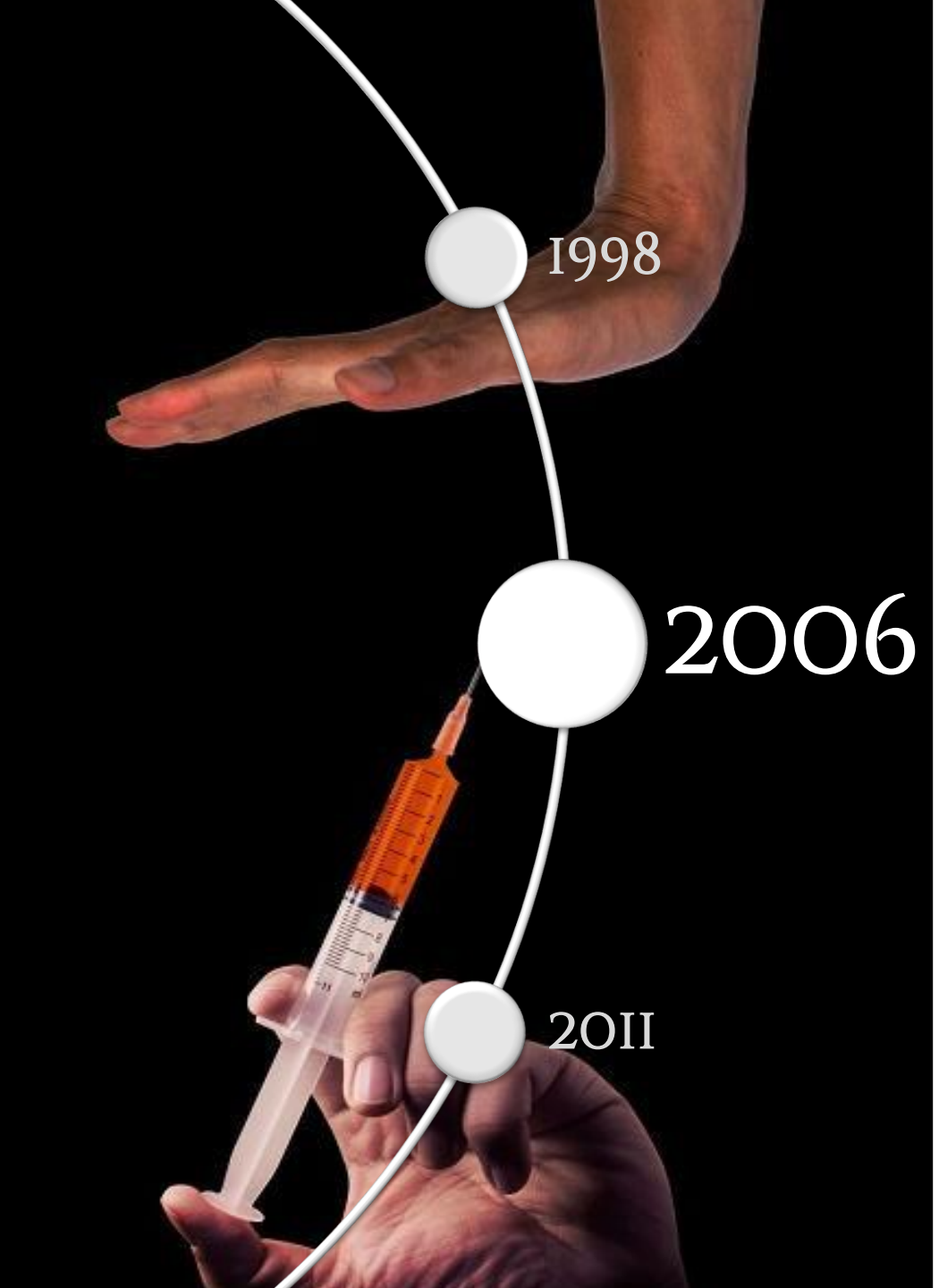

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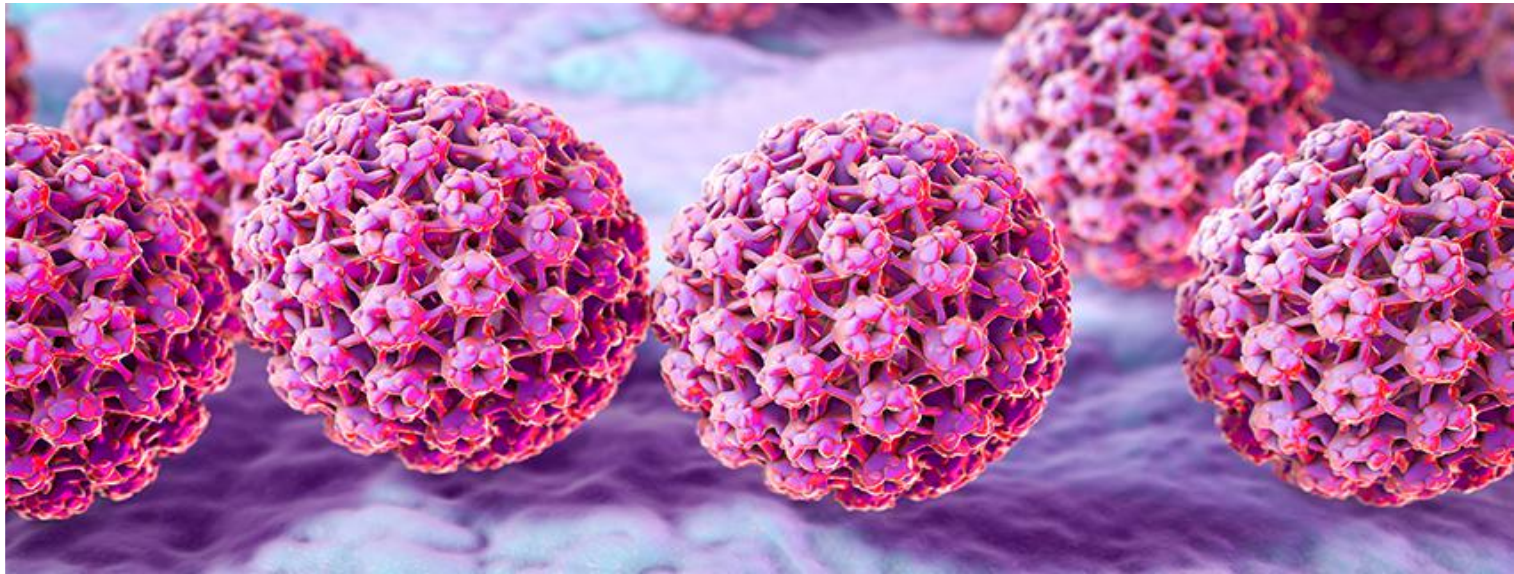
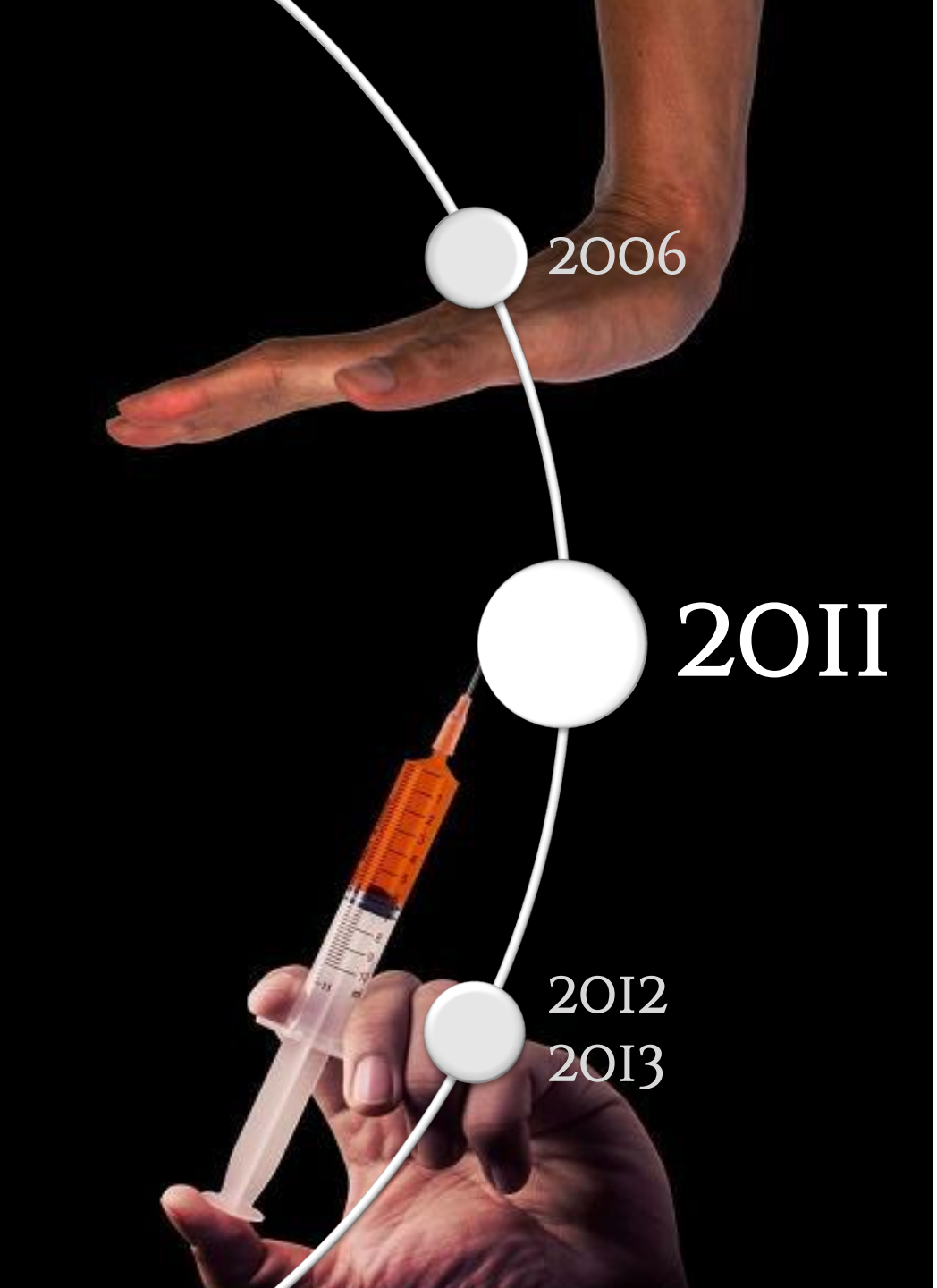
ACOMPANHOU MAIS DE 1,6 MILHÃO DE MULHERES, ENTRE 10 E 30 ANOS DE IDADE NA SUÉCIA, MOSTROU QUE A VACINA CONTRA O HPV, FOI CAPAZ DE REDUZIR EM 88% OS CASOS DE CÂNCER DE COLO DE ÚTERO NAQUELA POPULAÇÃO.





TRAMA DE ESPIONAGEM INTERNACIONAL

PARA CHEGAR A OSAMA BIN LADEN, A CIA ARQUITETOU UMA CAMPANHA CONTRA A PÓLIO NO PAQUISTÃO.

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CIA organised fake vaccination drive to get Osama bin Laden's family DNA

Senior Pakistani doctor who organised vaccine programme in Abbottabad arrested by ISI for working with US agents

▲ CIA organised fake vaccination programme in Abbottabad to try and find Osama bin Laden. Photograph: Md Nadeem/EPA

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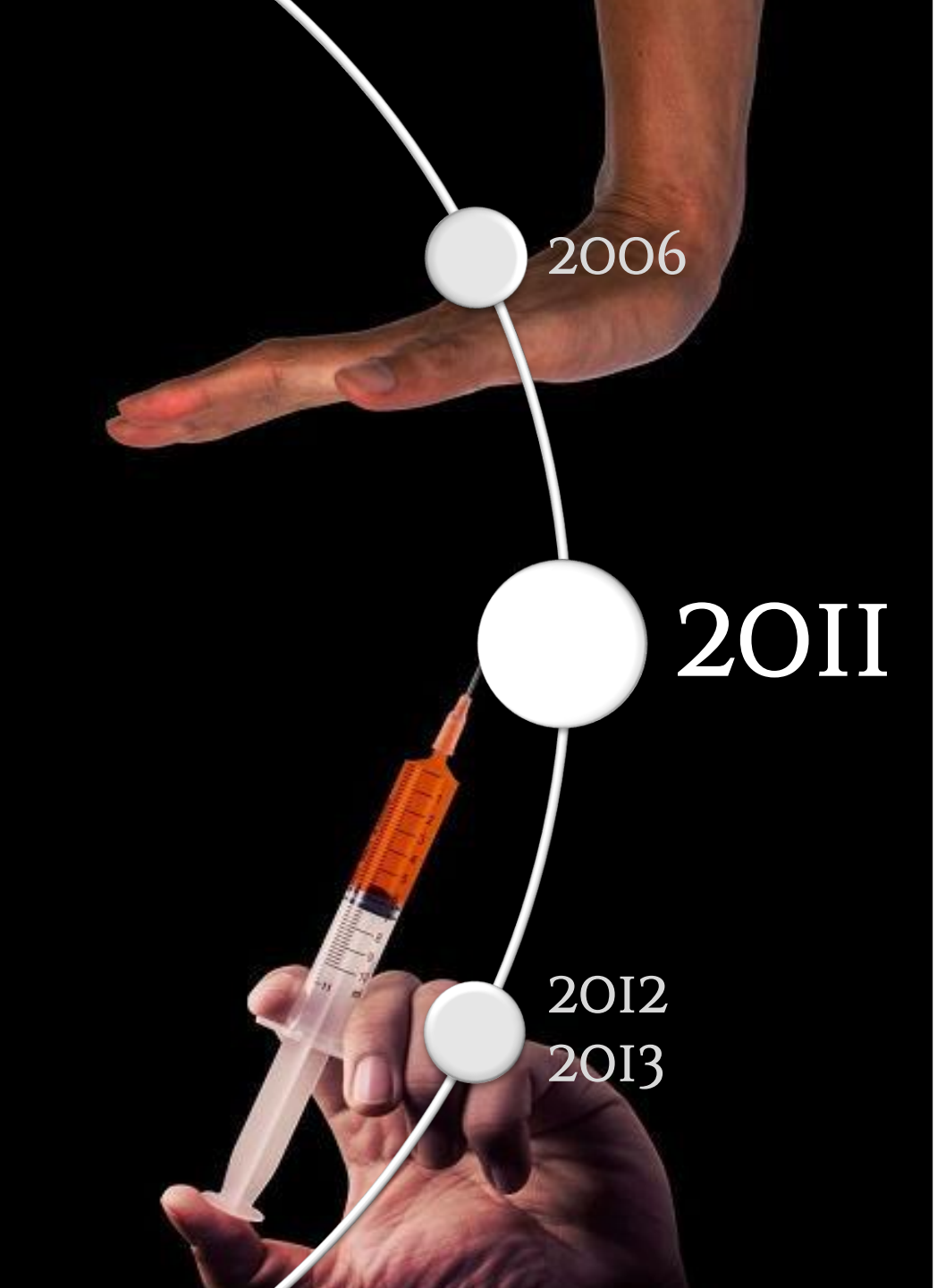

TRAMA DE ESPIONAGEM INTERNACIONAL

IMPULSIONOU UMA CAÇADA A PROFISSIONAIS DA SAÚDE, SOBRETUDO EM ÁREAS TRIBAIS NA FRONTEIRA DO PAÍS COM O AFEGANISTÃO.

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RSFLIX

THE LAST OF US

★ ★ ★ ★ ★ Ação / Aventura / Drama / Terror

Em um mundo devastado por um fungo mortal, um homem escolta uma adolescente que pode salvar a humanidade.

Top 14 em filmes e séries sobre Pandemia

10

11

12

RSFLIX

A HORA MAIS ESCURA

★ ★ ★ ★ ★ Ação / Thriller

Anos de investigação levam uma agente da CIA à operação que busca capturar o homem mais procurado do mundo Osama bin Laden.

Top 14 em filmes e séries sobre Pandemia

10

11

12

RSFLIX

AMERICAN MANHUNT

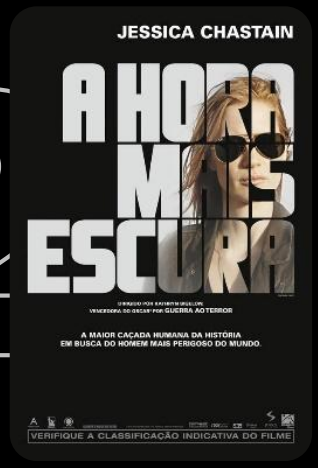
★ ★ ★ ★ ★ Documentário

Com imagens raras e entrevistas com informantes da CIA, este documentário eletrizante acompanha a caçada épica a Osama bin Laden.

Top 14 em filmes e séries sobre Pandemia

11

12



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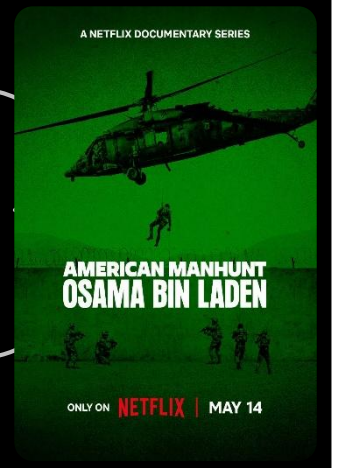

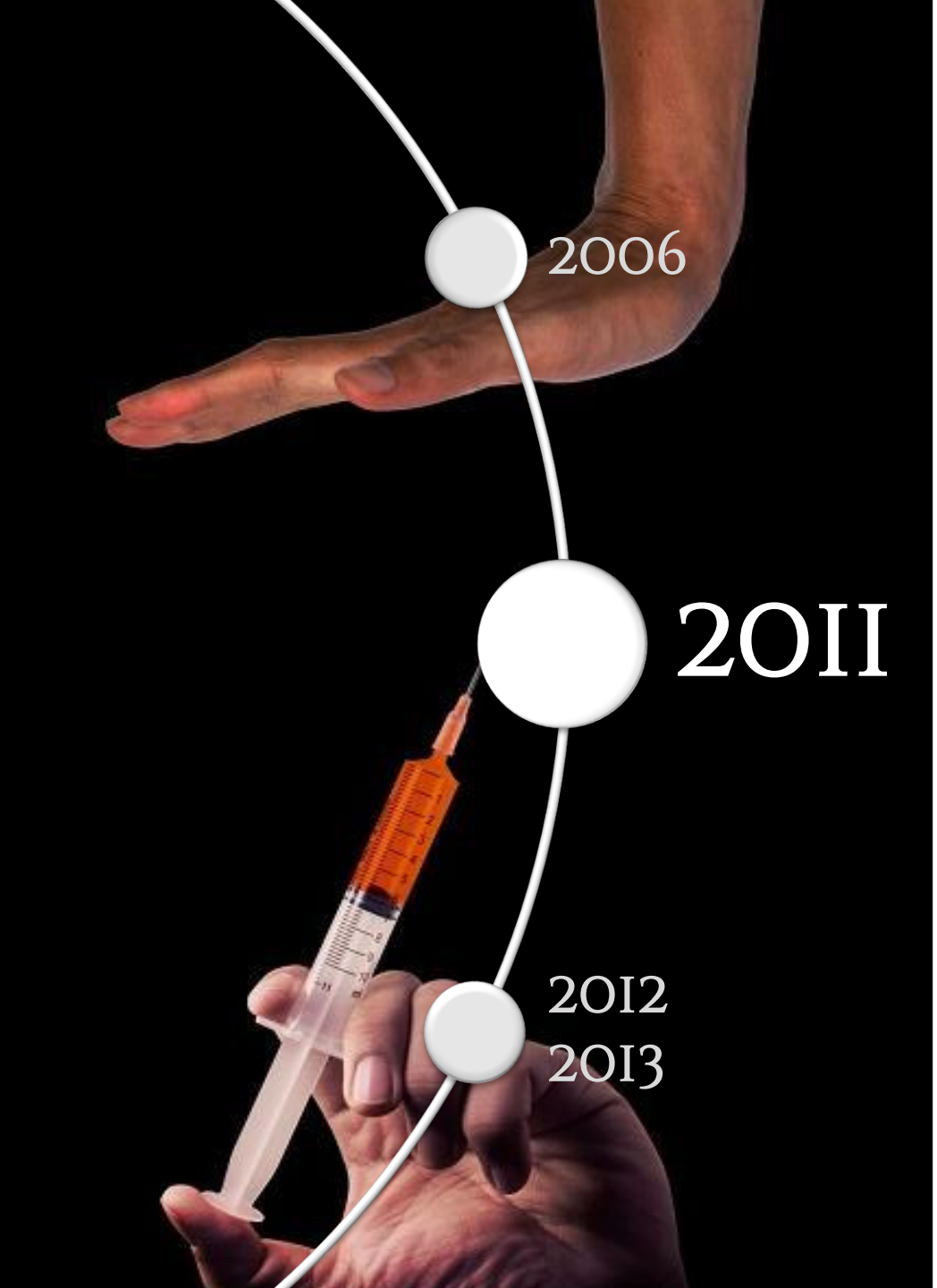

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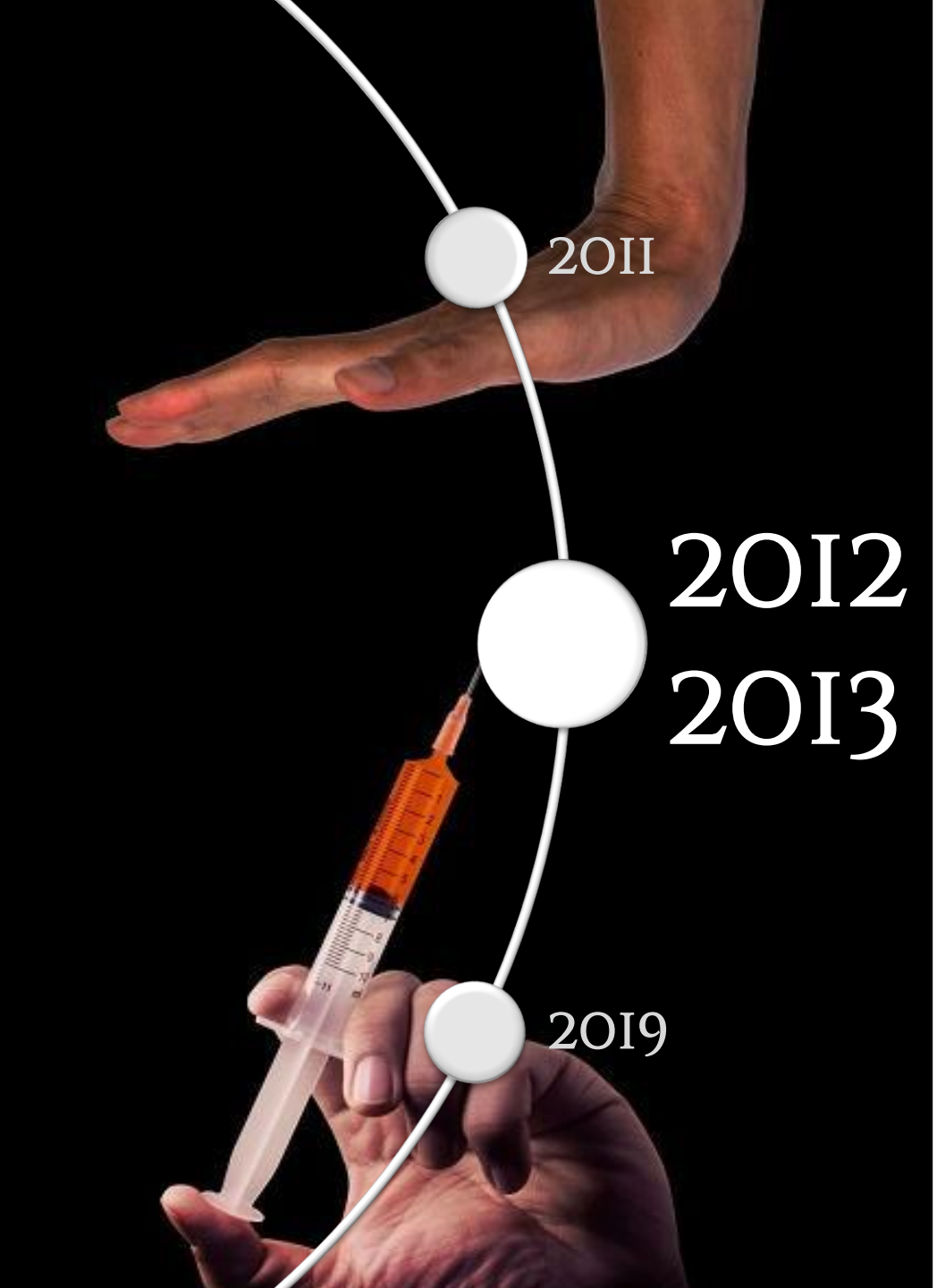

TALIBÃ

ESPALHOU QUE O OCIDENTE USAVA PROGRAMAS DE VACINAÇÃO PARA ATACAR MUÇULMANOS.

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Afghan Taliban bans polio vaccination teams from southern Helmand

Development marks attitude change of militants, who previously allowed medics into region but now suspect them of spying

▲ An Afghan health worker administers polio vaccine drops to a child. Photograph: AFP/Getty Images

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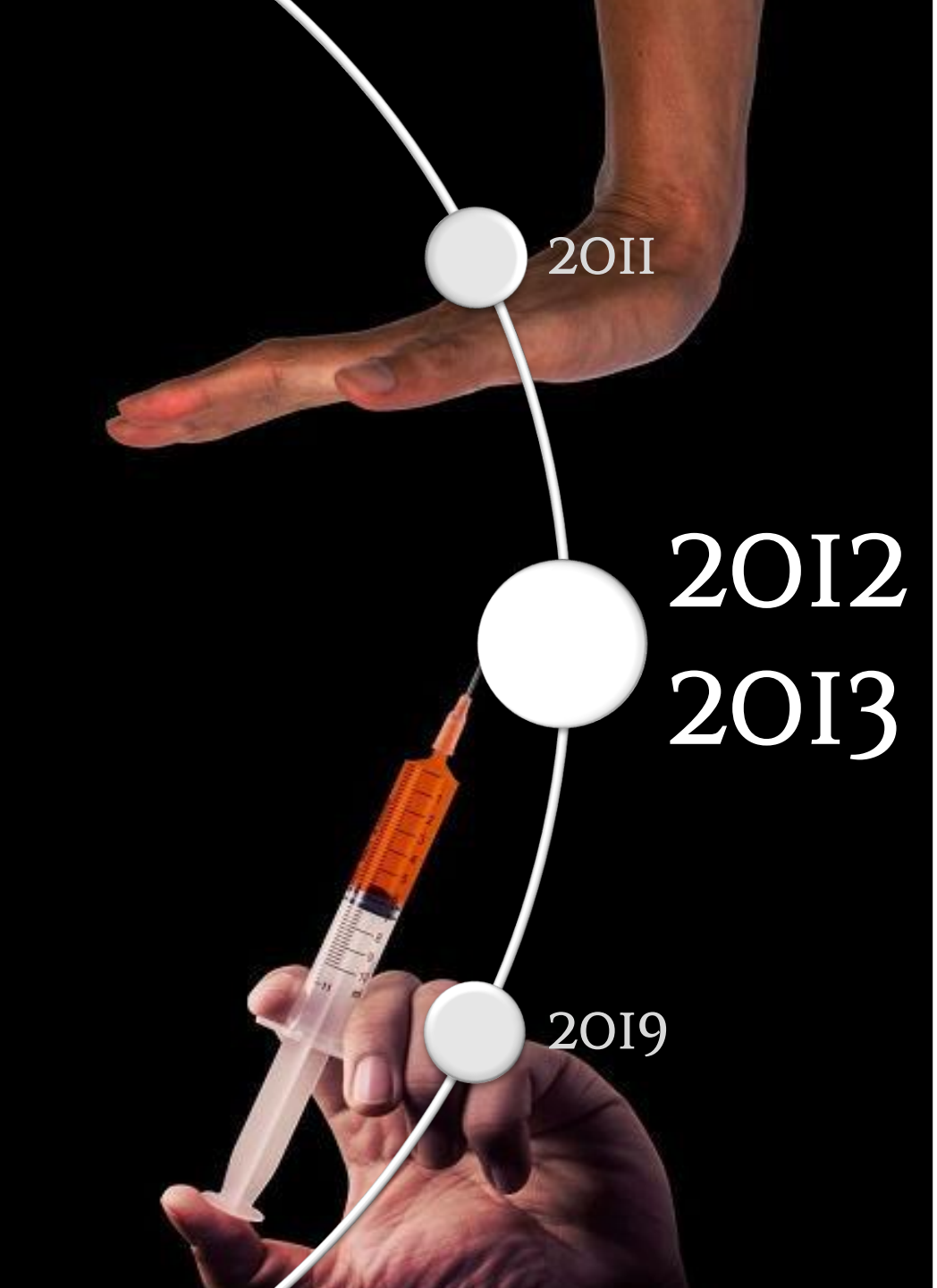

TALIBÃ

A BOATARIA INCLUIU VACINAS COM CARNE DE PORCO (VETADA PELO ISLÃ) E QUE PROVOCAVAM AIDS E ESTERILIDADE.

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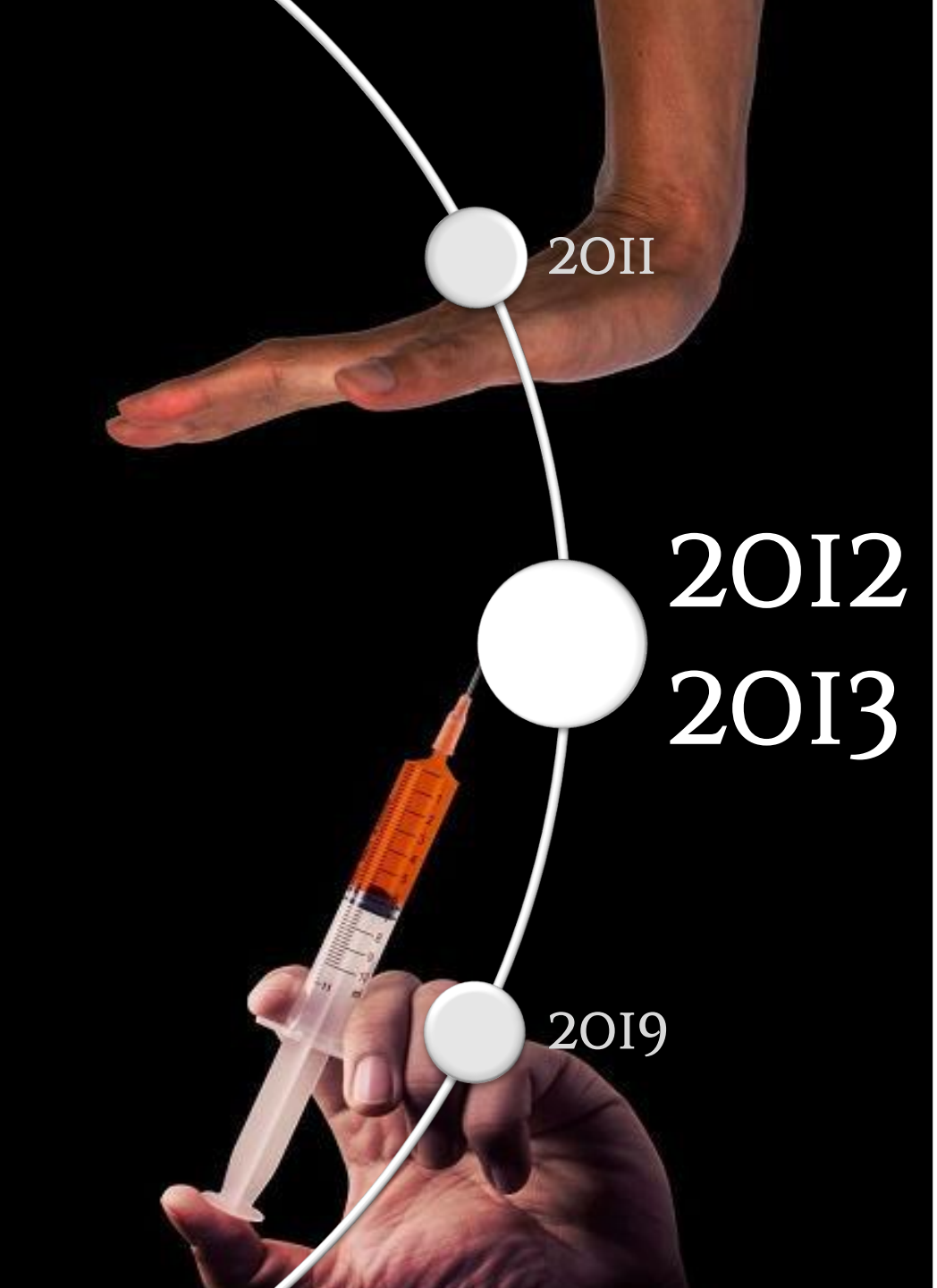

HUMAN RIGHTS WATCH

SALDO:
22 VACINADORES ASSASSINADOS E SURTO DE PÓLIO NO PAÍS.

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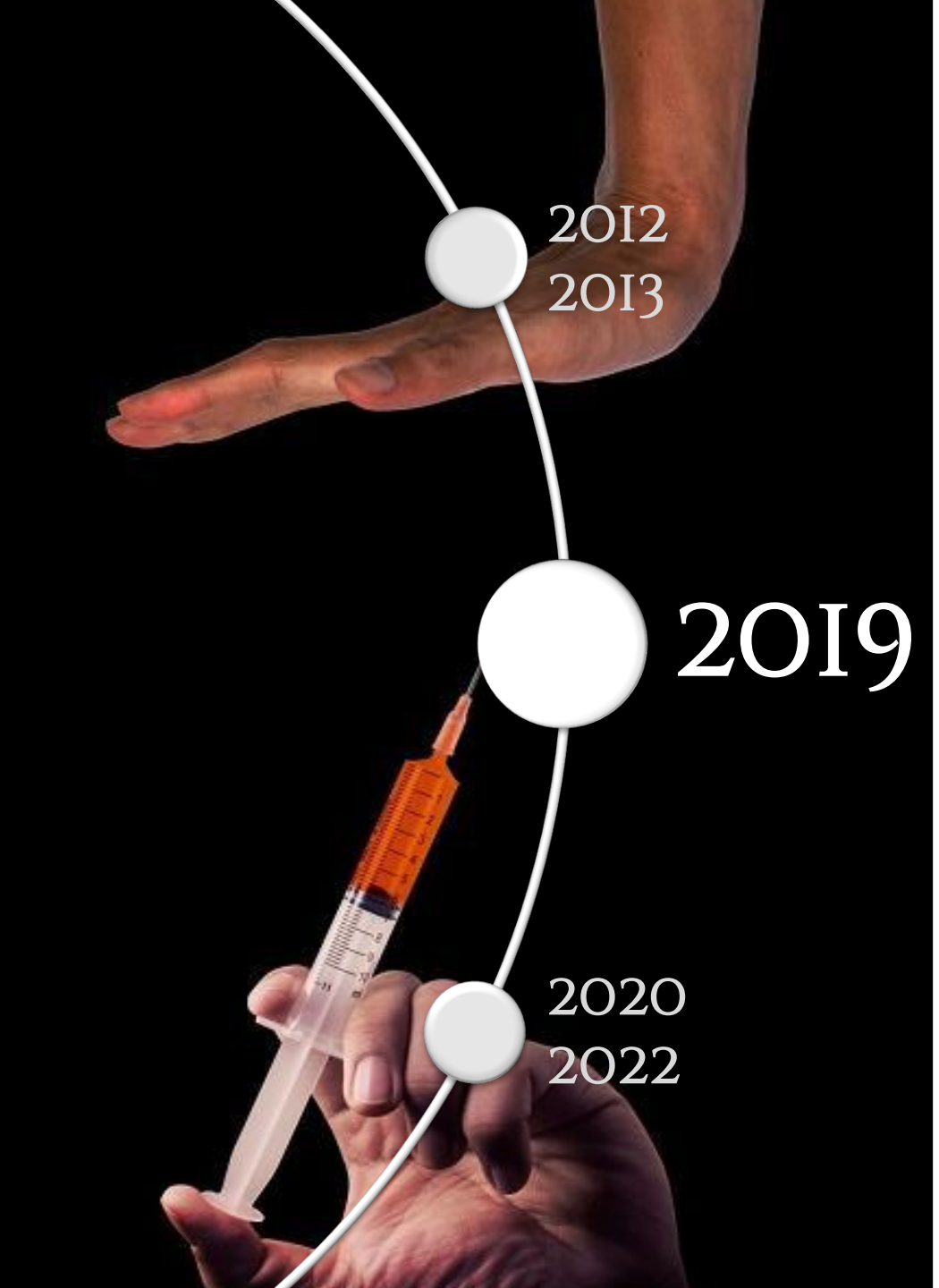
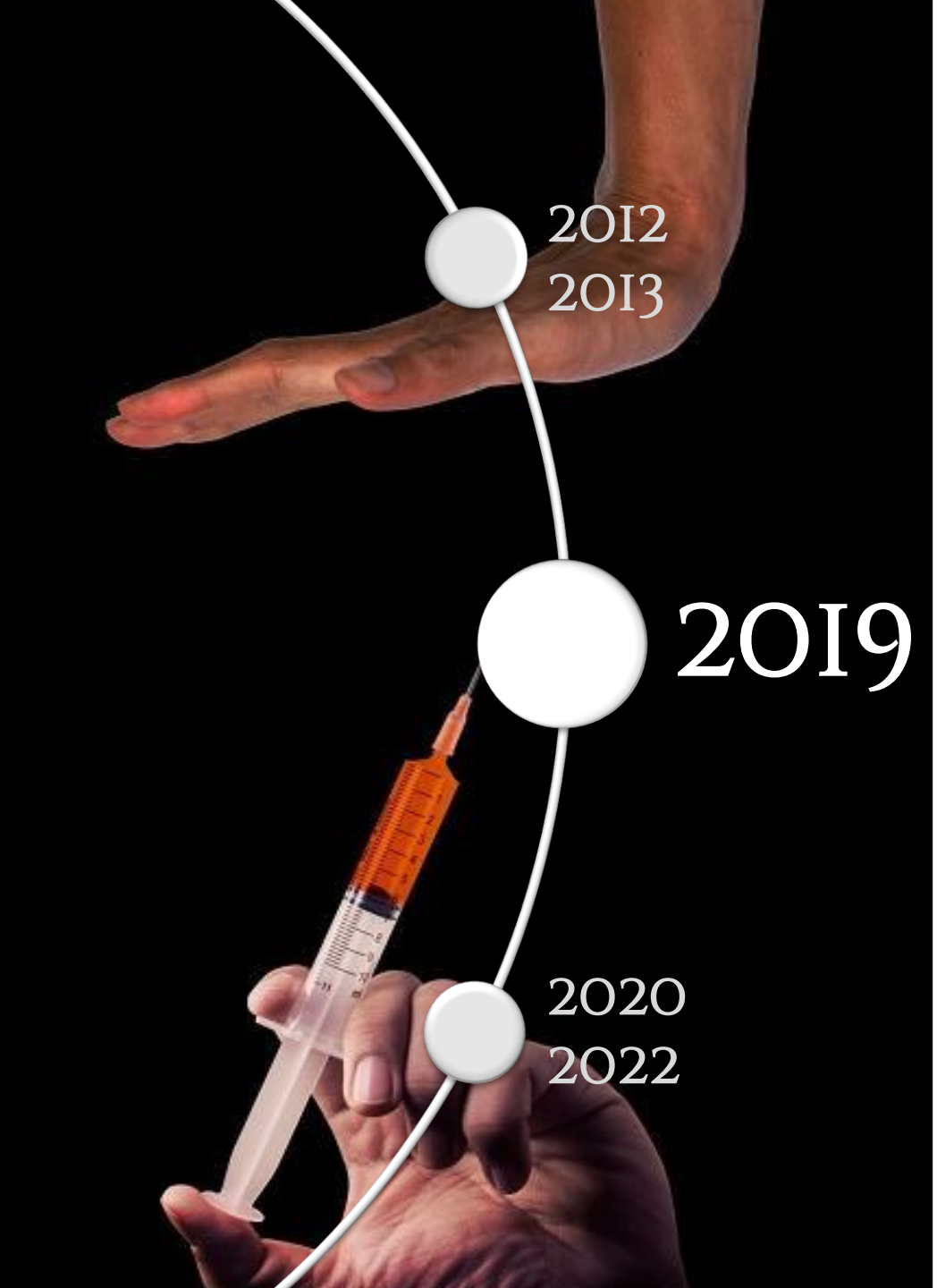


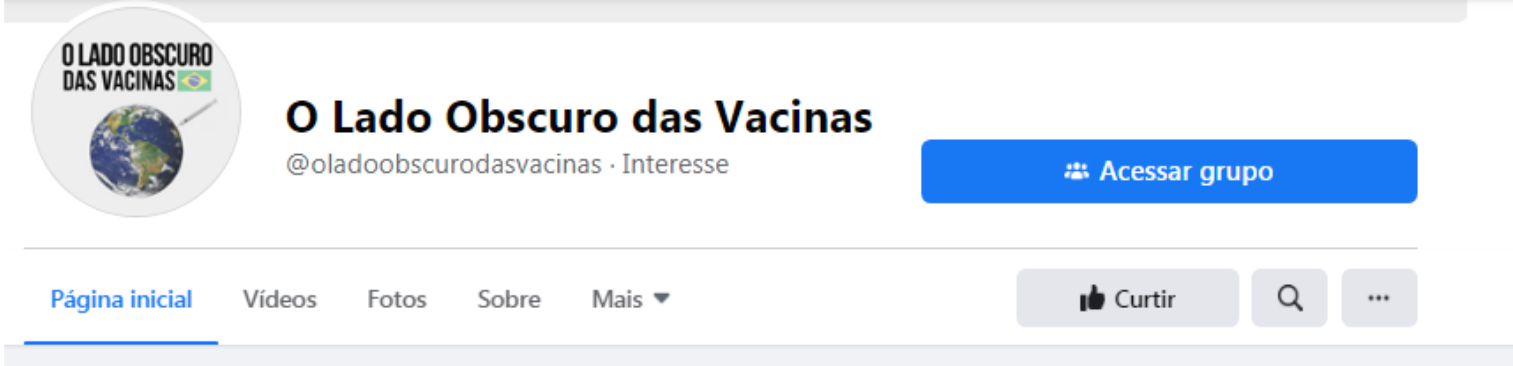
INTERNET

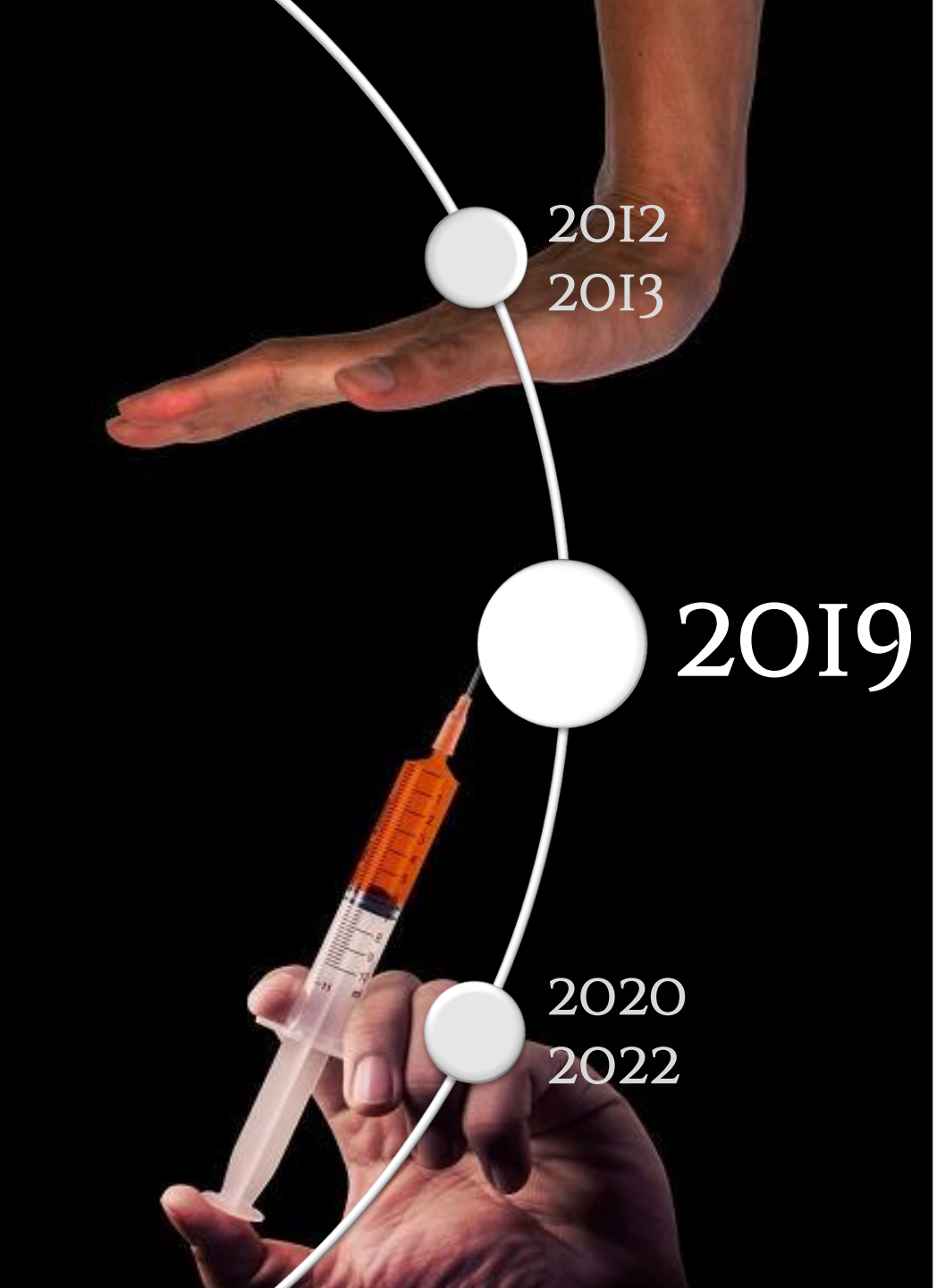
AMPLIFICADORA DESSAS VOZES.



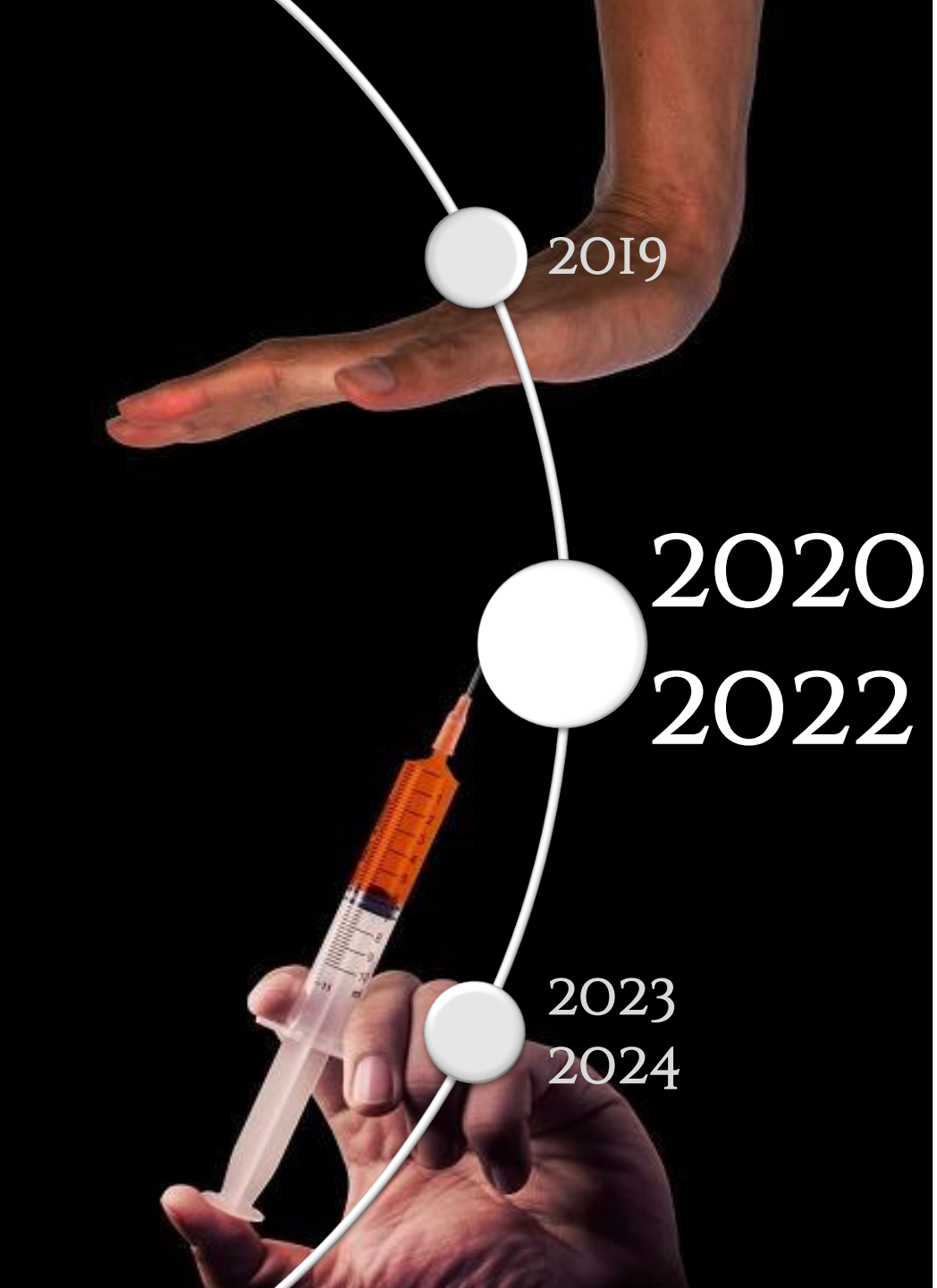
SOCIEDADE BRASILEIRA DE IMUNIZAÇÕES

YOUTUBE: 2,4 MILHÕES DE VISUALIZAÇÕES.
FACEBOOK: 23,5 MILHÕES DE VISUALIZAÇÕES.





Mesmo antes da pandemia, o medo dos imunizantes aparecia numa lista com as dez grandes ameaças à saúde.



CORONAVÍRUS DISEASE-19 (COVID-19)

Brazil's Amazonas state running out of oxygen amid COVID surge

Brazilian Health Minister Eduardo Pazuello said the hospital system in the city of Manaus is collapsing.

Short of oxygen, Brazilian state overwhelmed by Covid-19 despite ferocious first wave

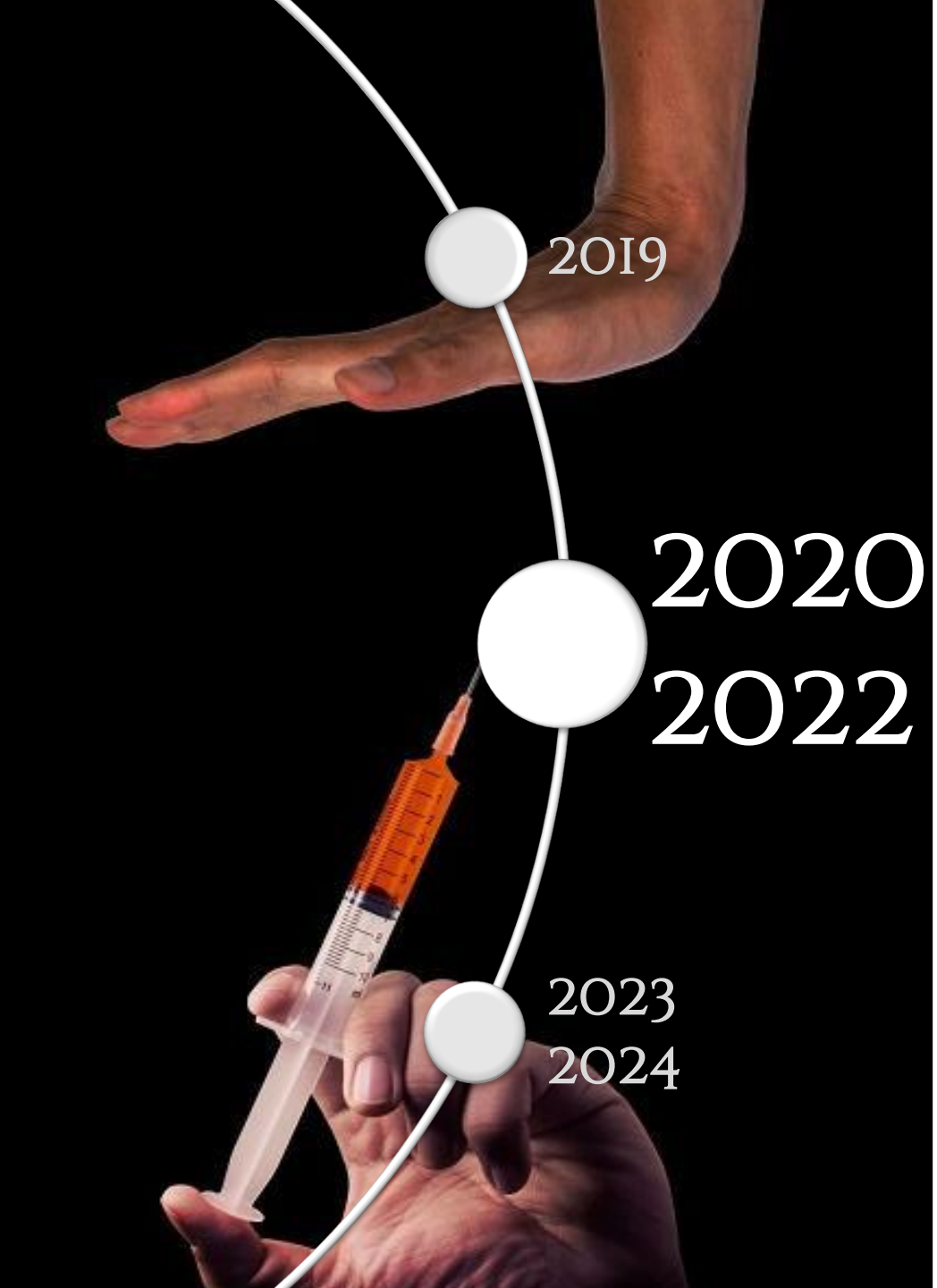
199

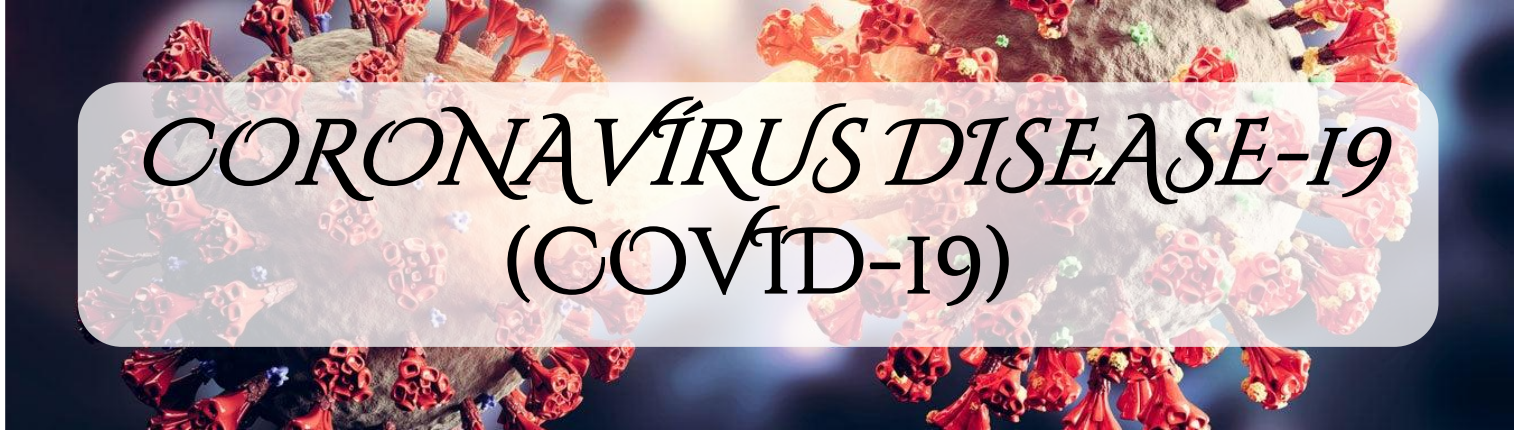
Covid-19: Brazil hospitals 'run out of oxygen' for virus patients

13 hours ago

Coronavirus pandemic

With oxygen running low, a health care system in Brazil nears collapse.





COLAPSO DO SISTEMA DE SAÚDE MUNDIAL

NORTE DO BRASIL: MANAUS | PORTUGAL | ÍNDIA.

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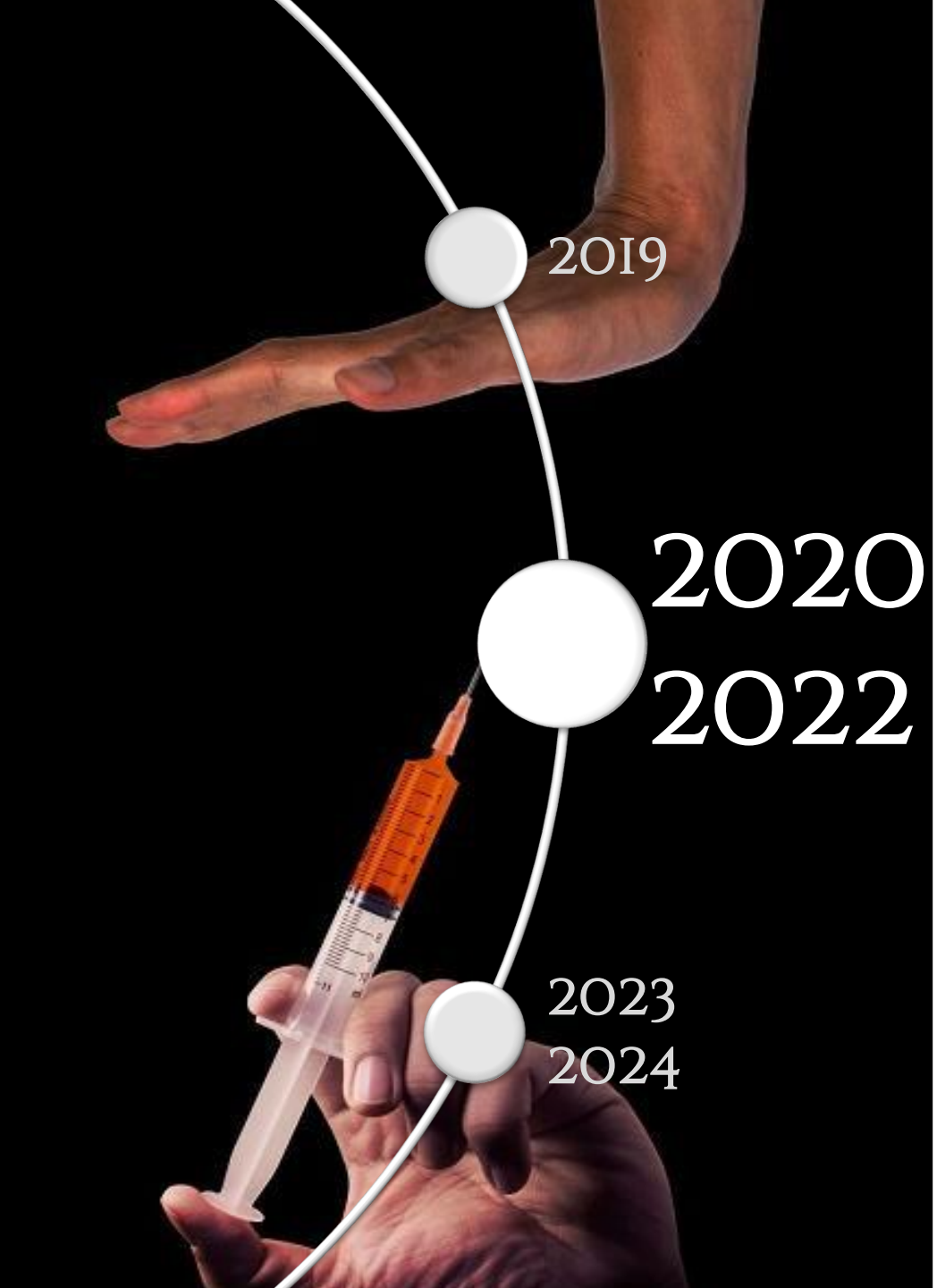
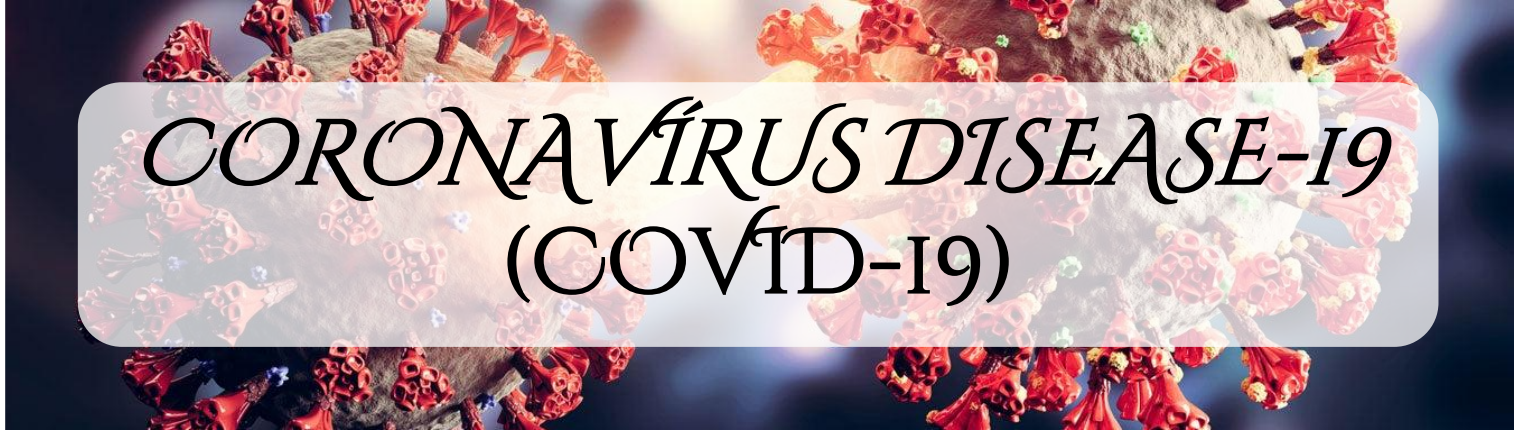




COLAPSO DO SISTEMA DE SAÚDE MUNDIAL

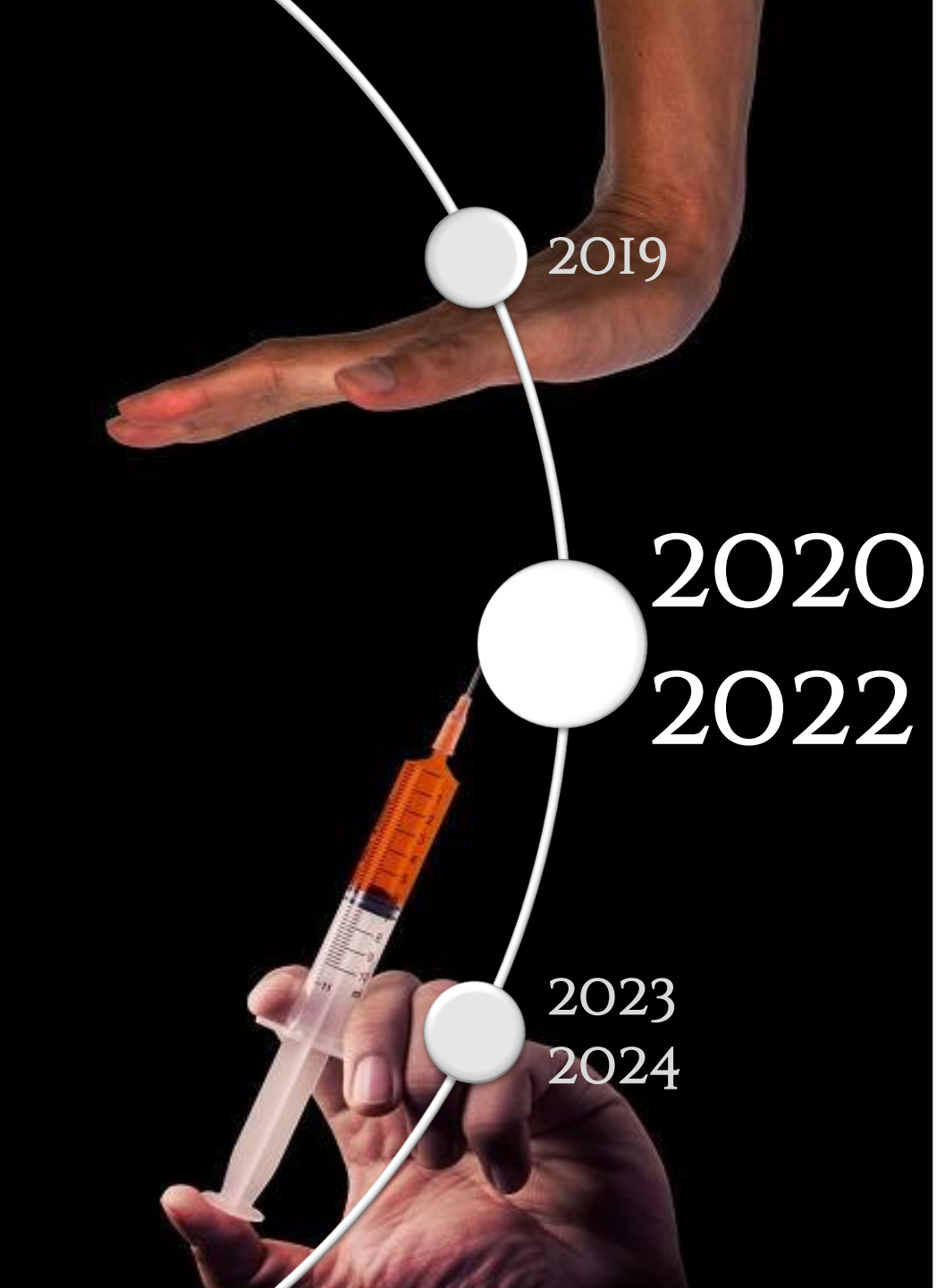
NORTE DO BRASIL: MANAUS | PORTUGAL | ÍNDIA.

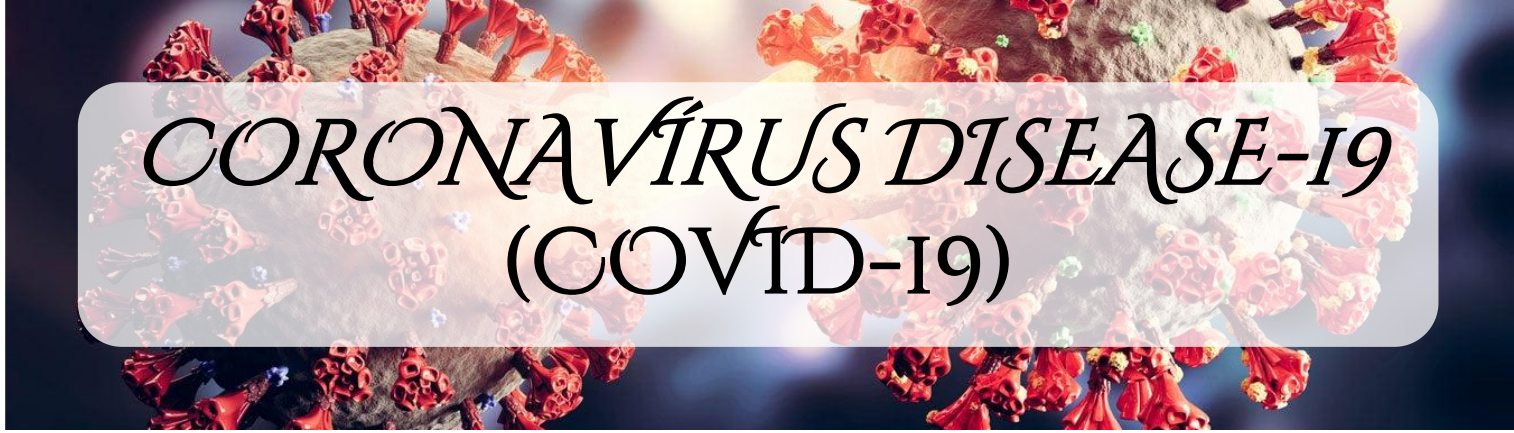
Portugal é estimulado a buscar ajuda internacional conforme mortes por Covid-19 batem recorde

Catarina Demony e Miguel Pereira

26 jan 2021 - 21h16 Compartilhar Exibir comentários

O governo de Portugal foi estimulado a transferir pacientes de Covid-19 para fora do país nesta terça-feira conforme as mortes pela doença atingiram uma alta recorde e o sistema de fornecimento de oxigênio de um grande hospital na região de Lisboa falhou em parte por conta do uso excessivo.





COLAPSO DO SISTEMA DE SAÚDE MUNDIAL

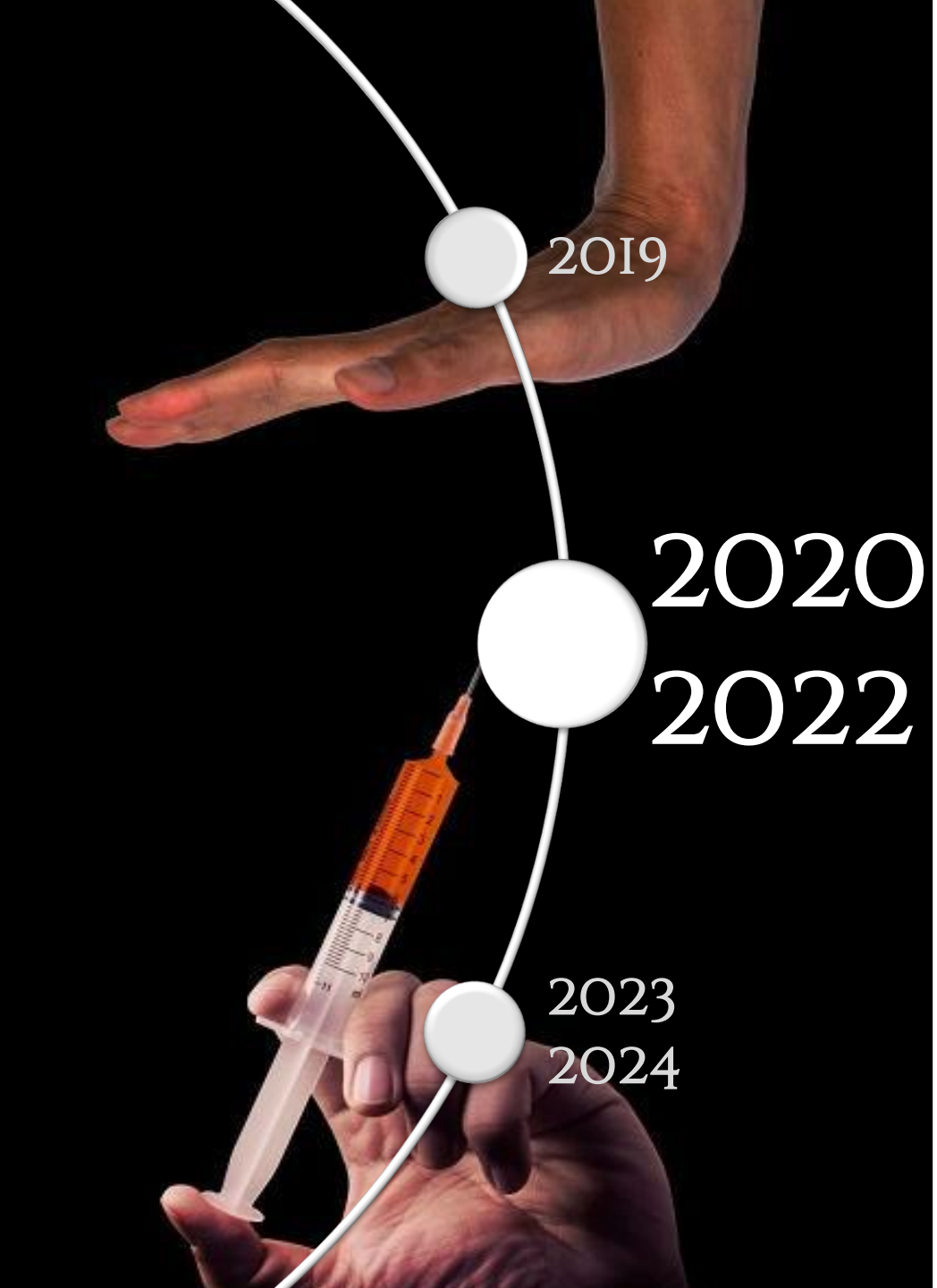
NORTE DO BRASIL: MANAUS | PORTUGAL | ÍNDIA.

A tragédia da covid-19 na Índia que não aparece nos números oficiais

Especialistas apontam que precariedade no sistema de registro de óbitos sugere que realidade é ainda mais dramática

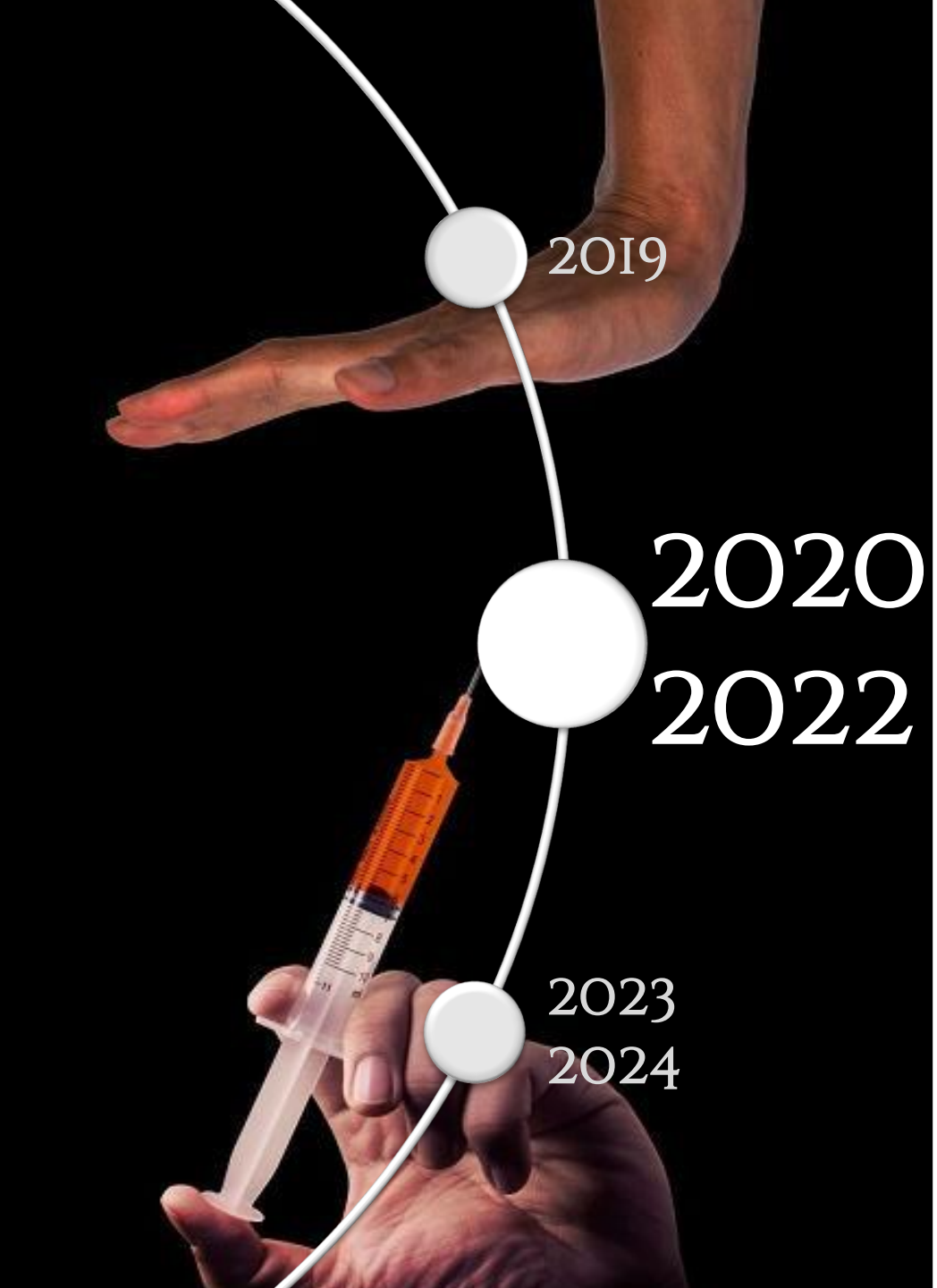
Murali Krishnan | DW
| 28 de abril de 2021 às 17:59

A Índia superou nesta quarta-feira (28) a marca de 200 mil mortos por covid-19, após registrar mais de 3 mil óbitos em 24 h pela primeira vez - Punit Paranjpe/AFP



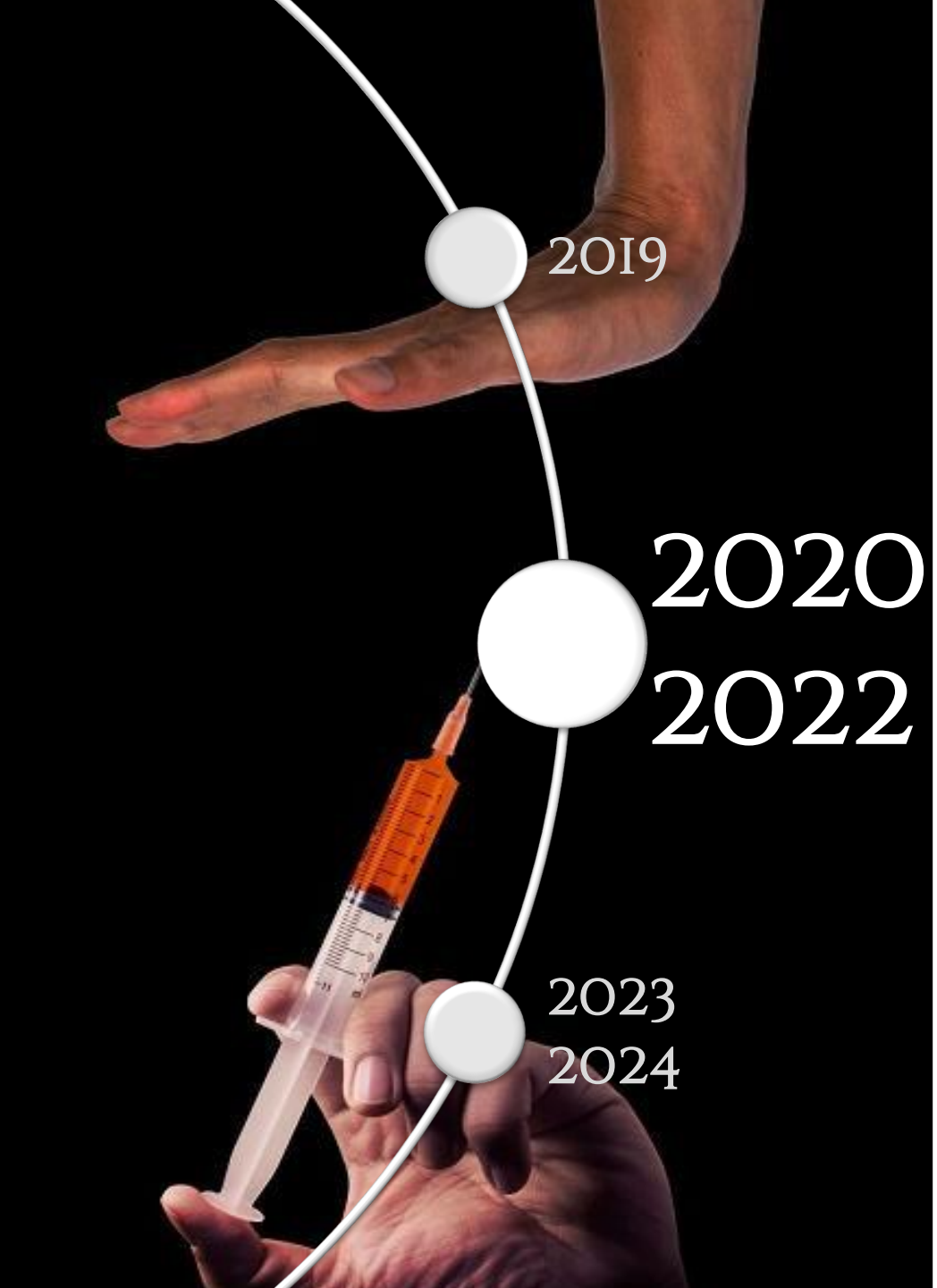
PESSOAS COMUNS

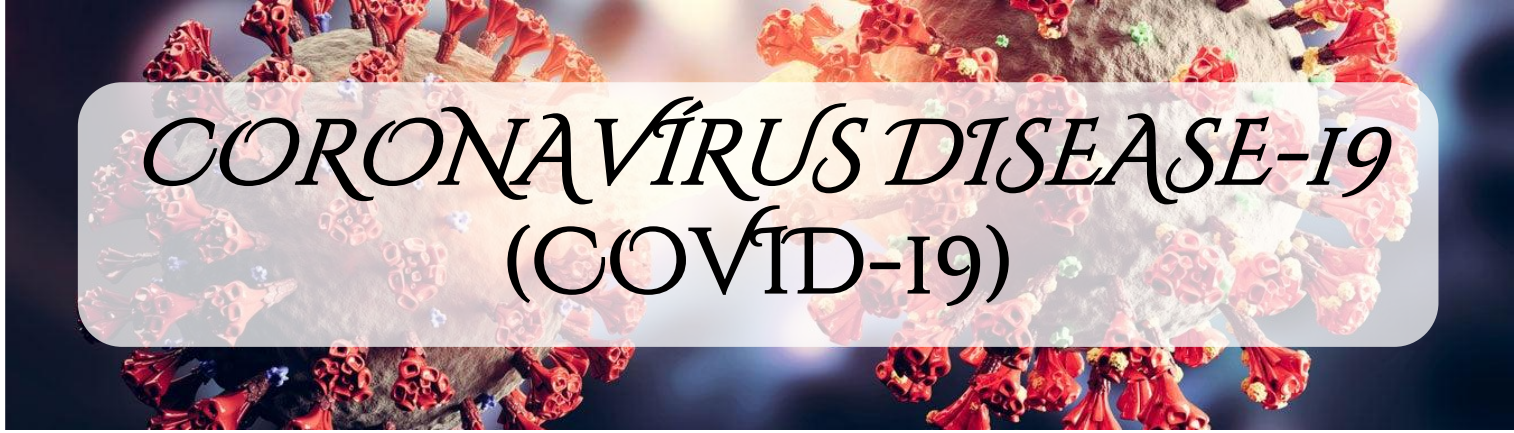
"Vou preferir adoecer, se for o caso, eu aceito a vida ou a morte".



PESSOAS COMUNS

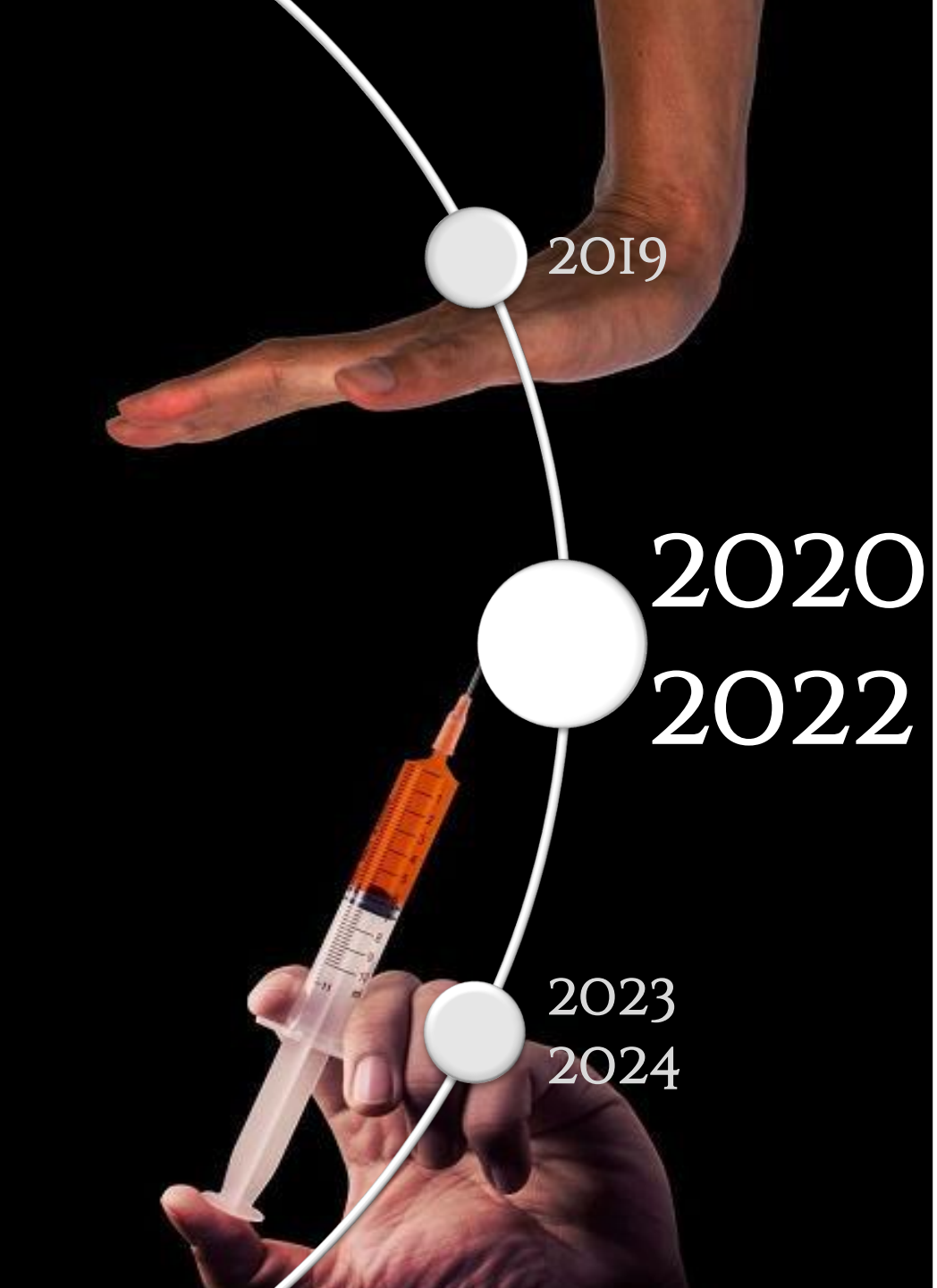
"AS VACINAS SAÍRAM MUITO RÁPIDO E PULARAM AS FASES DE PESQUISAS".

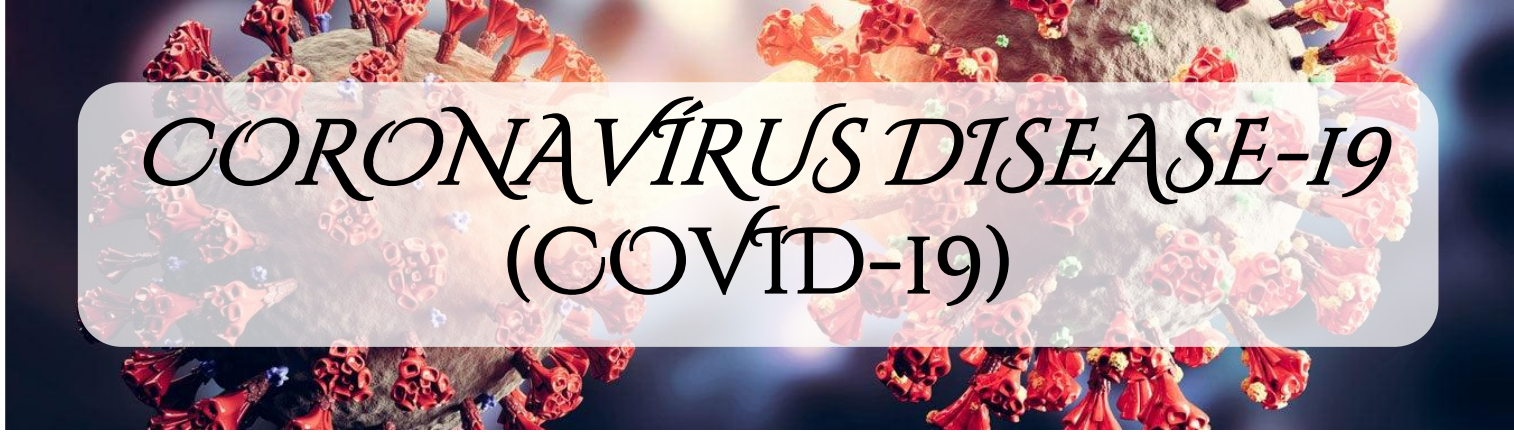




PESSOAS COMUNS

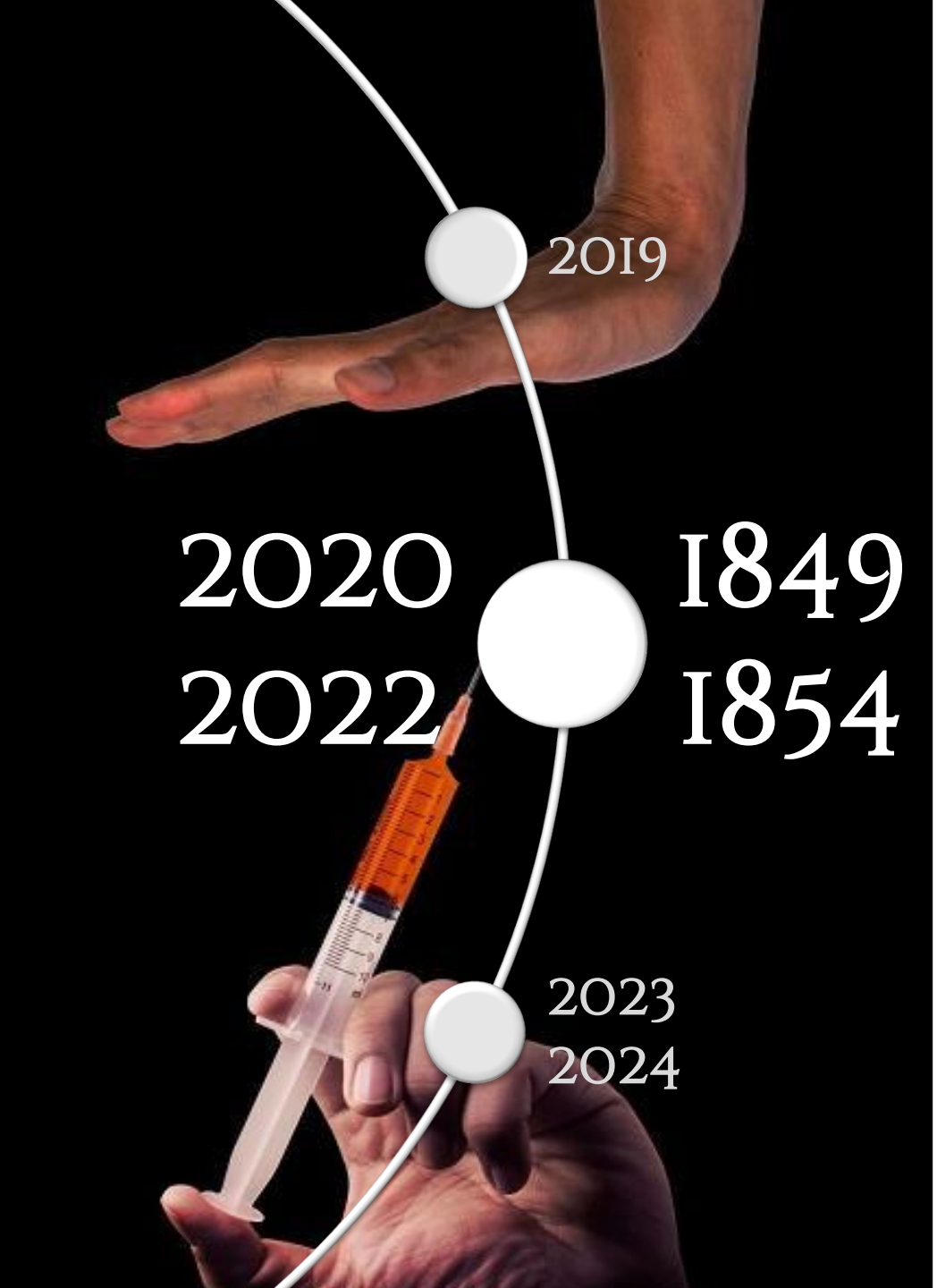
"NÃO PODEMOS VIVER COM A FALSA ILUSÃO DE QUE SENDO VACINADO VAMOS SOBREVIVER A TUDO".





PESSOAS COMUNS

"Não aguento mais ver números de casos e óbitos... Não existe necessidade".


2019
2020
2022
1849
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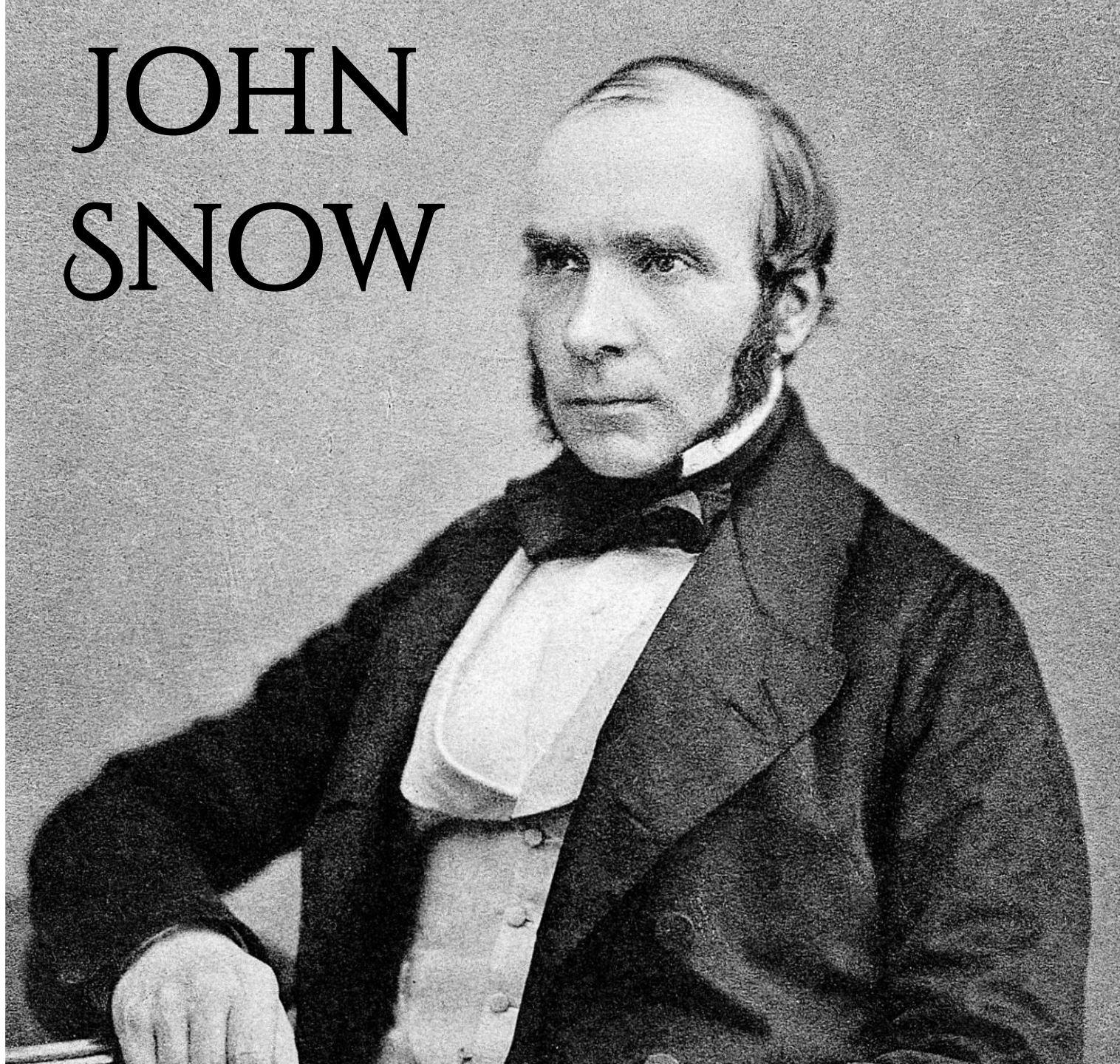
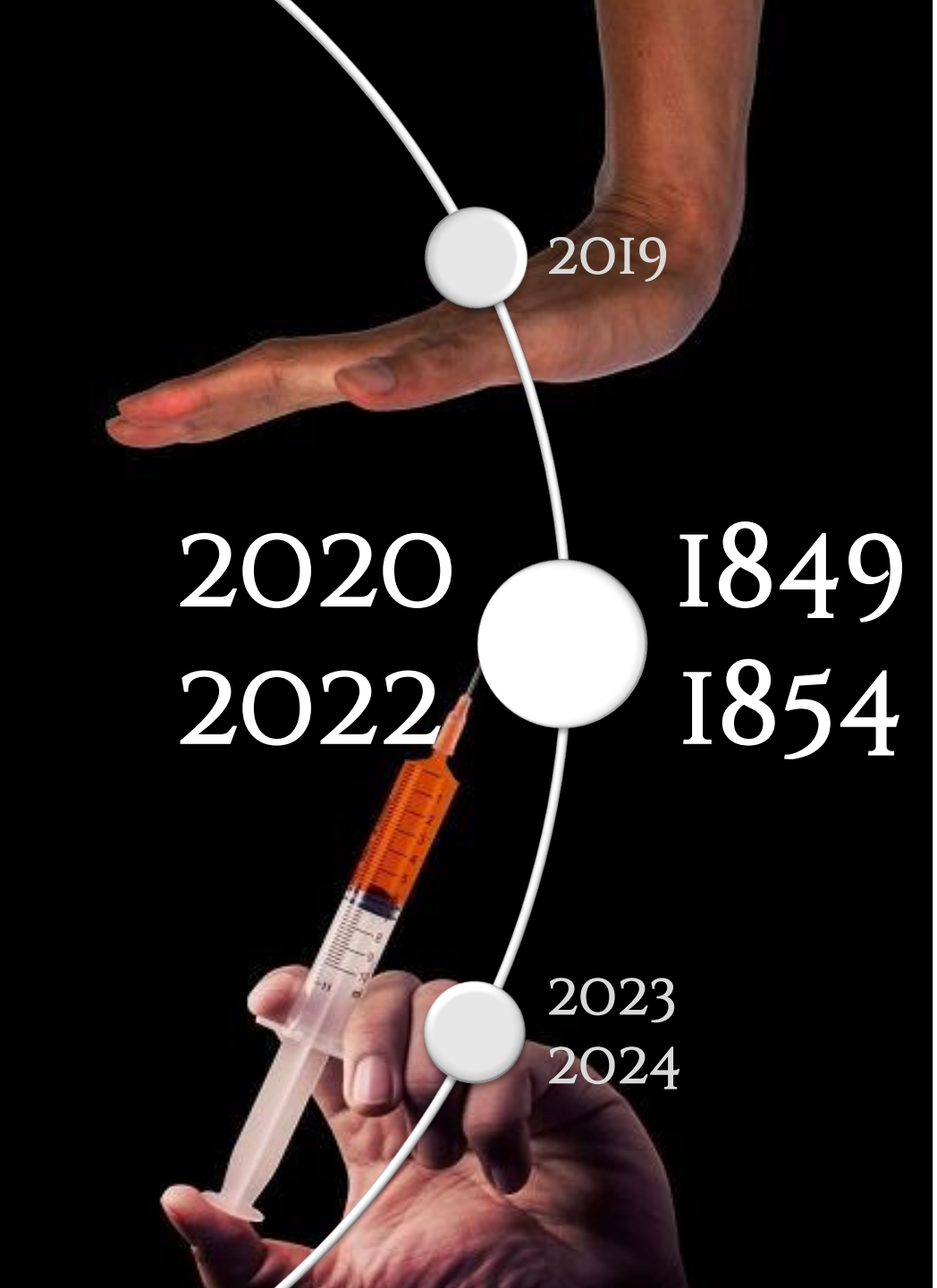
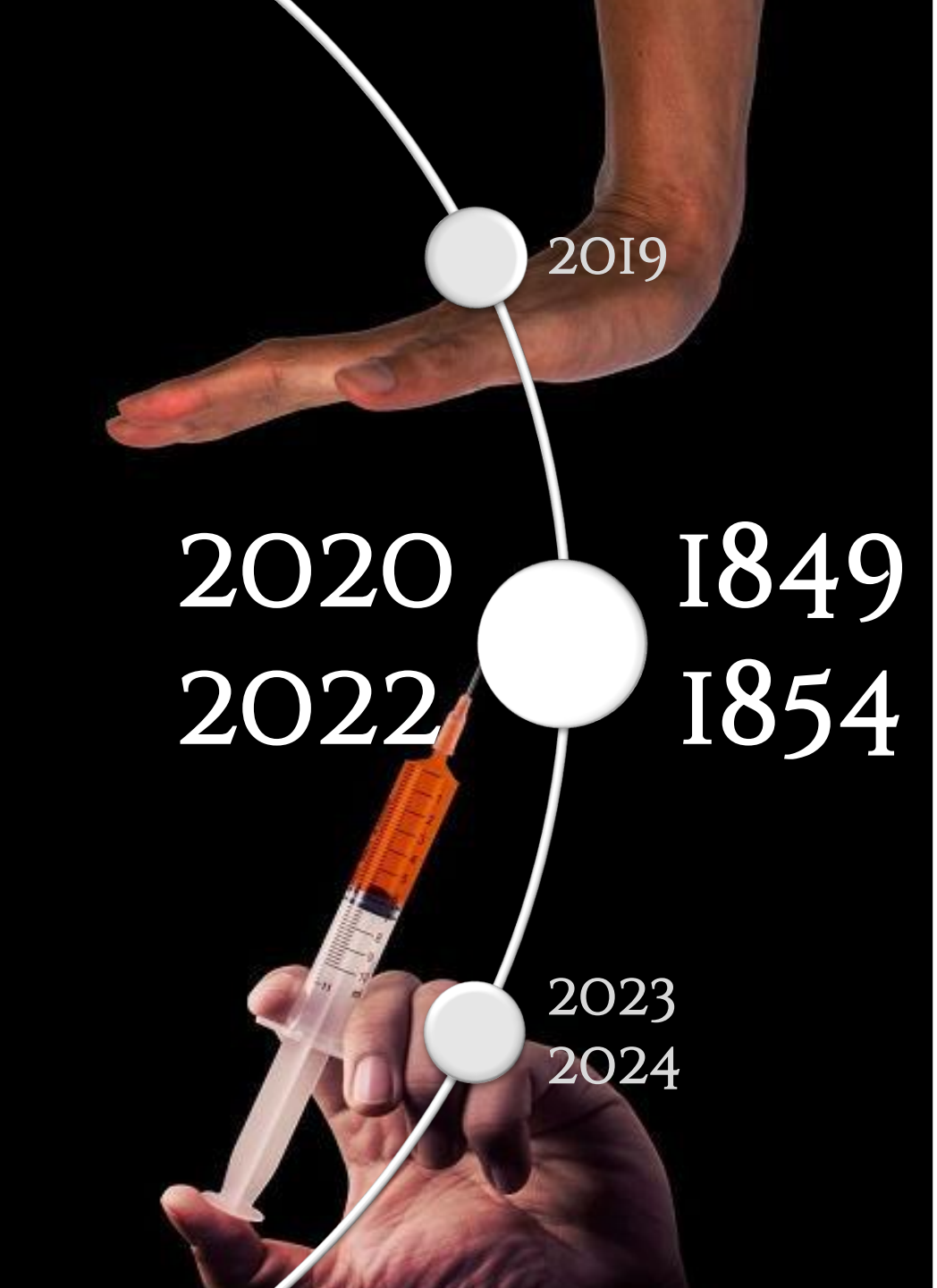

JOHN
SNOW



JOHN SNOW

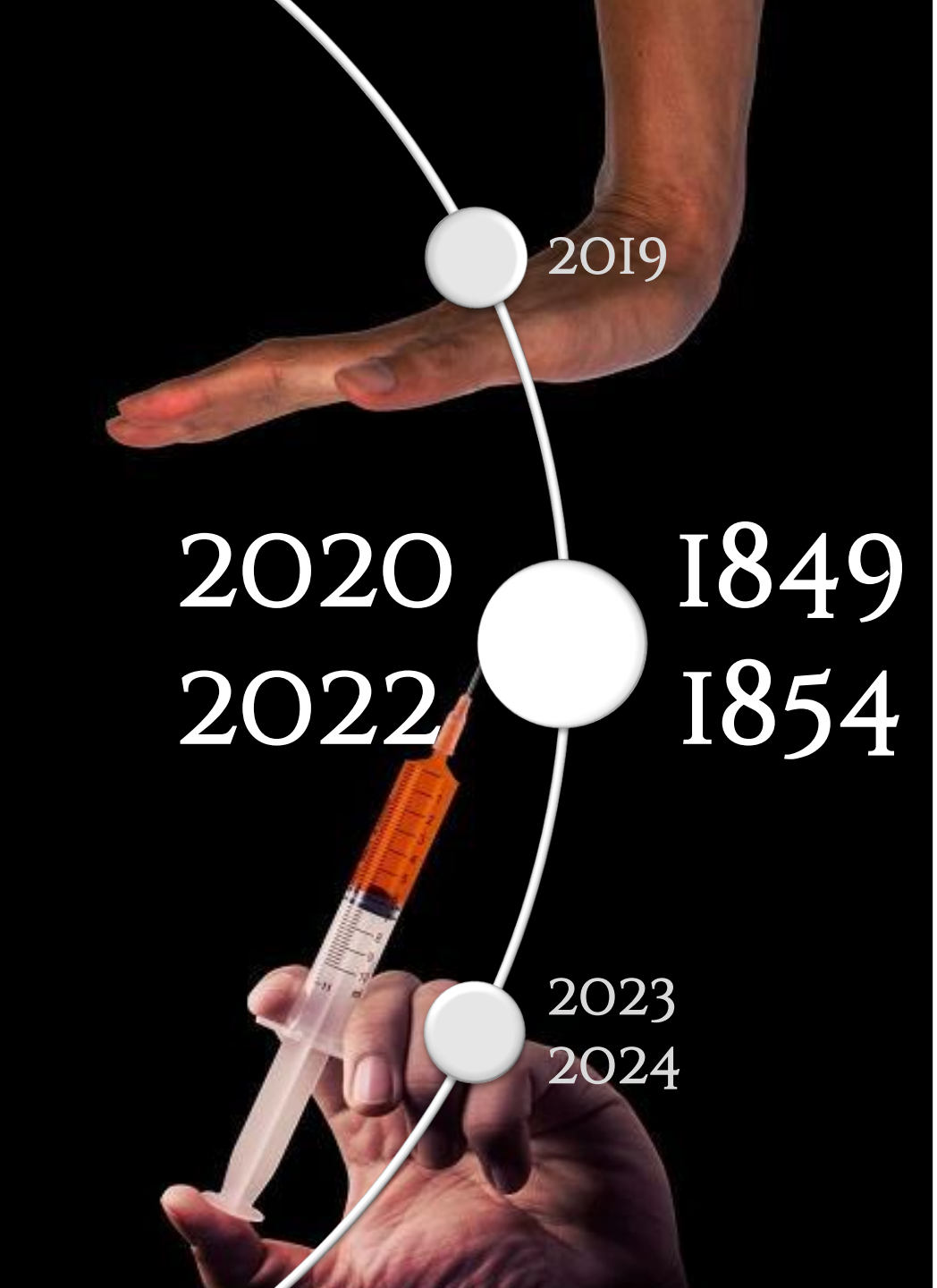
Médico e sanitarista britânico:

✓ Estudos pioneiros da epidemia de cólera em Londres.



JOHN SNOW

A COURT FOR KING CHOLERA.

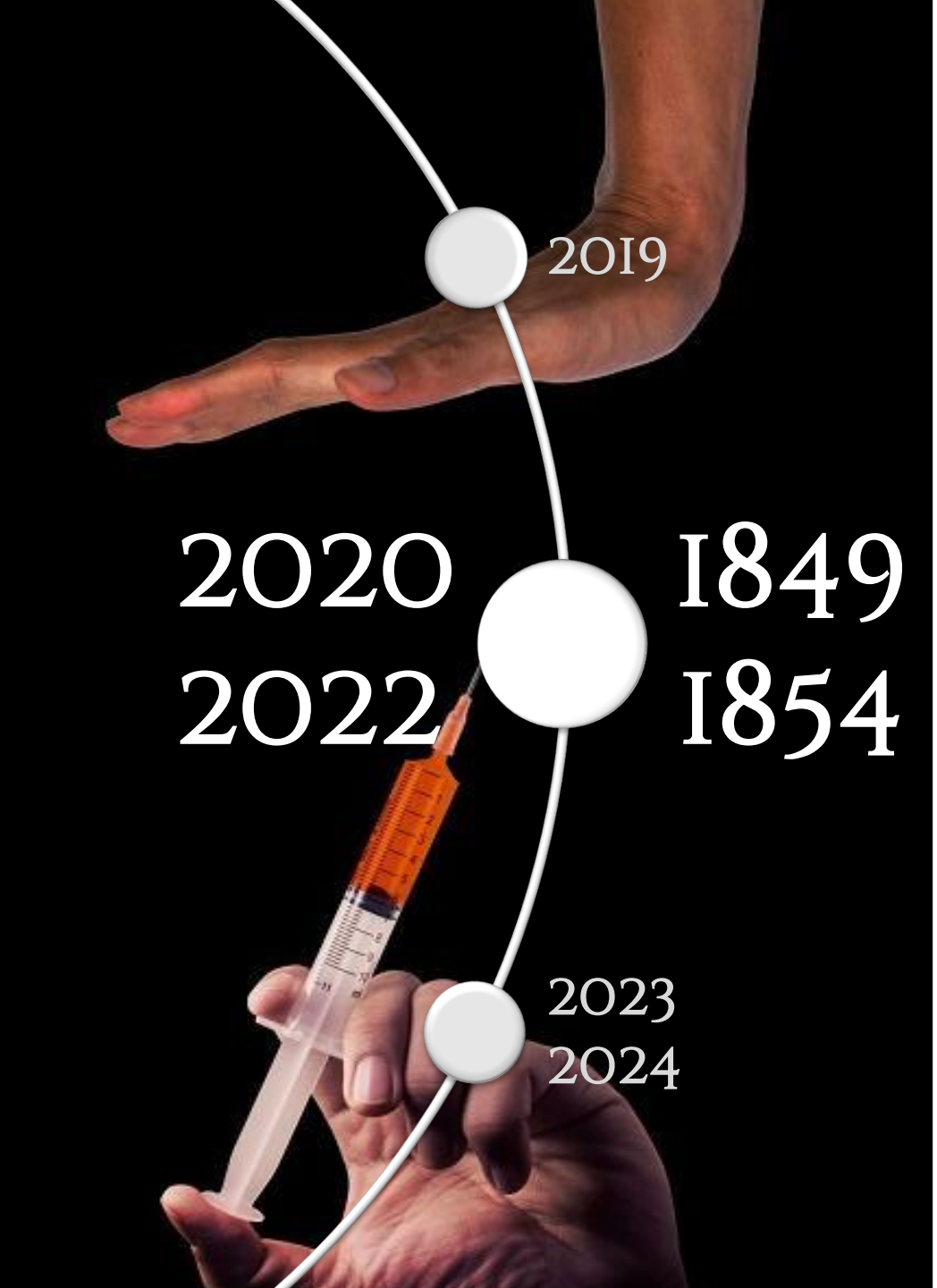


JOHN SNOW

DEMONSTROU O CONSUMO DE ÁGUA CONTAMINADA COMO RESPONSÁVEL PELA OCORRÊNCIA DA DOENÇA:

- ✓ 30 ANOS ANTES DE ROBERT KOCH ISOLAR E CULTIVAR O:
  - ➢ *VÍBRIO CHOLERAE.*

A COURT FOR KING CHOLERA.


2019
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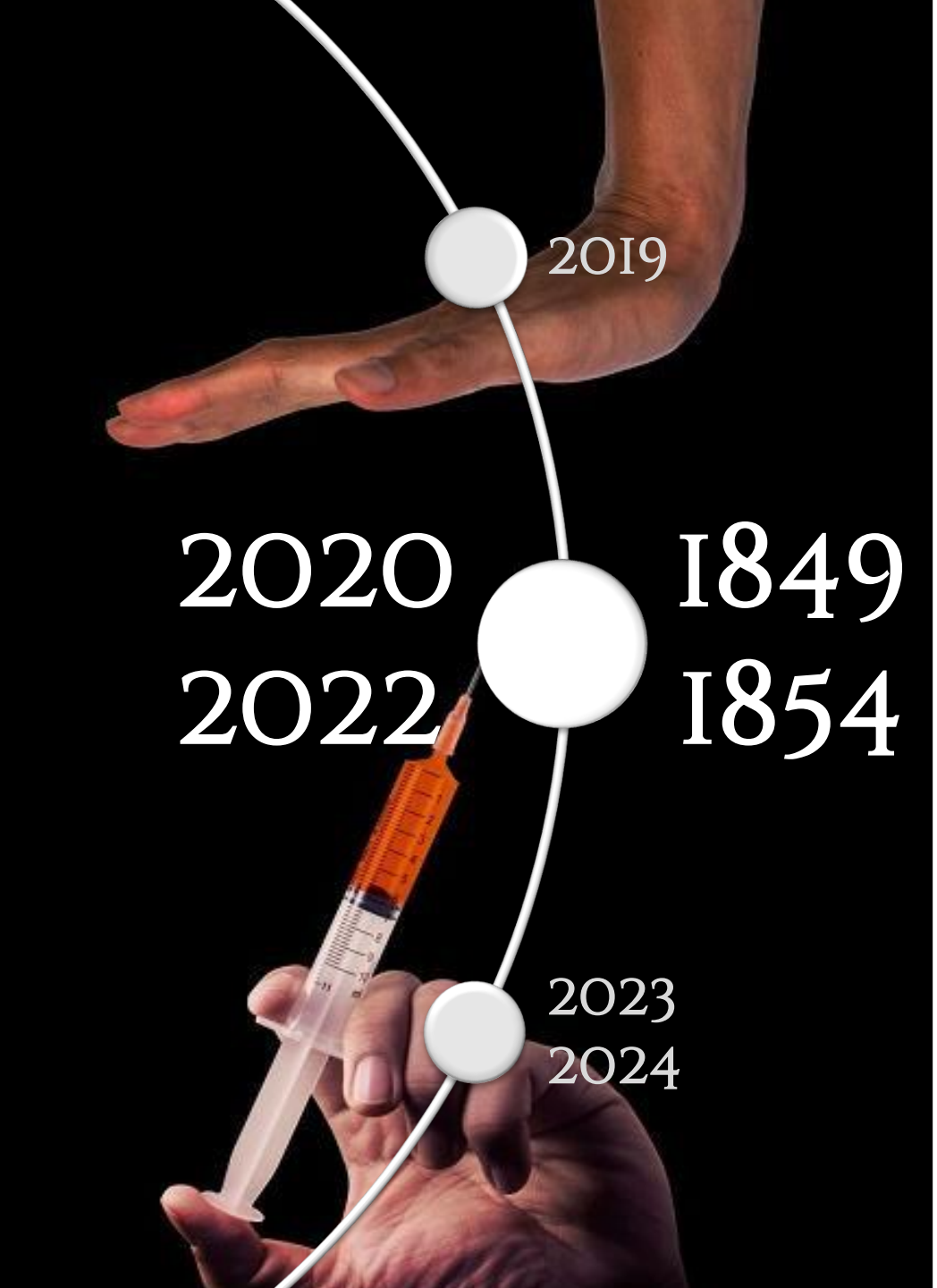
JOHN SNOW

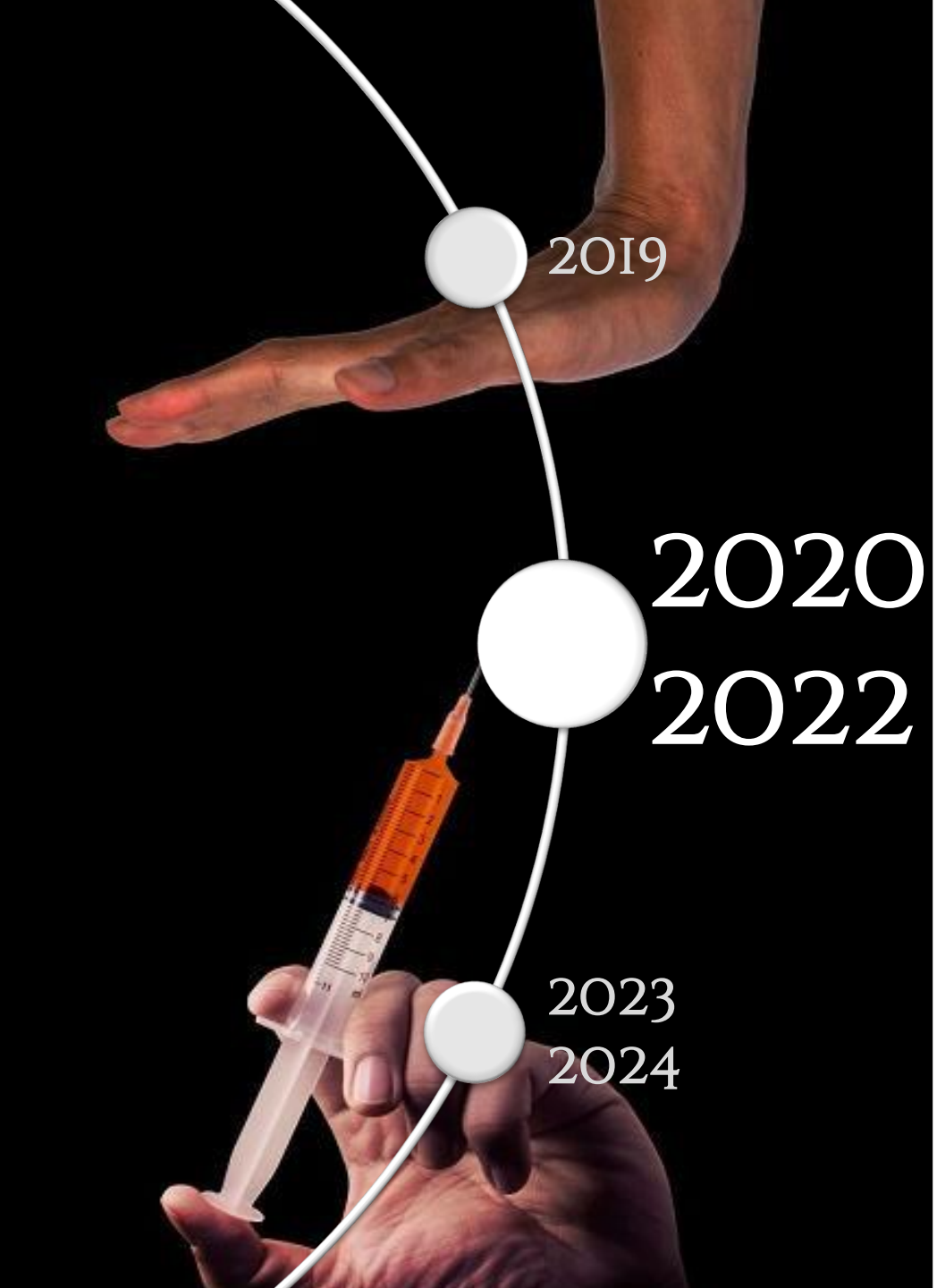
GREAT MARLBOROUGH STREET
WORK HOUSE
PUMP
GOLDEN SQUARE
BRIDLE STREET
PETER STREET
ARCHER STREET

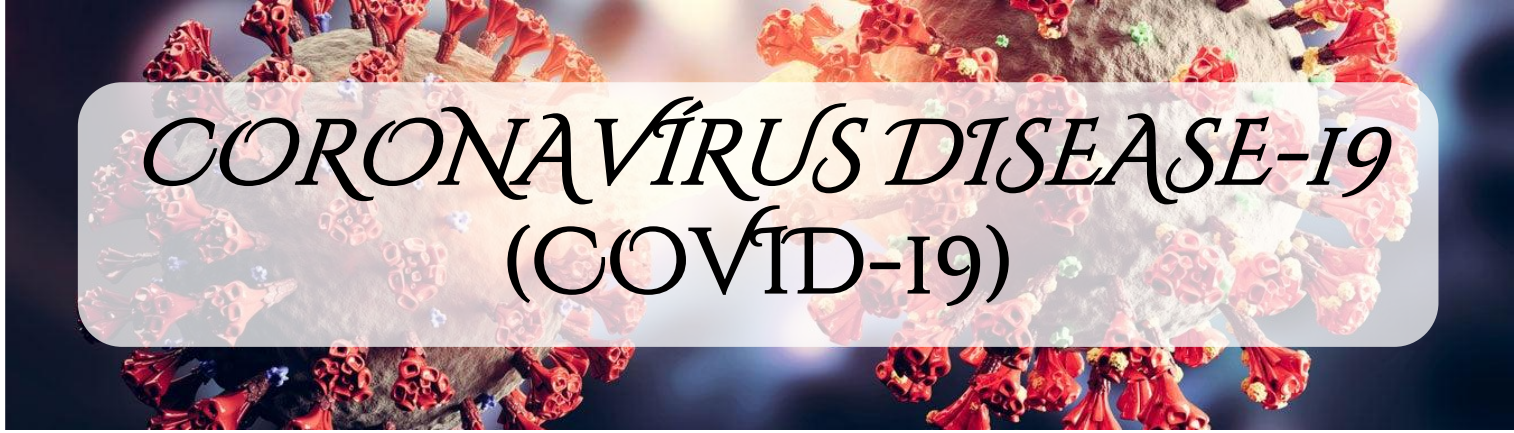


JOHN SNOW

MAPEAMENTO DOS:

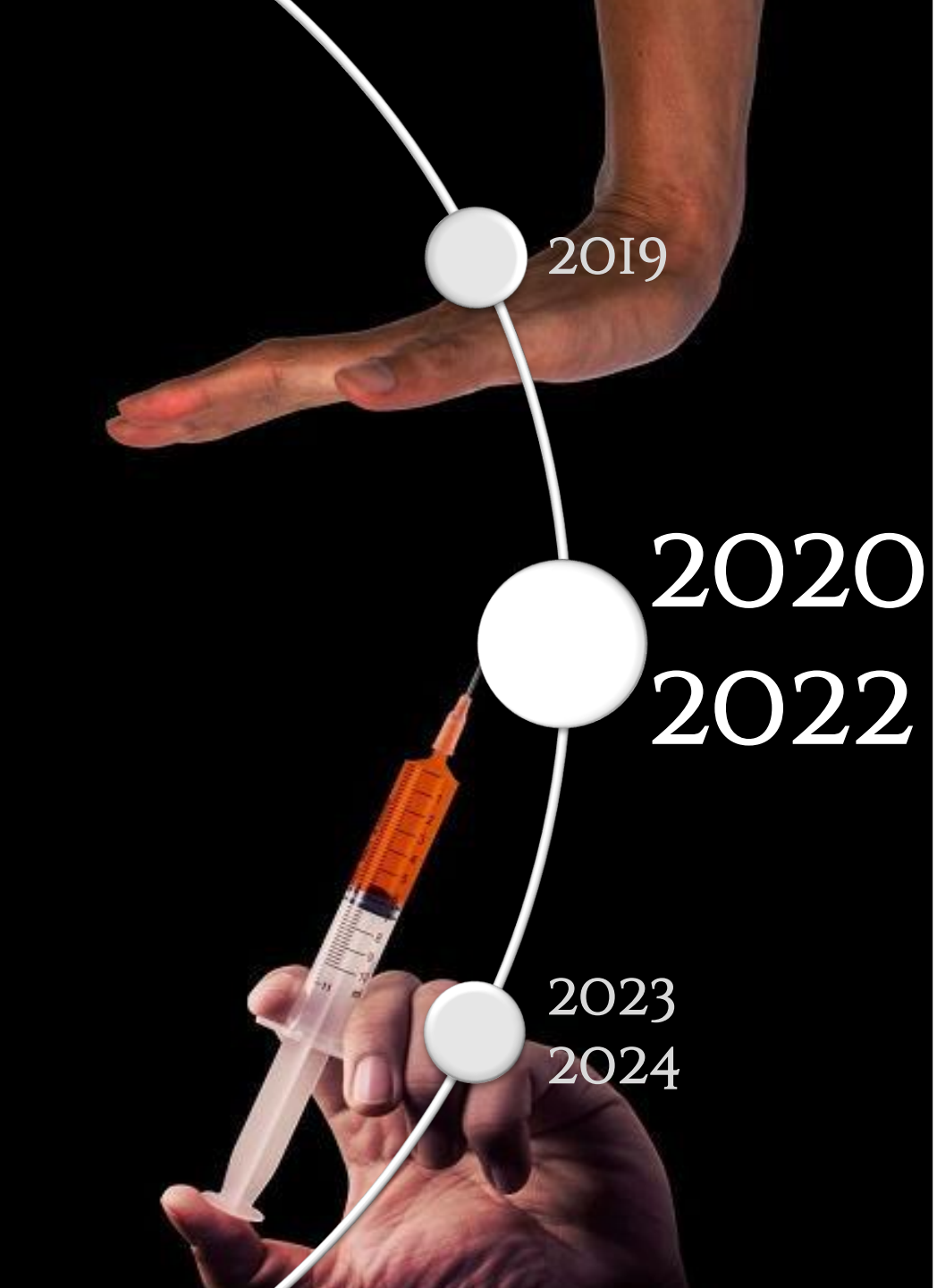
- ✓ CASOS E ÓBITOS;
- ✓ COMPORTAMENTO DA POPULAÇÃO COM O CONSUMO DE ÁGUA;
- ✓ ASPECTOS AMBIENTAIS DA LOCALIDADE.

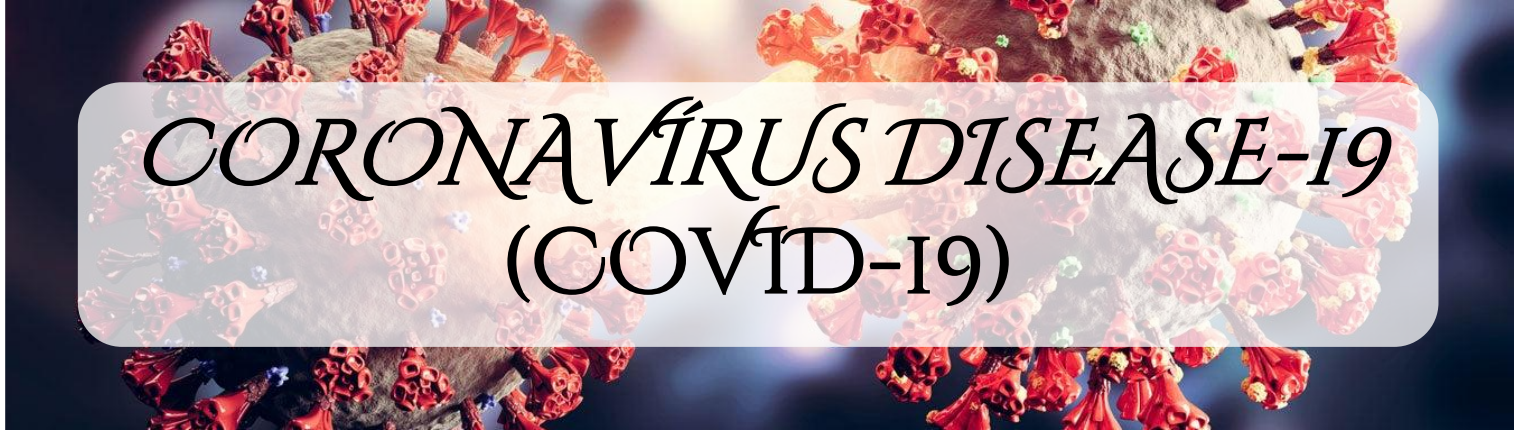




PESSOAS COMUNS

"Não aguento mais ver números de casos e óbitos... Não existe necessidade".





CAPACITAÇÃO DOS PROFISSIONAIS DA SAÚDE

SOB A TEMÁTICA DO SISTEMA DE INFORMAÇÕES QUE ESTEJAM VINCULADAS AO PNI.

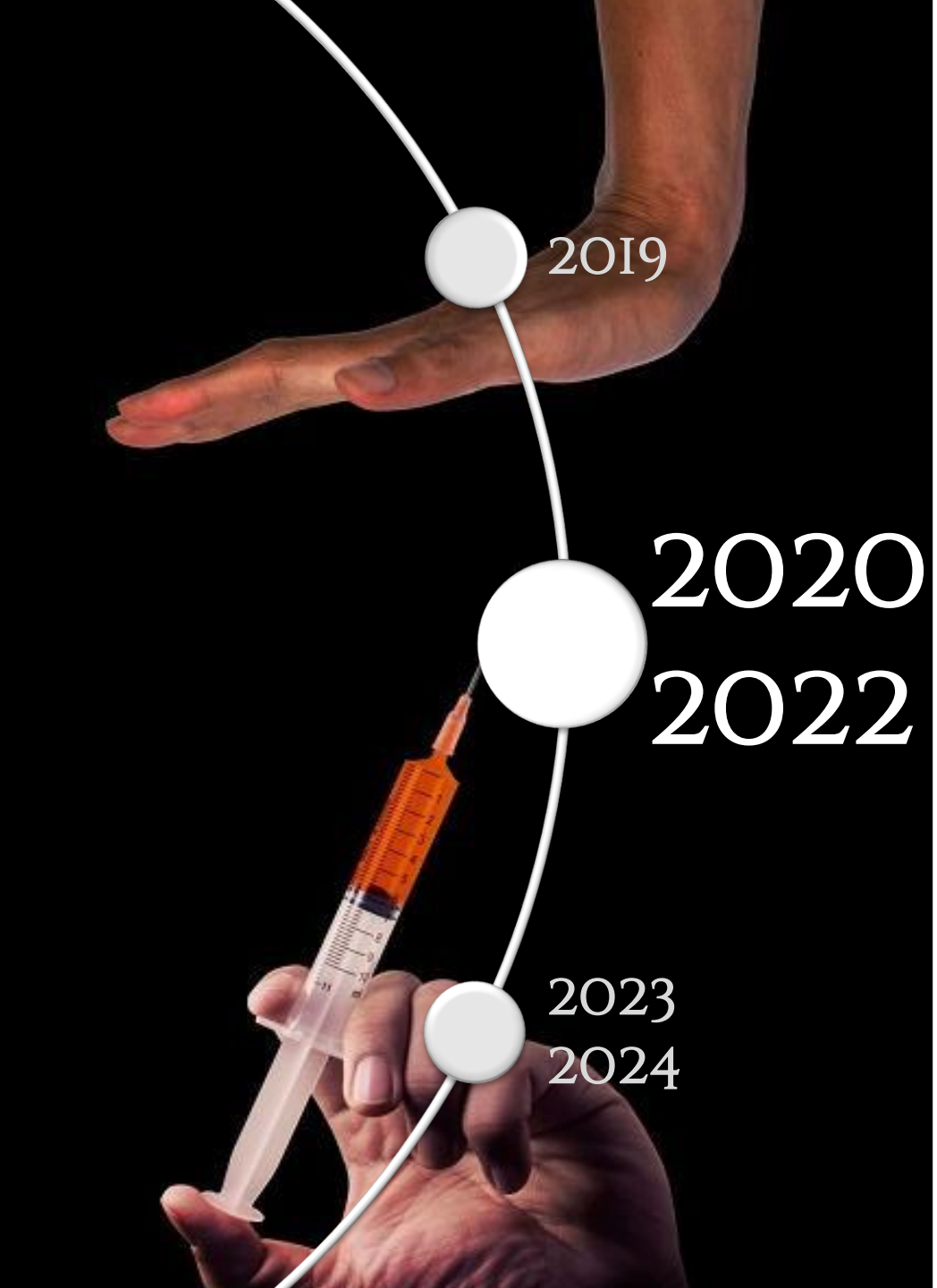
RIO DE JANEIRO

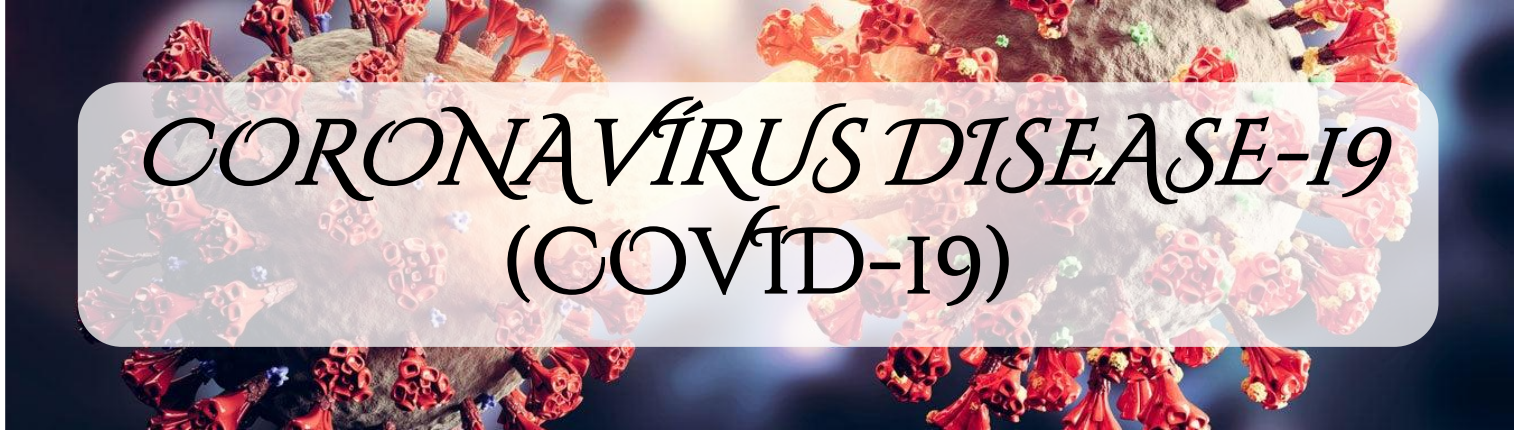
Imagens mostram idosos recebendo falsa aplicação de vacina contra a Covid-19 no RJ

Em Niterói, agulha foi inserida em idoso, mas líquido, não; no Rio, família alertou profissional que dose não tinha sido aplicada, e seringa precisou ser injetada pela 2ª vez; em Petrópolis, profissional espetou idosa com agulha, mas seringa estava vazia. Técnicas de enfermagem de Niterói e de Petrópolis foram afastadas.

Por Bom Dia Rio

15/02/2021 08h52 · Atualizado há 3 anos





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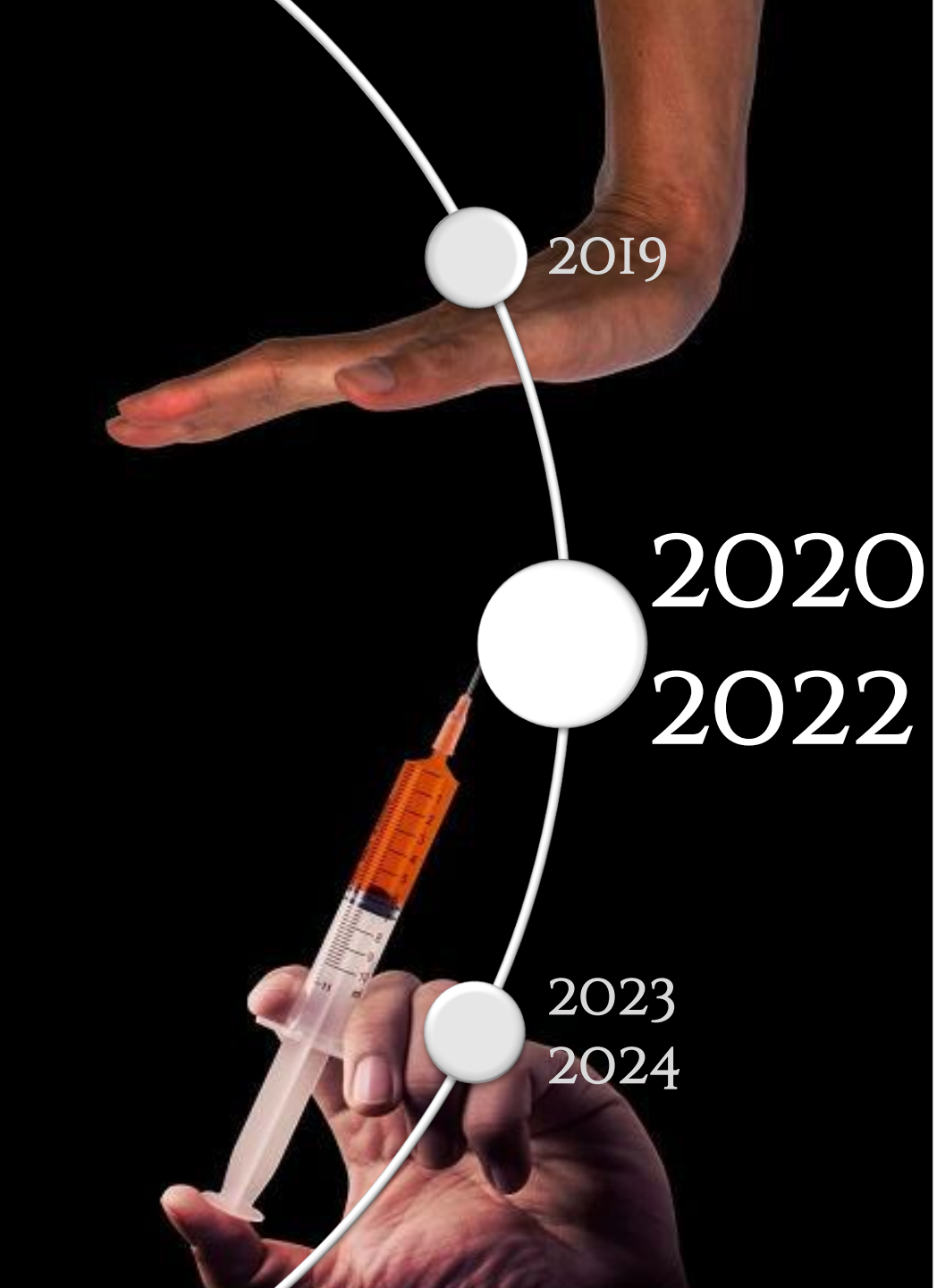
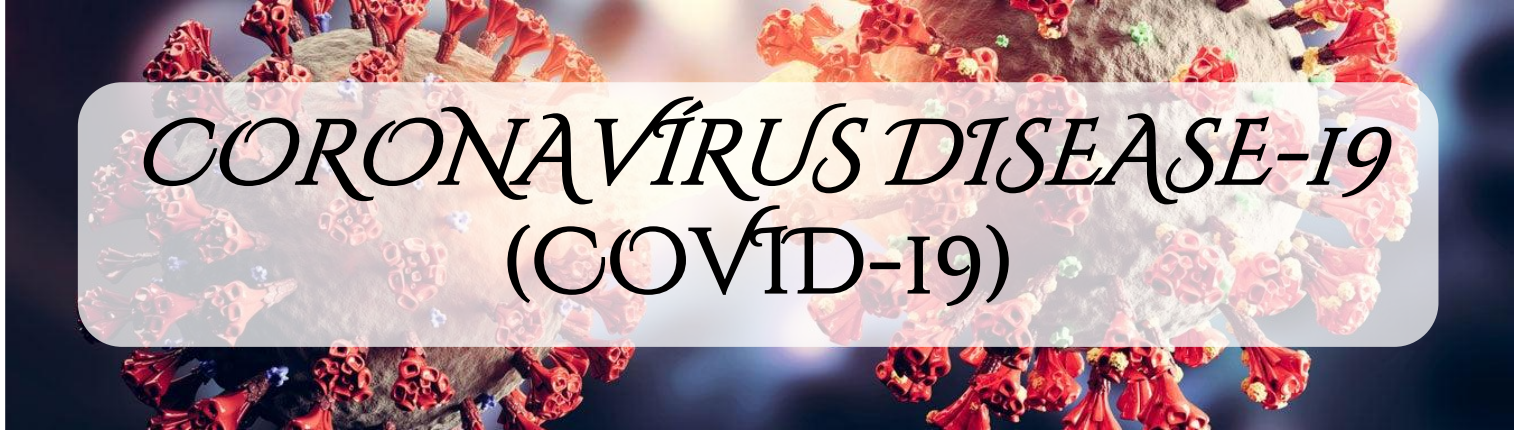
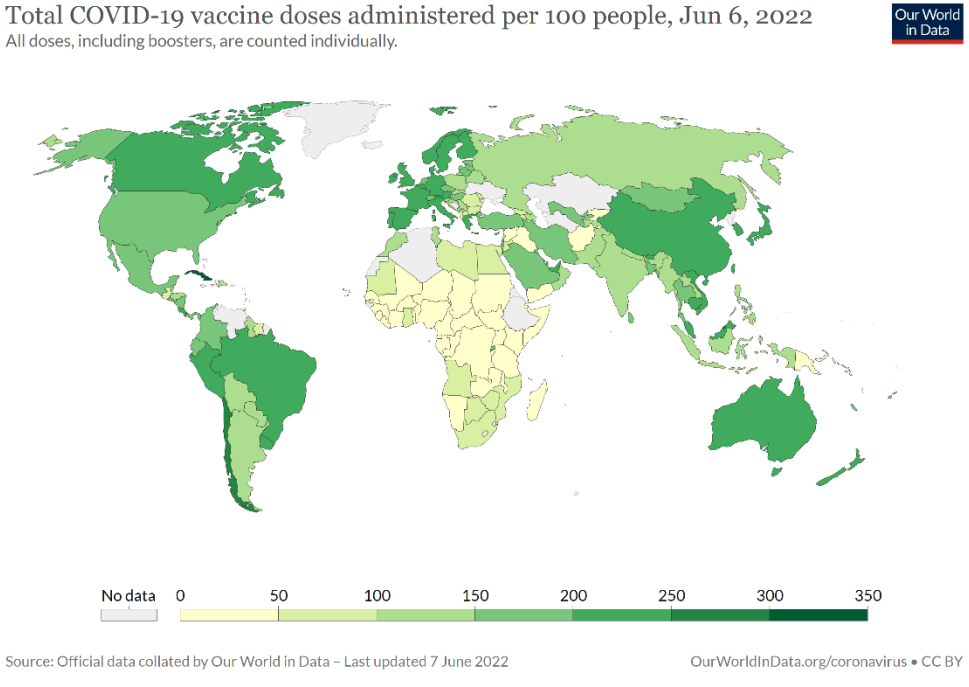


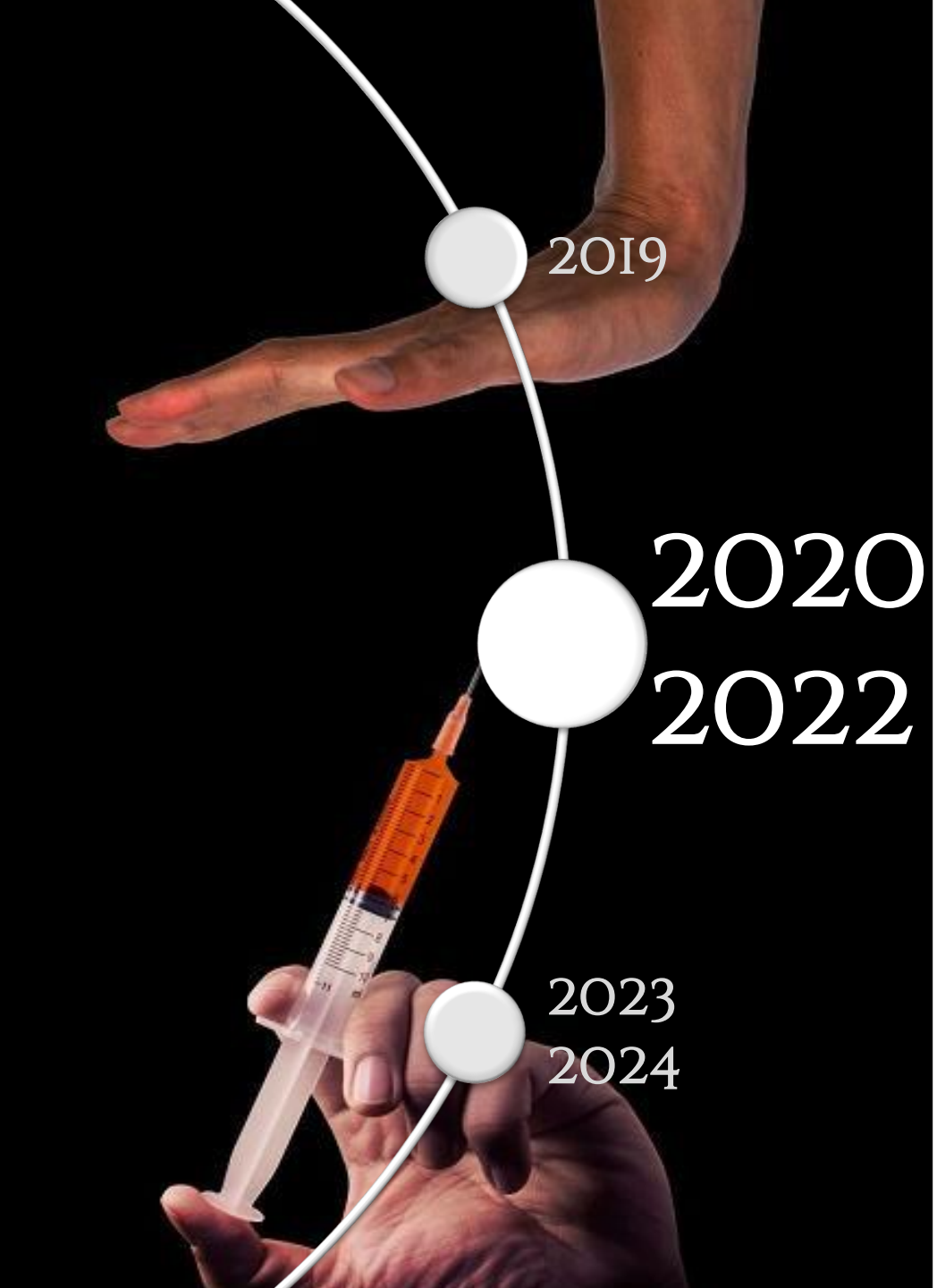


CORONAVÍRUS DISEASE-19 (COVID-19)

ESTRATÉGIAS DE DISTRIBUIÇÃO E ARMAZENAMENTO

INEQUIDADE VACINAL.

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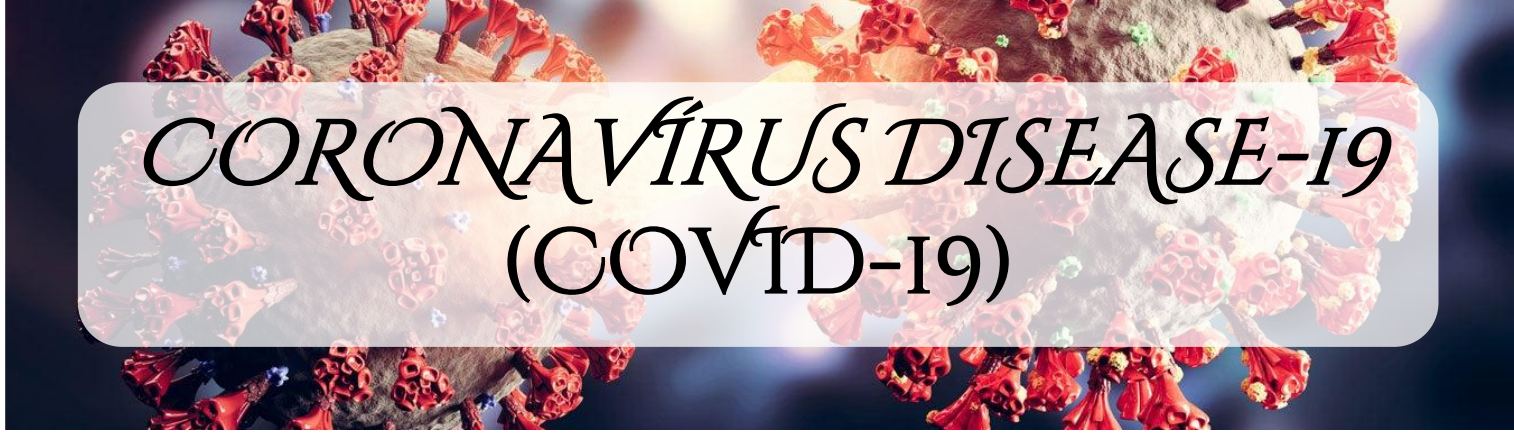
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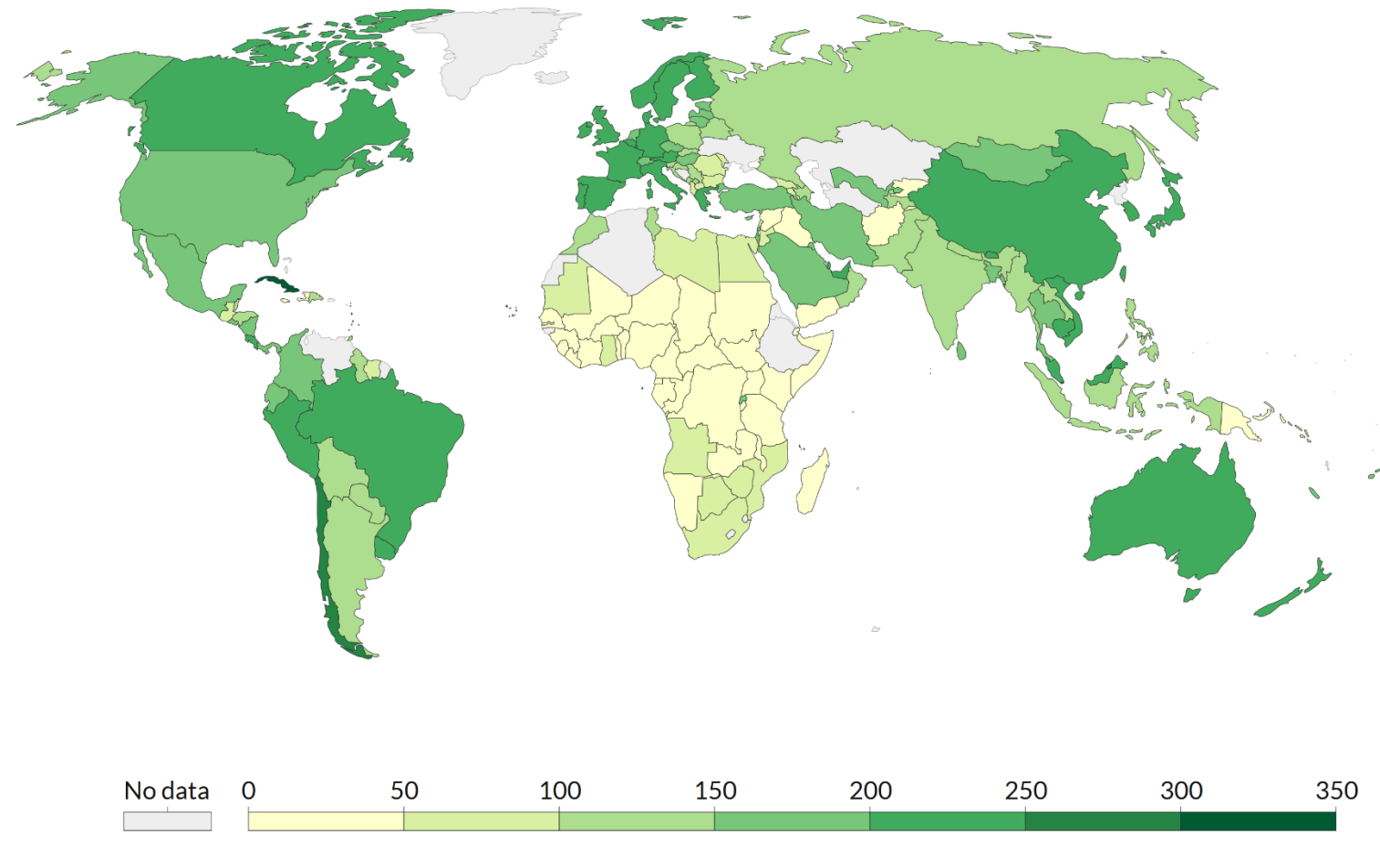
(COVID-19)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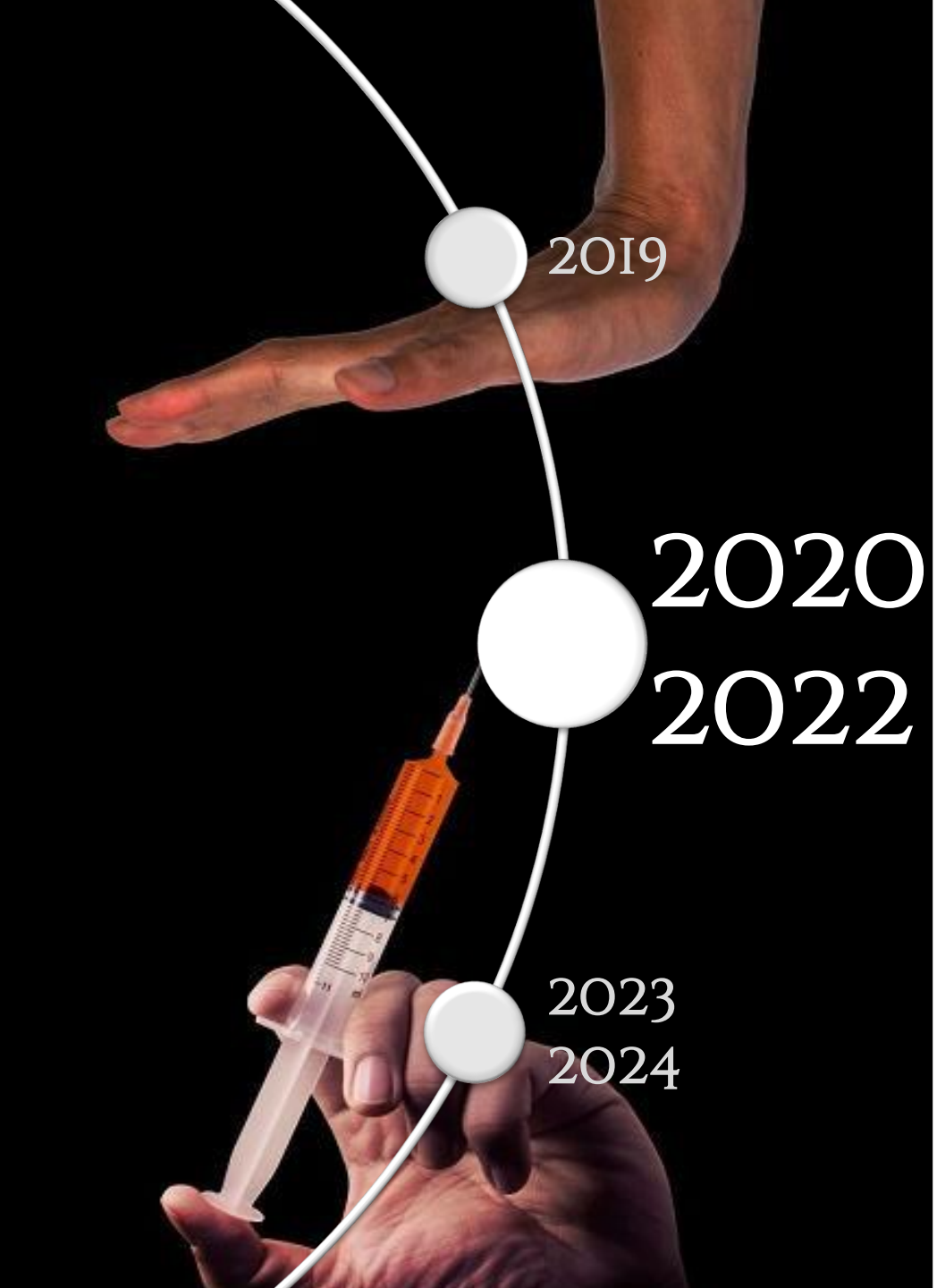
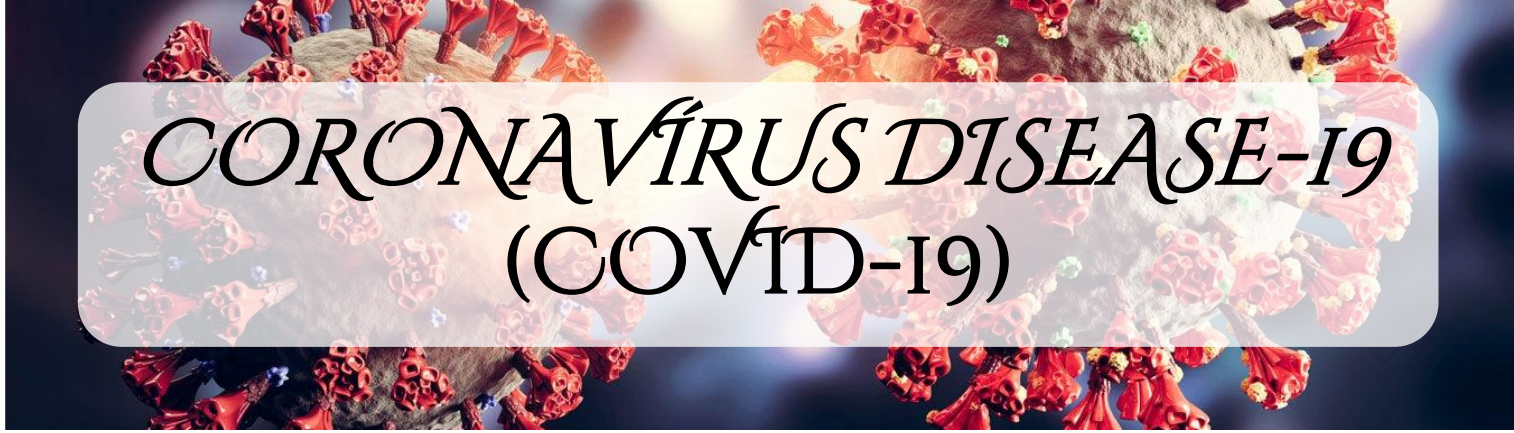




CORONAVÍRUS DISEASE-19 (COVID-19)

ESTRATÉGIAS DE DISTRIBUIÇÃO E ARMAZENAMENTO

INEQUIDADE VACINAL.

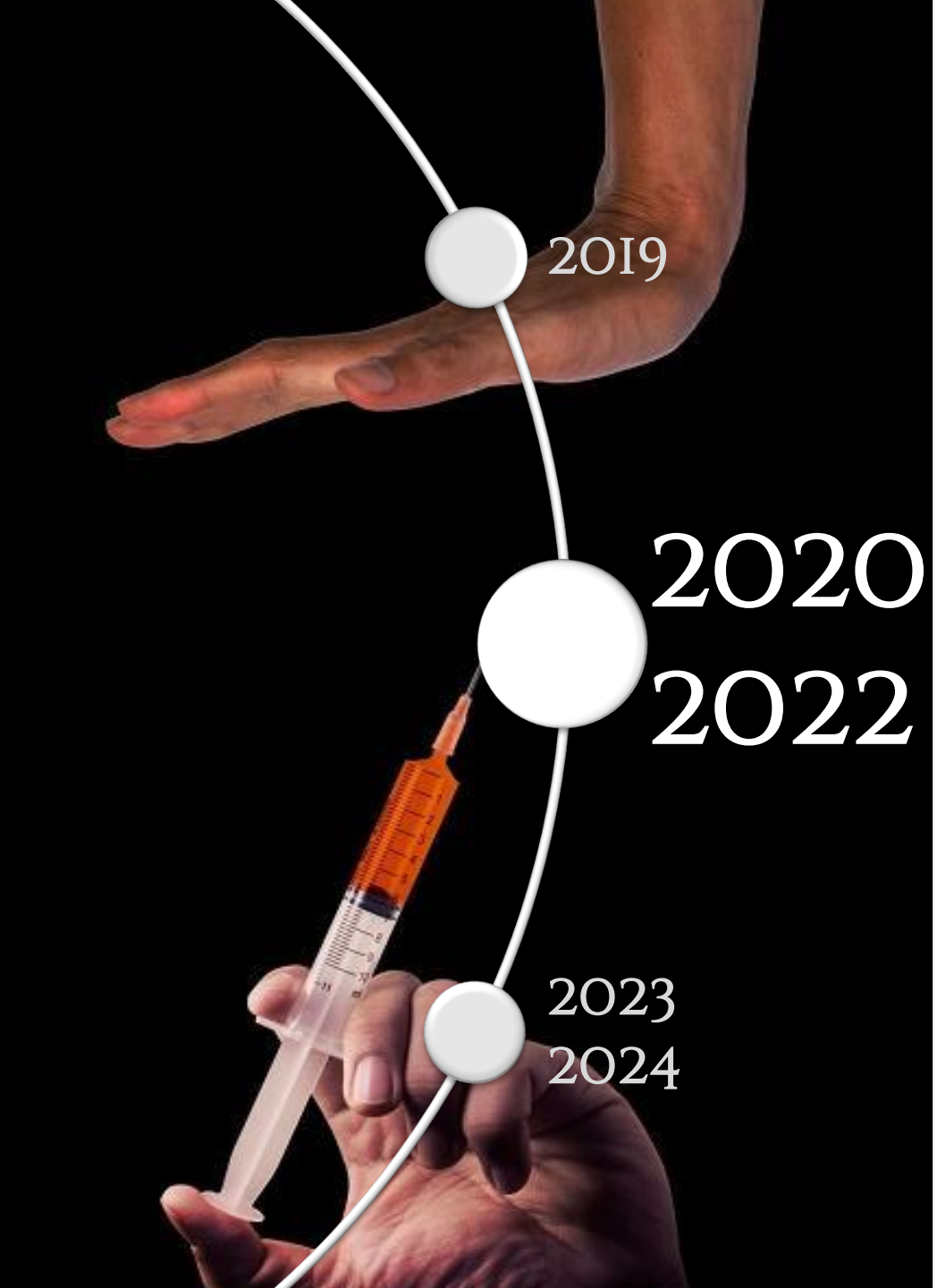
exame.

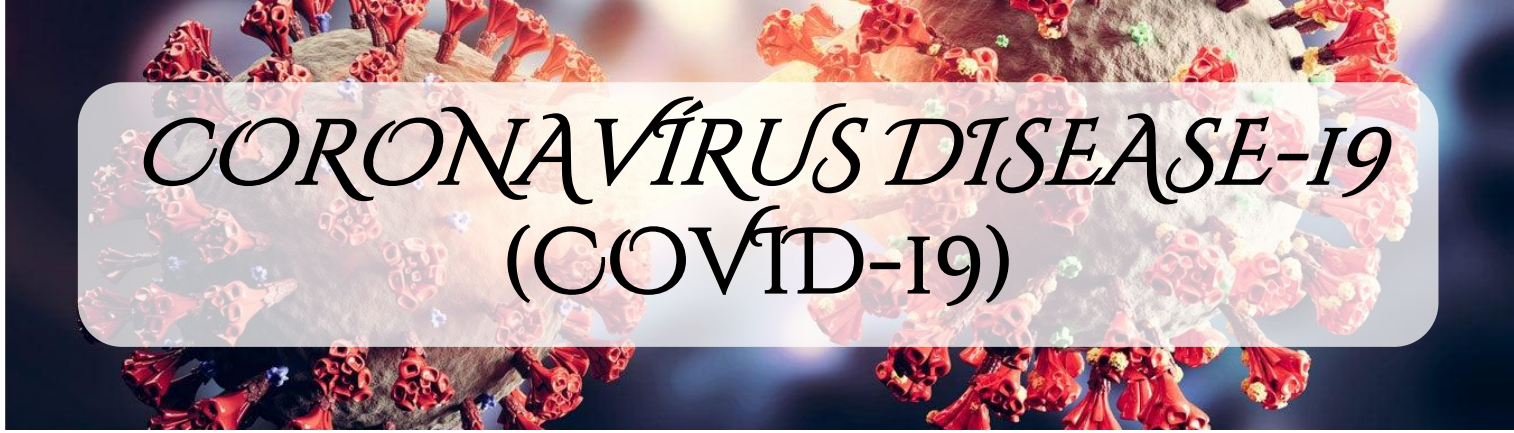
Home > Mundo

Países ricos compraram doses para vacinar 3 vezes a população

Nove em cada dez pessoas em 67 países pobres, por outro lado, não receberão vacina em 2021; fosso pode dimuinuir com posse de Joe Biden

Vacina AstraZeneca/Oxford (Paul Biris/Getty Images)





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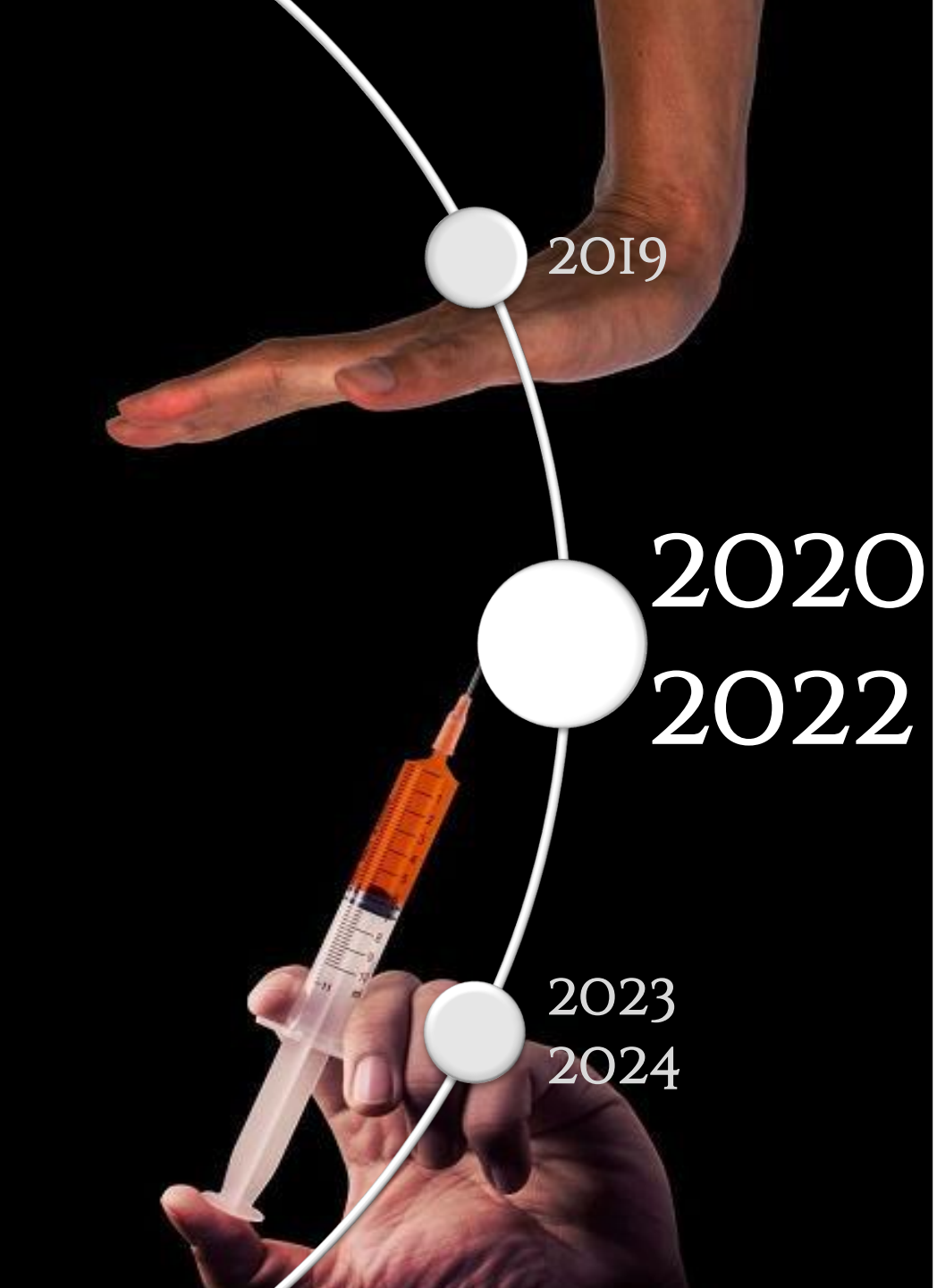
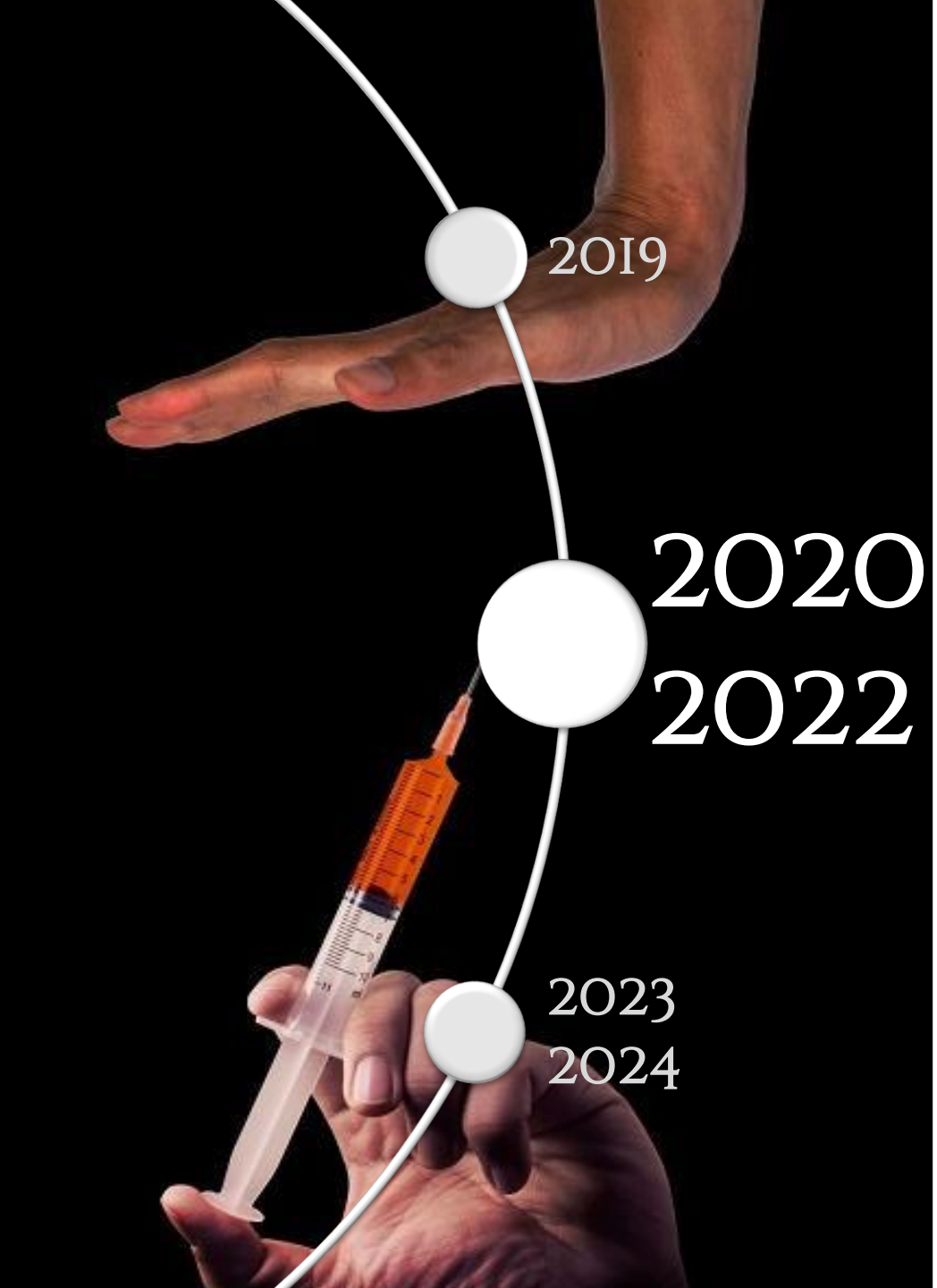


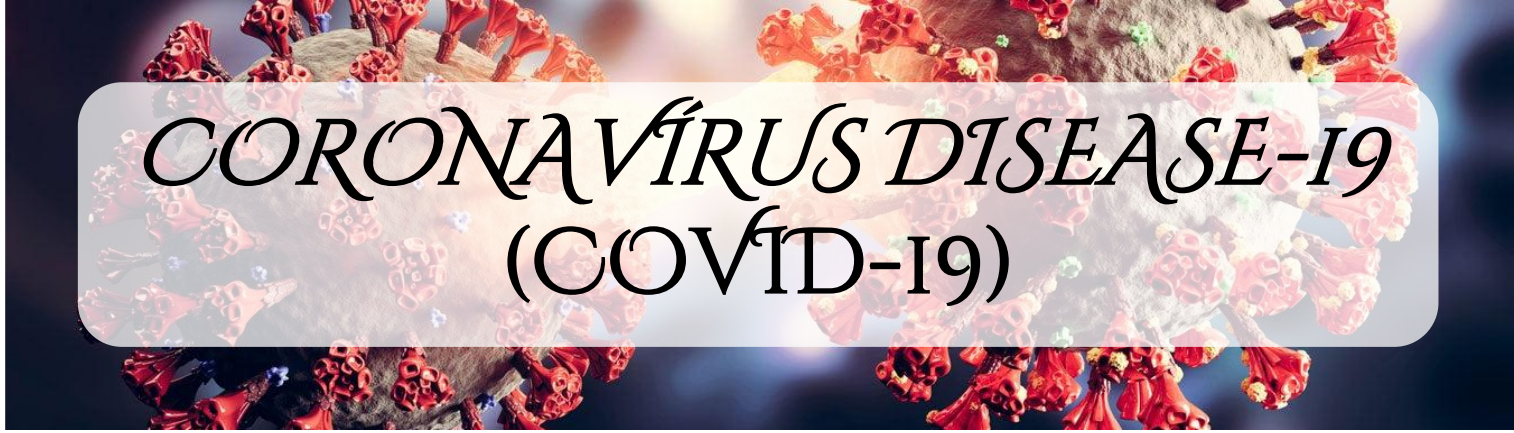
ESTRATÉGIAS DE DISTRIBUIÇÃO E ARMAZENAMENTO

ESTADOS ESGOTARAM TODAS AS PRIMEIRAS DOSES, SEM TER A EXPECTATIVA DE QUANDO A SEGUNDA DOSE IRIA CHEGAR.

Com vacinas esgotadas, Niterói suspende aplicação da primeira dose contra Covid-19 nesta quarta-feira

As vacinas em estoque são destinadas à segunda dose daqueles que tomaram a primeira





CORONAVÍRUS DISEASE-19 (COVID-19)

ESTRATÉGIAS DE DISTRIBUIÇÃO E ARMAZENAMENTO

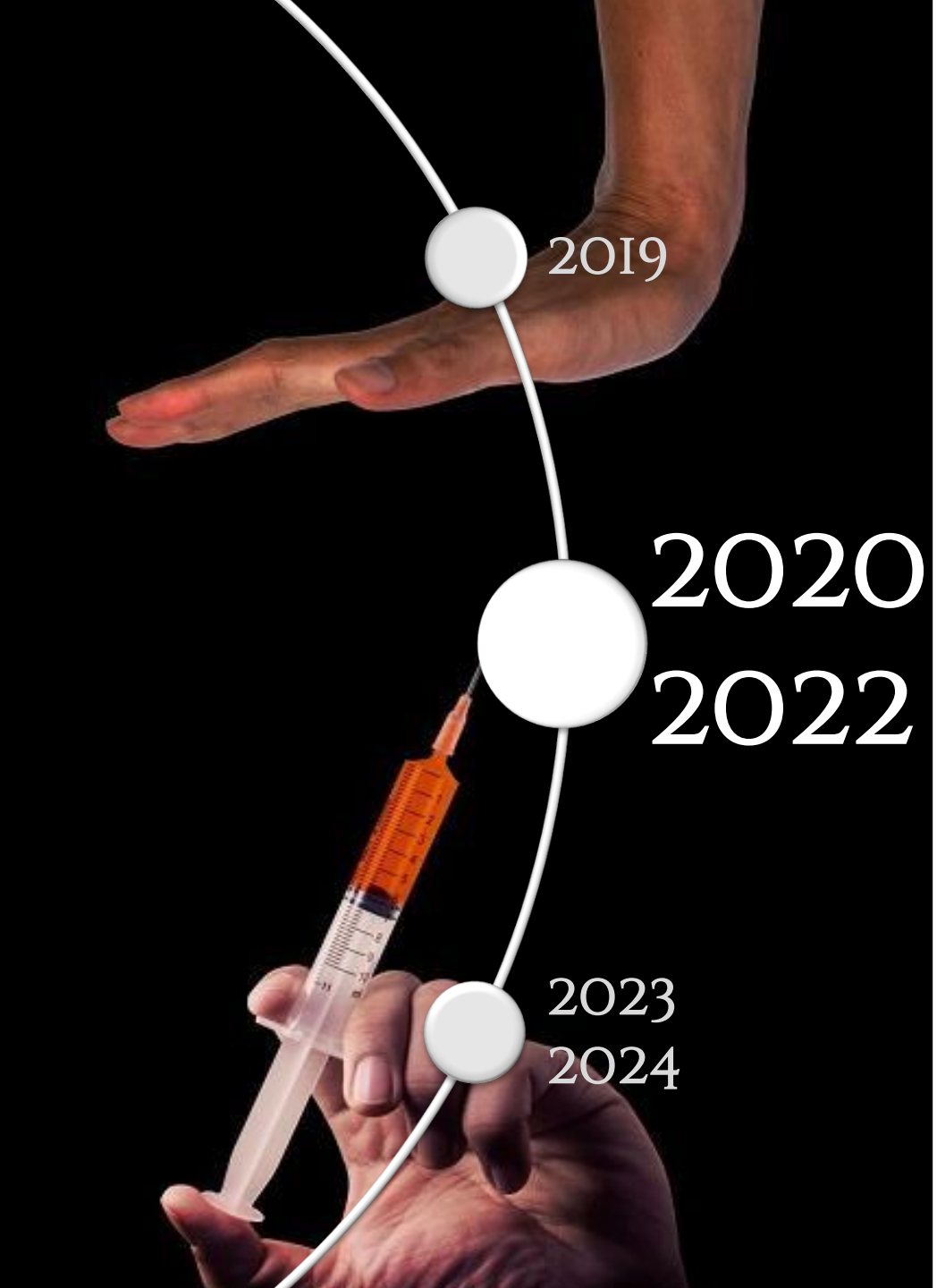
exame.

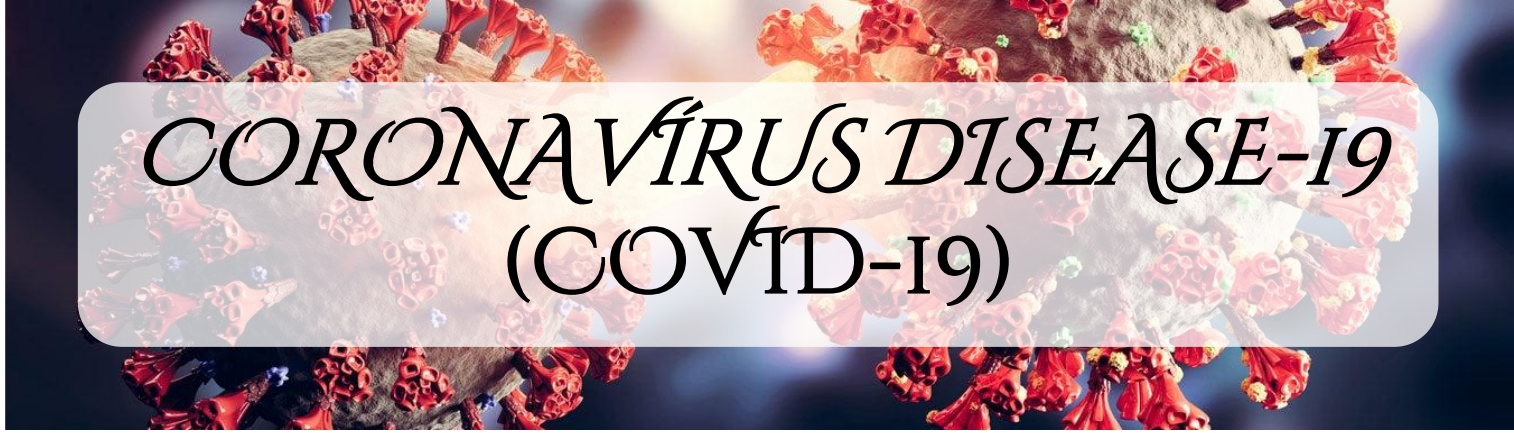
Home › Ciência

Pfizer pode produzir 1,3 bilhão de doses de vacina da covid-19 em 2021

Outras vacinas, como a de Oxford e a CoronaVac, também têm perspectiva de escala de produção alta

Pfizer: empresa prevê produção de 1,3 bilhão de doses em 2021 (Foto/Getty Images)





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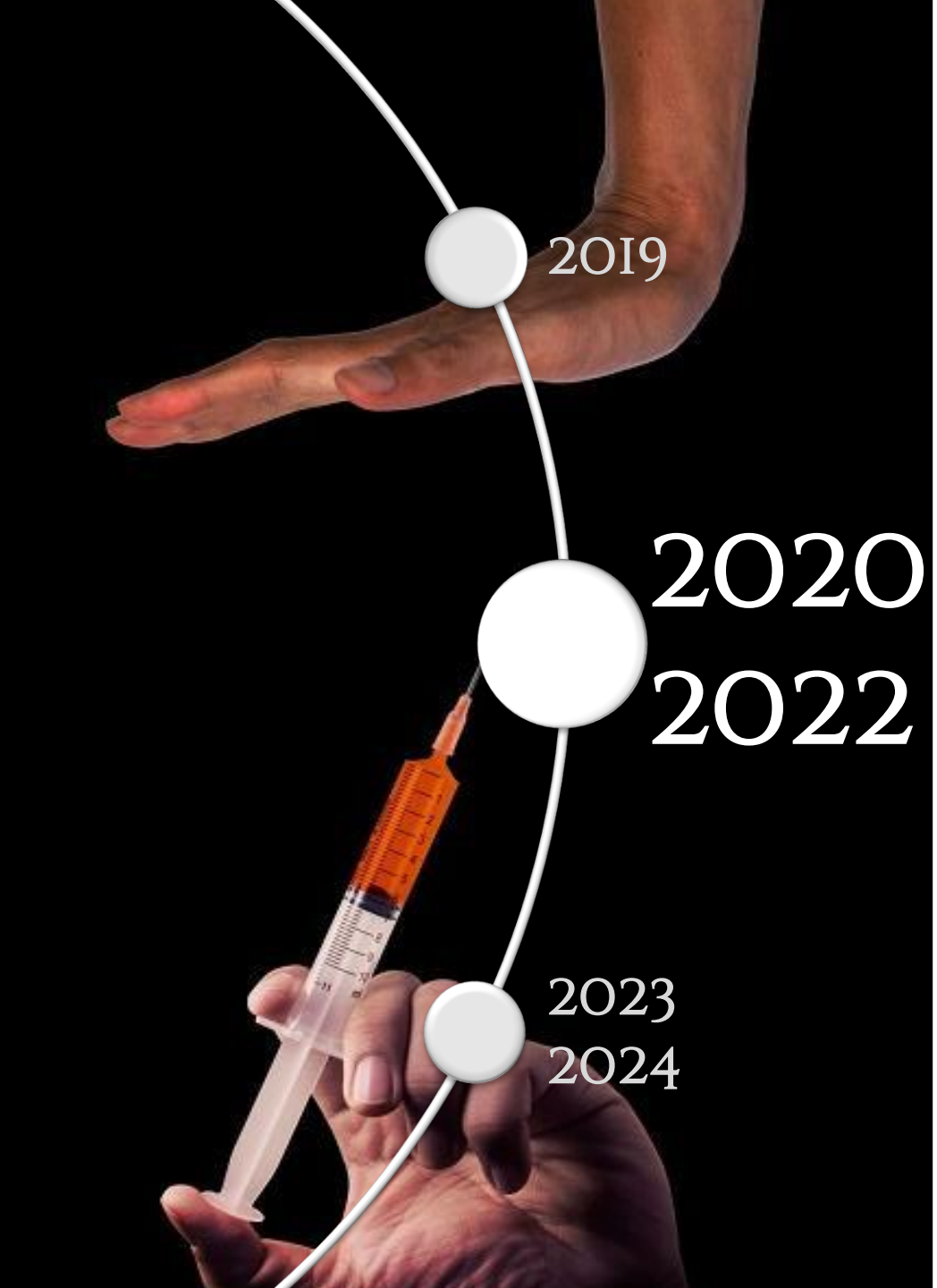
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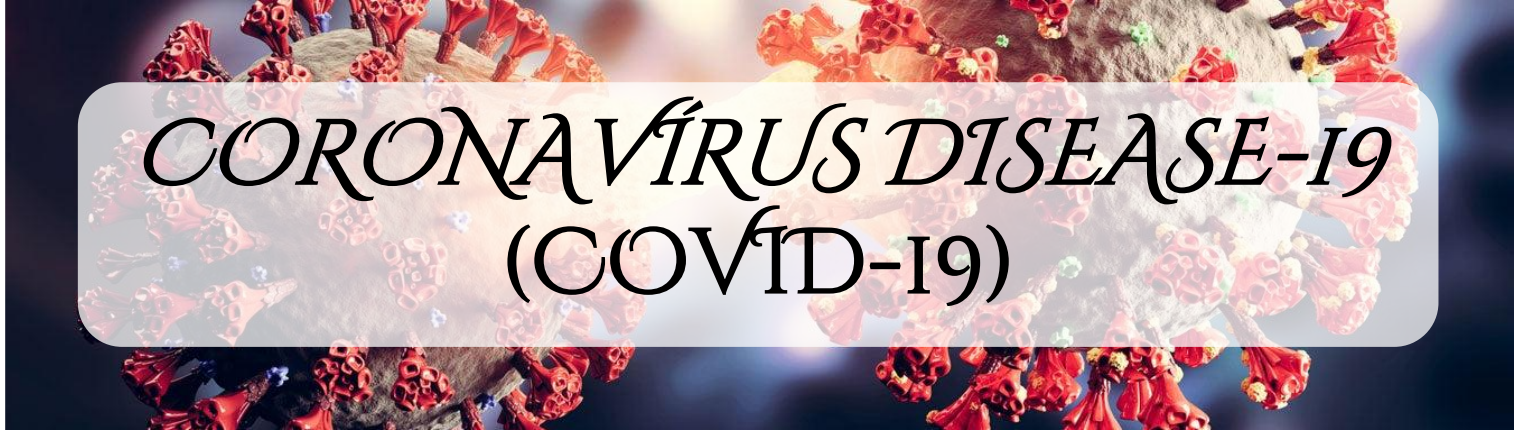
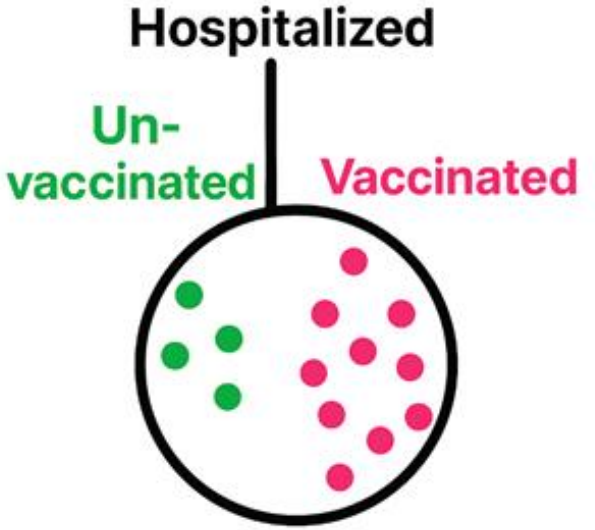
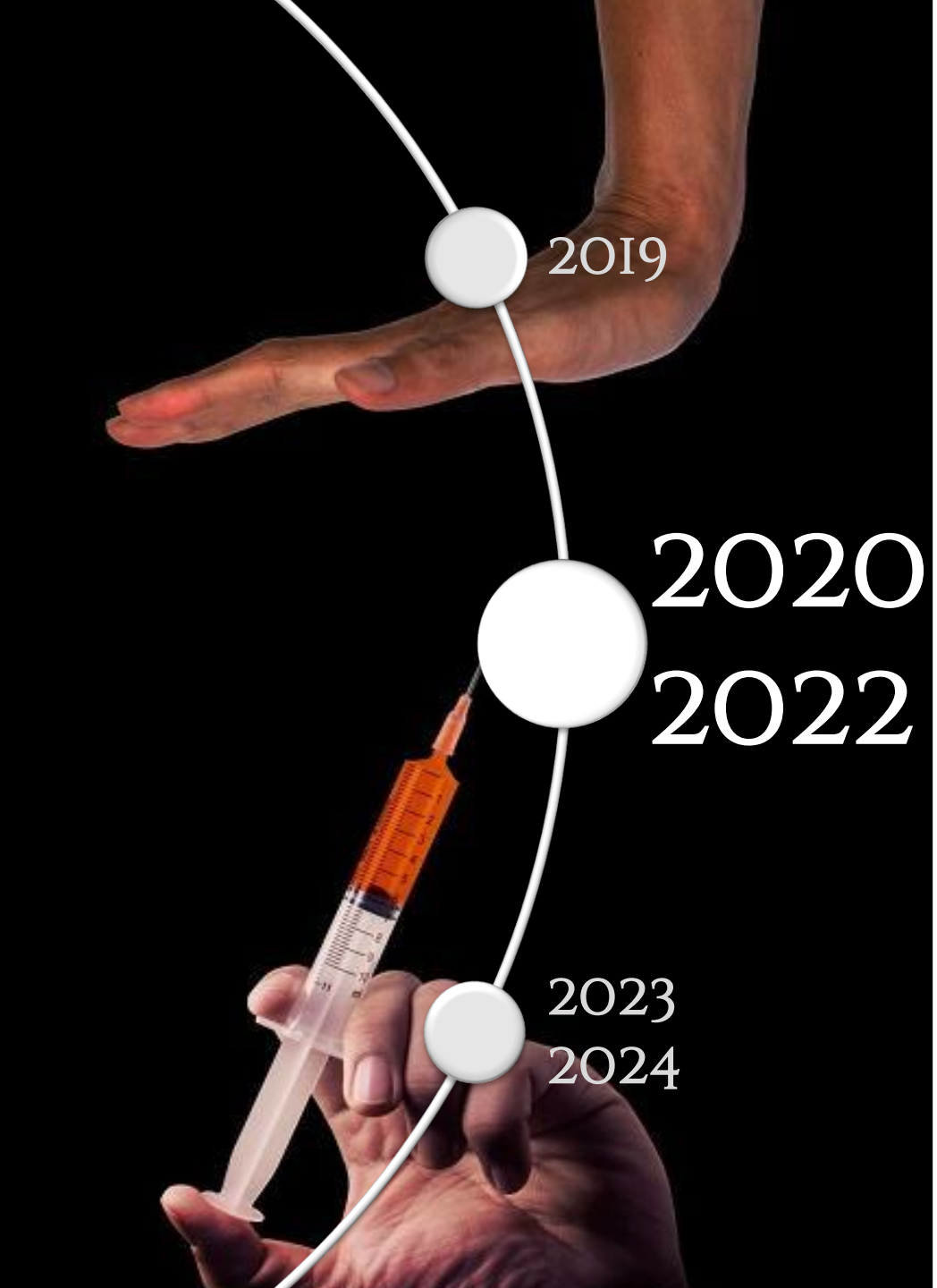

CORONAVÍRUS DISEASE-19
(COVID-19)


Hospitalized
Un-
vaccinated
Vaccinated

FAKE
NEWS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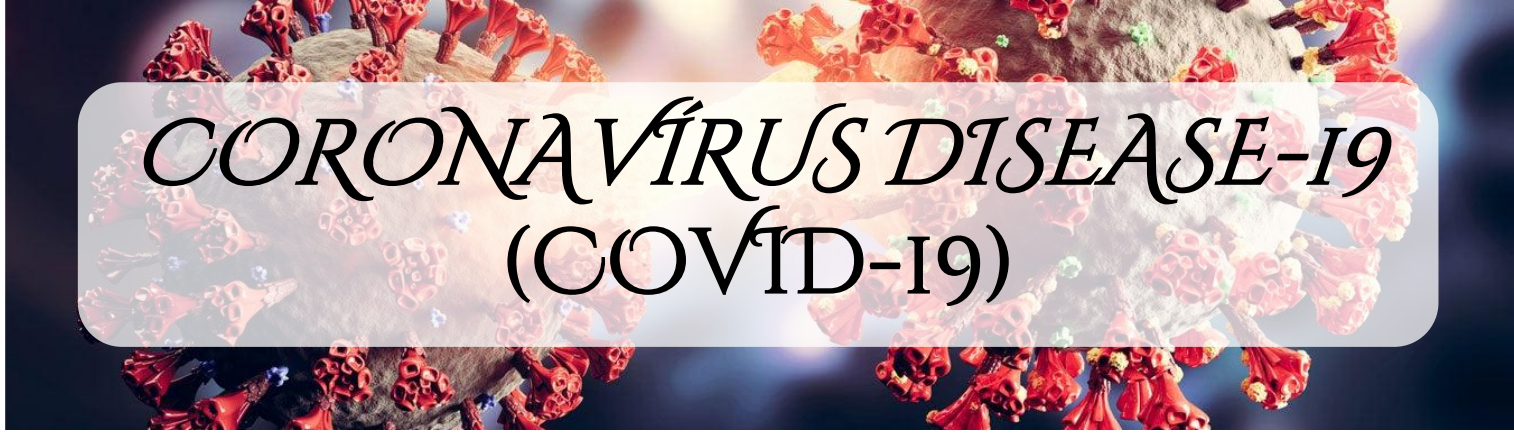
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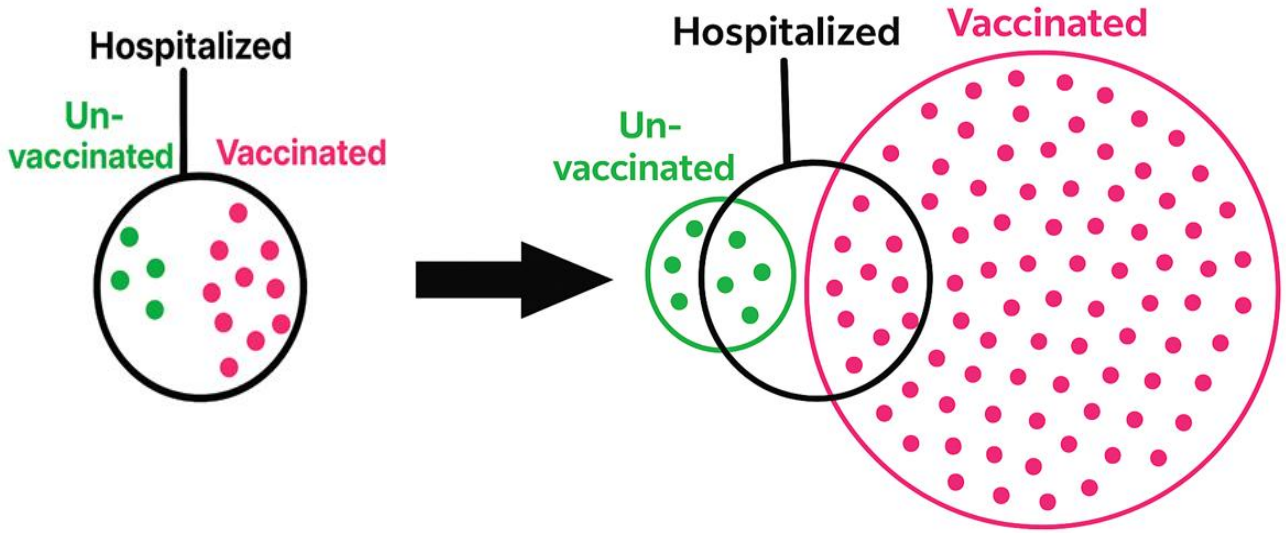
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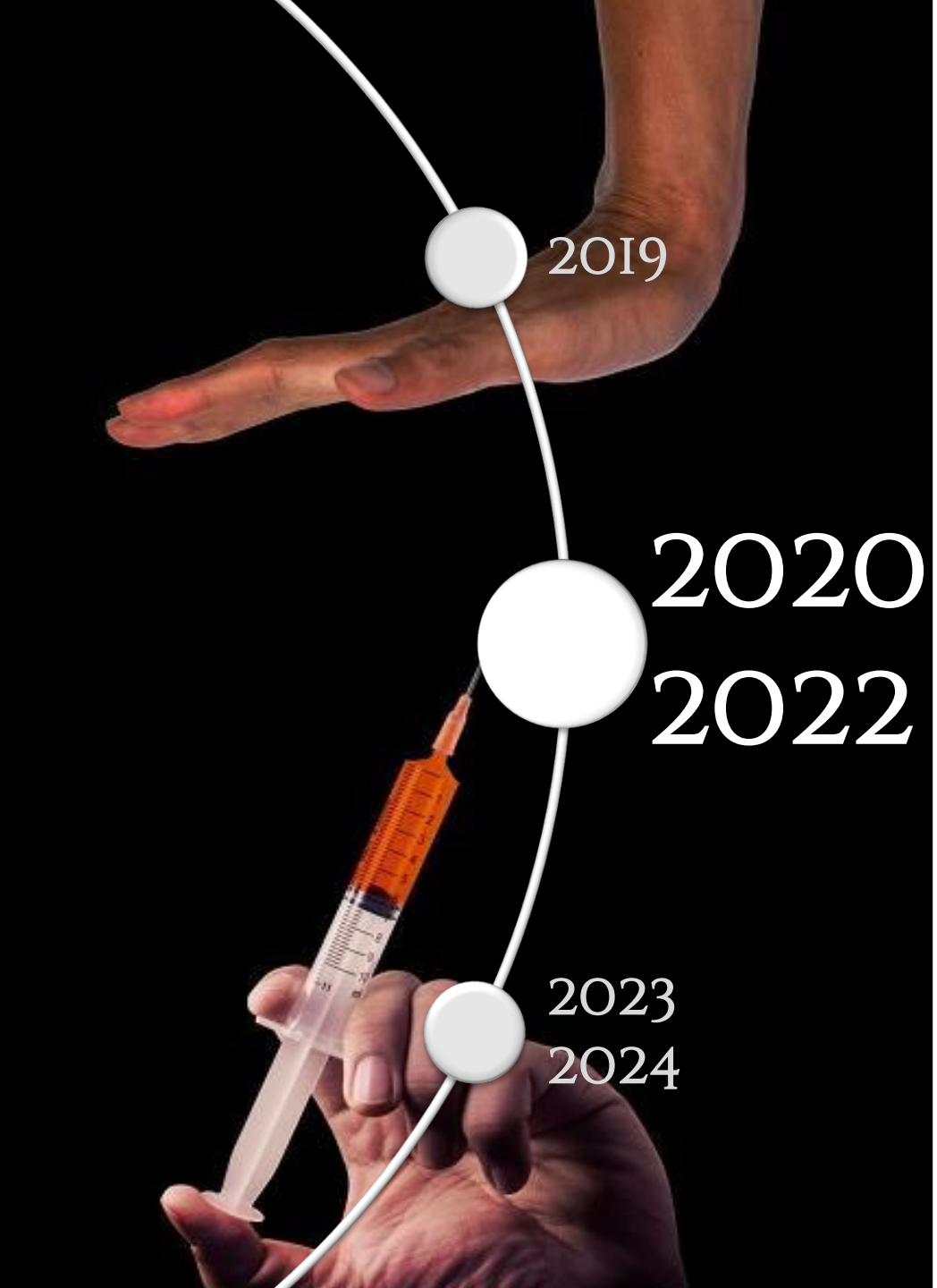
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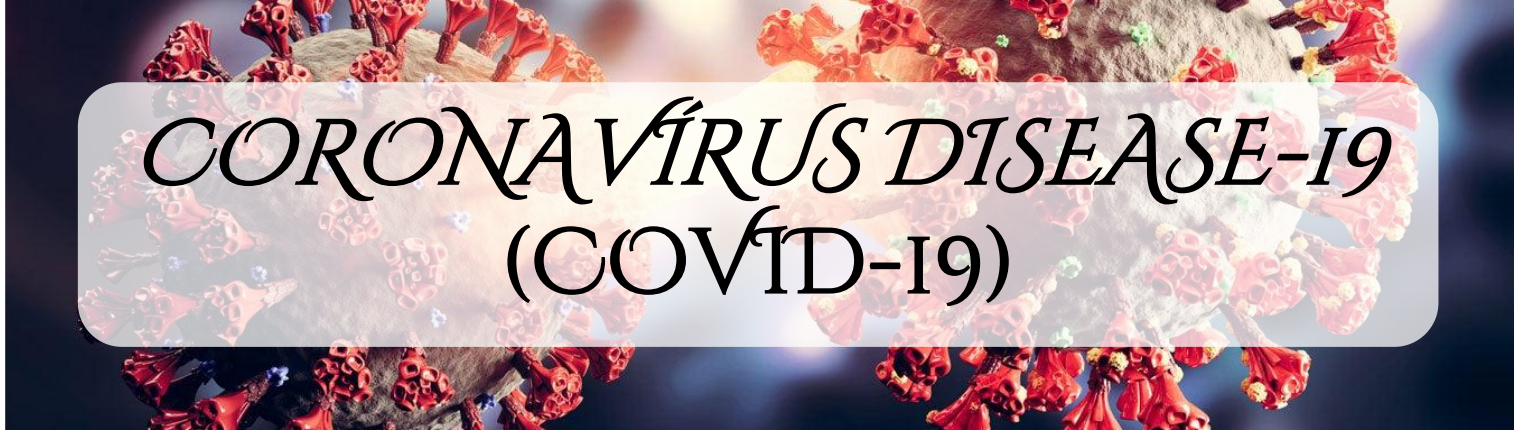
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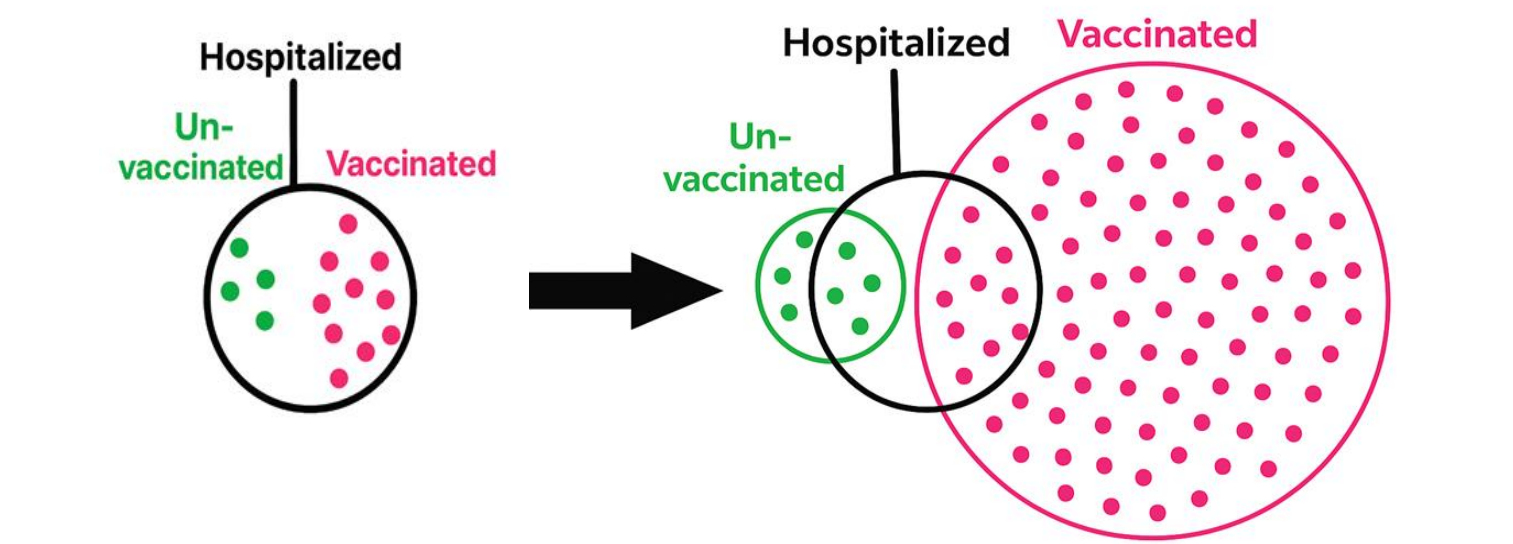
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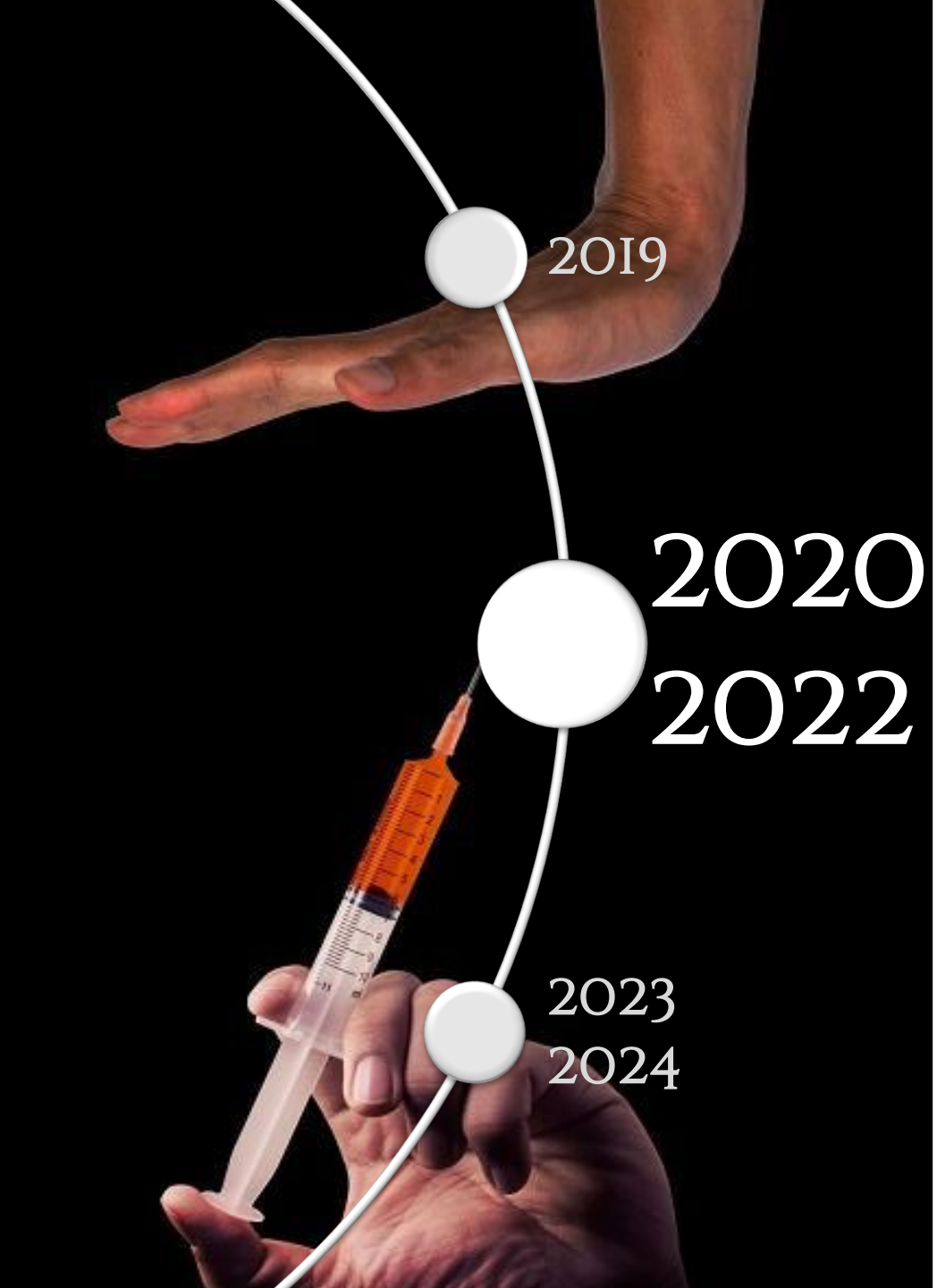
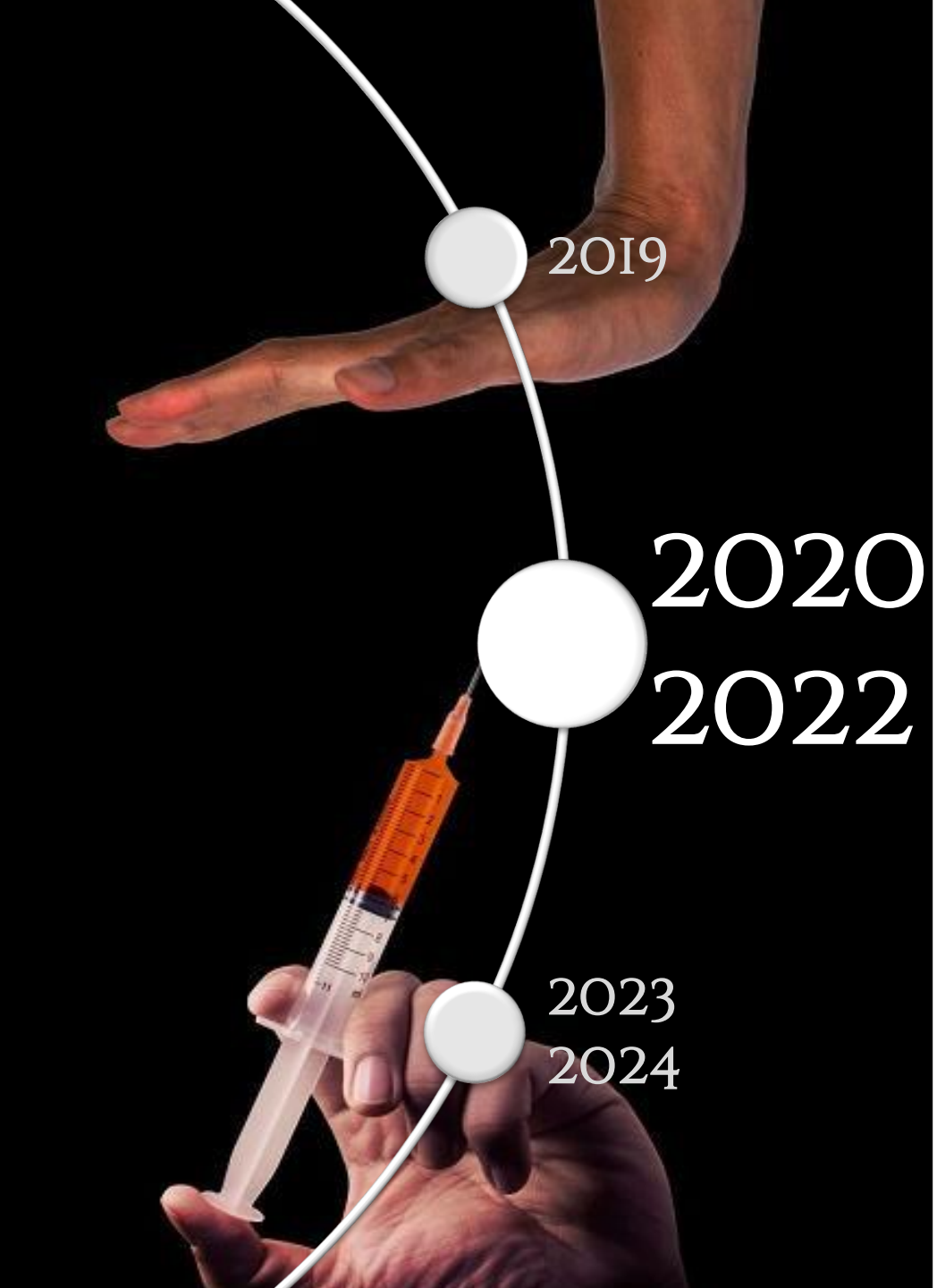
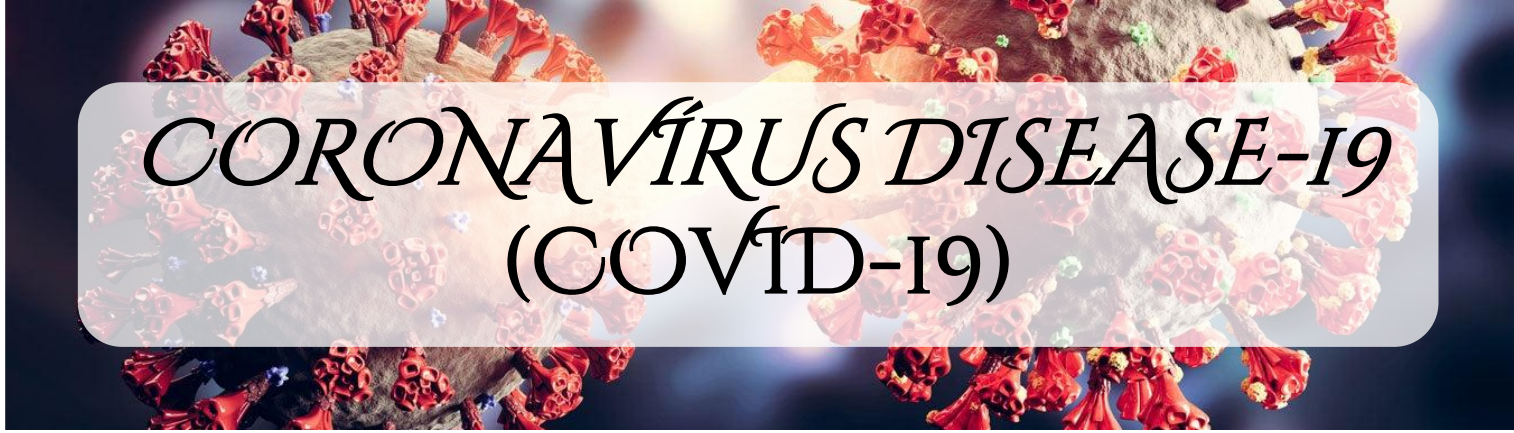


CORONAVÍRUS DISEASE-19 (COVID-19)

VIGILÂNCIA EPIDEMIOLÓGICA E MOLECULAR DE CASOS





CORONAVÍRUS DISEASE-19 (COVID-19)

MÁSCARAS MAIS EFICIENTES

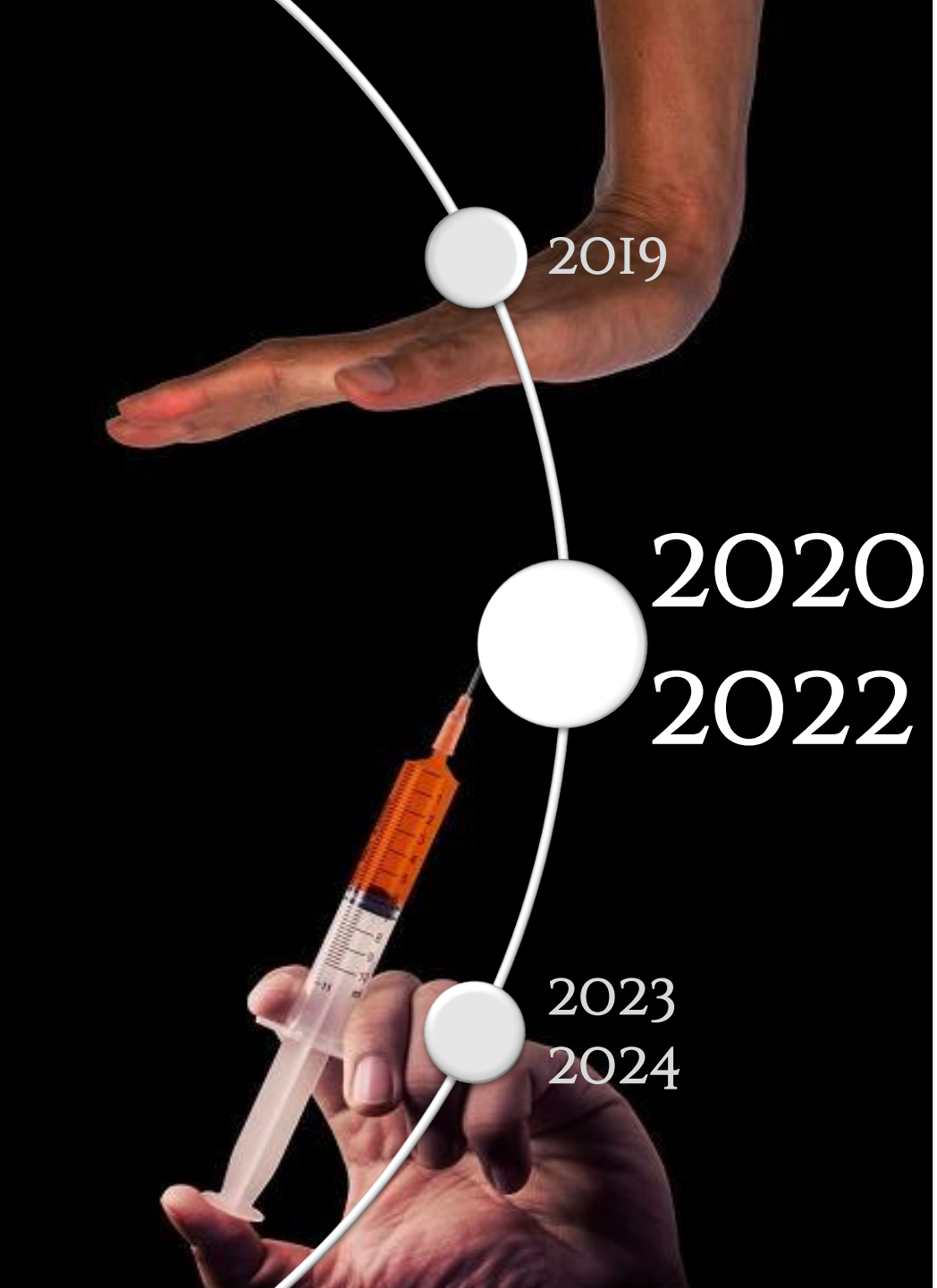
N95 E FPP2

PAÍSES NA EUROPA EXIGIRAM O USO DESSAS MÁSCARAS EM LOCAIS PÚBLICOS.

Bavaria mandates FFP2 masks, all of Germany may follow

Germany was mulling tougher anti-pandemic measures as authorities in Bavaria ordered highly protective FFP2 masks to be worn in shops and on public transport.

FONTE: HTTPS://WWW.DW.COM/EN/BAVARIA-MANDATES-FFP2-MASKS-ALL-OF-GERMANY-MAY-FOLLOW/



CORONAVÍRUS DISEASE-19 (COVID-19)

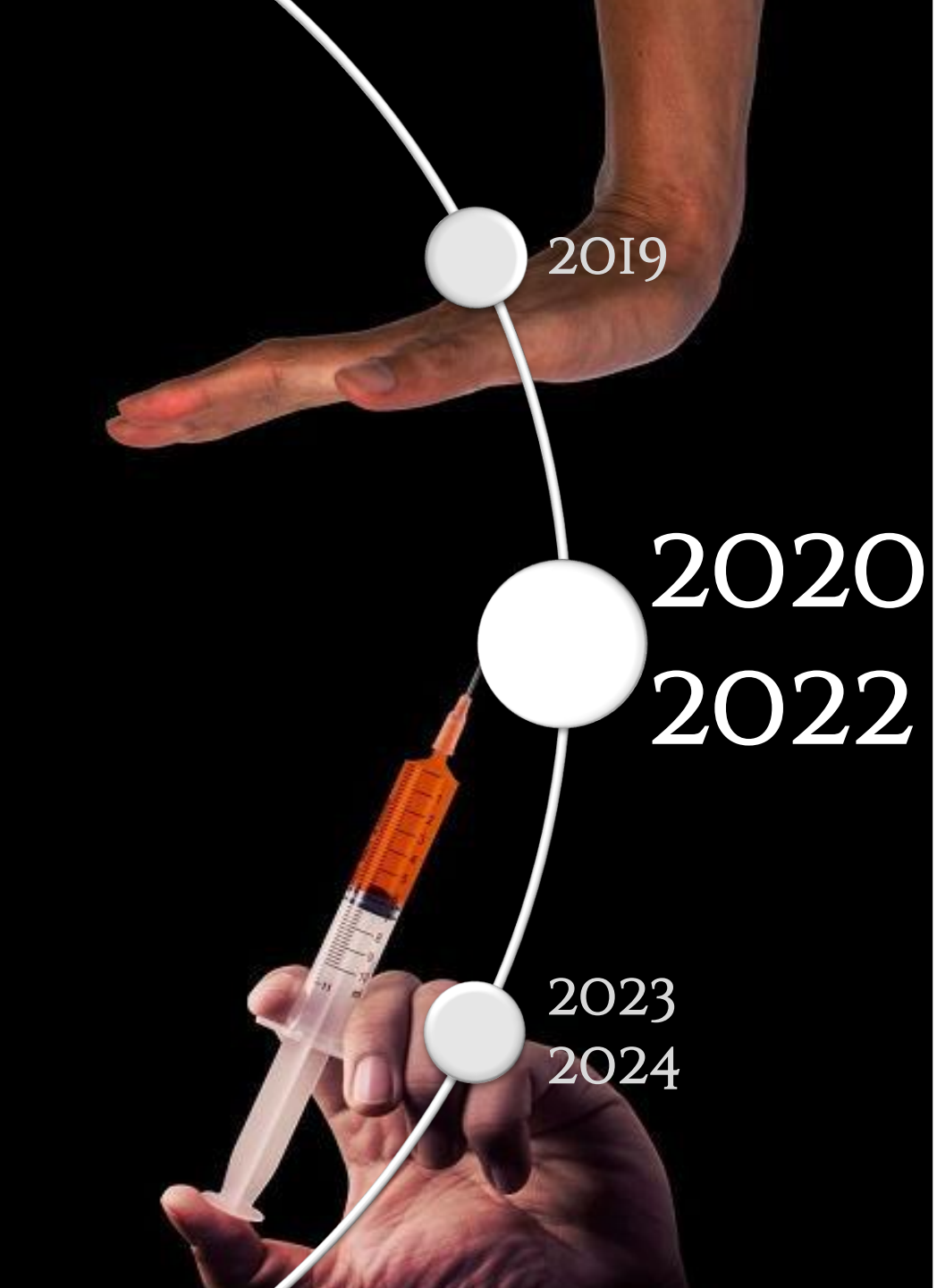
ANSIEDADE DA POPULAÇÃO

PESSOAS FURANDO A FILA DE PRIORIDADE;

Os 'fura-filas' da vacina: quem são os investigados em diferentes Estados do Brasil

Diversos relatos sobre 'fura-filas' durante a vacinação se tornaram alvos de investigações pelo país

FONTE: HTTPS://WWW.BBC.COM/PORTUGUESE/BRASIL-55744032



ANSIEDADE DA POPULAÇÃO

GOLPES EM APLICATIVOS E/OU MENSAGENS;

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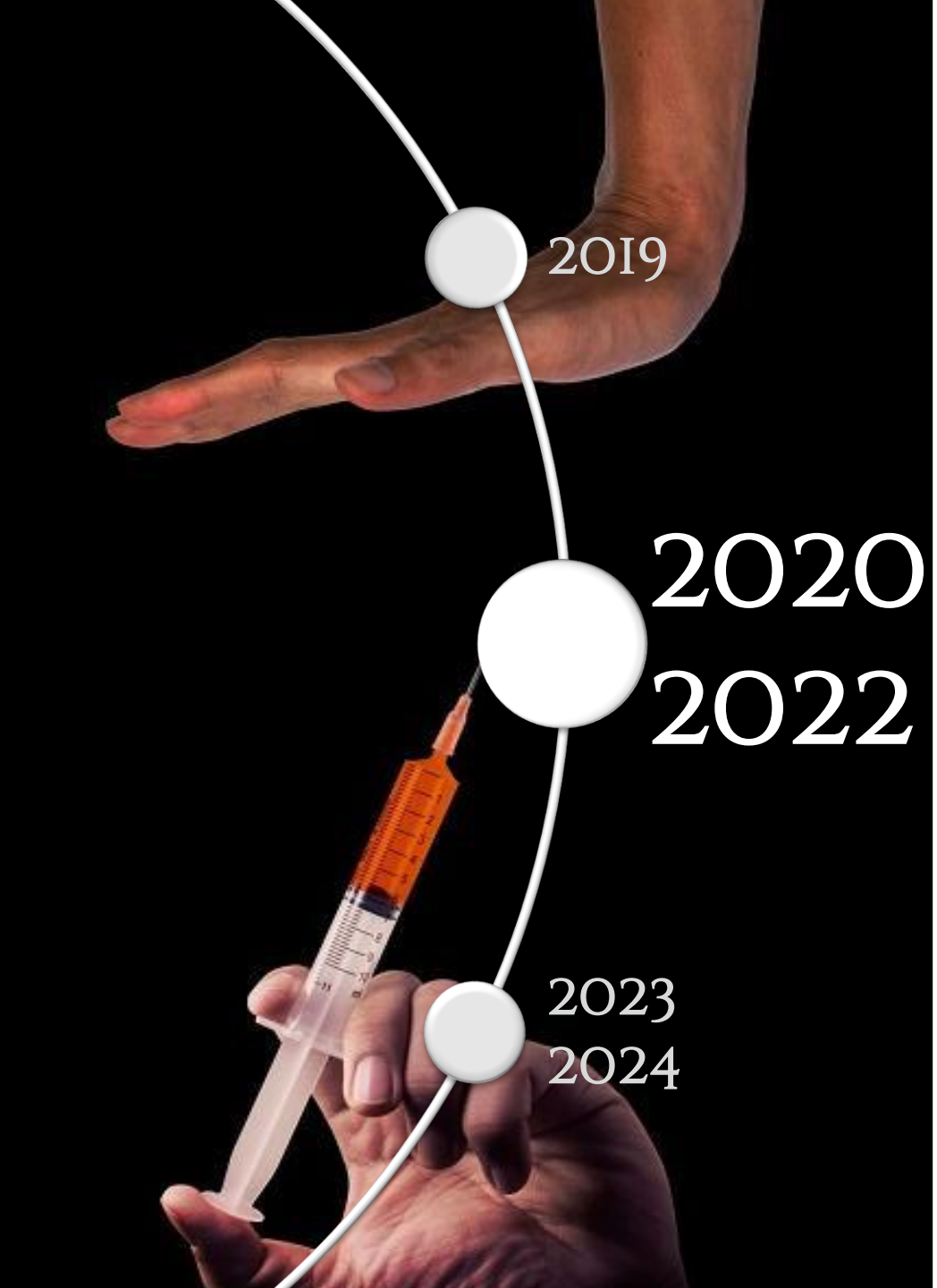
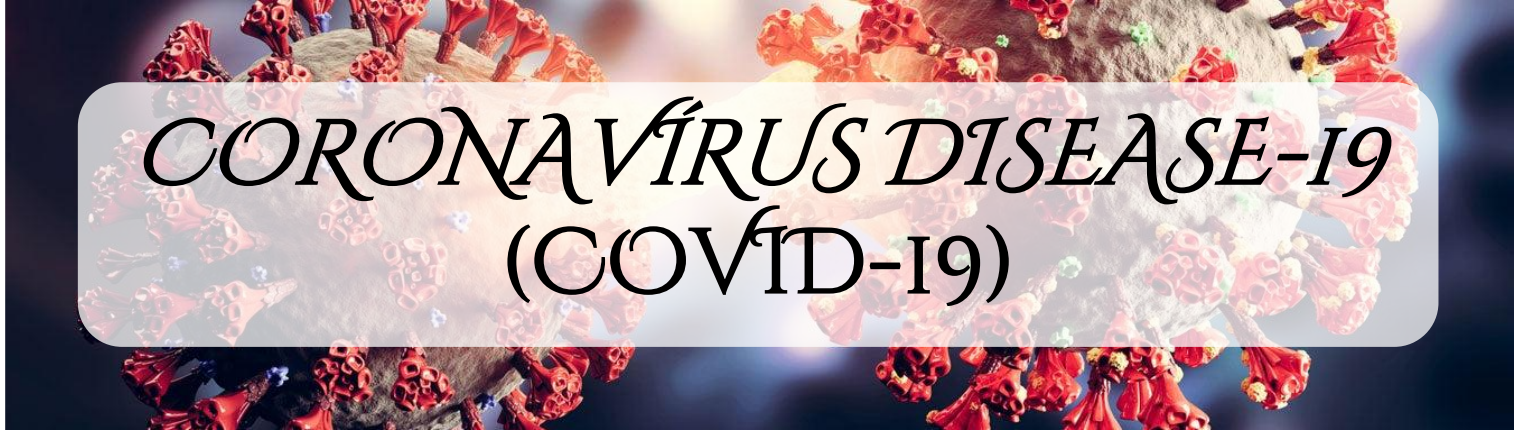


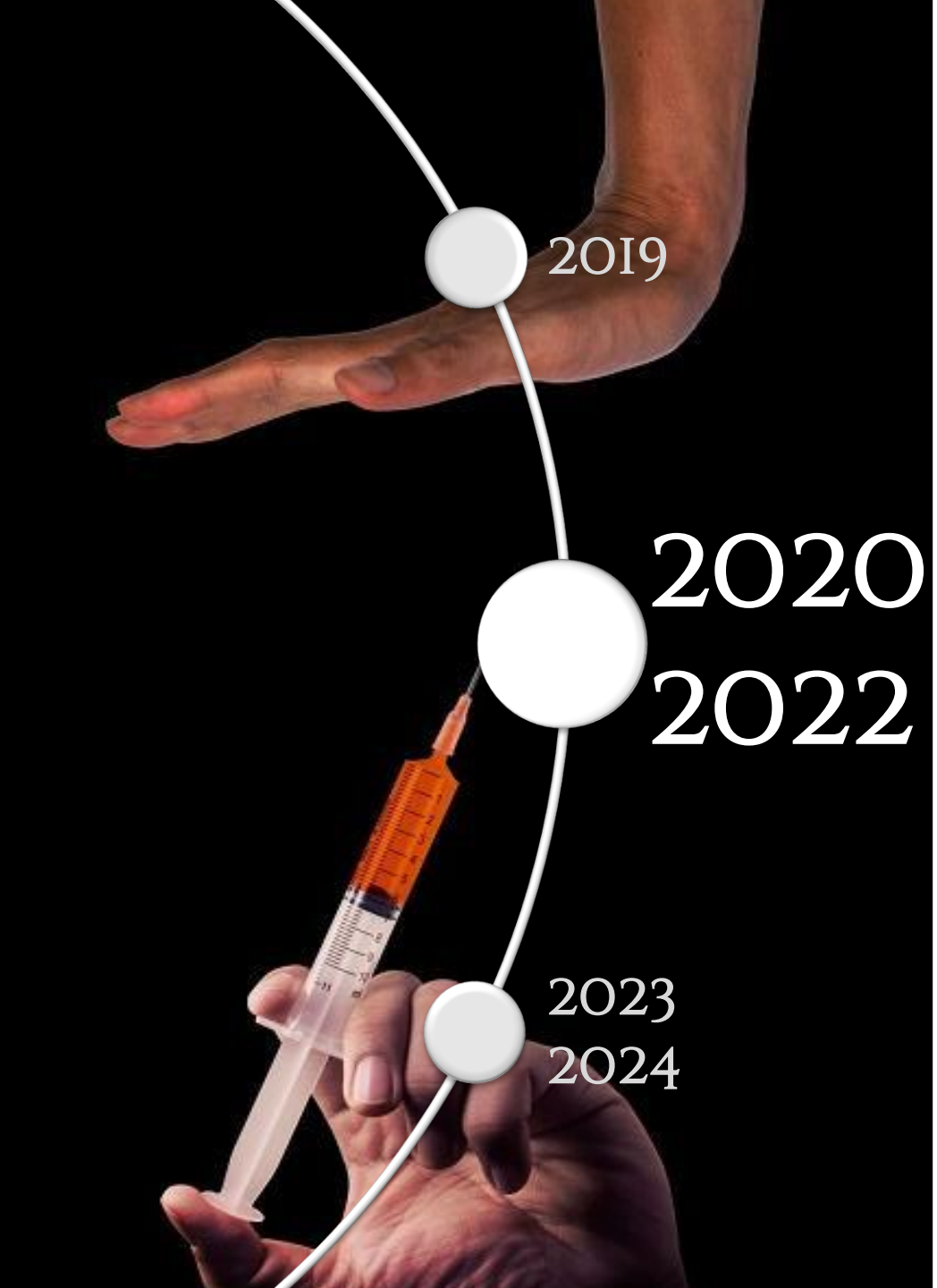


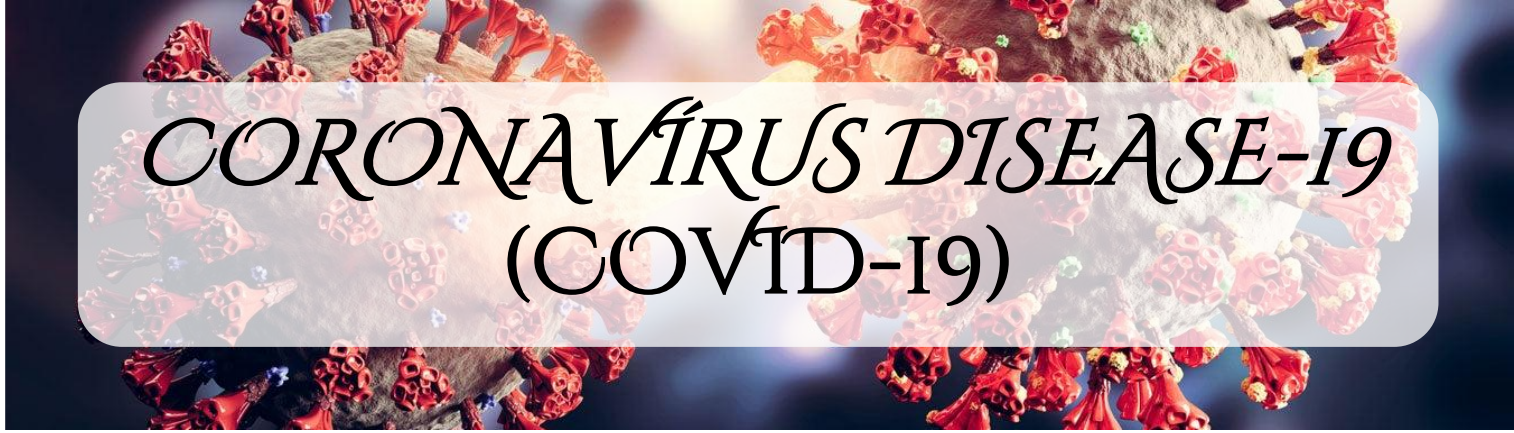
ANSIEDADE DA POPULAÇÃO

VACINAS SENDO VENDIDAS NAS RUAS POR R$ 50,00.

Em camelôs do Rio, já tem vacina falsa contra covid sendo vendida por R$ 50

Nas redes sociais, circulam fotos da vacina falsificada que estaria à venda em Madureira



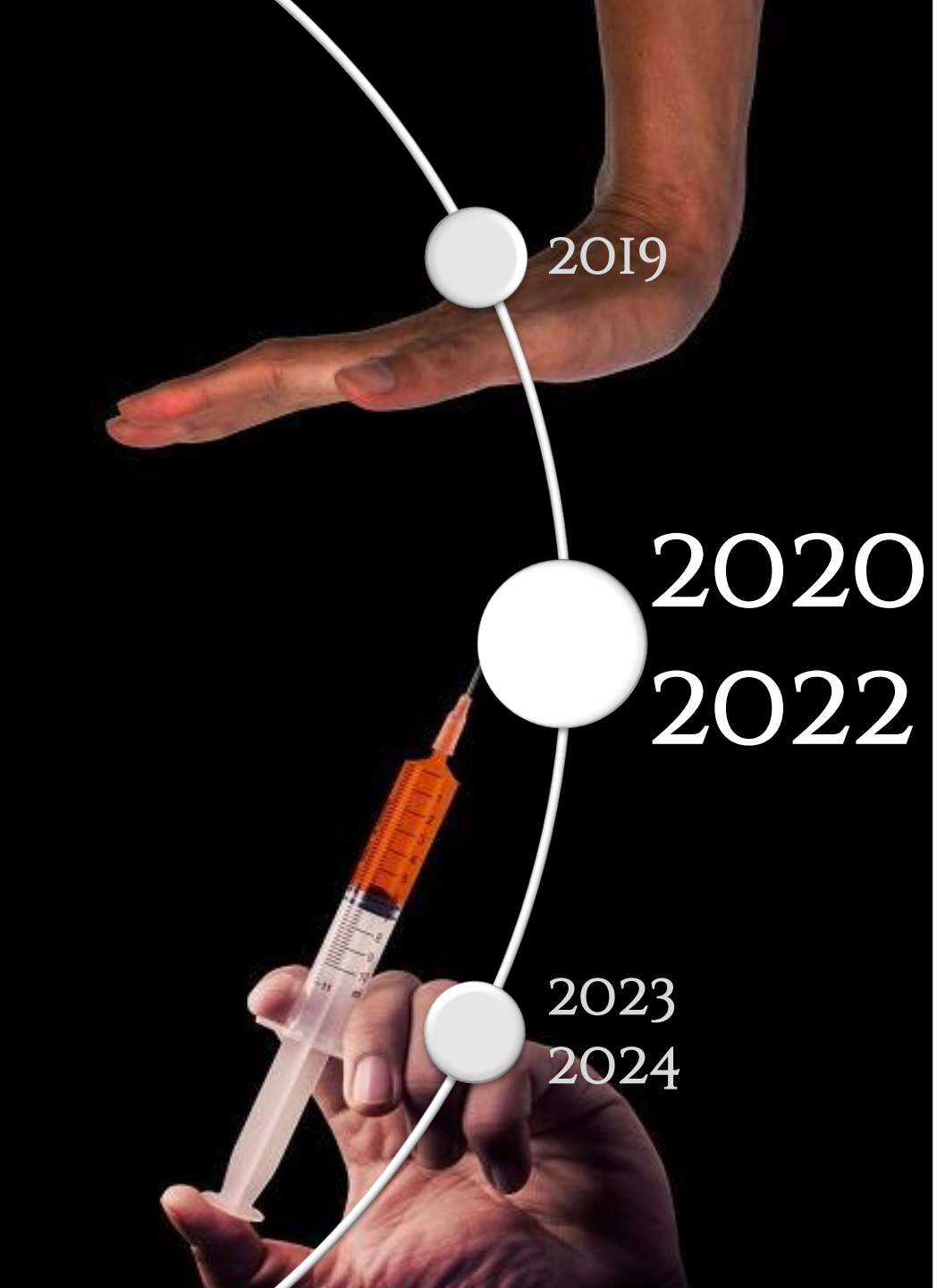


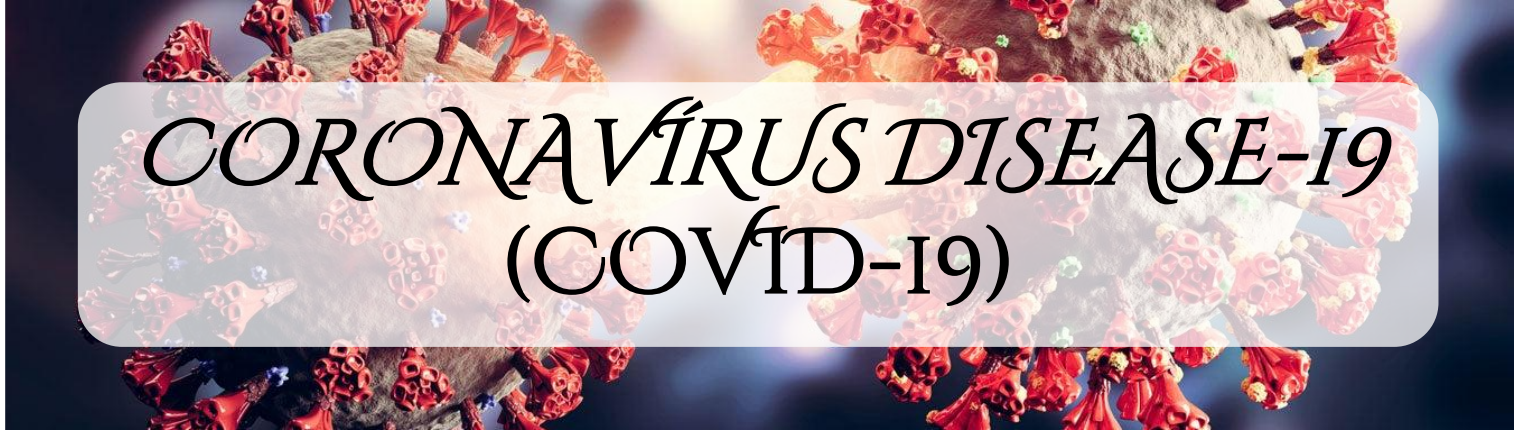
CORONAVÍRUS DISEASE-19 (COVID-19)

DESINFORMAÇÃO

TRATAMENTO PRECOCE INEFICAZ

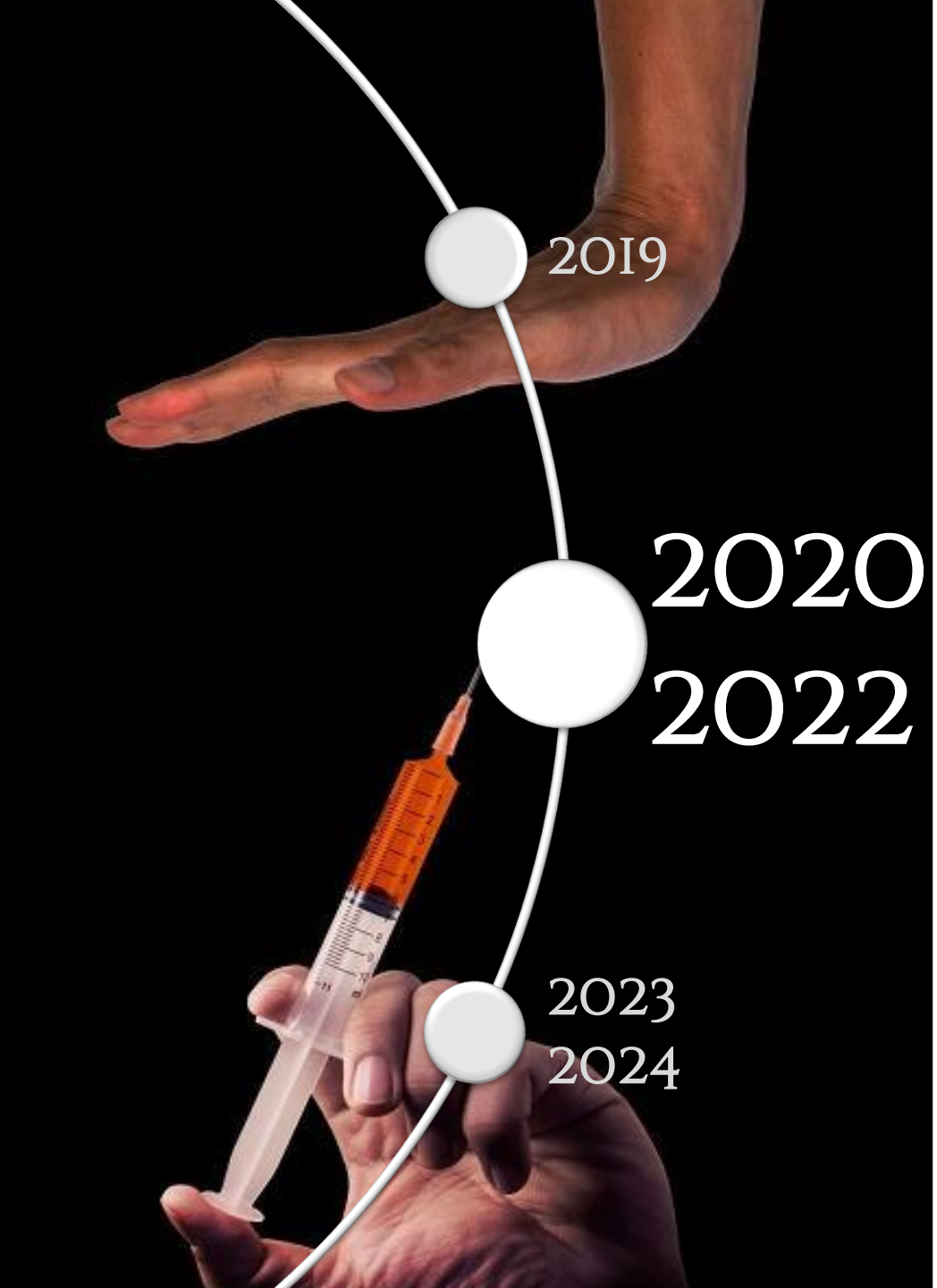
FALSA SENSAÇÃO DE O INDIVÍDUO ESTAR MEDICADO E ESTAR BEM E QUANDO CHEGAVAM AO HOSPITAL, JÁ ESTÃO NUM ESTADO CLÍNICO AVANÇADO.

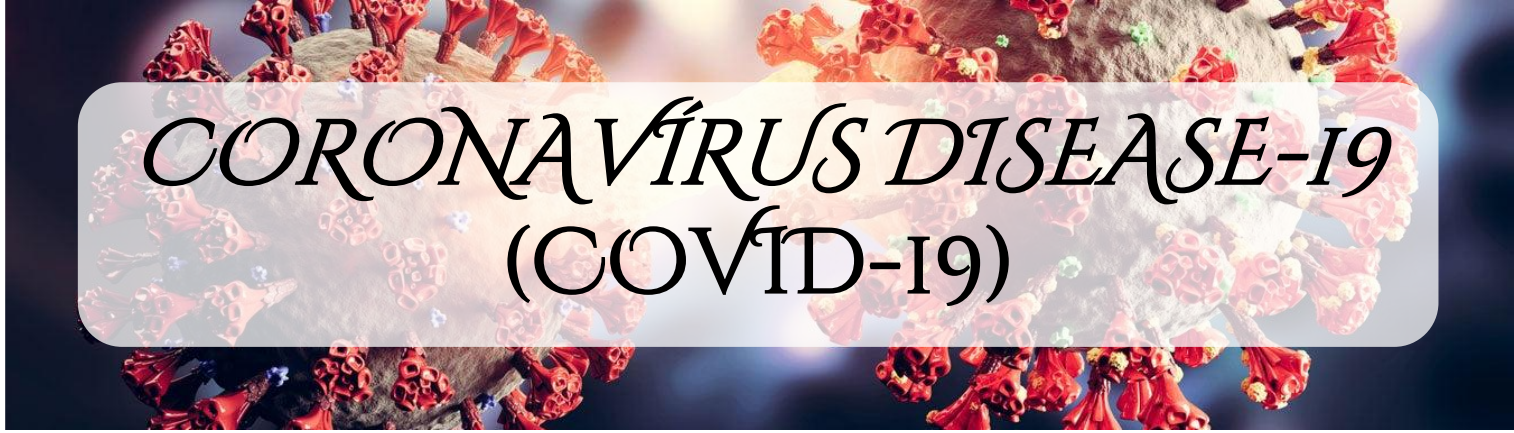




DESINFORMAÇÃO

TRATAMENTO PRECOCE INEFICAZ





CORONAVÍRUS DISEASE-19 (COVID-19)

DESINFORMAÇÃO

TRATAMENTO PRECOCE INEFICAZ

fda.gov/consumers/consumer-updates/why-you-should-not-use-ivermectin-treat-or-prevent-covid-19

FDA U.S. FOOD & DRUG ADMINISTRATION

Why You Should Not Use Ivermectin to Treat or Prevent COVID-19

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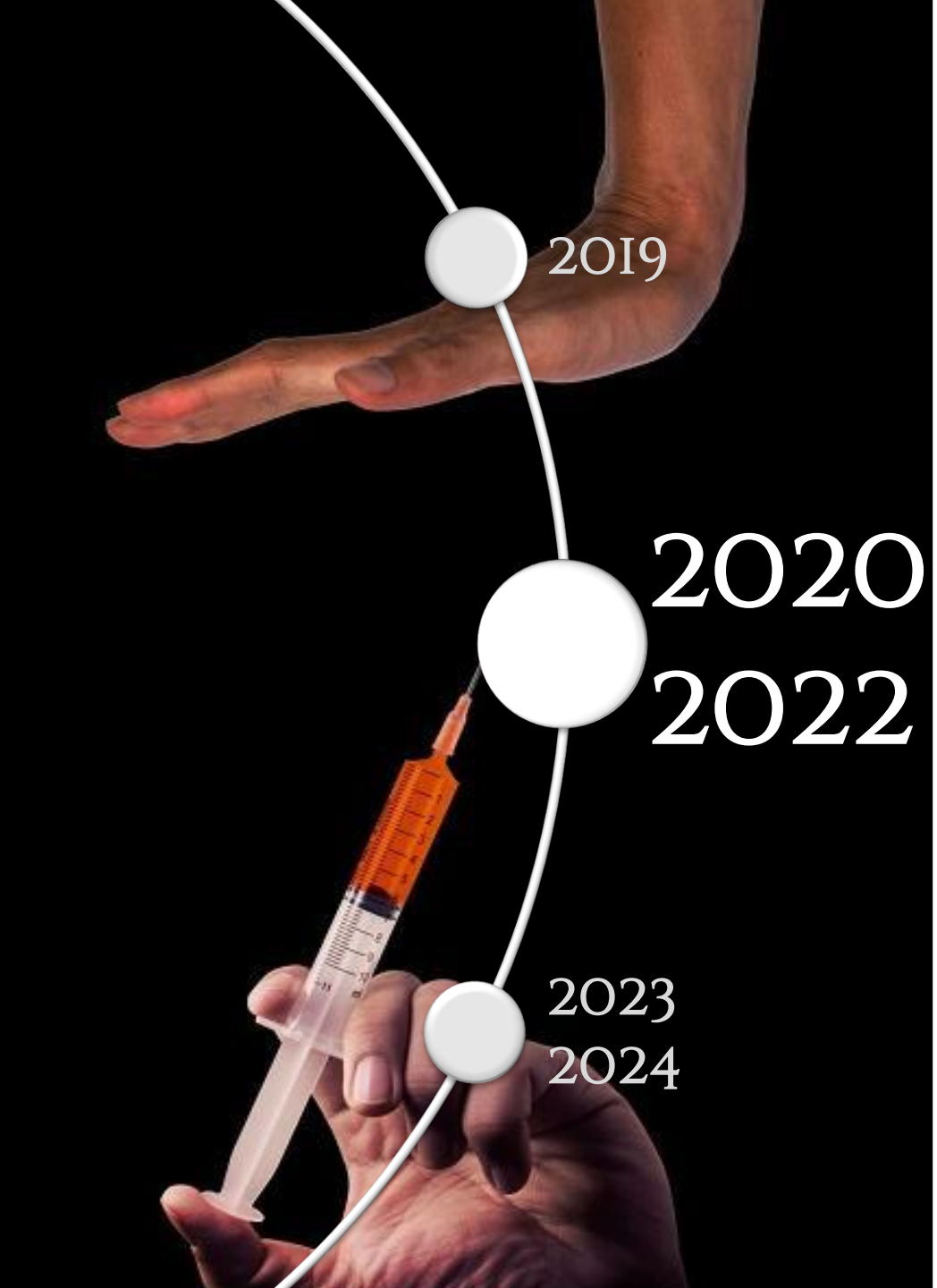
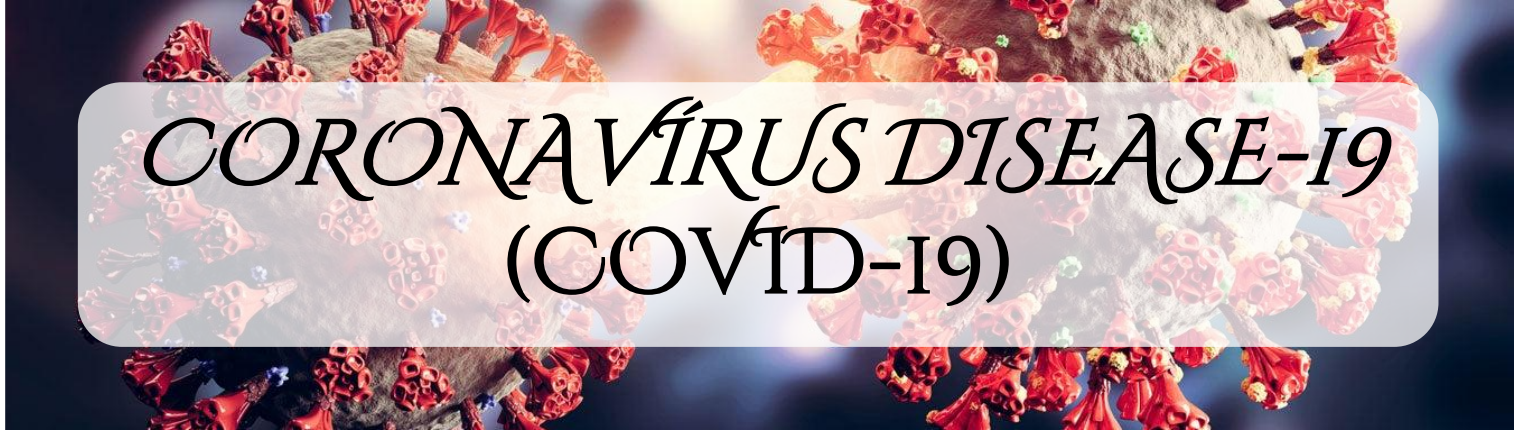




DESINFORMAÇÃO

TRATAMENTO PRECOCE INEFICAZ

MERCK

Search everything

Media > Company statements > Company statement

Merck Statement on Ivermectin use During the COVID-19 Pandemic

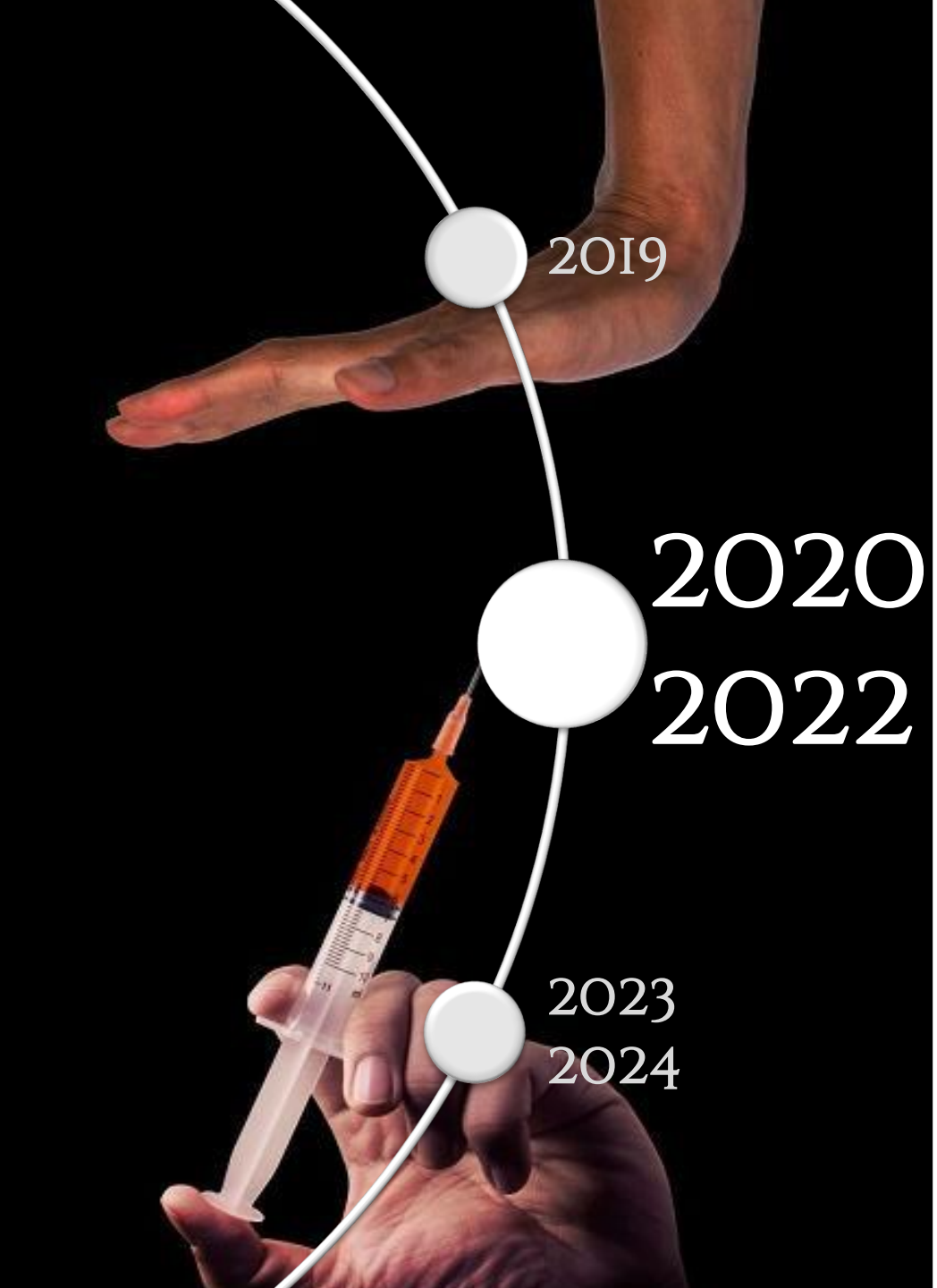
Print

February 4, 2021 11:45 am ET

KENILWORTH, N.J., Feb. 4, 2021 – Merck (NYSE: MRK), known as MSD outside the United States and Canada, today affirmed its position regarding use of ivermectin during the COVID-19 pandemic. Company scientists continue to carefully examine the findings of all available and emerging studies of ivermectin for the treatment of COVID-19 for evidence of efficacy and safety. It is important to note that, to-date, our analysis has identified:

- No scientific basis for a potential therapeutic effect against COVID-19 from pre-clinical studies;
- No meaningful evidence for clinical activity or clinical efficacy in patients with COVID-19 disease, and;
- A concerning lack of safety data in the majority of studies.

FONTE: HTTPS://WWW.MERCK.COM/NEWS/MERCK-STATEMENT-ON-IVERMECTIN-USE-DURING-THE-COVID-19-PANDEMIC/



DESINFORMAÇÃO

TRATAMENTO PRECOCE INEFICAZ

MERCK

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Merck Statement on Ivermectin use During the COVID-19 Pandemic

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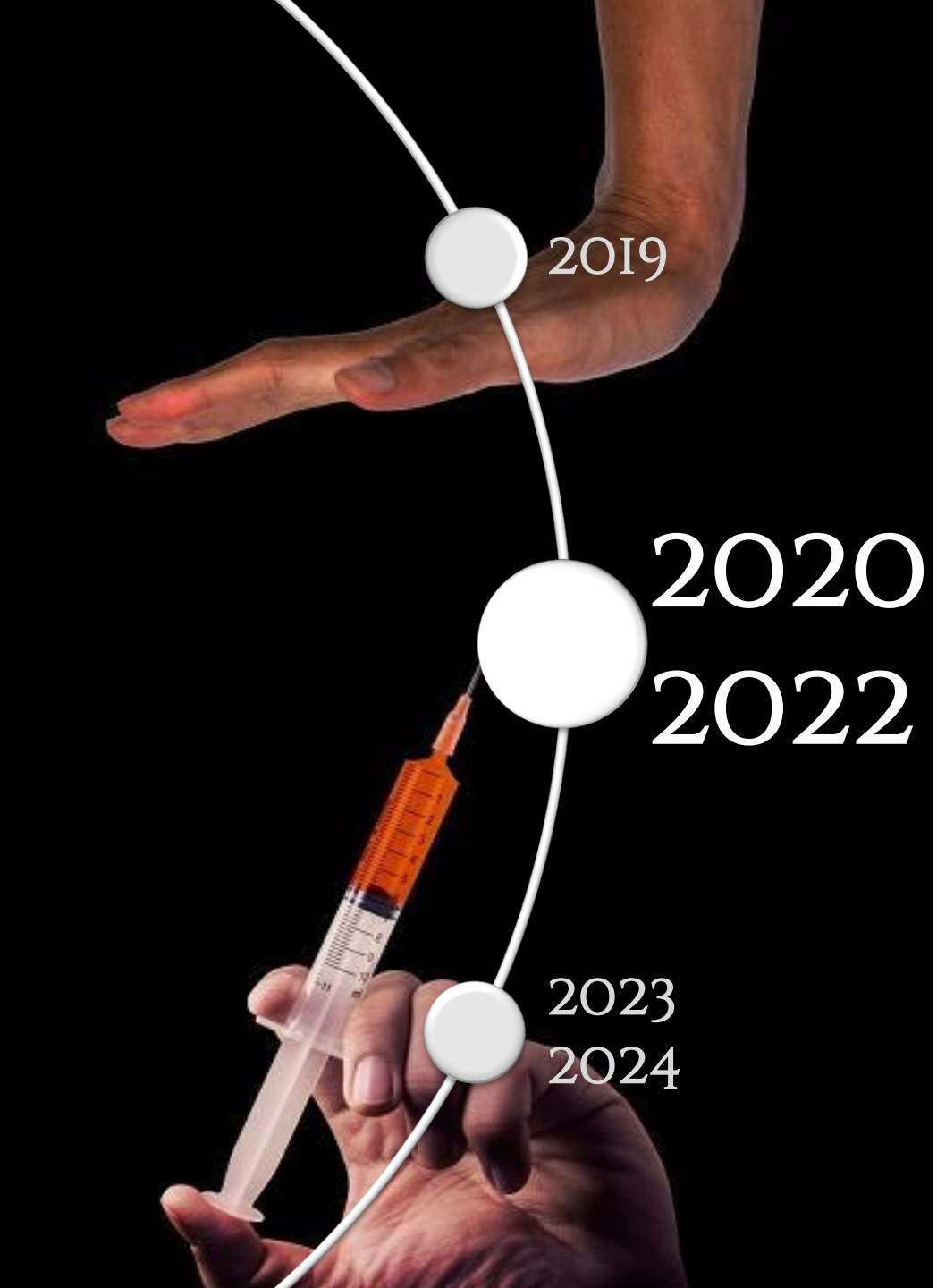
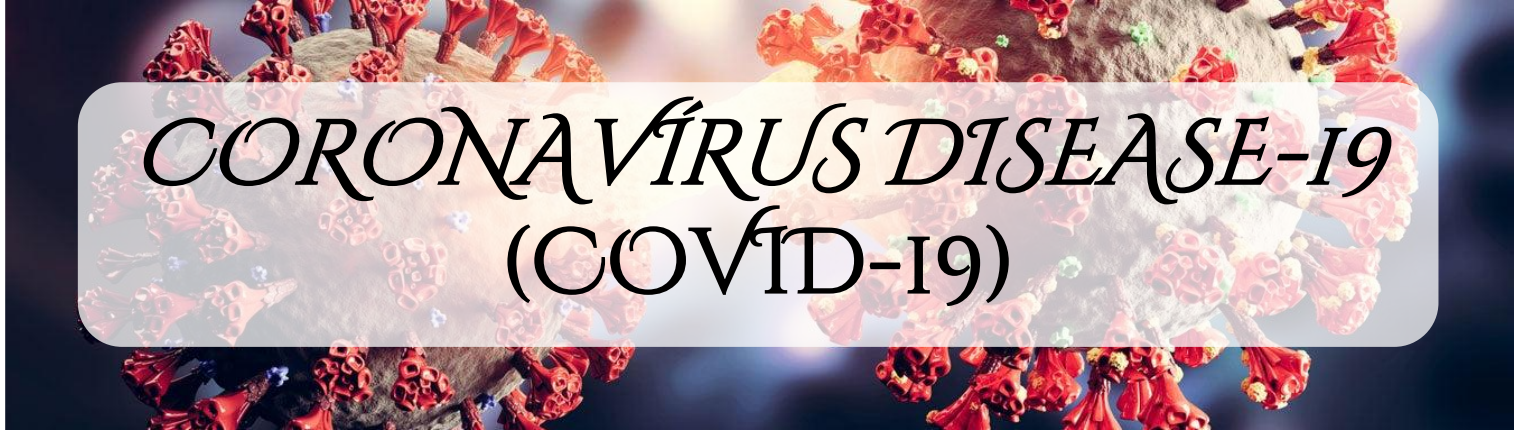




DESINFORMAÇÃO

TRATAMENTO PRECOCE INEFICAZ

SAÚDE • CORONAVÍRUS • SOCIEDADE

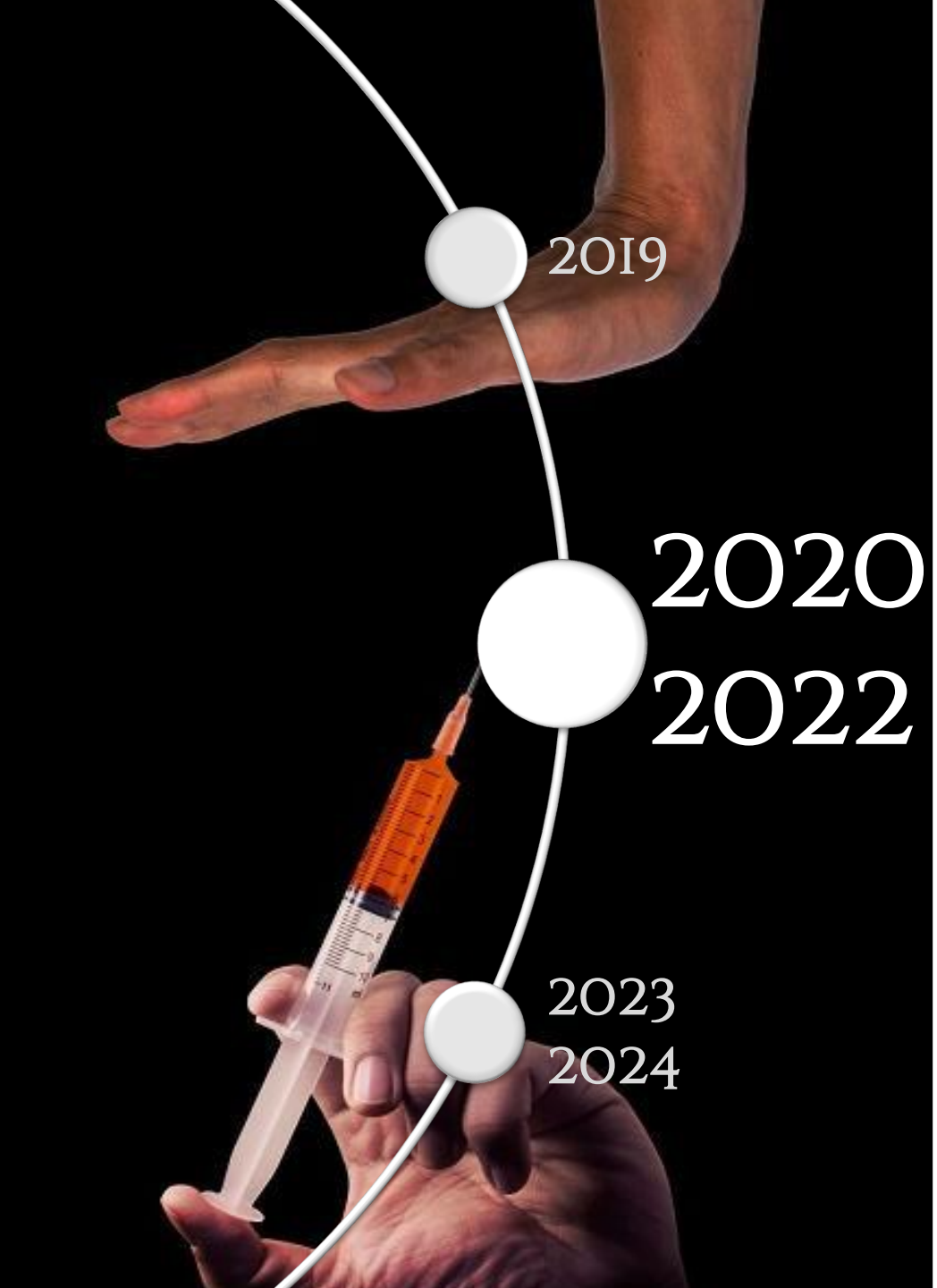
Governo planeja gastar R$ 250 milhões para distribuir remédios sem eficácia comprovada contra Covid-19

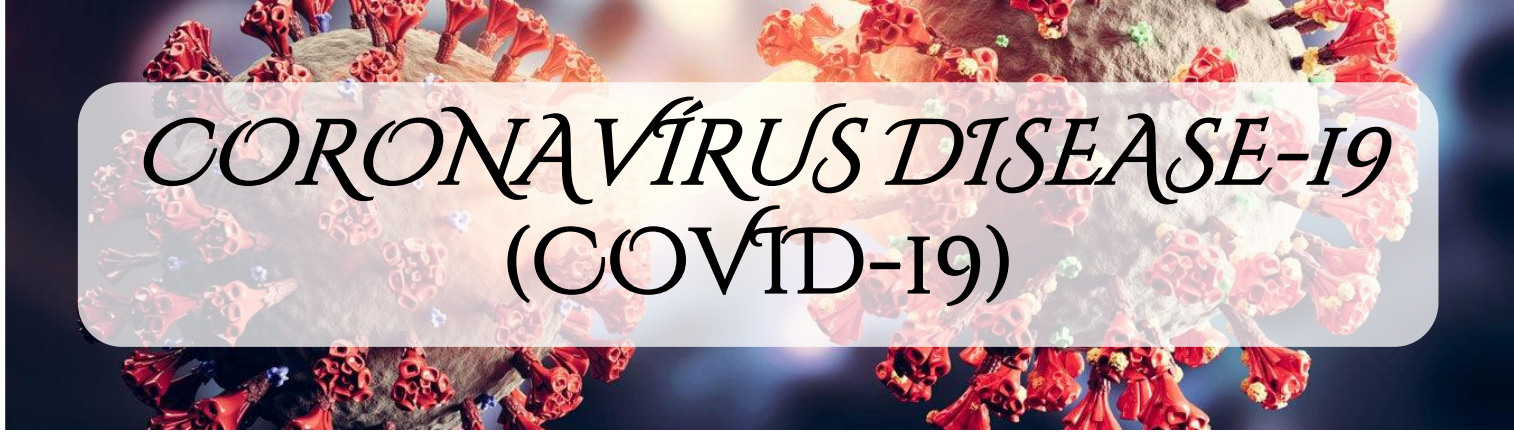
Farmacêutico segura pílulas de hidroxicloroquina, outro medicamento que já foi apontado, e depois descartado, para tratamento da Covid-19 Foto: LOUAI BESHARA / AFP

Departamento Jurídico do Ministério da Saúde deu aval para oferecer hidroxicloroquina e azitromicina no chamado 'kit Covid' pelo programa Farmácia Popular, diz jornal

O Globo
11/12/2020 - 11:46 / Atualizado em 11/12/2020 - 17:18

A partir das estimativas de laboratórios farmacêuticos , o valor previsto pela Saúde para o financiamento da distribuição de dois remédios sem eficácia contra a Covid-19 poderia, por exemplo, financiar 15,6 milhões de doses da vacina contra o Sars-CoV-2 desenvolvida pela AstraZeneca em parceria com a Universidade de Oxford (Reino Unido) ou 4,4 milhões de doses da CoronaVac, do laboratório chinês Sinovac Biotech, ambas testadas no Brasil.





ATRASOS

TESTES "ENCALHADOS" E VENCIDOS EM GUARULHOS

PRESTES A VENCER

Ministério da Saúde avalia ampliação da validade de testes de covid-19 armazenados e não distribuídos para rede pública

Pasta tem 7,1 milhões de exames RT-PCR encalhados em armazém

25/11/2020 - 13h13min
Atualizada em 25/11/2020 - 14h20min

COMPARTILHE:

2019

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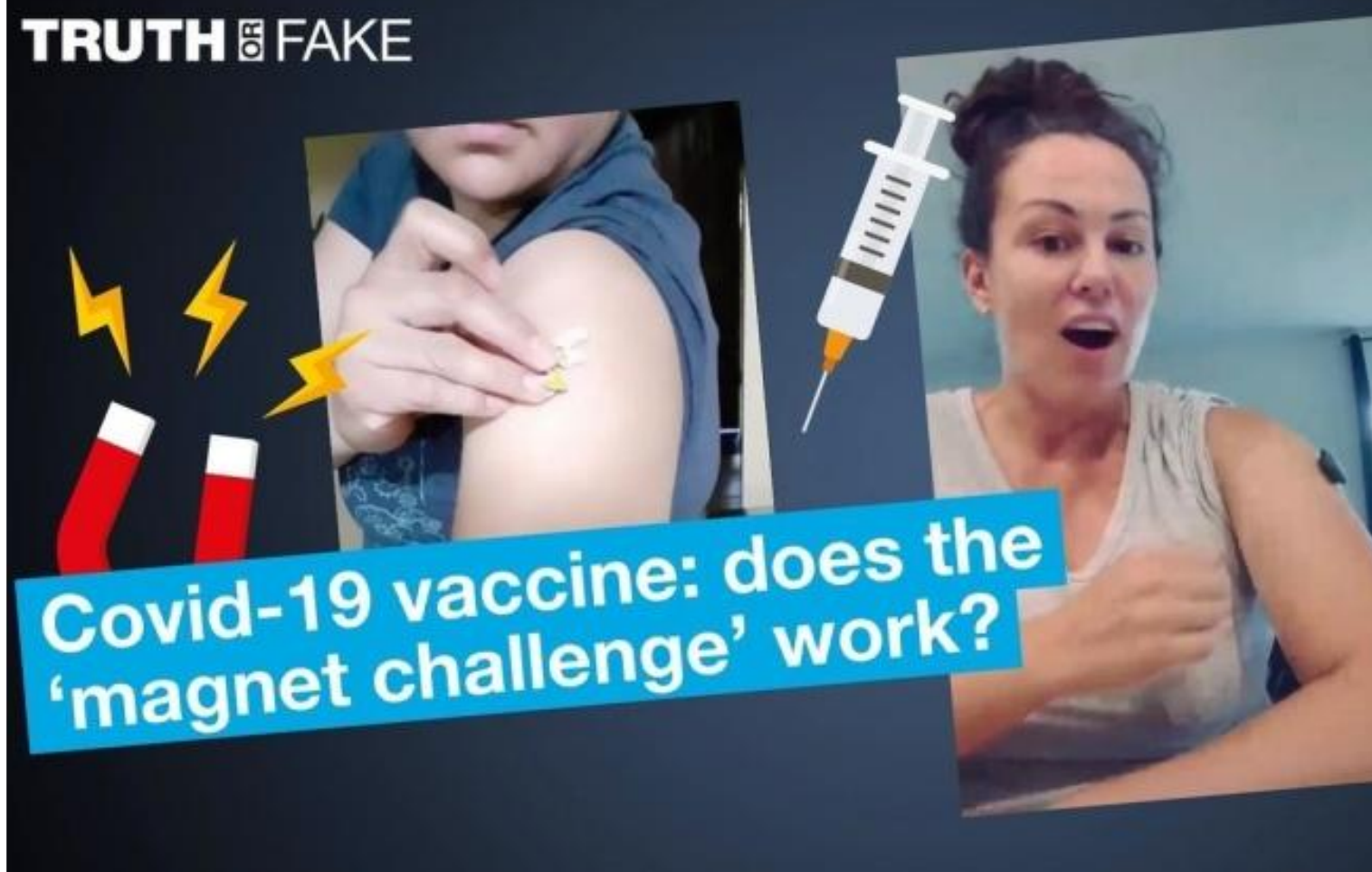
CORONAVÍRUS DISEASE-19
(COVID-19)



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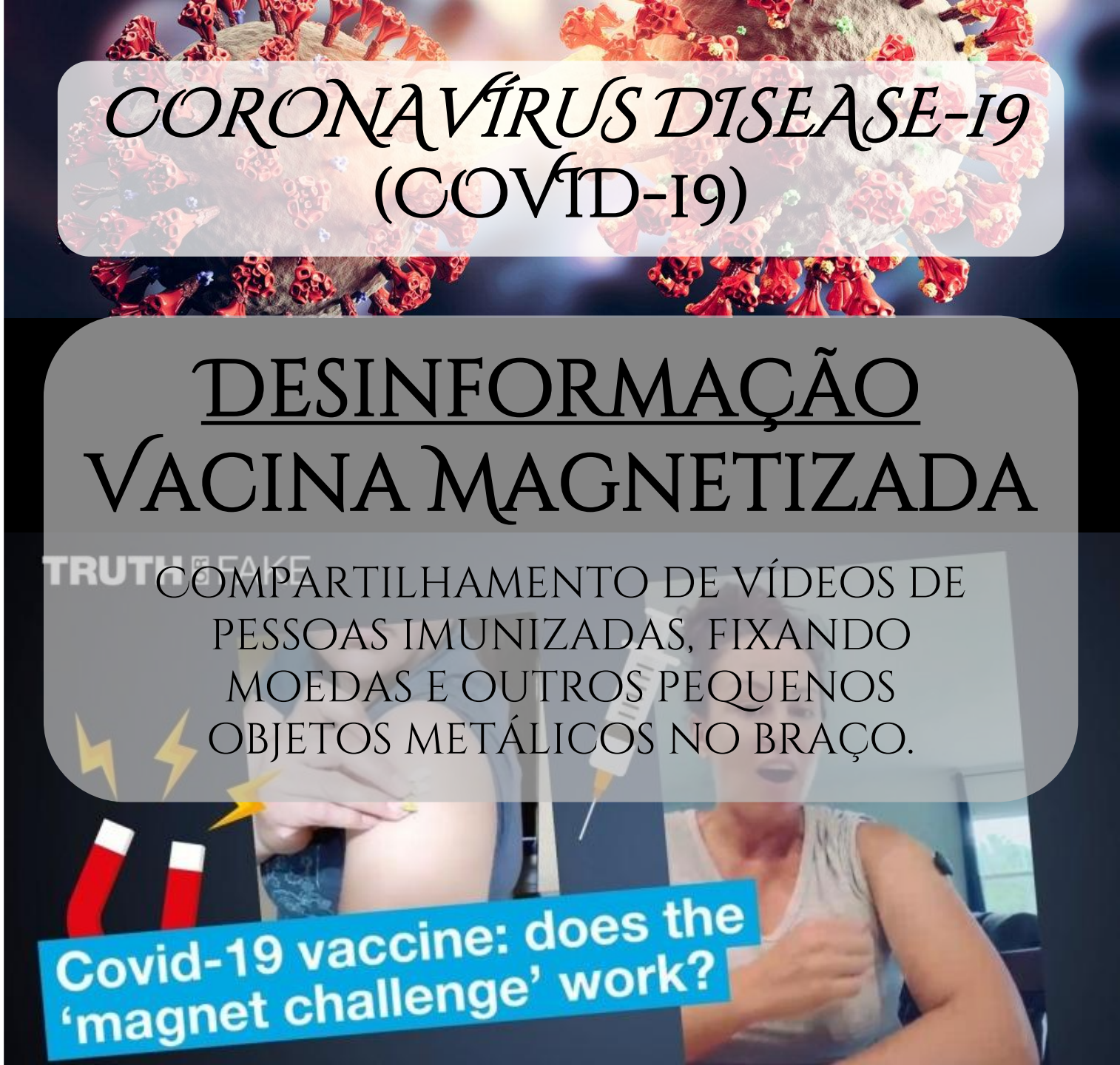
CORONAVÍRUS DISEASE-19 (COVID-19)

DESINFORMAÇÃO VACINA MAGNETIZADA

COMPARTILHAMENTO DE VÍDEOS DE PESSOAS IMUNIZADAS, FIXANDO MOEDAS E OUTROS PEQUENOS OBJETOS METÁLICOS NO BRAÇO.



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CORONAVÍRUS DISEASE-19 (COVID-19)

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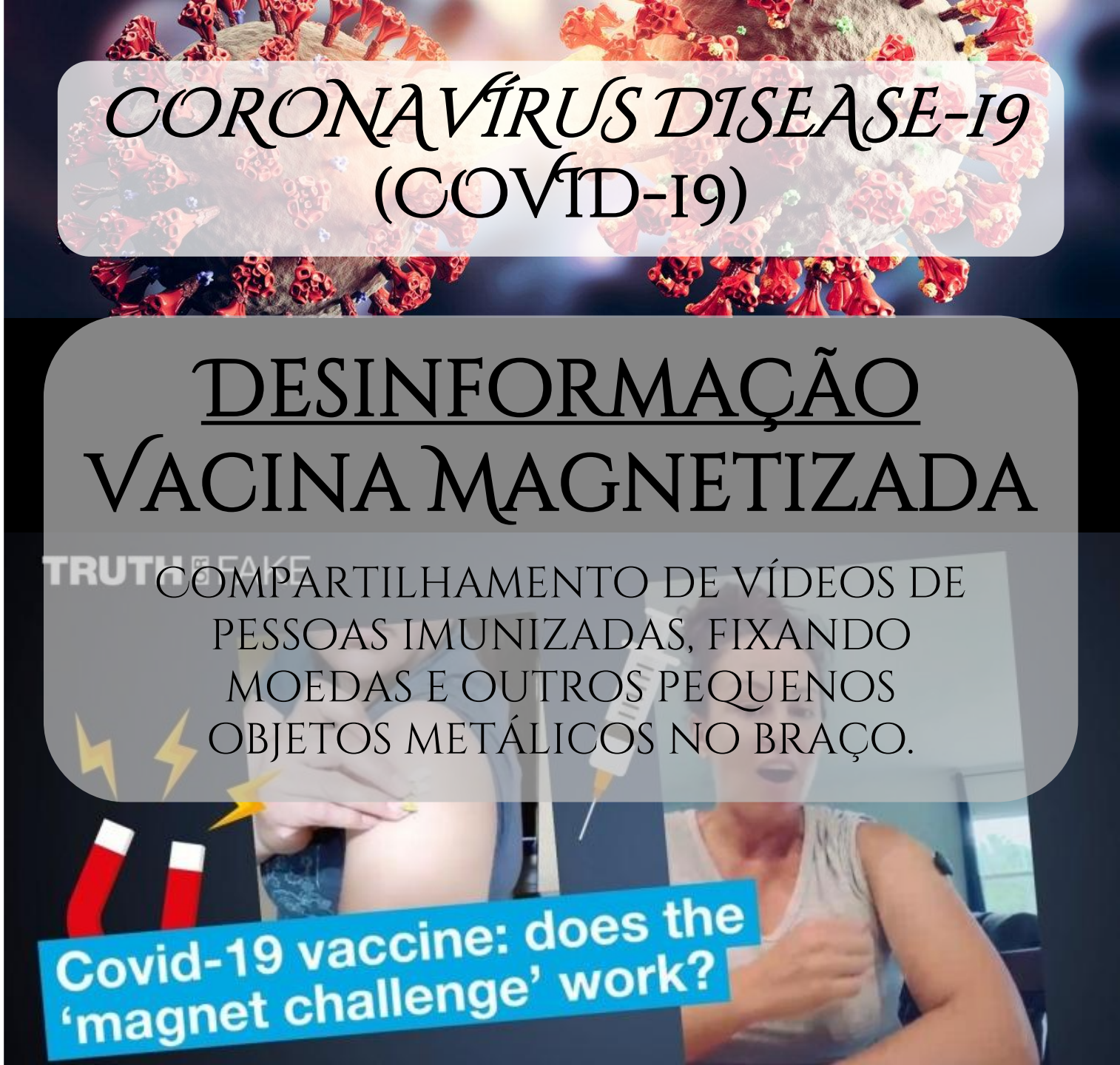

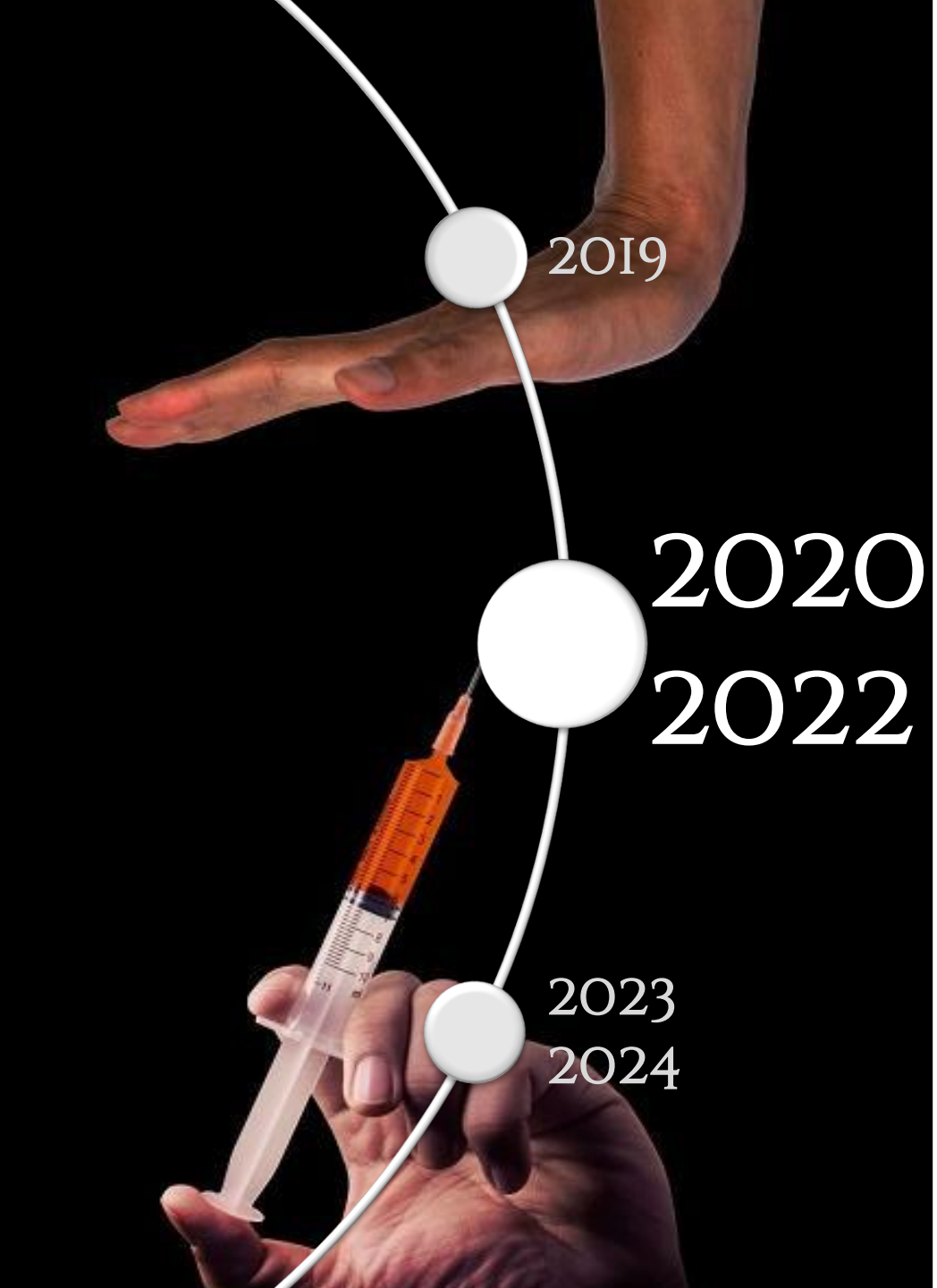
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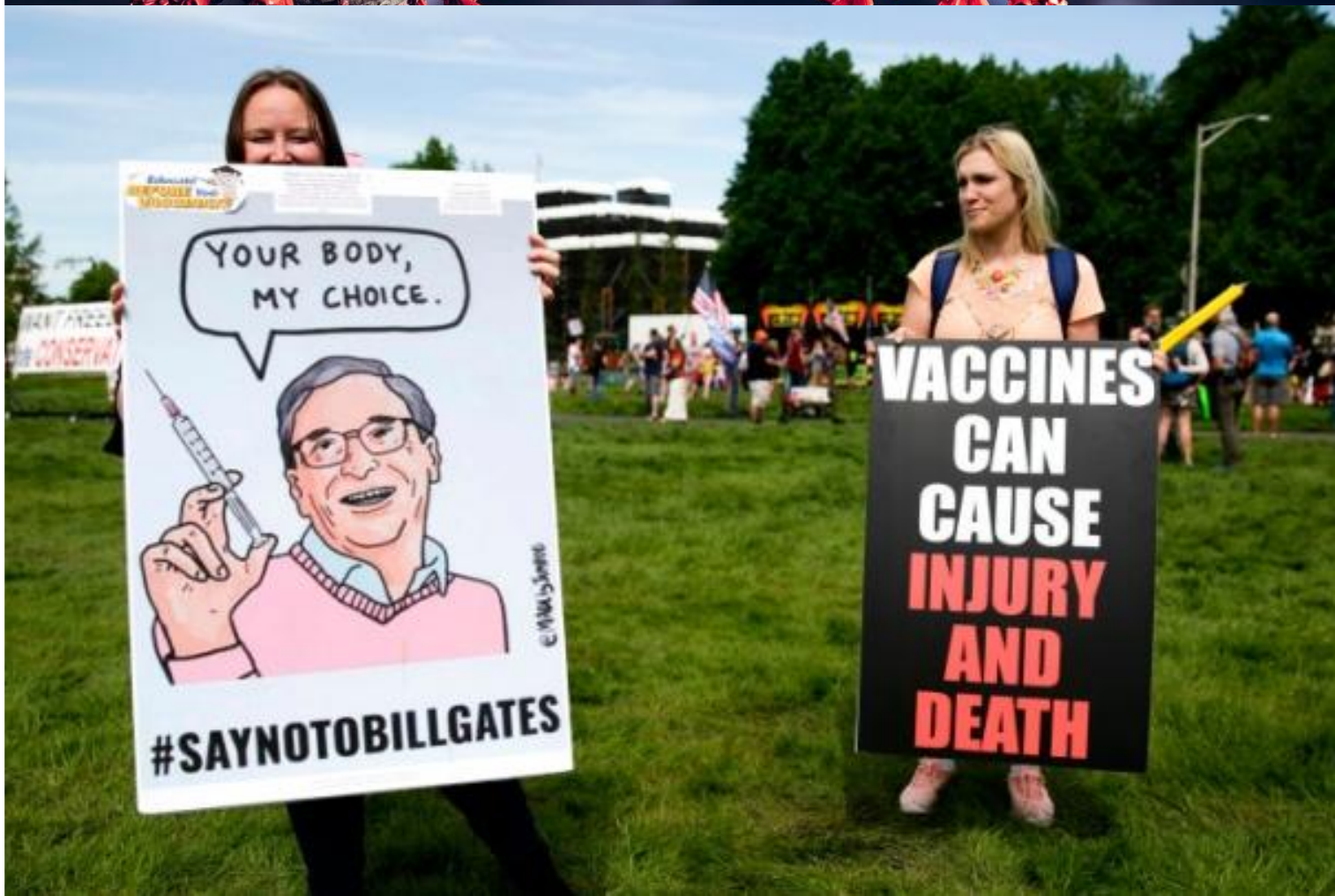
CORONAVÍRUS DISEASE-19
(COVID-19)


YOUR BODY,
MY CHOICE.
#SAYNOTOBILLGATES
VACCINES
CAN
CAUSE
INJURY
AND
DEATH

2019

2020
2022

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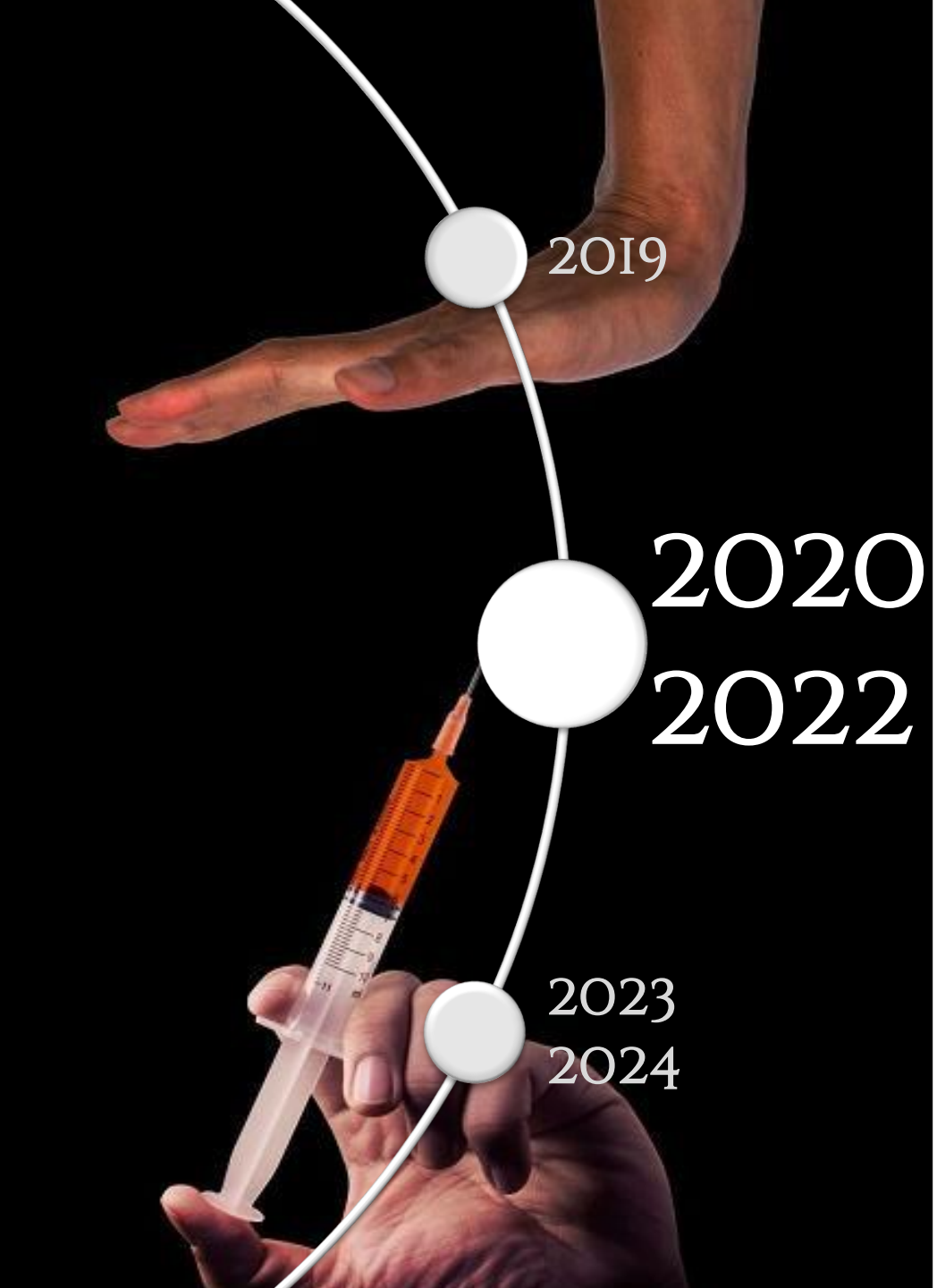
2024

CORONAVÍRUS DISEASE-19
(COVID-19)

BILL GATES

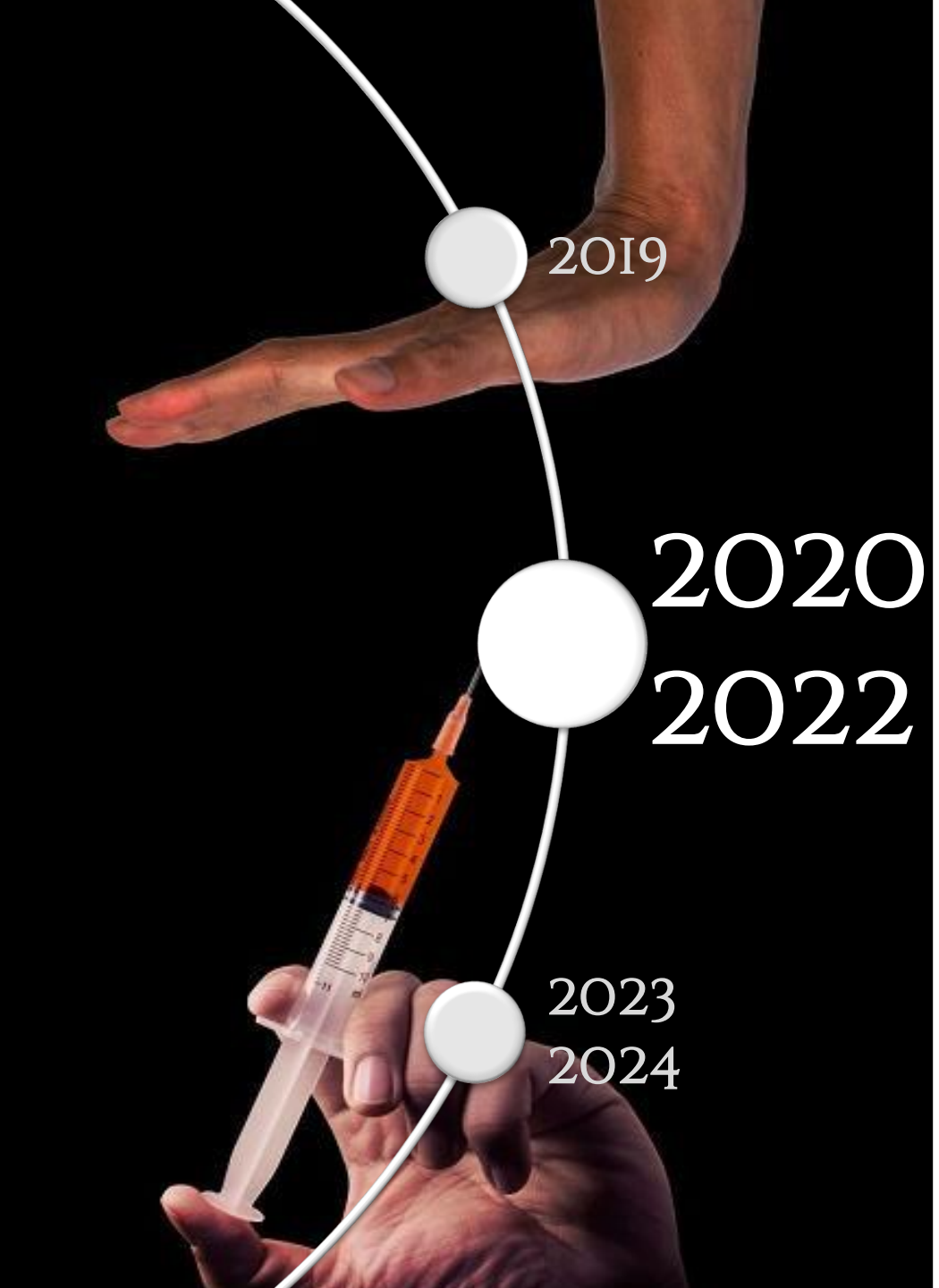
ESTARIA POR TRÁS DO USO DE MICROCHIPS DE IDENTIFICAÇÃO E NANO ROBÔS DE MONITORAMENTO E RASTREIO EM TEMPO REAL DOS CIDADÃOS, PELA CONEXÃO 5G.



BILL GATES

2014 – TED2015

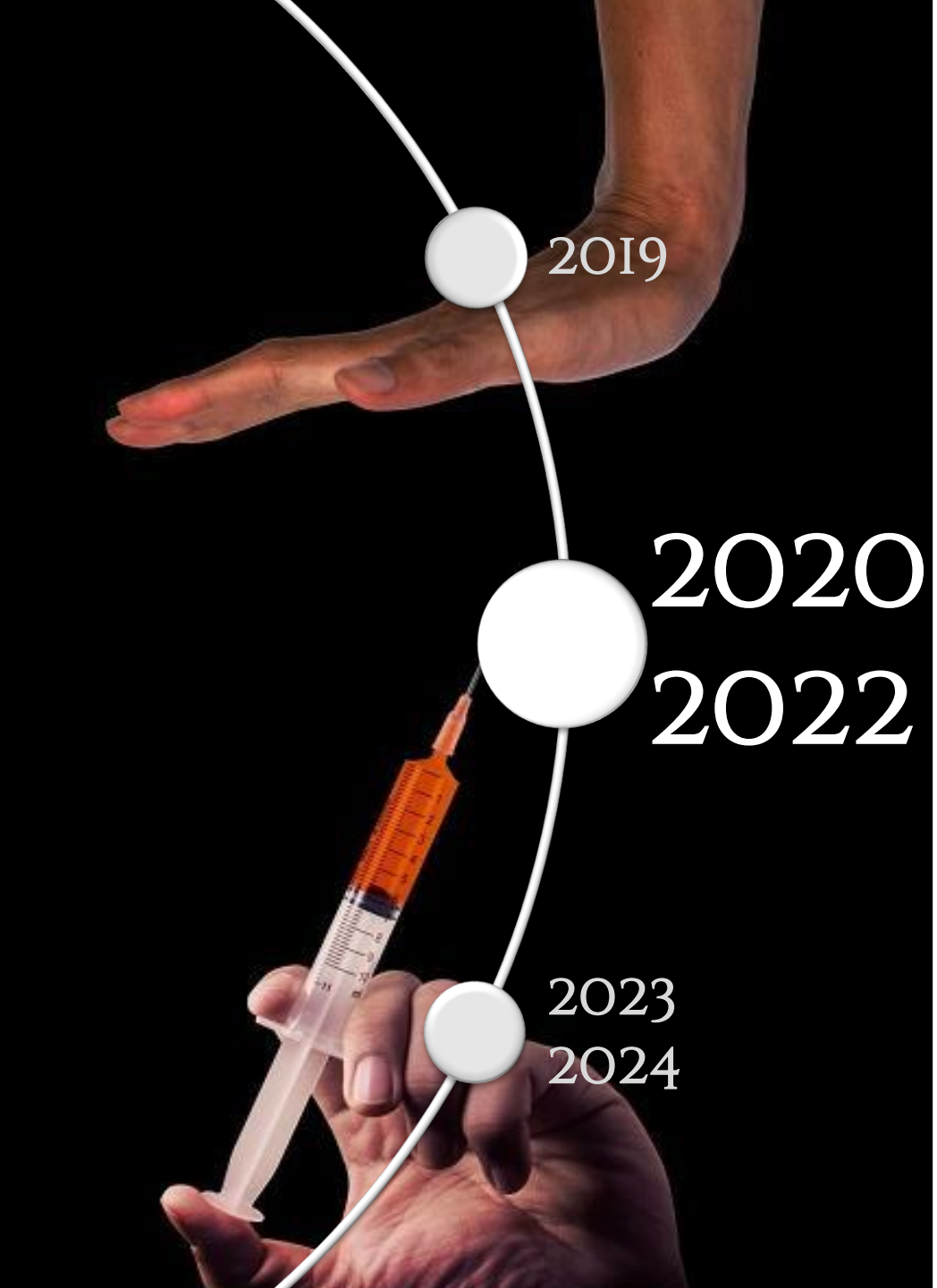
"THE NEXT OUTBREAK? WE'RE NOT READY"



BILL GATES

2014 – TED2015

"THE NEXT OUTBREAK? WE'RE NOT READY"



BILL GATES

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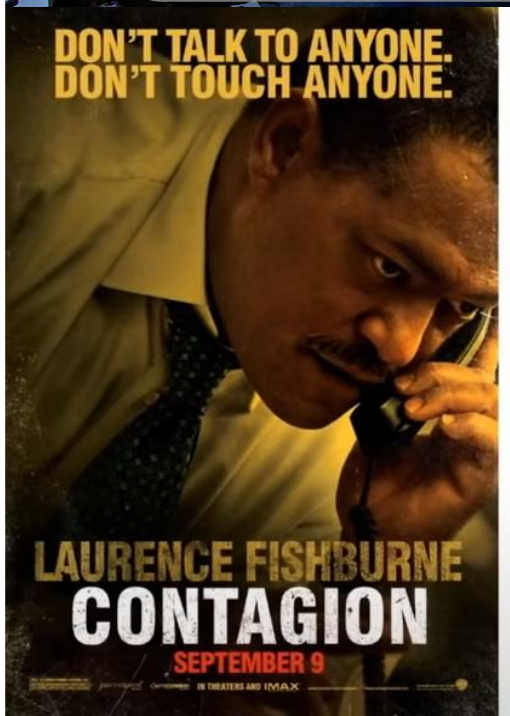
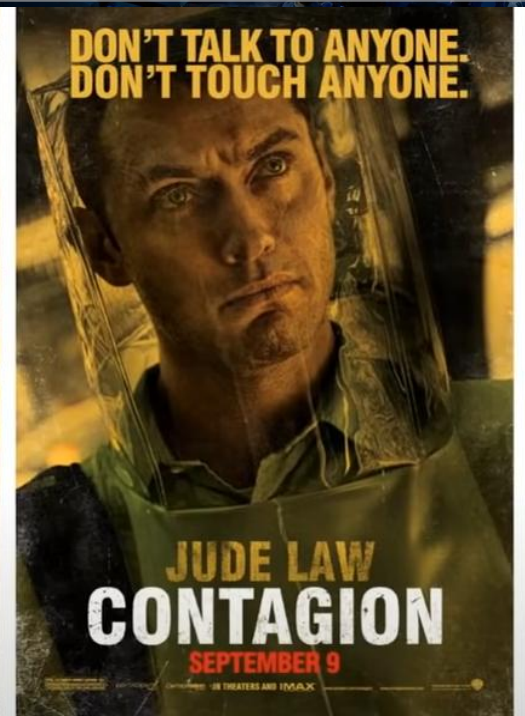




RSFLIX

AMERICAN MANHUNT

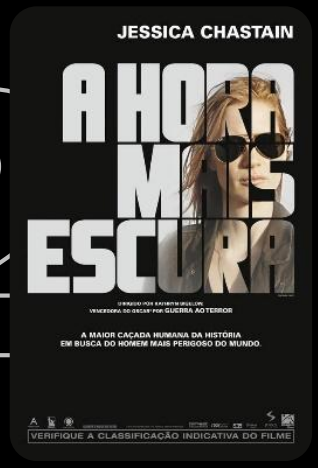
★ ★ ★ ★ ★ Documentário

Com imagens raras e entrevistas com informantes da CIA, este documentário eletrizante acompanha a caçada épica a Osama bin Laden.

Top 14 em filmes e séries sobre Pandemia

11

12



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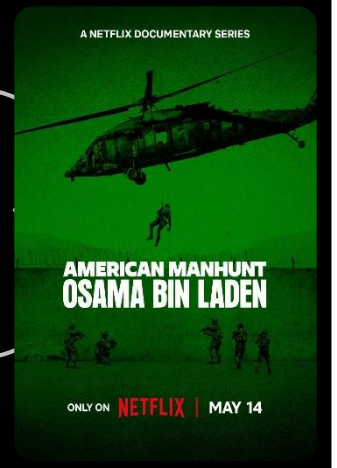
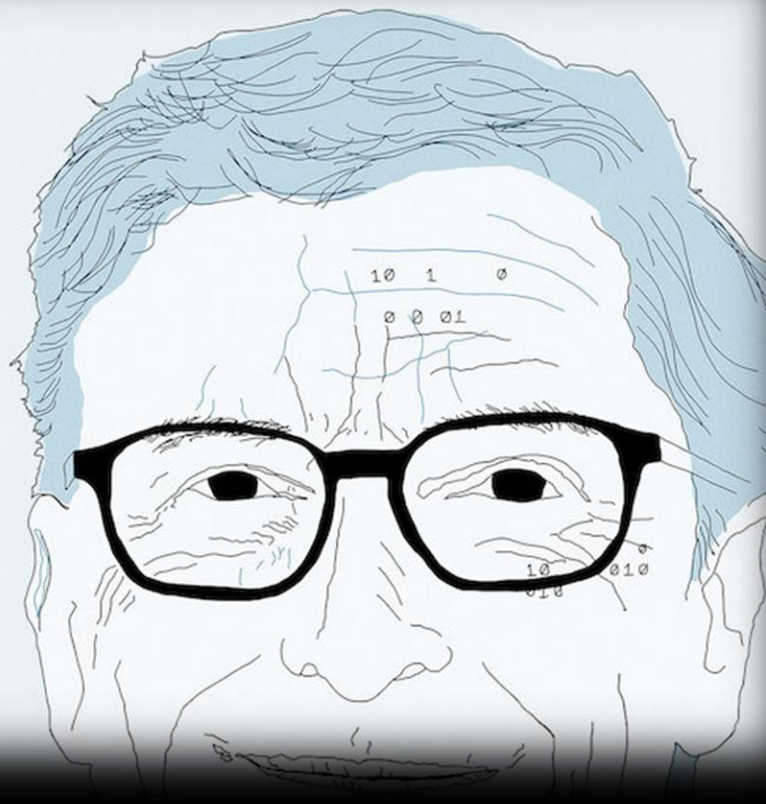


RSFLIX

O CÓDIGO BILL GATES

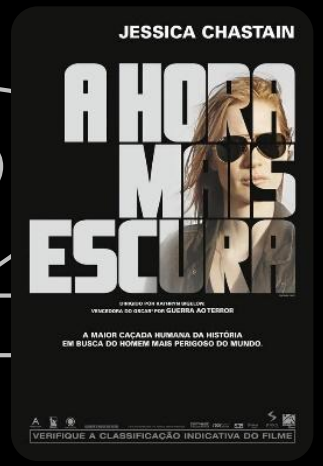
★ ★ ★ ★ ★ Documentário

Em uma jornada pela mente do bilionário Bill Gates, que retrata a sua história, a criação da Microsoft, os desafios pessoais e profissionais do casal e a filantropia da Fundação Bill e Melinda Gates.



Top 14 em filmes e séries sobre Pandemia

12



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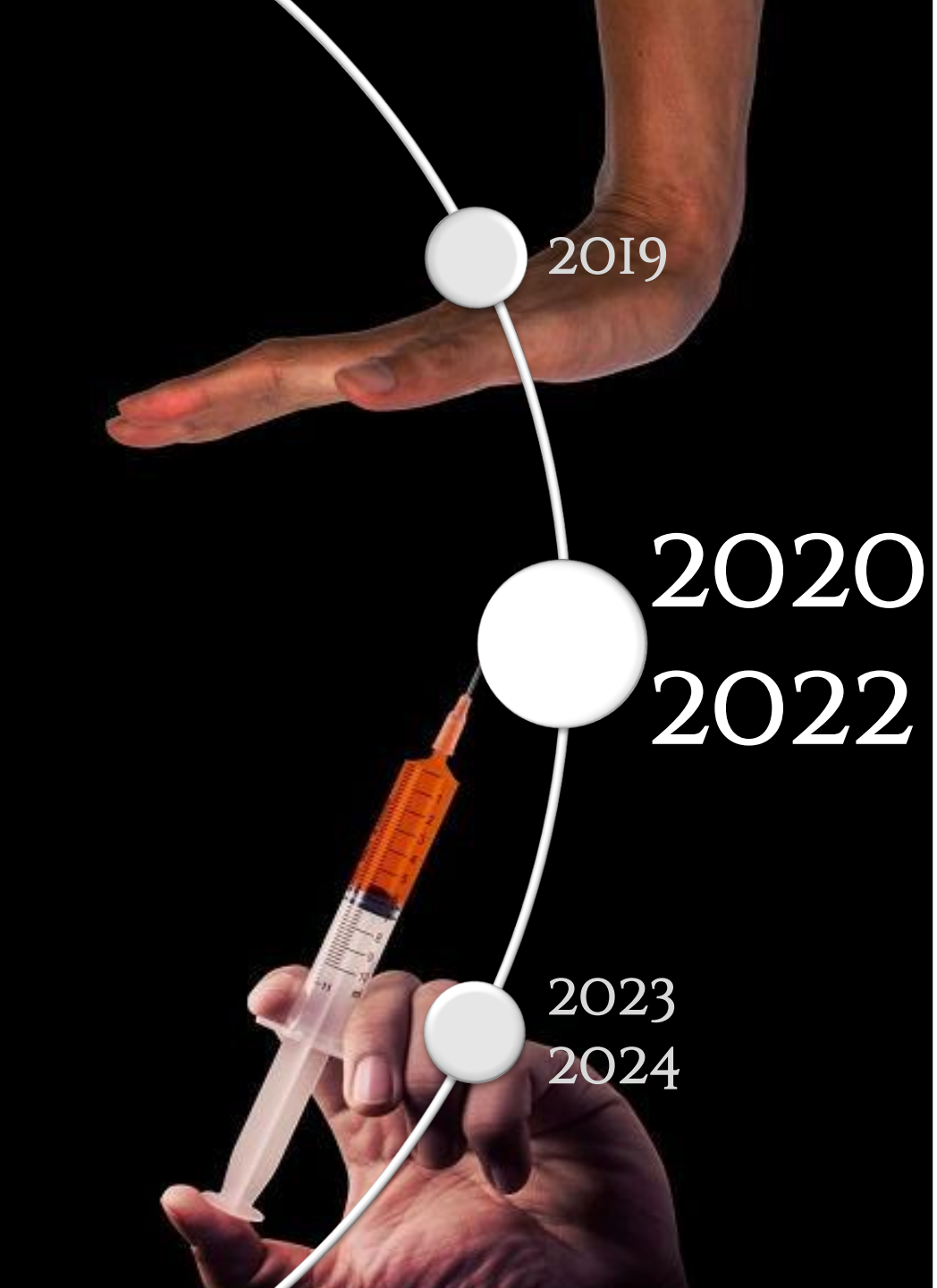

BILL GATES

2014 – TED2015

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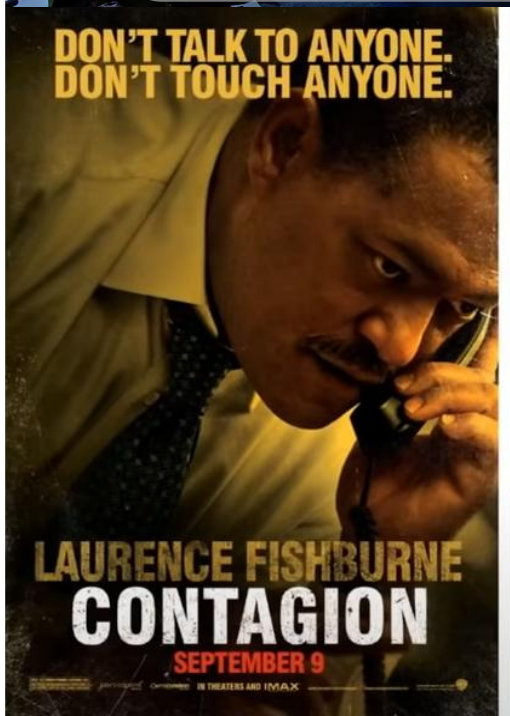

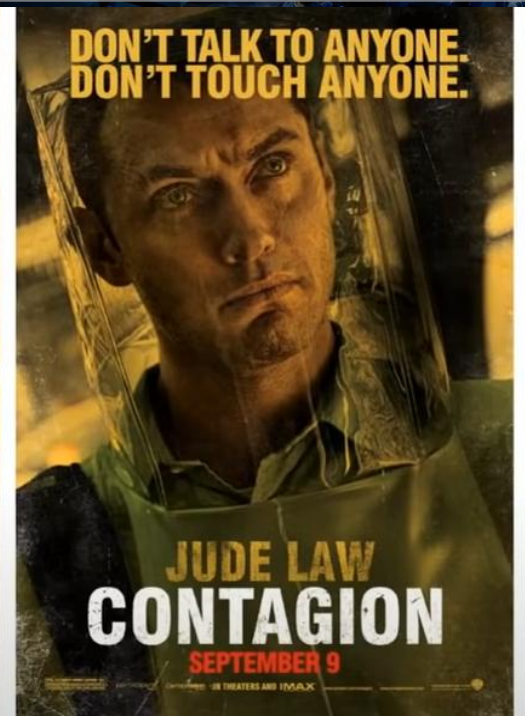

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TRATAMENTO EFICAZ!

INIBINDO A REPLICAÇÃO DO VÍRUS

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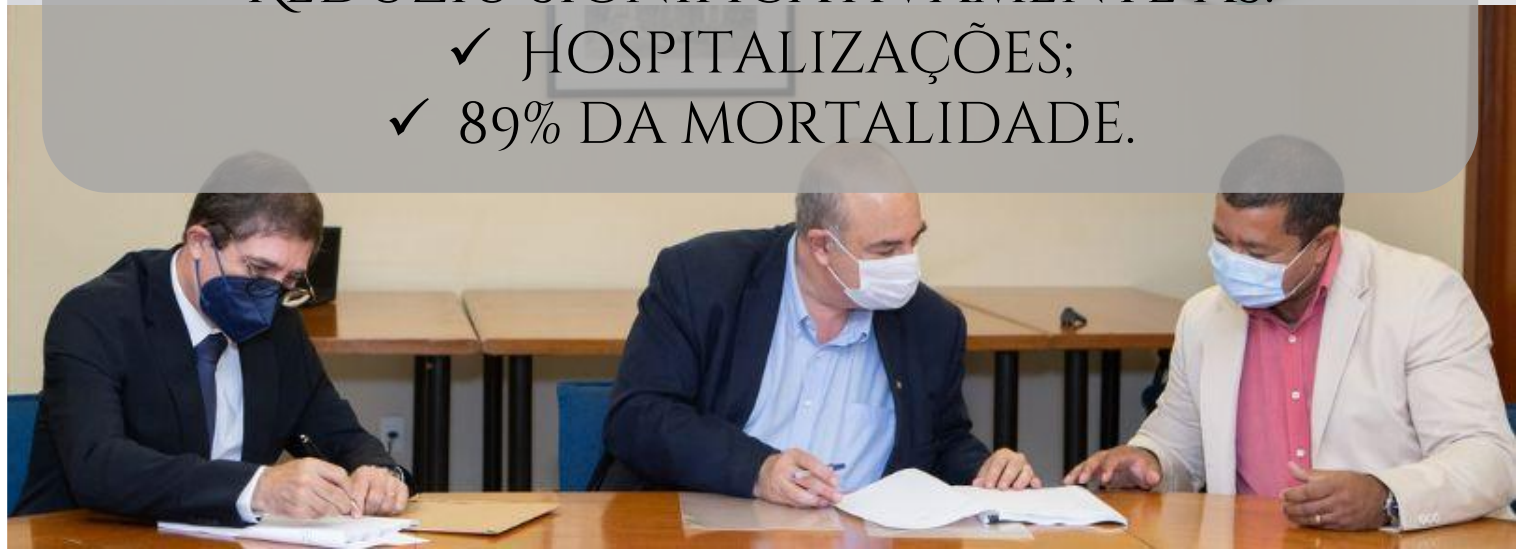
TRATAMENTO EFICAZ!

REDUZIU SIGNIFICATIVAMENTE AS:

- ✓ HOSPITALIZAÇÕES;
- ✓ 89% DA MORTALIDADE.



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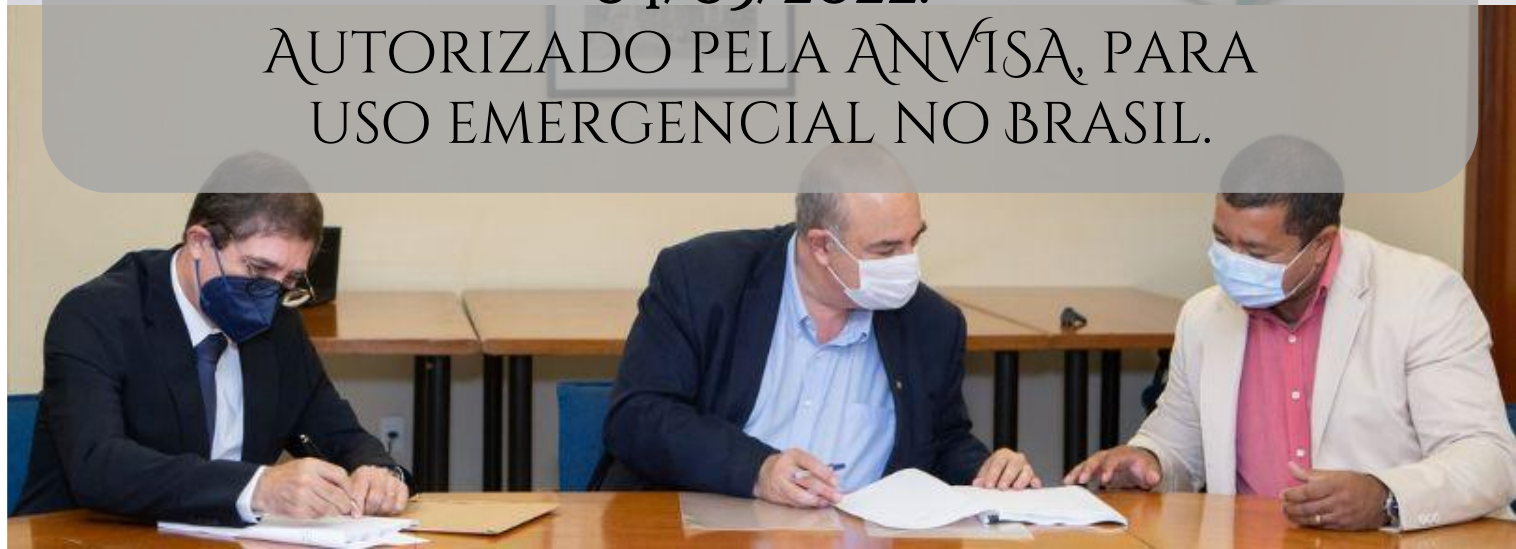
TRATAMENTO EFICAZ!

04/05/2022:
AUTORIZADO PELA ANVISA, PARA USO EMERGENCIAL NO BRASIL.



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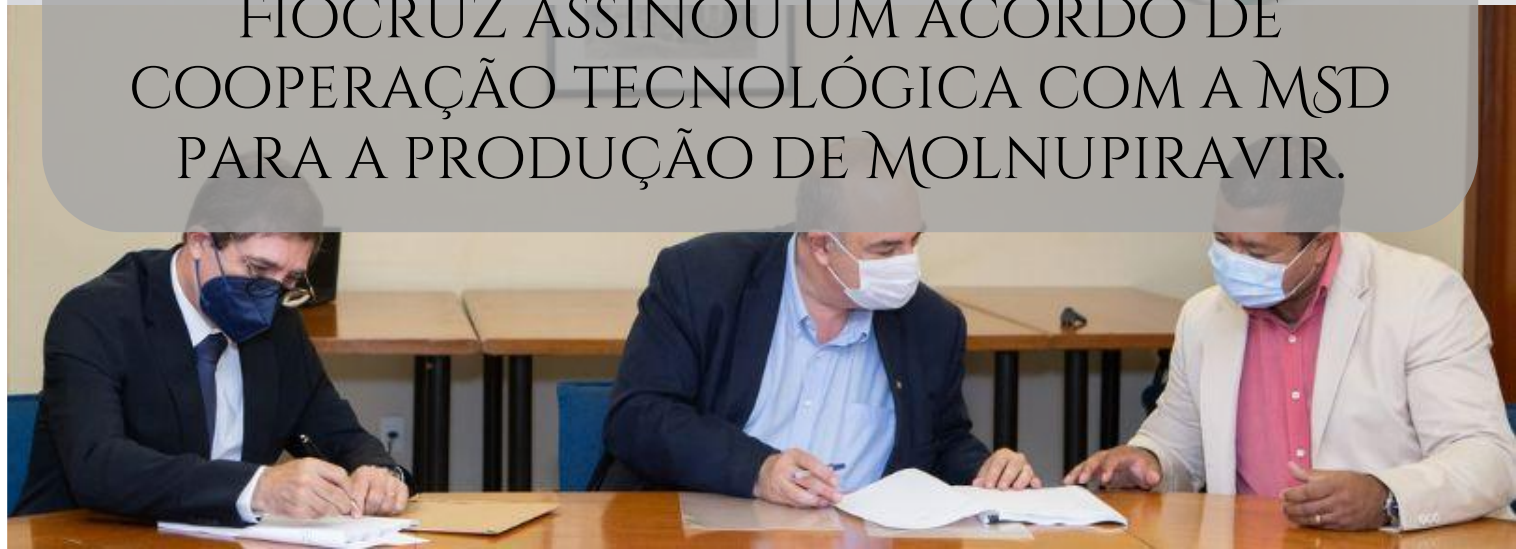
TRATAMENTO EFICAZ!

05/05/2022:

FIOCRUZ ASSINOU UM ACORDO DE COOPERAÇÃO TECNOLÓGICA COM A MSD PARA A PRODUÇÃO DE MOLNUPIRAVIR.



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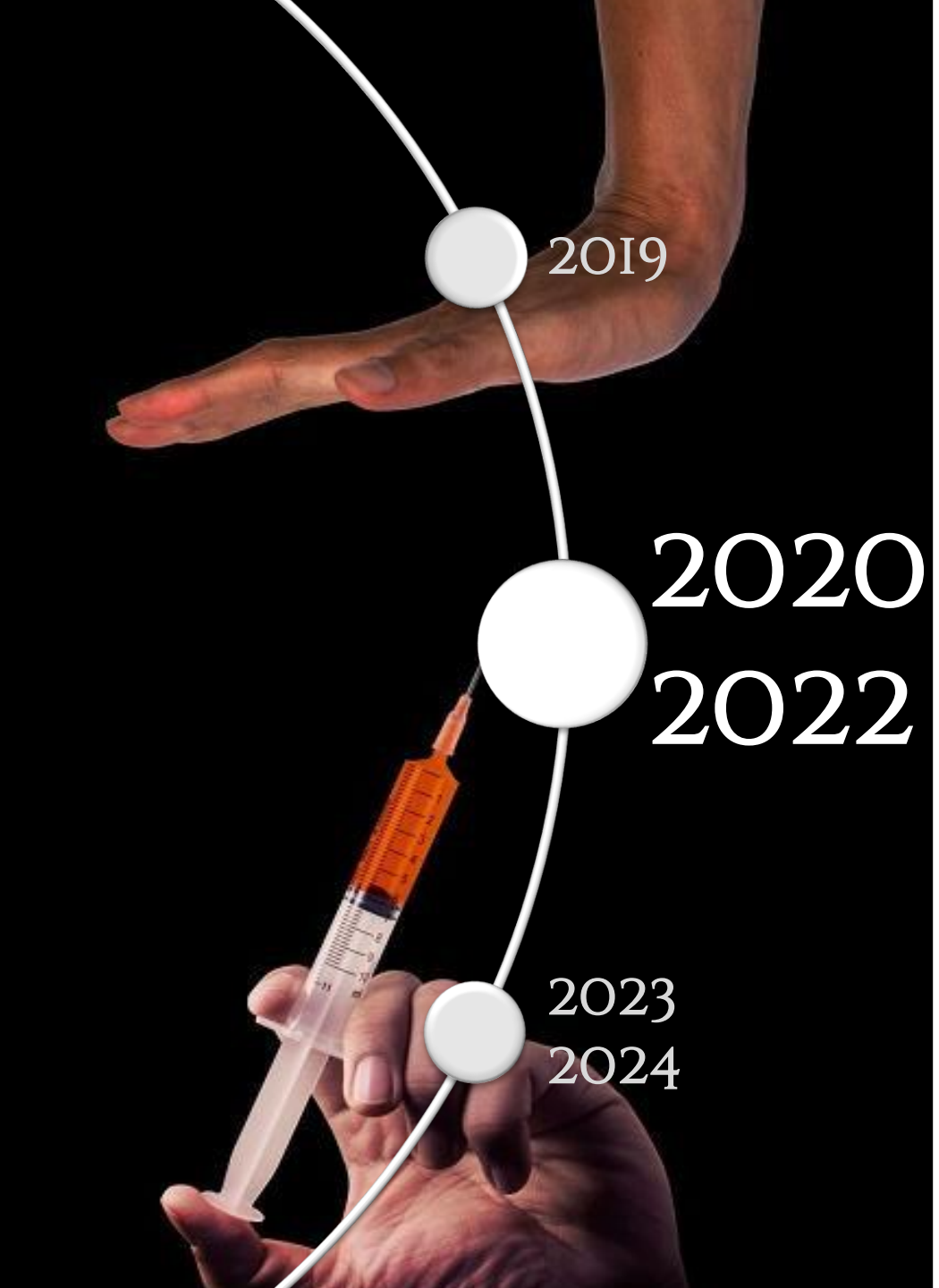
TRATAMENTO EFICAZ!

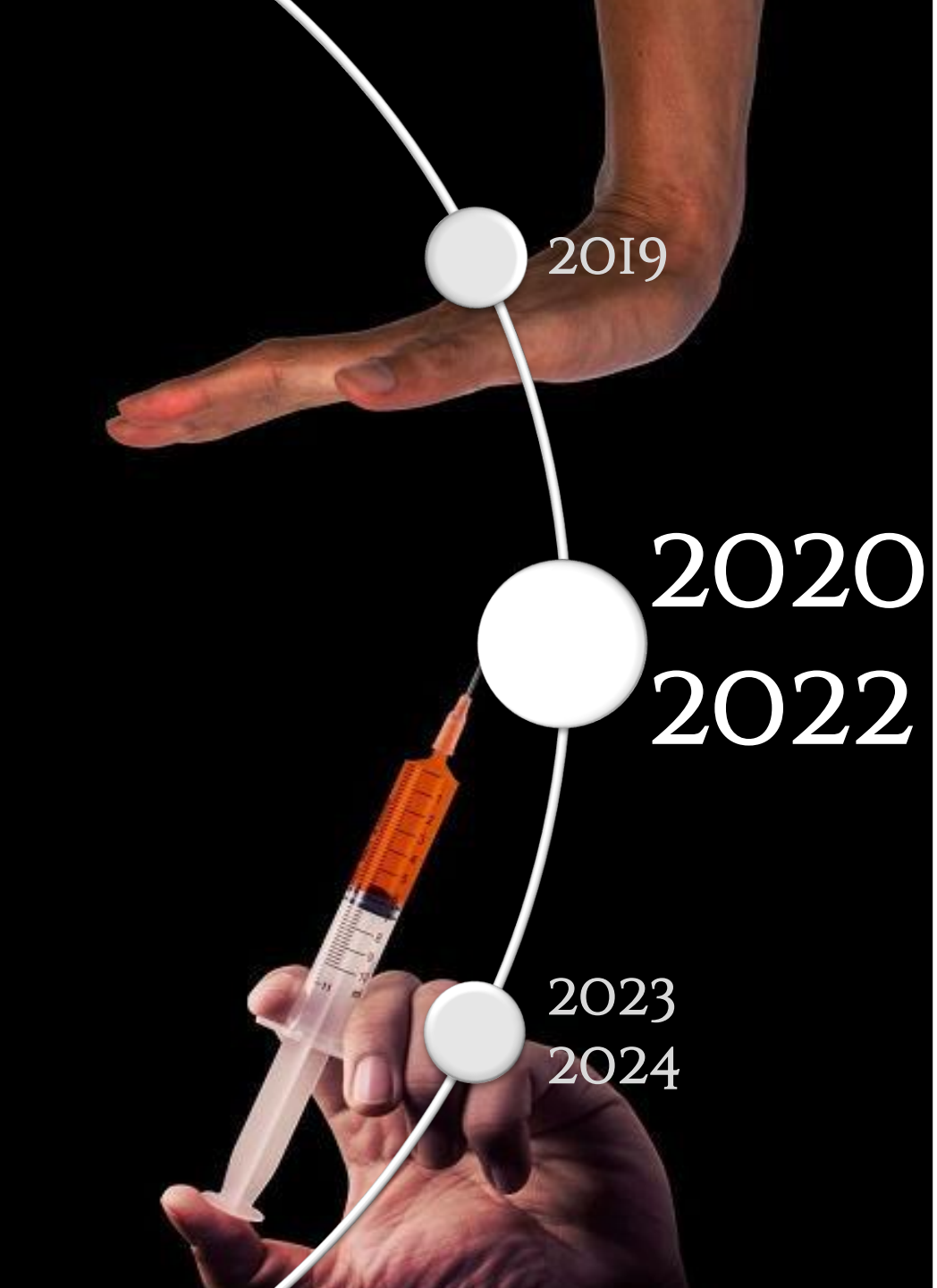
COMPOSTO PELOS ANTIVIRAIS NIRMATRELVIR E RITONAVIR, PROMOVE A INIBIÇÃO DA PROTEASE DO VÍRUS, IMPEDINDO QUE ELE COMPLETE SEU CICLO DE VIDA.



TRATAMENTO EFICAZ!

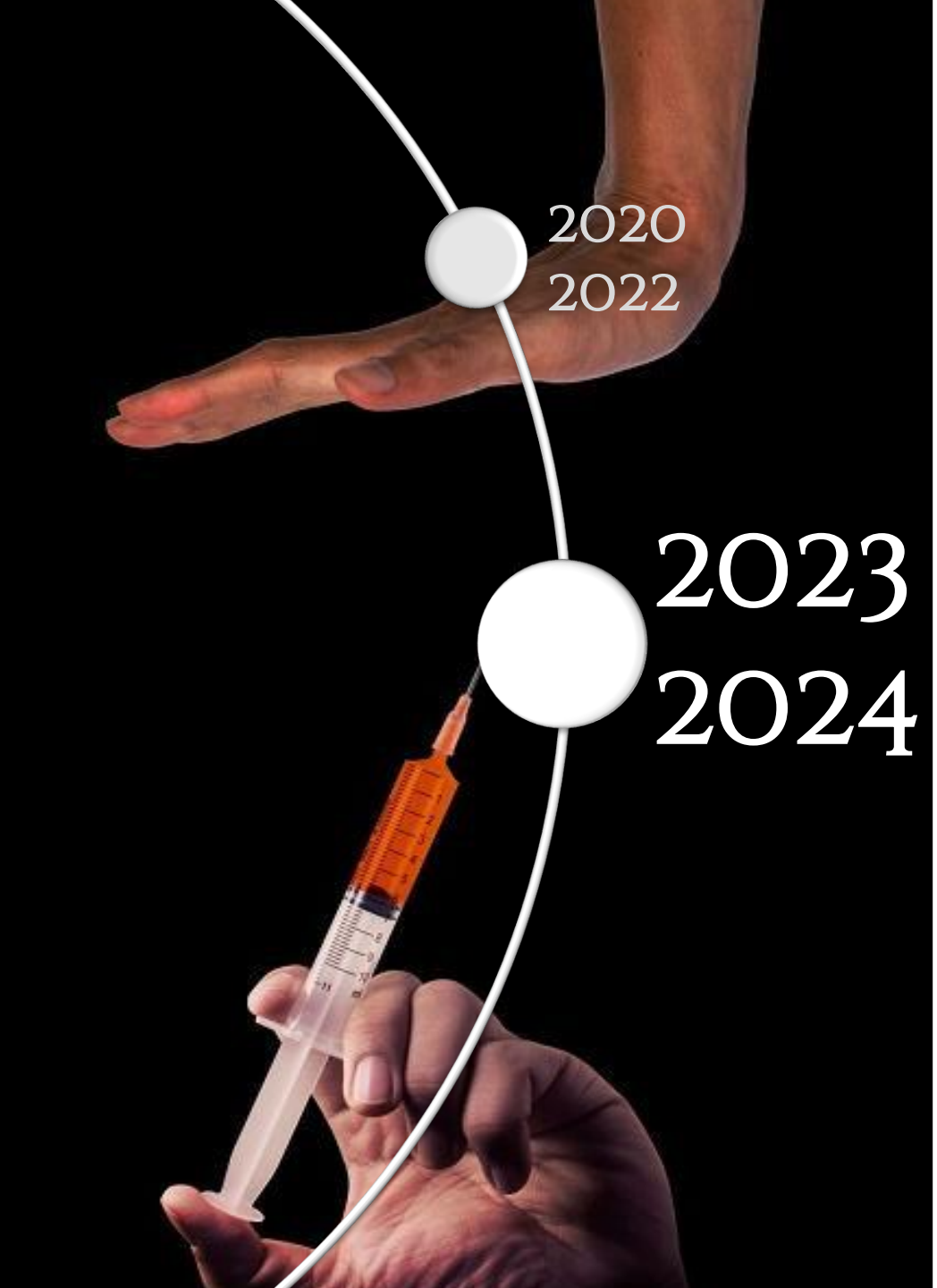
RESULTADOS PRELIMINARES:

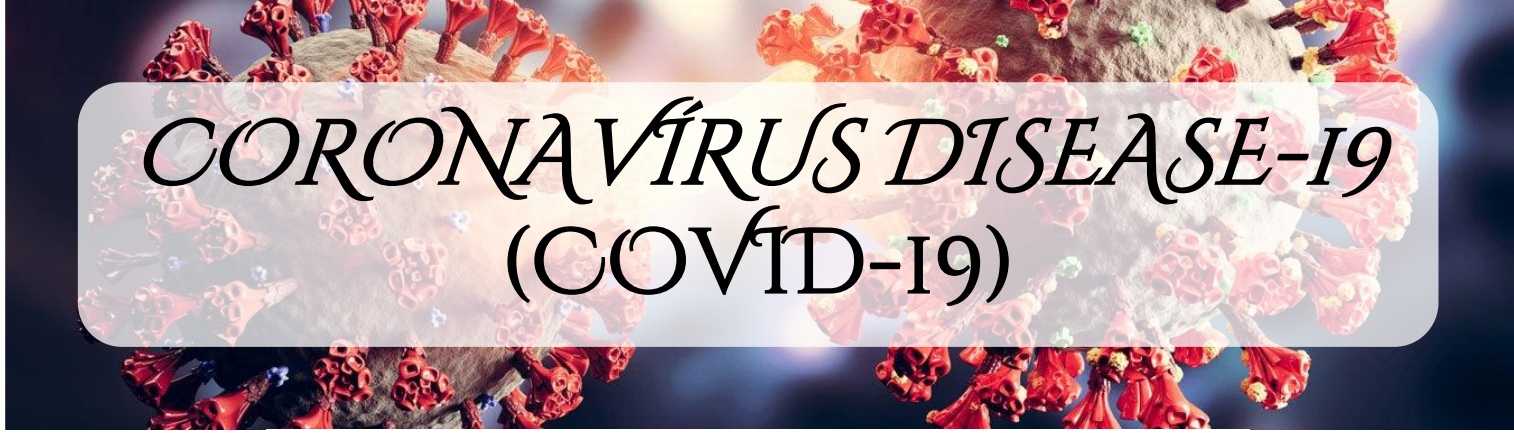
- ✓ ESTUDO DE FASE II E III REALIZADO COM 389 PACIENTES NÃO VACINADOS:
  - ➢ REDUÇÃO DE HOSPITALIZAÇÃO OU MORTE EM 89%.



TRATAMENTO EFICAZ!

30/03/2022:
AUTORIZADO PELA ANVISA, PARA
USO EMERGENCIAL NO BRASIL.





Cobertura da vacina bivalente contra a Covid está bem abaixo do esperado no país

Apenas 12% dos 63 milhões de brasileiros aptos a se vacinar procuraram os postos de saúde desde que a bivalente começou a ser aplicada, no fim de fevereiro.

Brasil vacinou 54% das crianças contra a polio

Ministério da Saúde está longe de atingir a meta de 95% das crianças menores de cinco anos vacinadas contra a poliomielite

MOVIMENTOS ANTI–VACINA

MOVIMENTOS ANTI–VACINA

MOVIMENTOS ANTI–VACINA

DESAFIOS E DEBATES ATUAIS

Pessoa Suscetível

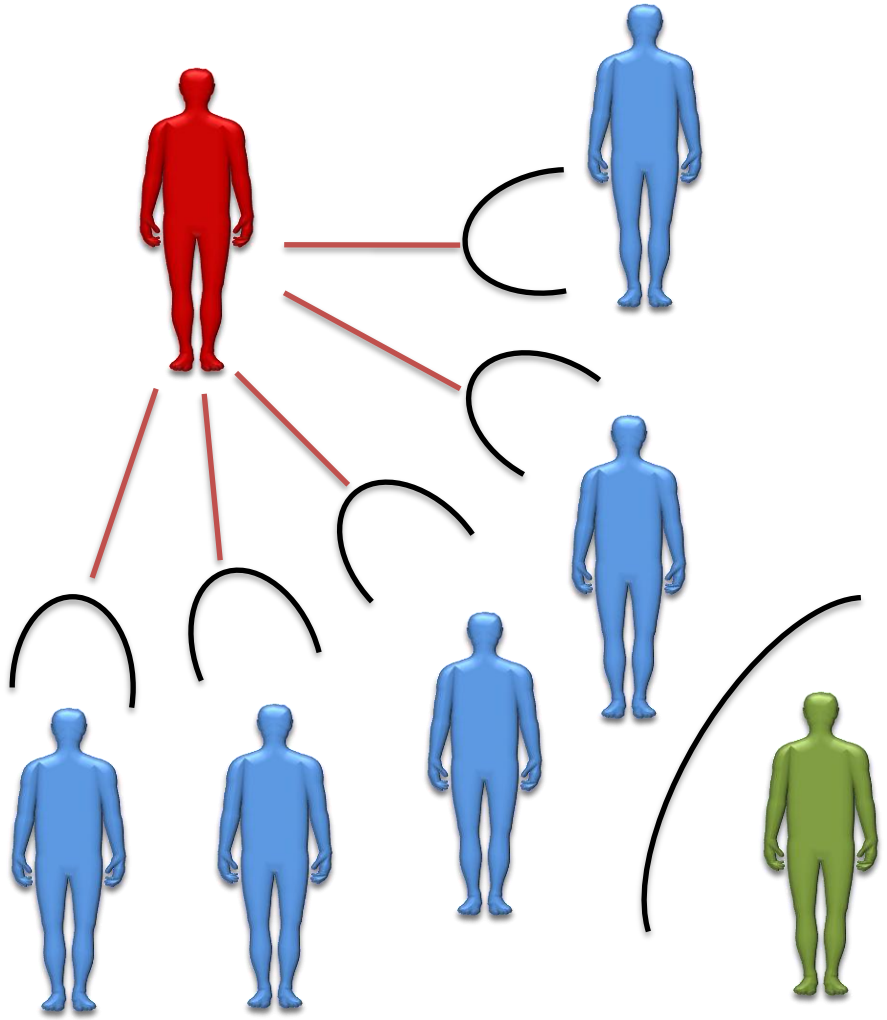
Impossibilitado de tomar a vacina ou mesmo tomando possui uma resposta imune menor.

- ✓ Idosos;
- ✓ Comorbidades;
- ✓ Imunodeprimidos;
- ✓ Crianças.

Pessoa Doente

Pessoa Suscetível

DESAFIOS E DEBATES ATUAIS



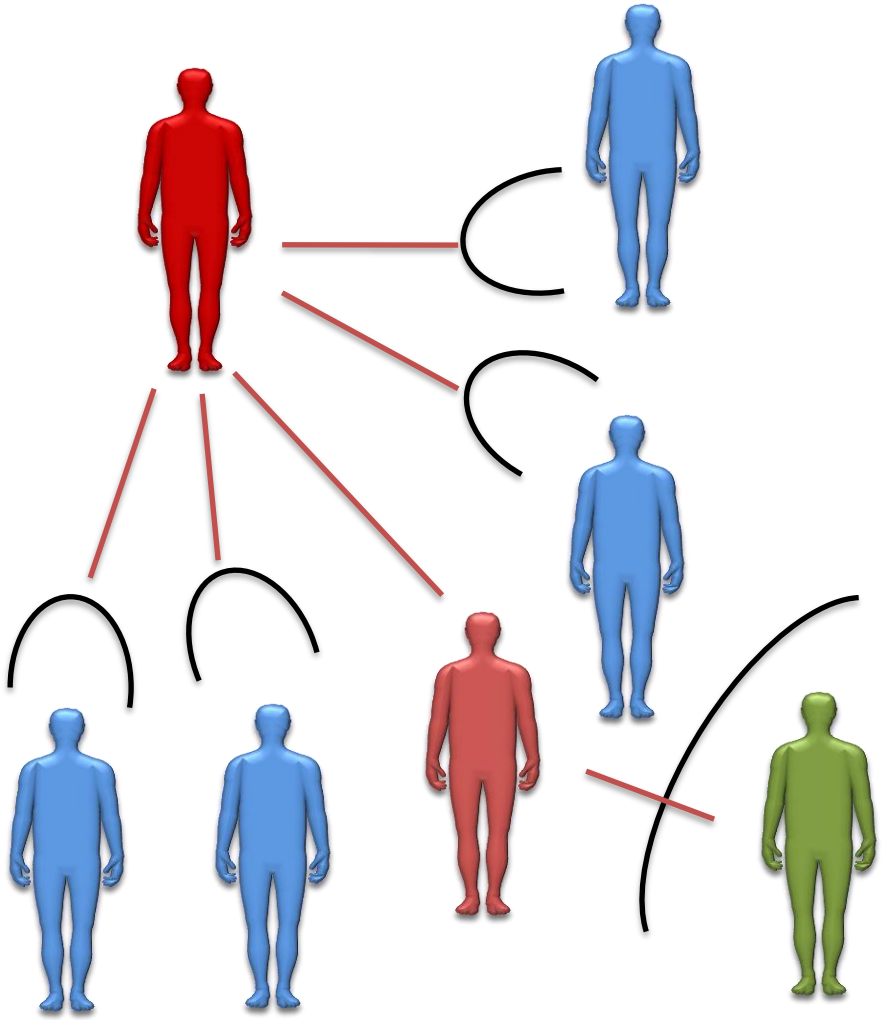
Pessoas Vacinadas

Se todas as outras que podem tomar a vacina, mesmo sem ser do grupo de risco, forem imunizadas:

- ✓ A resposta vacinal será ótima, consequentemente poderão criar uma “barreira” protegendo os suscetíveis.

- Pessoa Doente
- Pessoa Vacinada
- Pessoa Suscetível

DESAFIOS E DEBATES ATUAIS



Pessoa Não Vacinada

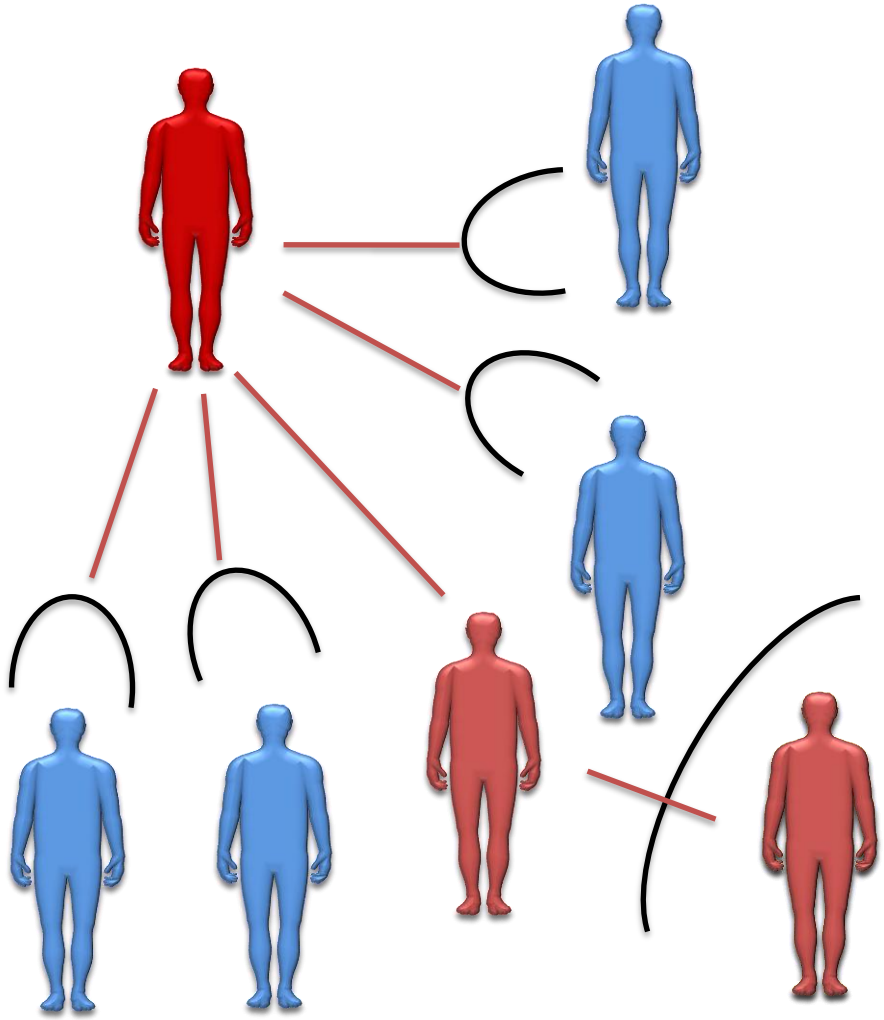
Se uma pessoa, que não é do grupo de risco para complicações, não tomar a vacina, “porque não precisa”:

- -

✓ Caso pegue a doença, ela pode passar muito mal ou não.

- Pessoa Doente
- Pessoa Vacinada
- Pessoa Suscetível

DESAFIOS E DEBATES ATUAIS



Pessoa Não Vacinada

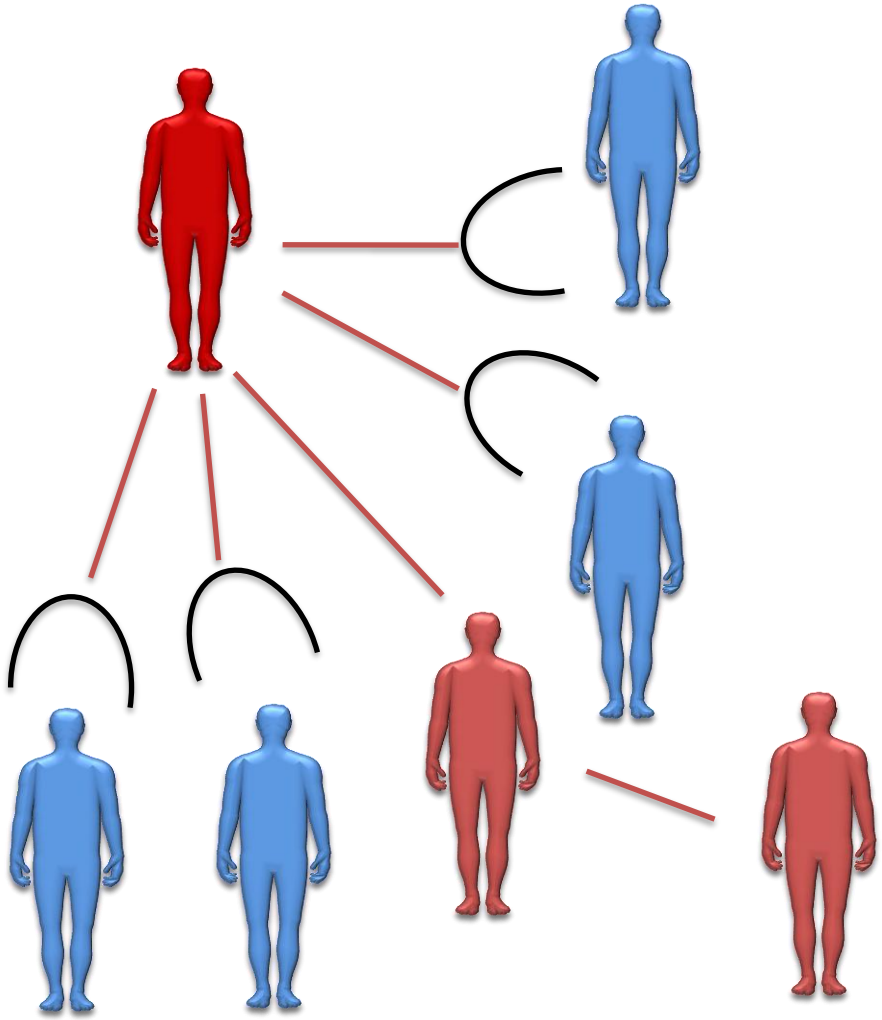
De todo jeito, acabará passando a doença para o suscetível:

- ✓ Alguém que não podia tomar a vacina;
- ✓ Ou que tomou, mas não teve uma resposta vacinal desejada.

➢ E nessa pessoa sim, pode causar um grande dano.

- Pessoa Doente
- Pessoa Vacinada
- Pessoa Suscetível

DESAFIOS E DEBATES ATUAIS



Resumindo

A vacina pode não impedir que a pessoa adoeça.

Mas com certeza ajudará para que o quadro clínico seja mais brando e com menor risco de evoluir com complicações.

- Pessoa Doente
- Pessoa Vacinada
- Pessoa Suscetível

RECADO FINAL

NÃO DEIXEM QUE DOENÇAS JÁ ERRADICADAS NO PAÍS VOLTEM!!!

VACINEM-SE!!!

GRADUAÇÃO EM FISIOTERAPIA

IMUNOLOGIA

IMUNO BIOLÓGICOS

SOROS E VACINAS

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